

Nobel Prize in Chemistry

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Preface

This volume collects the Nobel Prizes in Chemistry from 1901 to the present. Each chapter details the laureates, their prize-winning work, and the scientific context of their achievements.

Chapter 1

1901: Jacobus H. van 't Hoff

The Nobel Prize in Chemistry for the year 1901 was awarded to Jacobus H. van 't Hoff.

Motivation: in recognition of the extraordinary services he has rendered by the discovery of the laws of chemical dynamics and osmotic pressure in solutions

1.1 Jacobus H. van 't Hoff

Early Life and Education

Jacobus H. van 't Hoff was born on 1852-08-30 in Rotterdam, the Netherlands.

Career and Major Research

van 't Hoff held professorships at the University of Amsterdam and later at the University of Berlin. In 1874 he proposed the tetrahedral arrangement of the carbon valence bonds, founding the field of stereochemistry. His work on chemical dynamics, published in *études de dynamique chimique* (1884), established the temperature dependence of equilibrium constants, and his studies of osmotic pressure in dilute solutions revealed the analogy between solutions and gases.

Nobel Prize-winning Work

The work for which Jacobus H. van 't Hoff was awarded the Nobel Prize in Chemistry in 1901 was described by the Nobel Committee as: "in recognition of the extraordinary services he has rendered by the discovery of the laws of chemical dynamics and osmotic pressure in solutions".

This research fundamentally advanced the field by establishing the laws of chemical dynamics and osmotic pressure in solutions, providing a theoretical foundation for physical chemistry and demonstrating the analogy between solutions and gases.

Significance and Impact

The impact of Jacobus H. van 't Hoff's work extends far beyond the specific achievement recognized by the Nobel Prize. Jacobus H. van 't Hoff's laws of chemical dynamics and osmotic pressure theory laid the foundation for modern physical chemistry, influencing fields from thermodynamics to biochemistry. The van 't Hoff equation relating temperature dependence of equilibrium constants remains a cornerstone of chemical thermodynamics.

Later Career and Honors

Jacobus H. van 't Hoff continued his scientific work until 1911-03-01, passing away in Berlin, Germany. He received numerous honors including membership in major scientific academies and societies. Jacobus H. van

't Hoff was a founding member of the International Association of Chemical Societies and served as its first president.

Legacy

The legacy of Jacobus H. van 't Hoff endures through the lasting impact of his scientific contributions. The van 't Hoff factor in colligative properties, the van 't Hoff equation in thermodynamics, and the concept of stereochemical configuration remain fundamental to chemical education and research worldwide.

Modern Developments

The van 't Hoff equation and the van 't Hoff factor remain core tools in chemical thermodynamics and physical chemistry. The concept of osmotic pressure underpins modern membrane science, from desalination and reverse osmosis to drug delivery and the modeling of biological cells. His tetrahedral carbon model, introduced in 1874, evolved into modern stereochemistry and, through the work of van 't Hoff's successors, into today's understanding of molecular chirality, conformational analysis, and asymmetric catalysis.

Selected Publications

"Études de dynamique chimique" (1884) "L'équilibre chimique dans les systèmes gazeux ou dissous" (1884)
"The Role of Osmotic Pressure in the Analogy Between Solutions and Gases" (1887) "Lehrbuch der chemischen Dynamik" (1896)

Chapter Summary

The first Nobel Prize in Chemistry was awarded to Jacobus H. van 't Hoff for the discovery of the laws of chemical dynamics and osmotic pressure in solutions, work that established physical chemistry as a modern discipline.

Chapter 2

1902:Emil Fischer

The Nobel Prize in Chemistry for the year 1902 was awarded to Emil Fischer.

Motivation: in recognition of the extraordinary services he has rendered by his work on sugar and purine syntheses

2.1 Emil Fischer

Early Life and Education

Emil Fischer was born on 1852-10-09 in Euskirchen, Prussia (now Germany). Hermann Emil Louis Fischer was a German chemist and 1902 recipient of the Nobel Prize in Chemistry. He discovered the Fischer esterification. He also developed the Fischer projection, a symbolic way of drawing asymmetric carbon atoms. He also hypothesized the lock and key mechanism of enzyme action.

Career and Major Research

Fischer held professorships at Erlangen, Würzburg, and, from 1902, Berlin. He determined the structures of the sugars, proving their stereochemistry and synthesizing glucose and other monosaccharides, and he worked out the chemistry of purines, synthesizing caffeine and related compounds. His investigations of amino acids and peptides, including the synthesis of peptides, made him a founder of carbohydrate and protein chemistry.

Nobel Prize-winning Work

The work for which Emil Fischer was awarded the Nobel Prize in Chemistry in 1902 was described by the Nobel Committee as: "in recognition of the extraordinary services he has rendered by his work on sugar and purine syntheses".

This research fundamentally advanced the field by elucidating the structures of sugars and purines through brilliant synthetic work, including the development of the Fischer projection for representing stereochemistry.

Significance and Impact

The impact of Emil Fischer's work extends far beyond the specific achievement recognized by the Nobel Prize. Emil Fischer's elucidation of sugar structures and development of the Fischer projection became essential tools in organic chemistry and biochemistry. Emil Fischer's work on purines established the chemical basis for understanding nucleic acids.

Later Career and Honors

Emil Fischer continued his scientific work until 1919-07-15, passing away in Berlin, Germany. He received numerous honors including membership in major scientific academies and societies. Emil Fischer served as president of the German Chemical Society and was instrumental in establishing the Kaiser Wilhelm Society.

Legacy

The legacy of Emil Fischer endures through the lasting impact of his scientific contributions. The Fischer projection remains the standard method for depicting carbohydrate stereochemistry. Emil Fischer's work on enzyme specificity and the lock-and-key model laid the groundwork for modern biochemistry.

Modern Developments

Carbohydrate and purine chemistry founded by Emil Fischer remains central to biochemistry and the pharmaceutical industry. Sugar chemistry underlies glycobiology, vaccine development, and the synthesis of antiviral drugs, while Fischer's esterification and his lock-and-key model of enzyme action are still taught as foundational concepts. Modern glycomics and synthetic carbohydrate vaccines build directly on the analytical framework Fischer established more than a century ago.

Selected Publications

"Über die Chemie der Purine" (1882) "Synthese der Zuckergruppe" (1890) "Configuration der Zuckergruppe" (1891) "Untersuchungen über die Amidosäuren" (1895)

Chapter Summary

Emil Fischer was awarded the prize for his work on sugar and purine synthesis, which laid the foundation of carbohydrate chemistry and much of biochemistry.

Chapter 3

1903:Svante Arrhenius

The Nobel Prize in Chemistry for the year 1903 was awarded to Svante Arrhenius.

Motivation: in recognition of the extraordinary services he has rendered to the advancement of chemistry by his electrolytic theory of dissociation

3.1 Svante Arrhenius

Early Life and Education

Svante Arrhenius was born on 1859-02-19 in Vik, Sweden. Svante August Arrhenius was a Swedish scientist. Originally a physicist, but often referred to as a chemist, Arrhenius was one of the founders of the science of physical chemistry. In 1903, he received the Nobel Prize in Chemistry, becoming the first Swedish Nobel laureate. In 1905, he became the director of the Nobel Institute, where he remained until his death.

Career and Major Research

Arrhenius was professor of physics at Stockholm University and later director of the Nobel Institute for Physical Chemistry. In his 1884 doctoral thesis he proposed the theory of electrolytic dissociation, holding that salts in solution split into charged ions, and he formulated the equation relating reaction rate to temperature that bears his name. His ideas became central to physical chemistry and electrochemistry.

Nobel Prize-winning Work

The work for which Svante Arrhenius was awarded the Nobel Prize in Chemistry in 1903 was described by the Nobel Committee as: "in recognition of the extraordinary services he has rendered to the advancement of chemistry by his electrolytic theory of dissociation".

This research fundamentally advanced the field by introducing the theory of electrolytic dissociation, which explained the conductivity of electrolytes and revolutionized the understanding of acids, bases, and salts in solution.

Significance and Impact

The impact of Svante Arrhenius's work extends far beyond the specific achievement recognized by the Nobel Prize. The theory of electrolytic dissociation transformed the understanding of chemical reactions in solution and provided the framework for modern acid-base theory. Svante Arrhenius's work also pioneered the application of physical principles to chemical problems.

Later Career and Honors

Svante Arrhenius continued his scientific work until 1927-10-02, passing away in Stockholm, Sweden. He received numerous honors including membership in major scientific academies and societies. Svante Arrhenius became director of the Nobel Institute for Physical Chemistry in Stockholm and was a key figure in establishing the Nobel Prizes.

Legacy

The legacy of Svante Arrhenius endures through the lasting impact of his scientific contributions. The Arrhenius equation describing temperature dependence of reaction rates is ubiquitous in chemical kinetics. Svante Arrhenius's early work on the greenhouse effect anticipated modern climate science.

Modern Developments

The Arrhenius equation remains one of the most widely used relationships in chemistry, governing reaction rates in industrial catalysis, atmospheric chemistry, and materials science. Arrhenius's electrolytic dissociation theory is the basis of modern aqueous solution chemistry, pH theory, and electrochemistry. His broader work also anticipated concerns about carbon dioxide and climate change, linking his legacy to present-day climate science.

Selected Publications

"Recherches sur la conductibilité galvanique des électrolytes" (1884) "Über die Dissociation der in Wasser gelösten Stoffe" (1887) "Lehrbuch der chemischen Dynamik" (1896)

Chapter Summary

Svante Arrhenius received the prize for his electrolytic theory of dissociation, which explained how salts conduct electricity in solution.

Chapter 4

1904: Sir William Ramsay

The Nobel Prize in Chemistry for the year 1904 was awarded to Sir William Ramsay.

Motivation: in recognition of his services in the discovery of the inert gaseous elements in air, and his determination of their place in the periodic system

4.1 Sir William Ramsay

Early Life and Education

Sir William Ramsay was born on 1852-10-02 in Glasgow, Scotland. Sir William Ramsay was a Scottish chemist who discovered the noble gases and received the Nobel Prize in Chemistry in 1904 "in recognition of his services in the discovery of the inert gaseous elements in air" along with his collaborator, John William Strutt, 3rd Baron Rayleigh, who received the Nobel Prize in Physics that same year for their discovery of argon. After the two men identified argon, Ramsay investigated other atmospheric gases. His work in isolating argon, helium, neon, krypton, and xenon led to the development of a new section of the periodic table.

Career and Major Research

Ramsay was professor of chemistry at University College London for much of his career. Working with Lord Rayleigh, he isolated argon from air in 1894, and in 1895 he found helium in the mineral cleveite. With Morris Travers he then discovered neon, krypton, and xenon in liquid air, and in 1903 he identified helium produced by radioactive decay, completing a new group of the periodic table.

Nobel Prize-winning Work

The work for which Sir William Ramsay was awarded the Nobel Prize in Chemistry in 1904 was described by the Nobel Committee as: "in recognition of his services in the discovery of the inert gaseous elements in air, and his determination of their place in the periodic system".

This research fundamentally advanced the field by discovering the noble gases (argon, helium, neon, krypton, xenon) and determining their place in the periodic system, completing a major gap in the periodic table.

Significance and Impact

The impact of Sir William Ramsay's work extends far beyond the specific achievement recognized by the Nobel Prize. The discovery of the noble gases completed the periodic table and led to the development of a new group of elements with unique properties, influencing both theoretical chemistry and practical applications in lighting and cryogenics.

Later Career and Honors

Sir William Ramsay continued his scientific work until 1916-07-23, passing away in High Wycombe, United Kingdom. He received numerous honors including membership in major scientific academies and societies. Sir William Ramsay was knighted in 1902 and served as president of the Chemical Society and the British Association for the Advancement of Science.

Legacy

The legacy of Sir William Ramsay endures through the lasting impact of his scientific contributions. The noble gases, once considered chemically inert, now form compounds that have expanded the boundaries of chemistry. Sir William Ramsay's discovery completed a fundamental gap in the periodic system.

Modern Developments

Ramsay's isolation of the noble gases led to the modern understanding of the periodic table's final group and to practical applications ranging from neon lighting and argon-filled welding to helium-based cryogenics and MRI magnet cooling. Noble gas chemistry, long thought impossible, has since been demonstrated with compounds such as xenon fluorides. Industrial noble gas separation and gas-phase spectroscopy continue to depend on the methods Ramsay pioneered.

Selected Publications

"The Gases of the Atmosphere" (1896) "On a New Gas, Argon" (1895) "Helium, a Gaseous Constituent of Certain Minerals" (1895)

Chapter Summary

Sir William Ramsay was honored for the discovery of the inert gaseous elements of air, including argon, helium, neon, krypton, and xenon, completing a new group of the periodic table.

Chapter 5

1905: Adolf von Baeyer

The Nobel Prize in Chemistry for the year 1905 was awarded to Adolf von Baeyer.

Motivation: in recognition of his services in the advancement of organic chemistry and the chemical industry, through his work on organic dyes and hydroaromatic compounds

5.1 Adolf von Baeyer

Early Life and Education

Adolf von Baeyer was born on 1835-10-31 in Berlin, Prussia (now Germany). Johann Friedrich Wilhelm Adolf von Baeyer was a German chemist who synthesized indigo and developed a nomenclature for cyclic compounds. He was ennobled in the Kingdom of Bavaria in 1885 and was the 1905 recipient of the Nobel Prize in Chemistry.

Career and Major Research

Baeyer was professor at Munich from 1875 until his retirement. He synthesized indigo in 1880 and made it industrially available, and he established the structures of uric acid and other purines. He developed the strain theory of carbon ring compounds and contributed to the chemistry of hydroaromatic compounds, work of great importance to the dye industry.

Nobel Prize-winning Work

The work for which Adolf von Baeyer was awarded the Nobel Prize in Chemistry in 1905 was described by the Nobel Committee as: "in recognition of his services in the advancement of organic chemistry and the chemical industry, through his work on organic dyes and hydroaromatic compounds".

This research fundamentally advanced the field by synthesizing indigo and advancing the chemistry of cyclic compounds, including the development of systematic nomenclature for ring structures.

Significance and Impact

The impact of Adolf von Baeyer's work extends far beyond the specific achievement recognized by the Nobel Prize. Adolf von Baeyer's synthesis of indigo revolutionized the dye industry, and Adolf von Baeyer's systematic work on cyclic compounds established principles of ring strain and stereochemistry that remain fundamental.

Later Career and Honors

Adolf von Baeyer continued his scientific work until 1917-08-20, passing away in Starnberg, Germany. He received numerous honors including membership in major scientific academies and societies. Adolf von Baeyer was ennobled by the Kingdom of Bavaria and served as president of the German Chemical Society.

Legacy

The legacy of Adolf von Baeyer endures through the lasting impact of his scientific contributions. Adolf von Baeyer's strain theory for ring compounds and nomenclature systems for cyclic structures remain essential in organic chemistry. The von Baeyer name is associated with the Baeyer-Villiger oxidation, a key synthetic transformation.

Modern Developments

Baeyer's synthetic chemistry, including his synthesis of indigo, laid the groundwork for the modern dye and pigment industry, which still produces billions of dollars of colorants annually. The Baeyer strain theory and his structural investigations informed the development of cyclic molecules, including modern pharmaceuticals and agrochemicals. The Baeyer-Villiger oxidation, named after his student, remains a standard synthetic tool.

Selected Publications

"Synthese des Indigos" (1880) "Über die Constitution des Indigos" (1883) "Chemie der cyclischen Verbindungen" (1885)

Chapter Summary

Adolf von Baeyer received the prize for his work on organic dyes and hydroaromatic compounds, including his synthesis of indigo.

Chapter 6

1906: Henri Moissan

The Nobel Prize in Chemistry for the year 1906 was awarded to Henri Moissan.

Motivation: in recognition of the great services rendered by him in his investigation and isolation of the element fluorine, and for the adoption in the service of science of the electric furnace called after him

6.1 Henri Moissan

Early Life and Education

Henri Moissan was born on 1852-09-28 in Paris, France. Ferdinand Frédéric Henri Moissan was a French chemist and pharmacist who won the 1906 Nobel Prize in Chemistry for his work in isolating fluorine from its compounds. Among his other contributions, Moissan discovered moissanite and contributed to the development of the electric arc furnace. Moissan was one of the original members of the International Atomic Weights Committee.

Career and Major Research

Moissan was professor of inorganic chemistry at the Sorbonne. In 1886 he succeeded in isolating fluorine, the most reactive of the halogens, by electrolysis of anhydrous hydrogen fluoride, and he studied the reactions of this element. He also invented the electric arc furnace, which enabled the synthesis of refractory materials such as carbides and artificial diamonds.

Nobel Prize-winning Work

The work for which Henri Moissan was awarded the Nobel Prize in Chemistry in 1906 was described by the Nobel Committee as: "in recognition of the great services rendered by him in his investigation and isolation of the element fluorine, and for the adoption in the service of science of the electric furnace called after him".

This research fundamentally advanced the field by isolating elemental fluorine and developing the electric furnace for high-temperature chemistry, enabling the synthesis of artificial diamonds and new refractory materials.

Significance and Impact

The impact of Henri Moissan's work extends far beyond the specific achievement recognized by the Nobel Prize. The isolation of fluorine, the most reactive element, was a triumph of experimental technique. Henri Moissan's electric furnace enabled high-temperature synthesis and the production of artificial diamonds, opening new avenues in materials science.

Later Career and Honors

Henri Moissan continued his scientific work until 1907-02-20, passing away in Paris, France. He received numerous honors including membership in major scientific academies and societies. Henri Moissan was elected to the French Academy of Sciences and received numerous international honors.

Legacy

The legacy of Henri Moissan endures through the lasting impact of his scientific contributions. The Moissan process for fluorine production and the electric furnace technique enabled modern high-temperature chemistry and materials science. The Moissan crater on the Moon bears Henri Moissan's name.

Modern Developments

Moissan's isolation of fluorine led to the modern fluorine chemistry industry, including Teflon, hydrofluorocarbons, refrigerants, and fluorinated pharmaceuticals that now comprise a large fraction of prescription drugs. His invention of the electric arc furnace enabled high-temperature chemistry and the production of carbides, including silicon carbide, a key modern abrasive and semiconductor material.

Selected Publications

"Le fluor et ses composés" (1891) "Sur le diamant artificiel" (1893)

Chapter Summary

Henri Moissan was awarded the prize for his isolation of fluorine and for his development of the electric arc furnace.

Chapter 7

1907:Eduard Buchner

The Nobel Prize in Chemistry for the year 1907 was awarded to Eduard Buchner.

Motivation: for his biochemical researches and his discovery of cell-free fermentation

7.1 Eduard Buchner

Early Life and Education

Eduard Buchner was born on 1860-05-20 in Munich, Bavaria (now Germany). Eduard Buchner was a German chemist and expert on fermentation, awarded the 1907 Nobel Prize in Chemistry for his work on fermentation.

Career and Major Research

Buchner held professorships in Tübingen, Würzburg, and Berlin. In 1897 he showed that a cell-free extract of yeast, which he named zymase, could ferment sugar to alcohol and carbon dioxide, proving that fermentation is brought about by enzymes rather than by the living yeast cell. This discovery founded the study of enzymes outside the organism.

Nobel Prize-winning Work

The work for which Eduard Buchner was awarded the Nobel Prize in Chemistry in 1907 was described by the Nobel Committee as: "for his biochemical researches and his discovery of cell-free fermentation".

This research fundamentally advanced the field by making groundbreaking contributions that transformed chemical science.

Significance and Impact

The impact of Eduard Buchner's work extends far beyond the specific achievement recognized by the Nobel Prize. These contributions continue to influence chemical research and education today.

Later Career and Honors

Eduard Buchner continued his scientific work until 1917-08-13, passing away in Focsani, Romania. He received numerous honors including membership in major scientific academies and societies.

Legacy

The legacy of Eduard Buchner endures through the lasting impact of his scientific contributions. Eduard Buchner's work continues to be cited and built upon by researchers across the chemical sciences.

Modern Developments

Buchner's demonstration of cell-free fermentation proved that biological processes are chemical reactions catalyzed by enzymes, founding the field of biochemistry and enzymology. Today, industrial biocatalysis uses purified enzymes and engineered whole cells for everything from biofuels and food production to the synthesis of pharmaceuticals such as statins and semisynthetic insulin.

Selected Publications

Eduard Buchner's major publications include numerous papers in leading journals of the era, detailing the experimental and theoretical work summarized above.

Chapter Summary

Eduard Buchner was honored for his discovery of cell-free fermentation, showing that enzymes extracted from yeast could convert sugar to alcohol.

Chapter 8

1908:Ernest Rutherford

The Nobel Prize in Chemistry for the year 1908 was awarded to Ernest Rutherford.

Motivation: for his investigations into the disintegration of the elements, and the chemistry of radioactive substances

8.1 Ernest Rutherford

Early Life and Education

Ernest Rutherford was born on 1871-08-30 in Nelson, New Zealand. Ernest Rutherford, 1st Baron Rutherford of Nelson, was a New Zealand physicist and chemist who was a pioneering researcher in both atomic and nuclear physics. He has been described as "the father of nuclear physics" and "the greatest experimentalist since Michael Faraday." In 1908, he was awarded the Nobel Prize in Chemistry "for his investigations into the disintegration of the elements, and the chemistry of radioactive substances.

Career and Major Research

Rutherford was professor at McGill University in Montreal and at the University of Manchester before succeeding J.J. Thomson as director of the Cavendish Laboratory in Cambridge. He distinguished alpha and beta rays in radioactivity and, with Frederick Soddy, formulated the theory of radioactive disintegration. In 1911 he proposed the nuclear model of the atom, and in 1919 he achieved the first artificial transmutation of an element.

Nobel Prize-winning Work

The work for which Ernest Rutherford was awarded the Nobel Prize in Chemistry in 1908 was described by the Nobel Committee as: "for his investigations into the disintegration of the elements, and the chemistry of radioactive substances".

This research fundamentally advanced the field by making groundbreaking contributions that transformed chemical science.

Significance and Impact

The impact of Ernest Rutherford's work extends far beyond the specific achievement recognized by the Nobel Prize. These contributions continue to influence chemical research and education today.

Later Career and Honors

Ernest Rutherford continued his scientific work until 1937-10-19, passing away in Cambridge, United Kingdom. He received numerous honors including membership in major scientific academies and societies.

Legacy

The legacy of Ernest Rutherford endures through the lasting impact of his scientific contributions. Ernest Rutherford's work continues to be cited and built upon by researchers across the chemical sciences.

Modern Developments

Rutherford's work on radioactivity and atomic structure initiated nuclear physics and, through the concept of half-life, modern radiometric dating used in geology and archaeology. Nuclear medicine, radiotherapy for cancer, and carbon dating all trace their origins to Rutherford's laboratory discoveries. His nuclear atom model is the foundation of the modern quantum theory of matter.

Selected Publications

Ernest Rutherford's major publications include numerous papers in leading journals of the era, detailing the experimental and theoretical work summarized above.

Chapter Summary

Ernest Rutherford received the prize for his investigations into the disintegration of the elements and the chemistry of radioactive substances.

Chapter 9

1909: Wilhelm Ostwald

The Nobel Prize in Chemistry for the year 1909 was awarded to Wilhelm Ostwald.

Motivation: in recognition of his work on catalysis and for his investigations into the fundamental principles governing chemical equilibria and rates of reaction

9.1 Wilhelm Ostwald

Early Life and Education

Wilhelm Ostwald was born on 1853-09-02 in Riga, Russian Empire (now Latvia). Wilhelm Friedrich Ostwald was a Baltic German chemist and philosopher. Ostwald is credited with being one of the founders of the field of physical chemistry, with Jacobus Henricus van 't Hoff, Walther Nernst and Svante Arrhenius. He received the Nobel Prize in Chemistry in 1909 for his scientific contributions to the fields of catalysis, chemical equilibria and reaction velocities.

Career and Major Research

Ostwald was professor of physical chemistry at Leipzig, where he founded the *Zeitschrift für physikalische Chemie* with van 't Hoff and Arrhenius. He studied catalysis, chemical equilibria, and reaction velocities, and he developed the Ostwald process for the oxidation of ammonia to nitric acid. His textbooks established physical chemistry as a distinct discipline.

Nobel Prize-winning Work

The work for which Wilhelm Ostwald was awarded the Nobel Prize in Chemistry in 1909 was described by the Nobel Committee as: "in recognition of his work on catalysis and for his investigations into the fundamental principles governing chemical equilibria and rates of reaction".

This research fundamentally advanced the field by making groundbreaking contributions that transformed chemical science.

Significance and Impact

The impact of Wilhelm Ostwald's work extends far beyond the specific achievement recognized by the Nobel Prize. These contributions continue to influence chemical research and education today.

Later Career and Honors

Wilhelm Ostwald continued his scientific work until 1932-04-04, passing away in Leipzig, Germany. He received numerous honors including membership in major scientific academies and societies.

Legacy

The legacy of Wilhelm Ostwald endures through the lasting impact of his scientific contributions. Wilhelm Ostwald's work continues to be cited and built upon by researchers across the chemical sciences.

Modern Developments

Ostwald's physical chemistry established the field's core concepts, including catalysis, chemical equilibrium, and reaction kinetics. His dilution law and work on catalysis underpin modern industrial ammonia synthesis, automotive catalytic converters, and enzyme kinetics. Ostwald was also a founder of the science of color measurement, which remains important in materials and imaging science.

Selected Publications

Wilhelm Ostwald's major publications include numerous papers in leading journals of the era, detailing the experimental and theoretical work summarized above.

Chapter Summary

Wilhelm Ostwald was awarded the prize for his work on catalysis and on the fundamental principles governing chemical equilibria and rates of reaction.

Chapter 10

1910:Otto Wallach

The Nobel Prize in Chemistry for the year 1910 was awarded to Otto Wallach.

Motivation: in recognition of his services to organic chemistry and the chemical industry by his pioneer work in the field of alicyclic compounds

10.1 Otto Wallach

Early Life and Education

Otto Wallach was born on 1847-03-27 in Koenigsberg, Germany (now Kaliningrad, Russia). Otto Wallach was a German chemist and recipient of the 1910 Nobel Prize in Chemistry for his work on alicyclic compounds.

Career and Major Research

Wallach was professor of chemistry at Göttingen from 1876 to 1915. He carried out a systematic investigation of the terpenes, the volatile constituents of essential oils, determining their structures and establishing the isoprene rule. His work transformed the chemistry of terpenes and provided a basis for the synthetic perfume industry.

Nobel Prize-winning Work

The work for which Otto Wallach was awarded the Nobel Prize in Chemistry in 1910 was described by the Nobel Committee as: "in recognition of his services to organic chemistry and the chemical industry by his pioneer work in the field of alicyclic compounds".

This research fundamentally advanced the field by making groundbreaking contributions that transformed chemical science.

Significance and Impact

The impact of Otto Wallach's work extends far beyond the specific achievement recognized by the Nobel Prize. These contributions continue to influence chemical research and education today.

Later Career and Honors

Otto Wallach continued his scientific work until 1931-02-26, passing away in Göttingen, Germany. He received numerous honors including membership in major scientific academies and societies.

Legacy

The legacy of Otto Wallach endures through the lasting impact of his scientific contributions. Otto Wallach's work continues to be cited and built upon by researchers across the chemical sciences.

Modern Developments

Wallach's terpene and essential oil chemistry founded the modern fragrance and flavor industry, which synthesizes terpenes for perfumes, foods, and cosmetics. Terpenes are now recognized as a vast class of natural products with roles in plant defense, medicine (such as artemisinin-based antimalarials), and biofuel production. Wallach's name lives on in the Wallach rearrangement.

Selected Publications

Otto Wallach's major publications include numerous papers in leading journals of the era, detailing the experimental and theoretical work summarized above.

Chapter Summary

Otto Wallach received the prize for his pioneering research on the chemistry of terpenes and essential oils.

Chapter 11

1911:Marie Curie

The Nobel Prize in Chemistry for the year 1911 was awarded to Marie Curie.

Motivation: in recognition of her services to the advancement of chemistry by the discovery of the elements radium and polonium, by the isolation of radium and the study of the nature and compounds of this remarkable element

11.1 Marie Curie

Early Life and Education

Marie Curie was born on 1867-11-07 in Warsaw, Russian Empire (now Poland). Maria Salomea Skłodowska Curie, better known as Marie Curie was a Polish and naturalised-French physicist and chemist. She shared the 1903 Nobel Prize in Physics with her husband, Pierre Curie, "for their joint researches on the radioactivity phenomena discovered by Professor Henri Becquerel". She won the 1911 Nobel Prize in Chemistry "for the discovery of the elements radium and polonium, by the isolation of radium and the study of the nature and compounds of this remarkable element".

Career and Major Research

Curie worked at the Sorbonne in Paris, becoming the first woman to hold a professorship there. In 1898 she and her husband Pierre Curie discovered polonium and radium, and she later isolated pure radium metal, a feat of painstaking chemical work. She shared the 1903 Nobel Prize in Physics with Pierre Curie and Henri Becquerel for their studies of radioactivity.

Nobel Prize-winning Work

The work for which Marie Curie was awarded the Nobel Prize in Chemistry in 1911 was described by the Nobel Committee as: "in recognition of her services to the advancement of chemistry by the discovery of the elements radium and polonium, by the isolation of radium and the study of the nature and compounds of this remarkable element".

This research fundamentally advanced the field by making groundbreaking contributions that transformed chemical science.

Significance and Impact

The impact of Marie Curie's work extends far beyond the specific achievement recognized by the Nobel Prize. These contributions continue to influence chemical research and education today.

Later Career and Honors

Marie Curie continued her scientific work until 1934-07-04, passing away in Sallanches, France. She received numerous honors including membership in major scientific academies and societies.

Legacy

The legacy of Marie Curie endures through the lasting impact of her scientific contributions. Marie Curie's work continues to be cited and built upon by researchers across the chemical sciences.

Modern Developments

Marie Curie's discovery of radium and polonium established the science of radioactivity and gave medicine the tools of radiotherapy and radioactive tracers. Today, radiopharmaceuticals and techniques such as PET imaging and brachytherapy are direct descendants of her work. Her methods for isolating radioactive elements also laid the groundwork for the modern nuclear fuel cycle and radiation standards.

Selected Publications

Marie Curie's major publications include numerous papers in leading journals of the era, detailing the experimental and theoretical work summarized above.

Chapter Summary

Marie Curie was honored for the discovery of the elements radium and polonium, the isolation of radium, and the study of the nature and compounds of this remarkable element.

Chapter 12

1912: Victor Grignard and Paul Sabatier

The Nobel Prize in Chemistry for the year 1912 was awarded jointly to Victor Grignard and Paul Sabatier.

Motivation: for the discovery of the so-called Grignard reagent, which in recent years has greatly advanced the progress of organic chemistry (Victor Grignard) and for his method of hydrogenating organic compounds in the presence of finely disintegrated metals whereby the progress of organic chemistry has been greatly advanced in recent years (Paul Sabatier)

12.1 Victor Grignard

Early Life and Education

Victor Grignard was born on 1871-05-06 in Cherbourg, France. François Auguste Victor Grignard was a French chemist who won the Nobel Prize for his discovery of the eponymously named Grignard reagent and Grignard reaction, both of which are important in the formation of carbon-carbon bonds. He also wrote some of his experiments in his laboratory notebooks.

Career and Major Research

Grignard was professor at Nancy and later at Lyon. In 1900 he developed the organomagnesium compounds that bear his name, obtained by reacting magnesium with organic halides, and showed that these reagents combine readily with aldehydes and ketones to form alcohols. The Grignard reaction became one of the most versatile tools of organic synthesis.

Nobel Prize-winning Work

The work for which Victor Grignard was awarded the Nobel Prize in Chemistry in 1912 was described by the Nobel Committee as: "for the discovery of the so-called Grignard reagent, which in recent years has greatly advanced the progress of organic chemistry".

This research fundamentally advanced the field by making groundbreaking contributions that transformed chemical science.

Significance and Impact

The impact of Victor Grignard's work extends far beyond the specific achievement recognized by the Nobel Prize. These contributions continue to influence chemical research and education today.

Later Career and Honors

Victor Grignard continued his scientific work until 1935-12-13, passing away in Lyon, France. He received numerous honors including membership in major scientific academies and societies.

Legacy

The legacy of Victor Grignard endures through the lasting impact of his scientific contributions. Victor Grignard's work continues to be cited and built upon by researchers across the chemical sciences.

Modern Developments

Grignard reagents remain among the most widely used organometallic reagents in organic synthesis, employed in the manufacture of pharmaceuticals, agrochemicals, flavors, and polymers. The Grignard reaction is a staple of both academic laboratories and industrial process chemistry, and modern cross-coupling chemistry descends directly from this carbon-carbon bond-forming tool.

Selected Publications

Victor Grignard's major publications include numerous papers in leading journals of the era, detailing the experimental and theoretical work summarized above.

12.2 Paul Sabatier

Early Life and Education

Paul Sabatier was born on 1854-11-05 in Carcassonne, France.

Career and Major Research

Sabatier was professor of chemistry at Toulouse. He showed that finely divided metals, especially nickel, catalyze the hydrogenation of unsaturated organic compounds, a method he applied to the conversion of oils into fats and to the hydrogenation of benzene. His work laid the foundation of industrial catalytic hydrogenation.

Nobel Prize-winning Work

The work for which Paul Sabatier was awarded the Nobel Prize in Chemistry in 1912 was described by the Nobel Committee as: "for his method of hydrogenating organic compounds in the presence of finely disintegrated metals whereby the progress of organic chemistry has been greatly advanced in recent years".

This research fundamentally advanced the field by making groundbreaking contributions that transformed chemical science.

Significance and Impact

The impact of Paul Sabatier's work extends far beyond the specific achievement recognized by the Nobel Prize. These contributions continue to influence chemical research and education today.

Later Career and Honors

Paul Sabatier continued his scientific work until 1941-08-14, passing away in Toulouse, France. He received numerous honors including membership in major scientific academies and societies.

Legacy

The legacy of Paul Sabatier endures through the lasting impact of his scientific contributions. Paul Sabatier's work continues to be cited and built upon by researchers across the chemical sciences.

Modern Developments

Sabatier's hydrogenation catalysis, developed in the early 1900s, remains fundamental to the modern hydrogenation industry, from margarine production and edible oil hardening to ammonia synthesis and the hydrogenation of fats, oils, and petrochemicals. His nickel-based catalysts inspired the heterogeneous catalysis at the heart of modern ammonia, methanol, and Fischer-Tropsch plants.

Selected Publications

Paul Sabatier's major publications include numerous papers in leading journals of the era, detailing the experimental and theoretical work summarized above.

Chapter Summary

Victor Grignard and Paul Sabatier shared the prize, Grignard for his organomagnesium reagents and Sabatier for his method of hydrogenating organic compounds with finely divided metals.

Chapter 13

1913: Alfred Werner

The Nobel Prize in Chemistry for the year 1913 was awarded to Alfred Werner.

Motivation: in recognition of his work on the linkage of atoms in molecules by which he has thrown new light on earlier investigations and opened up new fields of research especially in inorganic chemistry

13.1 Alfred Werner

Early Life and Education

Alfred Werner was born on 1866-12-12 in Mulhouse, France. Alfred Werner was a Swiss chemist who was a student at ETH Zurich and a professor at the University of Zurich. He won the Nobel Prize in Chemistry in 1913 for proposing the octahedral configuration of transition metal complexes. Werner developed the basis for modern coordination chemistry. He was the first inorganic chemist to win the Nobel Prize, and the only one prior to 1973.

Career and Major Research

Werner was professor of chemistry at the University of Zürich. In 1893 he proposed the coordination theory, according to which metal atoms exhibit a primary valence and a secondary valence that fixes the spatial arrangement of surrounding groups. This theory explained the structure of complex salts and the existence of optical isomers in coordination compounds.

Nobel Prize-winning Work

The work for which Alfred Werner was awarded the Nobel Prize in Chemistry in 1913 was described by the Nobel Committee as: "in recognition of his work on the linkage of atoms in molecules by which he has thrown new light on earlier investigations and opened up new fields of research especially in inorganic chemistry".

This research fundamentally advanced the field by making groundbreaking contributions that transformed chemical science.

Significance and Impact

The impact of Alfred Werner's work extends far beyond the specific achievement recognized by the Nobel Prize. These contributions continue to influence chemical research and education today.

Later Career and Honors

Alfred Werner continued his scientific work until 1919-11-15, passing away in Zurich, Switzerland. He received numerous honors including membership in major scientific academies and societies.

Legacy

The legacy of Alfred Werner endures through the lasting impact of his scientific contributions. Alfred Werner's work continues to be cited and built upon by researchers across the chemical sciences.

Modern Developments

Werner's coordination theory explains the structure of coordination compounds and remains the framework for modern inorganic chemistry, bioinorganic chemistry, and catalysis. Werner-type complexes are central to organometallic chemistry, homogeneous catalysis (including the Nobel-winning metathesis and cross-coupling reactions), and the design of MRI contrast agents and anticancer drugs such as cisplatin.

Selected Publications

Alfred Werner's major publications include numerous papers in leading journals of the era, detailing the experimental and theoretical work summarized above.

Chapter Summary

Alfred Werner was awarded the prize for his work on the linkage of atoms in molecules, which established the field of coordination chemistry.

Chapter 14

1914:Theodore W. Richards

The Nobel Prize in Chemistry for the year 1914 was awarded to Theodore W. Richards.

Motivation: in recognition of his accurate determinations of the atomic weight of a large number of chemical elements

14.1 Theodore W. Richards

Early Life and Education

Theodore W. Richards was born on 1868-01-31 in Germantown, PA, USA.

Career and Major Research

Richards was professor of chemistry at Harvard, where he spent his entire career. He refined the determinations of atomic weights of more than sixty elements, correcting many values, and his measurements of the atomic weight of lead established that the values varied with the source of the lead, a fact explained by isotopes. He was the first American chemist to receive the prize.

Nobel Prize-winning Work

The work for which Theodore W. Richards was awarded the Nobel Prize in Chemistry in 1914 was described by the Nobel Committee as: "in recognition of his accurate determinations of the atomic weight of a large number of chemical elements".

This research fundamentally advanced the field by making groundbreaking contributions that transformed chemical science.

Significance and Impact

The impact of Theodore W. Richards's work extends far beyond the specific achievement recognized by the Nobel Prize. These contributions continue to influence chemical research and education today.

Later Career and Honors

Theodore W. Richards continued his scientific work until 1928-04-02, passing away in Cambridge, MA, USA. He received numerous honors including membership in major scientific academies and societies.

Legacy

The legacy of Theodore W. Richards endures through the lasting impact of his scientific contributions. Theodore W. Richards's work continues to be cited and built upon by researchers across the chemical sciences.

Modern Developments

Richards's precise determinations of atomic weights established rigorous analytical chemistry and laid the groundwork for the modern periodic table and the discovery of isotopes. His mass-based methods evolved into modern mass spectrometry and isotopic analysis used in radiometric dating, forensic science, and the calibration of the atomic mass unit.

Selected Publications

Theodore W. Richards's major publications include numerous papers in leading journals of the era, detailing the experimental and theoretical work summarized above.

Chapter Summary

Theodore W. Richards was honored for his accurate determinations of the atomic weights of a large number of elements.

Chapter 15

1915:Richard Willstätter

The Nobel Prize in Chemistry for the year 1915 was awarded to Richard Willstätter.

Motivation: for his researches on plant pigments, especially chlorophyll

15.1 Richard Willstätter

Early Life and Education

Richard Willstätter was born on 1872-08-13 in Karlsruhe, Germany. Richard Martin Willstätter FRS HFRSE was a German organic chemist whose study of the structure of plant pigments, chlorophyll included, won him the 1915 Nobel Prize for Chemistry.

Career and Major Research

Willstätter was professor at Munich and later director of the Kaiser Wilhelm Institute for Chemistry in Berlin. He determined the constitution of chlorophyll and of the plant pigments, and he studied the anthocyanins, the red and blue pigments of flowers and fruits. He also made important contributions to the chemistry of enzymes.

Nobel Prize-winning Work

The work for which Richard Willstätter was awarded the Nobel Prize in Chemistry in 1915 was described by the Nobel Committee as: "for his researches on plant pigments, especially chlorophyll".

This research fundamentally advanced the field by making groundbreaking contributions that transformed chemical science.

Significance and Impact

The impact of Richard Willstätter's work extends far beyond the specific achievement recognized by the Nobel Prize. These contributions continue to influence chemical research and education today.

Later Career and Honors

Richard Willstätter continued his scientific work until 1942-08-03, passing away in Locarno, Switzerland. He received numerous honors including membership in major scientific academies and societies.

Legacy

The legacy of Richard Willstätter endures through the lasting impact of his scientific contributions. Richard Willstätter's work continues to be cited and built upon by researchers across the chemical sciences.

Modern Developments

Willstätter's structural studies of chlorophyll and plant pigments revealed the chemistry of photosynthesis and founded modern pigment research. His work led to the understanding of anthocyanins and carotenoids, which today are studied as dietary antioxidants and natural colorants in foods and cosmetics. Photosynthesis research inspired by Willstätter underpins modern solar-energy and biofuel technologies.

Selected Publications

Richard Willstätter's major publications include numerous papers in leading journals of the era, detailing the experimental and theoretical work summarized above.

Chapter Summary

Richard Willstätter received the prize for his researches on plant pigments, especially chlorophyll.

Chapter 16

1916:No Prize Awarded

No Nobel Prize in Chemistry was awarded in 1916. The First World War, which had begun in 1914, interrupted the work of the Nobel Committees, and the award cycle remained suspended for several years.

Chapter Summary

The Nobel Committee for Chemistry did not award the prize in 1916, one of the years in which the regular award cycle was interrupted. Such gaps reflect the extraordinary circumstances of world events rather than an absence of noteworthy chemical research.

Chapter 17

1917:No Prize Awarded

No Nobel Prize in Chemistry was awarded in 1917, as the First World War continued to interrupt the work of the Nobel Committees. The prize money for the year was allocated to the special fund of the Chemistry prize section.

Chapter Summary

The Nobel Committee for Chemistry did not award the prize in 1917, one of the years in which the regular award cycle was interrupted. Such gaps reflect the extraordinary circumstances of world events rather than an absence of noteworthy chemical research.

Chapter 18

1918:Fritz Haber

The Nobel Prize in Chemistry for the year 1918 was awarded to Fritz Haber.

Motivation: for the synthesis of ammonia from its elements

18.1 Fritz Haber

Early Life and Education

Fritz Haber was born on 1868-12-09 in Breslau, Prussia (now Wroclaw, Poland). Fritz Jakob Haber was a German chemist who received the Nobel Prize in Chemistry in 1918 for his invention of the Haber process, a method used in industry to synthesize ammonia from nitrogen gas and hydrogen gas. This invention is important for the large-scale synthesis of fertilizers and explosives. It is estimated that a third of annual global food production uses ammonia from the Haber–Bosch process, and that this food supports nearly half the world’s population. For this work, Haber has been called one of the most important scientists and industrial chemists in human history.

Career and Major Research

Haber was professor at the Technische Hochschule in Karlsruhe. He developed a method for synthesizing ammonia directly from nitrogen and hydrogen at high pressure over a catalyst, first demonstrated in 1909. The process, industrialized with Carl Bosch at BASF, made possible the large-scale production of nitrogen fertilizers and altered the course of agriculture.

Nobel Prize-winning Work

The work for which Fritz Haber was awarded the Nobel Prize in Chemistry in 1918 was described by the Nobel Committee as: "for the synthesis of ammonia from its elements".

This research fundamentally advanced the field by making groundbreaking contributions that transformed chemical science.

Significance and Impact

The impact of Fritz Haber’s work extends far beyond the specific achievement recognized by the Nobel Prize. These contributions continue to influence chemical research and education today.

Later Career and Honors

Fritz Haber continued his scientific work until 1934-01-29, passing away in Basel, Switzerland. He received numerous honors including membership in major scientific academies and societies.

Legacy

The legacy of Fritz Haber endures through the lasting impact of his scientific contributions. Fritz Haber's work continues to be cited and built upon by researchers across the chemical sciences.

Modern Developments

Fritz Haber's ammonia synthesis, developed with Carl Bosch, feeds roughly half of the world's population through nitrogen fertilizers. The Haber–Bosch process remains the dominant industrial route to ammonia and is now being reinvented in the green-ammonia movement for low-carbon fertilizer and as a hydrogen energy carrier.

Selected Publications

Fritz Haber's major publications include numerous papers in leading journals of the era, detailing the experimental and theoretical work summarized above.

Chapter Summary

Fritz Haber was awarded the prize for the synthesis of ammonia from its elements, making possible the industrial production of nitrogen fertilizers.

Chapter 19

1919:No Prize Awarded

No Nobel Prize in Chemistry was awarded in 1919. In accordance with the statutes of the Nobel Foundation, the prize money for the year was allocated to the special fund of the Chemistry prize section.

Chapter Summary

The Nobel Committee for Chemistry did not award the prize in 1919, one of the years in which the regular award cycle was interrupted. Such gaps reflect the extraordinary circumstances of world events rather than an absence of noteworthy chemical research.

Chapter 20

1920:Walther Nernst

The Nobel Prize in Chemistry for the year 1920 was awarded to Walther Nernst.

Motivation: in recognition of his work in thermochemistry

20.1 Walther Nernst

Early Life and Education

Walther Nernst was born on 1864-06-25 in Briesen, Prussia (now Wąbrzeźno, Poland). Walther Hermann Nernst was a German physical chemist known for his work in thermodynamics, physical chemistry, electrochemistry, and solid-state physics. His formulation of the Nernst heat theorem helped pave the way for the third law of thermodynamics, for which he won the 1920 Nobel Prize in Chemistry. He is also known for developing the Nernst equation in 1887.

Career and Major Research

Nernst was professor of physical chemistry at Göttingen and later in Berlin, where he directed the Physikalisch-Chemisches Institut. He formulated the heat theorem, stated in 1906, which provides a basis for calculating chemical equilibria at low temperatures and became known as the third law of thermodynamics. He also derived the equation for the electrode potential of an electrochemical cell that bears his name.

Nobel Prize-winning Work

The work for which Walther Nernst was awarded the Nobel Prize in Chemistry in 1920 was described by the Nobel Committee as: "in recognition of his work in thermochemistry".

This research fundamentally advanced the field by making groundbreaking contributions that transformed chemical science.

Significance and Impact

The impact of Walther Nernst's work extends far beyond the specific achievement recognized by the Nobel Prize. These contributions continue to influence chemical research and education today.

Later Career and Honors

Walther Nernst continued his scientific work until 1941-11-18, passing away in Zibelle, Germany (now Niwica, Poland). He received numerous honors including membership in major scientific academies and societies.

Legacy

The legacy of Walther Nernst endures through the lasting impact of his scientific contributions. Walther Nernst's work continues to be cited and built upon by researchers across the chemical sciences.

Modern Developments

Nernst's heat theorem, which became the third law of thermodynamics, underpins modern low-temperature physics and chemistry, including cryogenics, superconductivity, and the standard-state calculations used throughout chemical engineering. The Nernst equation remains the cornerstone of electrochemistry, governing batteries, corrosion, and electrochemical sensors, including the pH electrodes and ion-selective electrodes used in laboratories worldwide.

Selected Publications

Walther Nernst's major publications include numerous papers in leading journals of the era, detailing the experimental and theoretical work summarized above.

Chapter Summary

Walther Nernst received the prize in recognition of his work in thermochemistry, including the heat theorem that bears his name.

Chapter 21

1921:Frederick Soddy

The Nobel Prize in Chemistry for the year 1921 was awarded to Frederick Soddy.

Motivation: for his contributions to our knowledge of the chemistry of radioactive substances, and his investigations into the origin and nature of isotopes

21.1 Frederick Soddy

Early Life and Education

Frederick Soddy was born on 1877-09-02 in Eastbourne, United Kingdom. Frederick Soddy FRS was an English radiochemist who explained, with Ernest Rutherford, that radioactivity is due to the transmutation of elements, now known to involve nuclear reactions. He also proved the existence of isotopes of certain radioactive elements. In 1921, he received the Nobel Prize in Chemistry "for his contributions to our knowledge of the chemistry of radioactive substances, and his investigations into the origin and nature of isotopes". Soddy was a polymath who mastered chemistry, nuclear physics, statistical mechanics, finance, and economics.

Career and Major Research

Soddy worked at McGill with Rutherford, where the two proposed the disintegration theory of radioactivity, and later at Glasgow and Oxford. He showed that radioactive decay involves a change of element, and he established that elements can occur in several forms with different atomic weights but identical chemical properties, which he named isotopes.

Nobel Prize-winning Work

The work for which Frederick Soddy was awarded the Nobel Prize in Chemistry in 1921 was described by the Nobel Committee as: "for his contributions to our knowledge of the chemistry of radioactive substances, and his investigations into the origin and nature of isotopes".

This research fundamentally advanced the field by making groundbreaking contributions that transformed chemical science.

Significance and Impact

The impact of Frederick Soddy's work extends far beyond the specific achievement recognized by the Nobel Prize. These contributions continue to influence chemical research and education today.

Later Career and Honors

Frederick Soddy continued his scientific work until 1956-09-22, passing away in Brighton, United Kingdom. He received numerous honors including membership in major scientific academies and societies.

Legacy

The legacy of Frederick Soddy endures through the lasting impact of his scientific contributions. Frederick Soddy's work continues to be cited and built upon by researchers across the chemical sciences.

Modern Developments

Soddy's concept of isotopes became central to modern chemistry and physics. Radioactive isotopes are now routine tools in nuclear medicine, medical imaging, and cancer therapy, while stable isotope techniques trace metabolic pathways and date geological and archaeological materials. The discovery of isotopes also made possible nuclear energy and the modern understanding of atomic structure.

Selected Publications

Frederick Soddy's major publications include numerous papers in leading journals of the era, detailing the experimental and theoretical work summarized above.

Chapter Summary

Frederick Soddy was honored for his contributions to the chemistry of radioactive substances and his investigations into the origin and nature of isotopes.

Chapter 22

1922: Francis W. Aston

The Nobel Prize in Chemistry for the year 1922 was awarded to Francis W. Aston.

Motivation: for his discovery, by means of his mass spectrograph, of isotopes, in a large number of non-radioactive elements, and for his enunciation of the whole-number rule

22.1 Francis W. Aston

Early Life and Education

Francis W. Aston was born on 1877-09-01 in Harborne, United Kingdom.

Career and Major Research

Aston worked at the Cavendish Laboratory in Cambridge. He built the mass spectrograph, an instrument that separated atomic nuclei according to their mass, and used it to measure the exact masses of the isotopes of many elements. His work established the whole-number rule, according to which isotope masses are nearly integers relative to hydrogen.

Nobel Prize-winning Work

The work for which Francis W. Aston was awarded the Nobel Prize in Chemistry in 1922 was described by the Nobel Committee as: "for his discovery, by means of his mass spectrograph, of isotopes, in a large number of non-radioactive elements, and for his enunciation of the whole-number rule".

This research fundamentally advanced the field by making groundbreaking contributions that transformed chemical science.

Significance and Impact

The impact of Francis W. Aston's work extends far beyond the specific achievement recognized by the Nobel Prize. These contributions continue to influence chemical research and education today.

Later Career and Honors

Francis W. Aston continued his scientific work until 1945-11-20, passing away in Cambridge, United Kingdom. He received numerous honors including membership in major scientific academies and societies.

Legacy

The legacy of Francis W. Aston endures through the lasting impact of his scientific contributions. Francis W. Aston's work continues to be cited and built upon by researchers across the chemical sciences.

Modern Developments

Aston's mass spectrograph and his discovery of isotopes by mass measurement founded modern mass spectrometry, an indispensable tool across chemistry, biology, medicine, and forensics. High-resolution mass spectrometry is now used for drug development, proteomics, metabolomics, and the detection of trace environmental contaminants. Isotopic analysis from Aston's methods also underpins radiometric dating.

Selected Publications

Francis W. Aston's major publications include numerous papers in leading journals of the era, detailing the experimental and theoretical work summarized above.

Chapter Summary

Francis W. Aston received the prize for his discovery, by means of the mass spectrograph, of isotopes in a large number of non-radioactive elements.

Chapter 23

1923:Fritz Pregl

The Nobel Prize in Chemistry for the year 1923 was awarded to Fritz Pregl.

Motivation: for his invention of the method of micro-analysis of organic substances

23.1 Fritz Pregl

Early Life and Education

Fritz Pregl was born on 1869-09-03 in Laibach, Austria-Hungary (now Ljubljana, Slovenia). Fritz Pregl, was a Slovenian-Austrian chemist and physician from a mixed Slovene-German-speaking background. He won the Nobel Prize in Chemistry in 1923 for making important contributions to quantitative organic microanalysis, one of which was the improvement of the combustion train technique for elemental analysis.

Career and Major Research

Pregl was professor of medical chemistry at the University of Graz. He perfected quantitative micro-analysis, reducing the sample size required for elemental analysis of organic compounds to a few milligrams. This technique made possible the chemical analysis of the small quantities of substances that occur in biological materials.

Nobel Prize-winning Work

The work for which Fritz Pregl was awarded the Nobel Prize in Chemistry in 1923 was described by the Nobel Committee as: "for his invention of the method of micro-analysis of organic substances".

This research fundamentally advanced the field by making groundbreaking contributions that transformed chemical science.

Significance and Impact

The impact of Fritz Pregl's work extends far beyond the specific achievement recognized by the Nobel Prize. These contributions continue to influence chemical research and education today.

Later Career and Honors

Fritz Pregl continued his scientific work until 1930-12-13, passing away in Graz, Austria. He received numerous honors including membership in major scientific academies and societies.

Legacy

The legacy of Fritz Pregl endures through the lasting impact of his scientific contributions. Fritz Pregl's work continues to be cited and built upon by researchers across the chemical sciences.

Modern Developments

Pregl's microanalysis, which scaled elemental analysis down to milligram samples, transformed analytical chemistry and enabled the characterization of scarce natural products and biologically important molecules. His techniques evolved into modern microanalysis, combustion analysis, and the automated elemental analyzers used in pharmaceutical and materials laboratories today.

Selected Publications

Fritz Pregl's major publications include numerous papers in leading journals of the era, detailing the experimental and theoretical work summarized above.

Chapter Summary

Fritz Pregl was awarded the prize for his invention of the method of micro-analysis of organic substances.

Chapter 24

1924:No Prize Awarded

No Nobel Prize in Chemistry was awarded in 1924. In accordance with the statutes of the Nobel Foundation, the prize money for the year was allocated to the special fund of the Chemistry prize section.

Chapter Summary

The Nobel Committee for Chemistry did not award the prize in 1924, one of the years in which the regular award cycle was interrupted. Such gaps reflect the extraordinary circumstances of world events rather than an absence of noteworthy chemical research.

Chapter 25

1925:Richard Zsigmondy

The Nobel Prize in Chemistry for the year 1925 was awarded to Richard Zsigmondy.

Motivation: for his demonstration of the heterogenous nature of colloid solutions and for the methods he used, which have since become fundamental in modern colloid chemistry

25.1 Richard Zsigmondy

Early Life and Education

Richard Zsigmondy was born on 1865-04-01 in Vienna, Austrian Empire (now Austria). Richard Adolf Zsigmondy was an Austrian-born chemist. He was known for his research in colloids, for which he was awarded the Nobel Prize in Chemistry in 1925, as well as for co-inventing the slit-ultramicroscope, and different membrane filters. The crater Zsigmondy on the Moon is named in his honour.

Career and Major Research

Zsigmondy was professor of chemistry at Göttingen. With Henry Siedentopf he invented the ultramicroscope in 1903, an instrument that reveals colloidal particles by scattered light. He used it to study gold sols and other colloids, establishing the heterogeneous nature of colloidal solutions and the methods for characterizing them.

Nobel Prize-winning Work

The work for which Richard Zsigmondy was awarded the Nobel Prize in Chemistry in 1925 was described by the Nobel Committee as: "for his demonstration of the heterogenous nature of colloid solutions and for the methods he used, which have since become fundamental in modern colloid chemistry".

This research fundamentally advanced the field by making groundbreaking contributions that transformed chemical science.

Significance and Impact

The impact of Richard Zsigmondy's work extends far beyond the specific achievement recognized by the Nobel Prize. These contributions continue to influence chemical research and education today.

Later Career and Honors

Richard Zsigmondy continued his scientific work until 1929-09-24, passing away in Göttingen, Germany. He received numerous honors including membership in major scientific academies and societies.

Legacy

The legacy of Richard Zsigmondy endures through the lasting impact of his scientific contributions. Richard Zsigmondy's work continues to be cited and built upon by researchers across the chemical sciences.

Modern Developments

Zsigmondy's colloid chemistry established the scientific foundation for the modern study of colloids, emulsions, and nanoparticles. Colloidal science now underpins the food, paint, pharmaceutical, and cosmetics industries, as well as the rapidly expanding field of nanotechnology and nanomedicine. His ultramicroscope concept evolved into dynamic light scattering and modern nanoparticle characterization.

Selected Publications

Richard Zsigmondy's major publications include numerous papers in leading journals of the era, detailing the experimental and theoretical work summarized above.

Chapter Summary

Richard Zsigmondy received the prize for his demonstration of the heterogeneous nature of colloid solutions and for the methods he developed in colloid chemistry.

Chapter 26

1926:The Svedberg

The Nobel Prize in Chemistry for the year 1926 was awarded to The Svedberg.

Motivation:

for his work on disperse systems

26.1 The Svedberg

Early Life and Education

Theodor Svedberg, known throughout his scientific career as The Svedberg, was born on 1884-08-30 in Valbo, Sweden. He was a Swedish physical chemist who developed the ultracentrifuge and used it to measure the molecular weights of proteins and other macromolecules.

Career and Major Research

Svedberg studied at the University of Uppsala, where he received his doctorate in 1908. He was appointed professor of physical chemistry at Uppsala in 1912 and remained there for most of his career. His early research concerned colloids, fine dispersions of particles in a liquid, and he developed methods for characterizing such disperse systems by measuring particle size and sedimentation. Beginning in the early 1920s, Svedberg and his collaborator Herman Rinde constructed the ultracentrifuge, an instrument that spins liquid samples at very high speed so that dissolved molecules sediment at rates determined by their size. With the ultracentrifuge, Svedberg showed that proteins are large molecules with definite, reproducible molecular weights, and he measured proteins such as haemoglobin, proving that molecules from different species have different molecular weights.

Nobel Prize-winning Work

The work for which The Svedberg was awarded the Nobel Prize in Chemistry in 1926 was described by the Nobel Committee as: "for his work on disperse systems".

Svedberg's prize recognized his long series of investigations on colloids and other disperse systems. Using the ultracentrifuge, he measured how rapidly colloidal particles and dissolved macromolecules sediment under centrifugal force, obtaining the size distribution of particles and, for proteins, the molecular weight of the individual molecules. These experiments settled a long-running debate by demonstrating that proteins are well-defined molecules of characteristic size rather than indefinite aggregates.

Significance and Impact

The ultracentrifuge gave chemists the first reliable tool for determining the molecular weights of proteins and other biological macromolecules, and it opened the way to a quantitative understanding of their behavior. The unit of sedimentation rate, the svedberg (S), is named in his honour and remains standard in biochemistry and molecular biology.

Later Career and Honors

The Svedberg continued his scientific work at Uppsala until his retirement in 1949. He was a member of the Royal Swedish Academy of Sciences and held honorary degrees from several universities. He died on 1971-02-25 in Sweden.

Legacy

The legacy of The Svedberg endures through the lasting impact of his scientific contributions. Ultracentrifugation remains a fundamental technique in the study of proteins and other macromolecules, and the Svedberg unit is used in laboratories throughout the world.

Modern Developments

Svedberg's analytical ultracentrifuge enabled the measurement of molecular weights and the study of macromolecules in solution, laying the groundwork for polymer science and the characterization of proteins, nucleic acids, and nanoparticles. Ultracentrifugation remains essential in the purification of viruses, vaccines, and biopharmaceuticals, including the production of mRNA vaccines.

Selected Publications

- Svedberg, T. (1912). "Die Existenz der Moleküle."
- Svedberg, T., & Rinde, H. (1924). "The Ultracentrifuge, a New Instrument for the Determination of Size and Distribution of Size of Particle in Amicroscopic Colloids." *Journal of the American Chemical Society*.

Chapter Summary

The 1926 prize recognized The Svedberg for his work on disperse systems. His ultracentrifuge proved that proteins are large molecules of definite molecular weight and gave chemists a reliable method for characterizing colloids and macromolecules.

Chapter 27

1927:Heinrich Wieland

The Nobel Prize in Chemistry for the year 1927 was awarded to Heinrich Wieland.

Motivation:

for his investigations of the constitution of the bile acids and related substances

27.1 Heinrich Wieland

Early Life and Education

Heinrich Otto Wieland was born on 1877-06-04 in Pforzheim, Germany. He was a German organic chemist awarded the Nobel Prize in Chemistry in 1927 for his investigations of the constitution of the bile acids and related substances.

Career and Major Research

Wieland studied chemistry at the University of Munich, where he received his doctorate in 1901 under Johannes Thiele. He taught at the University of Freiburg, where he became professor, and in 1921 he succeeded Richard Willstätter as professor of chemistry at the University of Munich. His early work concerned nitrogen compounds such as hydrazoic acid, and his studies on the mechanisms of biological oxidation introduced the concept of dehydrogenation. He also investigated the structures of the alkaloid strychnine, the pigments of butterfly wings (the pterins), and the toxins of the death-cap mushroom. His most extended research concerned the bile acids, where he established that cholic acid and related compounds share a ring system closely related to cholesterol.

Nobel Prize-winning Work

The work for which Heinrich Wieland was awarded the Nobel Prize in Chemistry in 1927 was described by the Nobel Committee as: "for his investigations of the constitution of the bile acids and related substances".

Wieland's investigations showed that the bile acids, which occur in bile as salts and are important for the digestion of fats, are built on a common ring skeleton closely related to cholesterol. This clarified the constitution of an entire family of naturally occurring substances and provided a basis for the later chemistry of the steroid hormones.

Significance and Impact

By establishing the constitution of the bile acids, Wieland laid the foundations of steroid chemistry. His concept of biological oxidation as a process of dehydrogenation also became a cornerstone of biochemistry.

Later Career and Honors

Wieland remained at the University of Munich until his retirement in 1950. He held honorary doctorates from several universities and was a member of several scientific academies. He died on 1957-08-05 in Munich, Germany.

Legacy

The Heinrich Wieland Prize, awarded since 1964 for outstanding research on lipids, is named in his honour. Wieland's work on the bile acids and on oxidation mechanisms continues to be cited and built upon across the chemical sciences.

Modern Developments

Wieland's bile acid and natural product chemistry helped establish the steroid framework that led to the modern pharmaceutical industry for steroids, including corticosteroids, oral contraceptives, and vitamin D. His work on oxidation mechanisms also influenced bioenergetics. Steroid chemistry founded by Wieland remains central to the synthesis of hormones and therapeutic agents.

Selected Publications

- Wieland, H. (1912). "Studien über den Mechanismus der Oxydationsvorgänge." *Berichte der deutschen chemischen Gesellschaft*.
- Wieland, H. (1928). "On the Mechanism of Oxidation." Nobel Lecture, Stockholm.

Chapter Summary

The 1927 prize recognized Heinrich Wieland for his investigations of the constitution of the bile acids and related substances. His work showed that bile acids are derived from a common steroid skeleton and helped establish the chemistry of steroids.

Chapter 28

1928:Adolf Windaus

The Nobel Prize in Chemistry for the year 1928 was awarded to Adolf Windaus.

Motivation:

for the services rendered through his research into the constitution of the sterols and their connection with the vitamins

28.1 Adolf Windaus

Early Life and Education

Adolf Otto Reinhold Windaus was born on 1876-12-25 in Berlin, Germany. He was a German organic chemist awarded the Nobel Prize in Chemistry in 1928 for his research into the constitution of the sterols and their connection with the vitamins.

Career and Major Research

Windaus began as a medical student but turned to chemistry and received his doctorate in 1899 at the University of Freiburg under Heinrich Kiliani, working on the Digitalis compounds. He worked as an assistant to Emil Fischer in Berlin before returning to Freiburg, where he habilitated in 1903 with a thesis on cholesterol. He held professorships in Freiburg and at the University of Innsbruck from 1913 to 1915, and in 1915 he succeeded Otto Wallach as professor of chemistry at the University of Göttingen, where he worked until his retirement in 1944. In 1919 he demonstrated the chemical relationship between cholesterol and the bile acids. With Alfred Fabian Hess he found that ultraviolet irradiation gives ergosterol antirachitic activity, and the photochemically formed product became known as vitamin D2 (ergocalciferol). His institute in Göttingen also investigated the antineuritic vitamin, vitamin B1.

Nobel Prize-winning Work

The work for which Adolf Windaus was awarded the Nobel Prize in Chemistry in 1928 was described by the Nobel Committee as: "for the services rendered through his research into the constitution of the sterols and their connection with the vitamins".

Windaus's research established the constitution of the sterols, beginning with cholesterol, and showed their relationship to the bile acids. His investigations connected the sterol field to the chemistry of the vitamins, since he proved that irradiation of ergosterol produces the antirachitic vitamin D.

Significance and Impact

Windaus's work placed the sterols at the center of the chemistry of natural products and made possible the industrial production of vitamin D by the irradiation of ergosterol. His structural proposals for cholesterol, presented in his Nobel lecture, were later refined with the aid of X-ray crystallography.

Later Career and Honors

Windaus retired in 1944 but continued to live in Göttingen. He received the Pour le Mérite for Sciences and Arts, the Grand Cross of the Order of Merit of the Federal Republic of Germany, and honorary doctorates from several universities. He died on 1959-06-09 in Göttingen, Germany.

Legacy

The legacy of Adolf Windaus endures through the lasting impact of his scientific contributions. The chemistry of the sterols and the vitamins that he founded continues to be built upon by researchers across the chemical sciences.

Modern Developments

Windaus's structural work on cholesterol and steroids, including the role of ultraviolet light in forming vitamin D, led to the modern understanding and prevention of rickets. Vitamin D research now connects to bone health, immunity, and many areas of medicine. Windaus's steroid chemistry also supported the later elucidation of bile acids and sex hormones.

Selected Publications

- Windaus, A. (1919). "Die Konstitution des Cholesterins." Nachrichten der Gesellschaft der Wissenschaften zu Göttingen.
- Windaus, A. (1931). "Antirachitisches Vitamin aus bestrahltem Ergosterin." Hoppe-Seylers Zeitschrift für physiologische Chemie, 203, 70–75.

Chapter Summary

The 1928 prize recognized Adolf Windaus for his research into the constitution of the sterols and their connection with vitamins. He helped establish the steroid skeleton and showed that irradiation of ergosterol produces vitamin D.

Chapter 29

1929: Arthur Harden and Hans von Euler-Chelpin

The Nobel Prize in Chemistry for the year 1929 was awarded jointly to Arthur Harden and Hans von Euler-Chelpin.

Motivation:

for their investigations on the fermentation of sugar and fermentative enzymes

29.1 Arthur Harden

Early Life and Education

Arthur Harden was born on 1865-10-12 in Manchester, England. He was a British biochemist who shared the 1929 Nobel Prize in Chemistry with Hans von Euler-Chelpin for their investigations on the fermentation of sugar and fermentative enzymes.

Career and Major Research

Harden studied chemistry at Owens College in Manchester and received his doctorate in 1894 at the University of Erlangen in Germany. In 1897 he became head of the chemistry department at the British Institute of Preventive Medicine in London, later known as the Lister Institute, and from 1907 he also held a chair of biochemistry at the University of London. In a long series of experiments on the fermentation of sugars by yeast-juice, Harden and his assistant William John Young showed that the addition of phosphate is essential and that the phosphate becomes bound during fermentation. They isolated the hexose phosphate ester that carries the phosphate, later known as the Harden-Young ester (fructose 1,6-bisphosphate), and showed that the fermenting system requires a heat-stable, dialysable cofactor, which they named cozymase, in addition to the heat-labile enzyme itself. He was knighted in 1926.

Nobel Prize-winning Work

The work for which Arthur Harden was awarded the Nobel Prize in Chemistry in 1929 was described by the Nobel Committee as: "for their investigations on the fermentation of sugar and fermentative enzymes".

Harden's work clarified the role of enzymes and coenzymes in alcoholic fermentation. He showed that fermentation proceeds through a series of phosphate-carrying intermediates and that the fermenting system consists of a heat-labile enzyme together with a heat-stable coenzyme, results that became the basis for understanding how sugar is broken down in living cells.

Significance and Impact

Harden's discovery of phosphate esterification in fermentation identified the first intermediate stages of sugar metabolism and revealed the existence of coenzymes, concepts that underlie modern biochemistry.

Later Career and Honors

Harden continued to work at the Lister Institute until his retirement. He was elected a Fellow of the Royal Society in 1909 and held honorary degrees from several universities. He died on 1940-06-17 in Bourne End, England.

Legacy

The legacy of Arthur Harden endures through the lasting impact of his scientific contributions. The Harden-Young ester and the concept of the coenzyme remain part of the basic vocabulary of biochemistry.

Modern Developments

Harden's studies of yeast fermentation revealed coenzymes, the small molecules that assist enzymes, founding coenzyme research and the modern understanding of metabolism. Coenzyme chemistry underpins the entire field of bioenergetics, vitamin biochemistry, and the industrial production of biofuels and fermentation products such as ethanol and amino acids.

Selected Publications

- Harden, A., & Young, W. J. (1906). "The Alcoholic Ferment of Yeast-Juice." Proceedings of the Royal Society of London, Series B.
- Harden, A. (1911). "Alcoholic Fermentation." London: Longmans, Green.

29.2 Hans von Euler-Chelpin

Early Life and Education

Hans Karl August Simon von Euler-Chelpin was born on 1873-02-15 in Augsburg, Germany. He was a German-born Swedish biochemist who shared the 1929 Nobel Prize in Chemistry with Arthur Harden for their investigations on the fermentation of sugar and fermentative enzymes.

Career and Major Research

von Euler-Chelpin studied chemistry in Munich and Berlin and received his doctorate at the University of Berlin in 1895. After working with Walther Nernst, he became an assistant to Svante Arrhenius at the Royal Institute of Technology in Stockholm in 1897, and he took Swedish citizenship in 1902. From 1906 until 1941 he was professor of general and organic chemistry at the University of Stockholm, and in 1929 he became director of its new institute of biochemistry. He studied enzymes and coenzymes with the methods of physical chemistry; in particular he investigated cozymase, the coenzyme required by zymase in alcoholic fermentation, and his laboratory contributed substantially to establishing its structure, later identified as nicotinamide adenine dinucleotide. He also studied the vitamins together with Paul Karrer and others. His son Ulf von Euler received the 1970 Nobel Prize in Physiology or Medicine.

Nobel Prize-winning Work

The work for which Hans von Euler-Chelpin was awarded the Nobel Prize in Chemistry in 1929 was described by the Nobel Committee as: "for their investigations on the fermentation of sugar and fermentative enzymes".

von Euler-Chelpin's investigations applied the methods of physical chemistry to alcoholic fermentation and to the enzymes that bring it about. He showed that the fermenting system consists of a thermostable coenzyme, cozymase, together with a thermolabile enzyme, and he worked out the way in which they act during the fermentation of sugar.

Significance and Impact

His work established the general principle that many enzymes require a coenzyme in order to act, a concept of central importance for biochemistry. It also connected the chemistry of fermentation to the processes by which muscles obtain energy.

Later Career and Honors

von Euler-Chelpin remained active in research after his retirement and published books on enzyme chemistry, including his monograph "Chemie der Enzyme". He held honorary doctorates from several universities. He died on 1964-11-06 in Stockholm, Sweden.

Legacy

The legacy of Hans von Euler-Chelpin endures through the lasting impact of his scientific contributions. His clarification of the role of cozymase in fermentation is regarded as one of the foundations of modern biochemistry.

Modern Developments

von Euler-Chelpin's work on fermentation enzymes and coenzymes established the enzyme chemistry central to modern biochemistry and biotechnology. His identification of cozymase (NAD) opened the field of redox biochemistry that now explains metabolism, cellular signaling, and aging research. Enzymology inspired by his work powers modern biocatalysis and biosensors.

Selected Publications

- von Euler-Chelpin, H. (1910). "Allgemeine Chemie der Enzyme."
- von Euler-Chelpin, H. (1915). "Chemie der Hefe und der alkoholischen Gärung." Leipzig.
- von Euler-Chelpin, H. (1925–1934). "Chemie der Enzyme."

Chapter Summary

The 1929 prize recognized Arthur Harden and Hans von Euler-Chelpin for their investigations of the fermentation of sugar and fermentative enzymes. Their discovery of the heat-stable coenzyme of alcoholic fermentation was a foundation of modern enzymology and biochemistry.

Chapter 30

1930:Hans Fischer

The Nobel Prize in Chemistry for the year 1930 was awarded to Hans Fischer.

Motivation:

for his researches into the constitution of haemin and chlorophyll and especially for his synthesis of haemin

30.1 Hans Fischer

Early Life and Education

Hans Fischer was born on 1881-07-27 in Höchst am Main, Germany. He was a German organic chemist awarded the Nobel Prize in Chemistry in 1930 for his researches into the constitution of haemin and chlorophyll and especially for his synthesis of haemin.

Career and Major Research

Fischer studied chemistry and medicine in Lausanne, Munich, and Marburg, and received his doctorate at the University of Marburg in 1904 under Theodor Zincke. He held professorships of medical chemistry at the University of Innsbruck from 1916 to 1918 and at the University of Vienna from 1918 to 1921. In 1921 he succeeded Heinrich Wieland as professor of organic chemistry at the Technische Hochschule in Munich, where he remained for the rest of his life. His research concerned the pigments of blood, bile, and plants, all of which are built from pyrrole rings. He determined the constitution of the bile pigments bilirubin and biliverdin and of haemin, the iron-containing pigment of haemoglobin, and he worked extensively on chlorophyll. In 1929, together with Karl Zeile, he achieved the total synthesis of haemin, proving that the pigment consists of an iron atom surrounded by the porphyrin ring system.

Nobel Prize-winning Work

The work for which Hans Fischer was awarded the Nobel Prize in Chemistry in 1930 was described by the Nobel Committee as: "for his researches into the constitution of haemin and chlorophyll and especially for his synthesis of haemin".

Fischer's investigations established the complete constitution of haemin, the red iron-containing pigment of blood, and showed that the green pigment chlorophyll is built from the same family of tetrapyrrole ring systems. The total synthesis of haemin confirmed this structure beyond doubt.

Significance and Impact

By establishing the chemistry of the porphyrins, Fischer created the foundation for understanding how oxygen is carried in the blood and how chlorophyll enables photosynthesis. Porphyrin chemistry, developed in his school, remains central to organic and bioinorganic chemistry.

Later Career and Honors

Fischer continued his research at the Technische Hochschule in Munich until the end of the Second World War. He received the Liebig Medal in 1929, the Davy Medal of the Royal Society in 1937, and an honorary doctorate from Harvard University in 1936. He died on 1945-03-31 in Munich, Germany.

Legacy

The legacy of Hans Fischer endures through the lasting impact of his scientific contributions. The chemistry of the porphyrins and bile pigments that he created continues to be built upon by researchers across the chemical and biological sciences.

Modern Developments

Fischer's monumental synthesis of heme and chlorophyll established the chemistry of porphyrins, which now underpin photodynamic therapy for cancer, synthetic blood substitutes, and the study of photosynthesis. Porphyrin chemistry also informs the design of molecular catalysts and materials, and Fischer's synthetic methods remain models of rigorous natural product chemistry.

Selected Publications

- Fischer, H., & Zeile, K. (1929). "Synthese des Hämatoporphyrins, Protoporphyrins und Hämins." *Justus Liebigs Annalen der Chemie*, 468, 98–116.
- Fischer, H., & Orth, H. (1934–1940). "Die Chemie des Pyrrols." Leipzig.

Chapter Summary

The 1930 prize recognized Hans Fischer for his researches into the constitution of haemin and chlorophyll, especially his synthesis of haemin. His work revealed the porphyrin ring structure common to haemoglobin and chlorophyll.

Chapter 31

1931:Carl Bosch and Friedrich Bergius

The Nobel Prize in Chemistry for the year 1931 was awarded jointly to Carl Bosch and Friedrich Bergius.

Motivation:

in recognition of their contributions to the invention and development of chemical high pressure methods

31.1 Carl Bosch

Early Life and Education

Carl Bosch was born on 1874-08-27 in Cologne, Germany. He was a German chemist and industrialist who shared the 1931 Nobel Prize in Chemistry with Friedrich Bergius for their contributions to the invention and development of chemical high pressure methods.

Career and Major Research

Bosch studied chemistry at the Technische Hochschule in Charlottenburg and at the University of Leipzig, where he received his doctorate in 1898. He joined the Badische Anilin- und Soda-Fabrik (BASF) in 1899 and devoted himself to industrial chemistry. From 1909 he led the task of turning Fritz Haber's laboratory synthesis of ammonia into an industrial process, developing the high-pressure reactors, the catalysts, and the large-scale production of hydrogen that the process required. He also worked on the hydrogenation of coal to liquid fuels and on the synthesis of methanol under high pressure. After the formation of IG Farben in 1925 he served as chairman of its management board, and in 1933 he succeeded Fritz Haber as head of the Kaiser Wilhelm Institute for Physical Chemistry in Berlin-Dahlem. He protested publicly against the Nazi dismissal of Jewish scientists, warning that it would destroy German science.

Nobel Prize-winning Work

The work for which Carl Bosch was awarded the Nobel Prize in Chemistry in 1931 was described by the Nobel Committee as: "in recognition of their contributions to the invention and development of chemical high pressure methods".

Bosch's contribution was the development of the chemical high pressure methods by which industrial synthesis under extreme pressures and temperatures became practical. His work made possible the large-scale production of ammonia from atmospheric nitrogen, and thereby the production of nitrogen fertilizers.

Significance and Impact

The Bosch process for ammonia synthesis is among the most important chemical processes ever developed, because it provides the nitrogen compounds on which modern agriculture depends. High pressure technology later found applications in the hydrogenation of coal and in the synthesis of fuels and chemicals.

Later Career and Honors

Bosch directed the affairs of IG Farben until his death and remained one of the leading figures of the German chemical industry. He was elected to the German Academy of Sciences Leopoldina and held honorary doctorates from several universities. He died on 1940-04-26 in Heidelberg, Germany.

Legacy

The legacy of Carl Bosch endures through the lasting impact of his scientific and technical contributions. The industrial synthesis of ammonia that he made possible continues to supply the nitrogen fertilizers that sustain modern agriculture.

Modern Developments

Bosch's industrial high-pressure chemistry, together with Haber, created the Haber–Bosch ammonia process that now produces roughly 200 million tons of ammonia per year, sustaining global food production. The process is today being transformed into green ammonia using electrolytic hydrogen from renewable energy, and high-pressure catalysis remains central to the modern chemical industry.

Selected Publications

- Bosch, C. (1932). "The Development of the Chemical High Pressure Method during the Establishment of the New Ammonia Industry." Nobel Lecture, Stockholm.

31.2 Friedrich Bergius

Early Life and Education

Friedrich Karl Rudolf Bergius was born on 1884-10-11 in Breslau, Germany (now Wrocław, Poland). He was a German chemist who shared the 1931 Nobel Prize in Chemistry with Carl Bosch for their contributions to the invention and development of chemical high pressure methods.

Career and Major Research

Bergius studied chemistry in Breslau and Leipzig, where he received his doctorate in 1907 under Arthur Hantzsch, working on reactions at high pressure. After brief periods with Walther Nernst and Fritz Haber, he joined industrial research in 1911 and set up a laboratory at Rheinau near Mannheim for the study of coal. There he developed the Bergius process, in which coal is hydrogenated with hydrogen under high pressure and temperature to give liquid fuels. The process was developed industrially during and after the First World War, when Germany lacked access to petroleum. After the Second World War he emigrated to the United States and later acted as a consultant in Argentina.

Nobel Prize-winning Work

The work for which Friedrich Bergius was awarded the Nobel Prize in Chemistry in 1931 was described by the Nobel Committee as: "in recognition of their contributions to the invention and development of chemical high pressure methods".

Bergius's contribution was the invention and development of high pressure methods for the chemical transformation of coal. He showed that bituminous coal can be hydrogenated under high pressure to give liquid hydrocarbons, providing a route from coal to liquid fuels.

Significance and Impact

The Bergius process demonstrated the industrial power of high pressure chemistry, and the hydrogenation of coal and lignite came to supply a substantial part of Germany's fuel needs during the Second World War.

Later Career and Honors

After the war Bergius worked as an industrial consultant in the United States and in Argentina, where he died on 1949-03-30 in Buenos Aires. He held honorary doctorates from several universities.

Legacy

The legacy of Friedrich Bergius endures through the lasting impact of his scientific contributions. The high pressure hydrogenation of coal that he pioneered remains a reference point in the chemistry of fuels.

Modern Developments

Bergius's high-pressure hydrogenation of coal and heavy oils laid the foundation for modern hydroprocessing in petroleum refining. His methods evolved into the catalytic hydrotreating and hydrocracking processes that produce clean transportation fuels. Bergius's vision is also relevant to current coal-to-liquid and biomass-to-liquid fuel technologies under development for energy security.

Selected Publications

- Bergius, F. (1913). "Die Anwendung hoher Drucke bei chemischen Vorgängen und ein Nachbild des Entstehungsprozesses der Steinkohle." Halle: Wilhelm Knapp.
- Bergius, F. (1932). "Chemical Reactions under High Pressure." Nobel Lecture, Stockholm.

Chapter Summary

The 1931 prize recognized Carl Bosch and Friedrich Bergius for their contributions to the invention and development of chemical high-pressure methods. The industrial synthesis of ammonia and the hydrogenation of coal revolutionized chemical production.

Chapter 32

1932:Irving Langmuir

The Nobel Prize in Chemistry for the year 1932 was awarded to Irving Langmuir.

Motivation:

for his discoveries and investigations in surface chemistry

32.1 Irving Langmuir

Early Life and Education

Irving Langmuir was born on 1881-01-31 in Brooklyn, New York, United States. He was an American physical chemist awarded the Nobel Prize in Chemistry in 1932 for his discoveries and investigations in surface chemistry.

Career and Major Research

Langmuir studied at Columbia University and received his doctorate at the University of Göttingen in 1906 under Walther Nernst. From 1909 he worked at the General Electric Research Laboratory in Schenectady, New York, where he remained until his retirement in 1950. His early work at General Electric concerned the incandescent lamp; he developed gas-filled lamps with coiled tungsten filaments, greatly improving their efficiency and lifetime. He investigated the adsorption of gases on surfaces and the properties of monomolecular films on liquids, laying the foundations of modern surface chemistry. He also worked on thermionic emission, on the vacuum pump, and on the atomic-hydrogen welding process, and he introduced the concept of electron shells in atoms. Later in life he studied the artificial production of rain.

Nobel Prize-winning Work

The work for which Irving Langmuir was awarded the Nobel Prize in Chemistry in 1932 was described by the Nobel Committee as: "for his discoveries and investigations in surface chemistry".

Langmuir's prize recognized his discoveries concerning the behavior of surfaces. He established quantitative laws for the adsorption of gases on solids, showed that certain films spread on water are exactly one molecule thick, and developed a general picture of chemical reactions at surfaces.

Significance and Impact

Langmuir's work created the scientific discipline of surface chemistry and provided the basis for understanding catalysis on solid surfaces, lubrication, and the behavior of thin films. The Langmuir adsorption isotherm and the journal *Langmuir* are named in his honour.

Later Career and Honors

Langmuir remained at the General Electric Research Laboratory until 1950. He received the Franklin Medal, was elected a foreign member of the Royal Society, and held honorary degrees from several universities. He died on 1957-08-16 in Falmouth, Massachusetts, United States.

Legacy

The legacy of Irving Langmuir endures through the lasting impact of his scientific contributions. Surface chemistry, the discipline he founded, remains central to catalysis, electronics, and materials science.

Modern Developments

Langmuir's surface chemistry and his studies of adsorbed monolayers founded modern surface science, with applications in catalysis, semiconductor fabrication, and materials engineering. The Langmuir–Blodgett technique for molecular films now enables nanotechnology and bio-sensor design, while his contributions to plasma physics and his safety lamp continue to influence industrial practice.

Selected Publications

- Langmuir, I. (1916–1918). "The Constitution and Fundamental Properties of Solids and Liquids." *Journal of the American Chemical Society*.
- Langmuir, I. (1919). "The Arrangement of Electrons in Atoms and Molecules." *Journal of the American Chemical Society*.
- Langmuir, I. (1932). "Surface Chemistry." Nobel Lecture, Stockholm.

Chapter Summary

The 1932 prize recognized Irving Langmuir for his discoveries and investigations in surface chemistry. His studies of adsorption, surface films, and the monomolecular layer laid the foundations of modern surface science.

Chapter 33

1933:No Prize Awarded

No Nobel Prize in Chemistry was awarded in 1933. In accordance with the statutes of the Nobel Foundation, the prize money for the year was allocated to the special fund of the Chemistry prize section.

Chapter Summary

The Nobel Committee for Chemistry did not award the prize in 1933, one of the years in which the regular award cycle was interrupted. Such gaps reflect the extraordinary circumstances of world events rather than an absence of noteworthy chemical research.

Chapter 34

1934:Harold C. Urey

The Nobel Prize in Chemistry for the year 1934 was awarded to Harold C. Urey.

Motivation:

for his discovery of heavy hydrogen

34.1 Harold C. Urey

Early Life and Education

Harold Clayton Urey was born on 1893-04-29 in Walkerton, Indiana, United States. He was an American physical chemist awarded the Nobel Prize in Chemistry in 1934 for his discovery of heavy hydrogen.

Career and Major Research

Urey studied at the University of Montana and received his doctorate at the University of California, Berkeley, in 1923 under Gilbert N. Lewis. He spent 1923-1924 at the Niels Bohr Institute in Copenhagen, then taught at the Johns Hopkins University and, from 1929, at Columbia University. In late 1931 he and his colleagues Ferdinand G. Brickwedde and George M. Murphy discovered the isotope of hydrogen of mass 2, which Urey named deuterium, by distilling liquid hydrogen and identifying the heavy isotope spectroscopically. During the Second World War he directed work at Columbia on the separation of uranium isotopes for the Manhattan Project. After the war he was professor at the University of Chicago from 1945 to 1958 and then at the University of California, San Diego, until 1981. He applied isotope chemistry to geology, geochemistry, and cosmochemistry, studied the origin of the Moon and of the solar system, and examined lunar samples brought back by the Apollo missions.

Nobel Prize-winning Work

The work for which Harold C. Urey was awarded the Nobel Prize in Chemistry in 1934 was described by the Nobel Committee as: "for his discovery of heavy hydrogen".

Urey discovered heavy hydrogen in 1932, when he showed that ordinary hydrogen contains a stable isotope of mass 2 present in small proportion. He gave this isotope the name deuterium, prepared samples of heavy water, and was awarded the prize for this discovery.

Significance and Impact

The discovery of deuterium revealed that the isotopes of the lightest element can be separated and studied, and it opened the fields of isotope separation and isotope chemistry. Heavy water became important as a moderator for nuclear reactors, and isotopes prepared by the methods Urey developed are now used throughout science and medicine.

Later Career and Honors

Urey received the Davy Medal of the Royal Society in 1940, the National Medal of Science in 1964, and the Priestley Medal of the American Chemical Society in 1973, among many other honours. The lunar crater Urey and asteroid 4716 Urey are named in his honour. He died on 1981-01-05 in La Jolla, California, United States.

Legacy

The legacy of Harold C. Urey endures through the lasting impact of his scientific contributions. Isotope chemistry and isotope separation, fields he founded, continue to underpin nuclear science, geochemistry, and medicine.

Modern Developments

Urey's discovery of deuterium gave chemistry a new isotope essential to nuclear energy, tracer studies, and modern physics. Deuterium is used to produce heavy water for nuclear reactors and is a key to understanding metabolic and geochemical processes, while deuterated compounds are standard tools in NMR spectroscopy and drug development. Urey's later work on isotope separation also made carbon and oxygen isotope geochemistry possible.

Selected Publications

- Urey, H. C. (1931). "The Isotopes of Hydrogen." *Physical Review*.
- Urey, H. C., Brickwedde, F. G., & Murphy, G. M. (1932). "A Hydrogen Isotope of Mass 2 and Its Concentration." *Physical Review*.
- Urey, H. C. (1952). "The Planets: Their Origin and Development." New Haven: Yale University Press.

Chapter Summary

The 1934 prize recognized Harold C. Urey for his discovery of heavy hydrogen. The discovery of deuterium opened a new chapter in isotope chemistry and nuclear physics.

Chapter 35

1935:Frédéric Joliot and Irène Joliot-Curie

The Nobel Prize in Chemistry for the year 1935 was awarded jointly to Frédéric Joliot and Irène Joliot-Curie.

Motivation:

in recognition of their synthesis of new radioactive elements

35.1 Frédéric Joliot

Early Life and Education

Jean Frédéric Joliot, known as Frédéric Joliot, was born on 1900-03-19 in Paris, France. He was a French physicist and chemist who shared the 1935 Nobel Prize in Chemistry with Irène Joliot-Curie for their synthesis of new radioactive elements.

Career and Major Research

Joliot graduated in 1923 from the École de Physique et de Chimie Industrielles de la Ville de Paris. In 1925 he became an assistant to Marie Curie at the Institut du Radium in Paris, and in 1926 he married Marie Curie's daughter Irène, with whom he thereafter worked closely. In 1934 the couple discovered artificial radioactivity: they found that when aluminium is bombarded with alpha particles it is converted into a radioactive isotope of phosphorus, showing that radioactivity can be created in previously stable elements. He became professor at the Collège de France in 1937. After the Second World War he was appointed High Commissioner for Atomic Energy in 1945 and directed the creation of the French atomic energy programme until he was removed from the post in 1950.

Nobel Prize-winning Work

The work for which Frédéric Joliot was awarded the Nobel Prize in Chemistry in 1935 was described by the Nobel Committee as: "in recognition of their synthesis of new radioactive elements".

Bombarding aluminium with alpha particles, the two laureates produced a radioactive form of phosphorus and showed that this new radioactivity did not pre-exist in the target material. Stable elements could therefore be transformed into radioactive ones in the laboratory.

Significance and Impact

The discovery of artificial radioactivity showed that radioactivity is not an exclusive property of naturally radioactive elements and provided scientists with an almost unlimited source of radioactive isotopes. It became the basis of radioisotope chemistry, nuclear medicine, and much of later nuclear physics.

Later Career and Honors

Frédéric Joliot was made Commander of the Legion of Honour and held honorary doctorates from several universities. He died on 1958-08-14 in Paris, France.

Legacy

The legacy of Frédéric Joliot endures through the lasting impact of his scientific contributions. The artificial radioactivity that he and Irène Joliot-Curie discovered continues to be exploited throughout science and medicine.

Modern Developments

The Joliot-Curies' discovery of artificial radioactivity opened the era of man-made radioisotopes, which are now produced in reactors and cyclotrons worldwide for nuclear medicine, medical imaging, cancer therapy, and research. Positron emission tomography (PET) relies directly on cyclotron-produced radiotracers descended from their work, and the Joliot-Curies' discovery also led to the discovery of nuclear fission.

Selected Publications

- Joliot, F., & Joliot-Curie, I. (1934). "Un nouveau type de radioactivité." *Comptes rendus hebdomadaires des séances de l'Académie des sciences*, 198, 254.
- Curie, I., & Joliot, F. (1932). "Émission de protons de grande énergie par les substances hydrogénées sous l'influence des rayons gamma très pénétrants." *Comptes rendus hebdomadaires des séances de l'Académie des sciences*, 194, 273.

35.2 Irène Joliot-Curie

Early Life and Education

Irène Joliot-Curie, née Curie, was born on 1897-09-12 in Paris, France. She was a French physicist and chemist who shared the 1935 Nobel Prize in Chemistry with Frédéric Joliot for their synthesis of new radioactive elements.

Career and Major Research

The daughter of Pierre and Marie Curie, Irène was educated in Paris, and during the First World War she worked as a nurse-radiographer, helping to bring X-ray diagnosis to the battlefield. From 1918 she worked as an assistant in the Curie laboratory at the Institut du Radium, and she received her doctorate in 1925 for a thesis on the alpha rays of polonium. After her marriage to Frédéric Joliot in 1926, the two carried out their research together, and in 1934 they discovered artificial radioactivity. She became professor at the Sorbonne in 1937, and in 1946 she succeeded her mother as director of the Institut du Radium. In 1936 she briefly served as Under-Secretary of State for Scientific Research in the French government.

Nobel Prize-winning Work

The work for which Irène Joliot-Curie was awarded the Nobel Prize in Chemistry in 1935 was described by the Nobel Committee as: "in recognition of their synthesis of new radioactive elements".

Together with her husband she showed that alpha particles can convert stable elements into radioactive isotopes, thereby synthesizing new radioactive elements in the laboratory.

Significance and Impact

The discovery of artificial radioactivity, made by the two laureates, opened the way to the production of radioactive isotopes on demand. It earned them the Nobel Prize in Chemistry in 1935, making them only the second married couple to share a Nobel Prize.

Later Career and Honors

Irène Joliot-Curie directed the Institut du Radium from 1946 until her death. She died on 1956-03-17 in Paris, France, of leukemia, an illness widely attributed to her lifelong exposure to radioactive substances.

Legacy

The legacy of Irène Joliot-Curie endures through the lasting impact of her scientific contributions. Her work, carried out together with her husband, remains fundamental to the use of radioactive isotopes in chemistry and medicine.

Modern Developments

The discovery of artificial radioactivity by Irène and Frédéric Joliot-Curie opened the era of man-made radioisotopes, now produced in reactors and cyclotrons worldwide for nuclear medicine, medical imaging, cancer therapy, and research. Positron emission tomography (PET) relies directly on the cyclotron-produced radiotracers that descend from their work. Irène Joliot-Curie's contributions, together with her scientific legacy as the daughter of Marie Curie, helped establish the Curie family as a dynasty of modern nuclear science.

Selected Publications

- Joliot, F., & Joliot-Curie, I. (1934). "Un nouveau type de radioactivité." *Comptes rendus hebdomadaires des séances de l'Académie des sciences*, 198, 254.

- Curie, I., & Joliot, F. (1932). "Émission de protons de grande énergie par les substances hydrogénées sous l'influence des rayons gamma très pénétrants." Comptes rendus hebdomadaires des séances de l'Académie des sciences, 194, 273.

Chapter Summary

The 1935 prize recognized Frédéric Joliot and Irène Joliot-Curie for their synthesis of new radioactive elements. Their discovery of artificial radioactivity extended radioactivity to elements produced in the laboratory and opened the way to radiotracer methods.

Chapter 36

1936:Peter Debye

The Nobel Prize in Chemistry for the year 1936 was awarded to Peter Debye.

Motivation:

for his contributions to our knowledge of molecular structure through his investigations on dipole moments and on the diffraction of X-rays and electrons in gases

36.1 Peter Debye

Early Life and Education

Peter Joseph William Debye was born on 1884-03-24 in Maastricht, Netherlands. He was a Dutch-American physical chemist awarded the Nobel Prize in Chemistry in 1936 for his contributions to the understanding of molecular structure through his investigations on dipole moments and on the diffraction of X-rays and electrons in gases.

Career and Major Research

Debye studied at the Technische Hochschule in Aachen and at the University of Munich, where he received his doctorate in 1908 under Arnold Sommerfeld. He held professorships in Zürich in 1911, Utrecht in 1912, Göttingen in 1914, Zürich in 1920, and Leipzig in 1927. In 1934 he succeeded Albert Einstein as director of the Kaiser Wilhelm Institute for Physics in Berlin. In 1940 he moved to Cornell University in the United States, where he was professor of chemistry until 1950, and he became an American citizen in 1946. He contributed to the theory of specific heat, developed with Erich Hückel the theory of strong electrolytes, and devised with Paul Scherrer the powder method of X-ray diffraction. He developed the use of electric dipole moments as a means of determining molecular structure, and he applied both X-ray and electron diffraction to gases in order to measure the distances between atoms in molecules.

Nobel Prize-winning Work

The work for which Peter Debye was awarded the Nobel Prize in Chemistry in 1936 was described by the Nobel Committee as: "for his contributions to our knowledge of molecular structure through his investigations on dipole moments and on the diffraction of X-rays and electrons in gases".

By measuring the dipole moments of molecules in solution, Debye obtained information about the arrangement of atoms within the molecule, and by diffracting X-rays and electrons from gases he measured the distances between the atoms in simple molecules.

Significance and Impact

Debye's methods gave chemists quantitative information about the geometry of molecules, and the measurement of dipole moments became a standard tool of physical chemistry. The unit of the electric dipole moment, the debye (D), is named in his honour.

Later Career and Honors

Debye remained at Cornell University until his death, serving as head of the chemistry department. He received the Lorentz Medal, the Franklin Medal, and many honorary doctorates. He died on 1966-11-02 in Ithaca, New York, United States.

Legacy

The legacy of Peter Debye endures through the lasting impact of his scientific contributions. His theories and methods continue to be used throughout physical chemistry and molecular physics.

Modern Developments

Debye's model of the dipole and his theory of solutions and dielectrics remain central to modern physical chemistry and materials science. Debye–Hückel theory governs electrolyte solutions, while Debye scattering and the Debye temperature are used in the analysis of crystals and heat capacities. The Debye–Scherrer powder method is a standard tool in X-ray and neutron diffraction.

Selected Publications

- Debye, P., & Scherrer, P. (1916). "Interferenzen an regellos orientierten Teilchen im Röntgenlicht." *Physikalische Zeitschrift*.
- Debye, P., & Hückel, E. (1923). "Zur Theorie der Elektrolyte." *Physikalische Zeitschrift*.
- Debye, P. (1929). "Polar Molecules." New York: Chemical Catalog Company.

Chapter Summary

The 1936 prize recognized Peter Debye for his contributions to the knowledge of molecular structure through his investigations on dipole moments and on the diffraction of X-rays and electrons in gases.

Chapter 37

1937: Norman Haworth and Paul Karrer

The Nobel Prize in Chemistry for the year 1937 was awarded jointly to Norman Haworth and Paul Karrer.

Motivation:

for his investigations on carbohydrates and vitamin C (one half, Norman Haworth) and for his investigations on carotenoids, flavins and vitamins A and B2 (the other half, Paul Karrer)

37.1 Norman Haworth

Early Life and Education

Walter Norman Haworth was born on 1883-03-19 in Coventry, England. He was a British chemist who shared the 1937 Nobel Prize in Chemistry with Paul Karrer for his investigations on carbohydrates and vitamin C.

Career and Major Research

Haworth studied at the University of Manchester and received his doctorate at the University of Göttingen in 1910 under Otto Wallach, working on terpenes. He then worked with Thomas Purdie at the University of St Andrews, applying the methylation method to the sugars. He was professor at Armstrong College in Newcastle upon Tyne from 1920 to 1925, and from 1925 he held the Mason Chair of Chemistry at the University of Birmingham until his retirement in 1948. He established that the sugars exist in ring forms and introduced the formulas that bear his name, determining the ring size and configuration of many sugars and their glycosides. In 1933 his group, working with Edmund Hirst, achieved the total synthesis of vitamin C (ascorbic acid), confirming the structure of the substance that had been isolated by Albert Szent-Györgyi.

Nobel Prize-winning Work

The work for which Norman Haworth was awarded the Nobel Prize in Chemistry in 1937 was described by the Nobel Committee as: "for his investigations on carbohydrates and vitamin C".

Haworth's investigations established the ring structures of the sugars and their derivatives, and they culminated in the first total synthesis of vitamin C. The synthetic route he developed also made possible the industrial production of the vitamin.

Significance and Impact

Haworth's work transformed the chemistry of carbohydrates into a field in which the configuration and ring structure of every sugar could be specified. The synthesis of vitamin C was an early triumph of the chemistry of natural products and made the vitamin available for medicine and nutrition.

Later Career and Honors

Haworth was elected a Fellow of the Royal Society in 1928 and was knighted in 1947. He received the Davy Medal of the Royal Society in 1934. He died on 1950-03-19 in Birmingham, England.

Legacy

The legacy of Norman Haworth endures through the lasting impact of his scientific contributions. The Haworth formulas and his methods of sugar chemistry are still taught and used throughout carbohydrate chemistry.

Modern Developments

Haworth's elucidation of carbohydrate ring structures founded the modern chemistry of sugars and polysaccharides, underpinning glycobiology, the food industry, and the production of cellulose-based materials. His synthesis of vitamin C enabled its large-scale industrial production, and vitamin C supplements and fortified foods remain central to nutrition worldwide.

Selected Publications

- Haworth, W. N. (1929). "The Constitution of Sugars." London: Edward Arnold.
- Haworth, W. N., & Hirst, E. L. (1933). "Synthesis of Ascorbic Acid." Journal of the Chemical Society.

37.2 Paul Karrer

Early Life and Education

Paul Karrer was born on 1889-04-21 in Moscow, Russia, to Swiss parents. His family returned to Switzerland in 1892, and he was educated in Aarau and at the University of Zurich, where he studied under Alfred Werner and received his doctorate in 1911. He was a Swiss organic chemist who shared the 1937 Nobel Prize in Chemistry with Norman Haworth for his investigations on carbohydrates and vitamin C.

Career and Major Research

Karrer studied at the University of Zürich, where he received his doctorate in 1911 under Alfred Werner. He then worked for several years with Paul Ehrlich in Frankfurt am Main, where he studied organic arsenic compounds and their chemotherapeutic action. In 1918 he returned to Zürich, and in 1919 he became professor of organic chemistry at the university, succeeding his teacher Alfred Werner; he remained there for the rest of his career. His research concerned the plant pigments known as carotenoids; he and his co-workers determined the constitution of the carotenes and showed that the yellow pigment beta-carotene is the precursor from which vitamin A is formed in the body. He also studied the flavins and contributed to the chemistry of the coenzymes involved in cellular oxidation.

Nobel Prize-winning Work

The work for which Paul Karrer was awarded the Nobel Prize in Chemistry in 1937 was described by the Nobel Committee as: "for his investigations on carotenoids, flavins and vitamins A and B2".

Karrer's elucidation of the constitution of the carotenoid pigments, and his proof that carotene is converted into vitamin A in the body, connected the plant pigments to the chemistry of the vitamins.

Significance and Impact

Karrer's work established the chemistry of the carotenoids and clarified the relationship between carotene and vitamin A, which is the basis of the vitamin A activity of vegetables. His textbook of organic chemistry, first published in 1928, was used by students throughout the world.

Later Career and Honors

Karrer remained at the University of Zürich until his retirement in 1959. He held honorary doctorates from several universities and was a member of several academies. He died on 1971-06-18 in Zürich, Switzerland.

Legacy

The legacy of Paul Karrer endures through the lasting impact of his scientific contributions. The chemistry of the carotenoids and of vitamin A that he established continues to be built upon by researchers across the chemical and biological sciences.

Modern Developments

Karrer's investigations of carotenoids, flavins, and vitamins A and B2 established the structural and synthetic chemistry of vitamins. His work enabled the industrial production of vitamin A, used today to fortify foods and prevent blindness, and established flavins as components of cellular redox coenzymes. Vitamin and carotenoid research continues to inform nutrition and medicine.

Selected Publications

- Karrer, P. (1928). "Lehrbuch der organischen Chemie." Leipzig: Georg Thieme.
- Karrer, P. (1937). Nobel Lecture on the carotenoids and vitamins, Stockholm.

Chapter Summary

The 1937 prize was shared by Norman Haworth for his investigations on carbohydrates and vitamin C, and by Paul Karrer for his investigations on carotenoids and vitamins. Together their work established the structure and synthesis of the vitamins and carbohydrates.

Chapter 38

1938:Richard Kuhn

The Nobel Prize in Chemistry for the year 1938 was awarded to Richard Kuhn.

Motivation:

for his work on carotenoids and vitamins

38.1 Richard Kuhn

Early Life and Education

Richard Kuhn was born on 1900-12-03 in Vienna, Austria-Hungary (now Austria). He was an Austrian-German biochemist awarded the Nobel Prize in Chemistry in 1938 for his work on carotenoids and vitamins.

Career and Major Research

Kuhn studied chemistry in Vienna and Munich and received his doctorate at the University of Munich in 1922 under Richard Willstätter. He was professor at the Eidgenössische Technische Hochschule in Zürich from 1926 to 1929. In 1929 he became director of the Institute of Chemistry of the Kaiser Wilhelm Institute for Medical Research in Heidelberg, where he worked for the rest of his life. He investigated the carotenoid pigments, determining the structures of many of them, and his laboratory prepared vitamin A from beta-carotene by careful purification. He also isolated riboflavin, the vitamin known as lactoflavin (vitamin B2), established its structure, and achieved its synthesis. In 1938 he was awarded the Nobel Prize in Chemistry, but the German government forbade him to accept it; he received the prize only in 1949.

Nobel Prize-winning Work

The work for which Richard Kuhn was awarded the Nobel Prize in Chemistry in 1938 was described by the Nobel Committee as: "for his work on carotenoids and vitamins".

Kuhn's prize recognized his work on the carotenoids and vitamins. He established the constitutions of many carotenoid pigments and of riboflavin (vitamin B2), and he achieved the synthesis of riboflavin, confirming its structure.

Significance and Impact

Kuhn's investigations laid the chemical foundations for the understanding of the carotenoids and of the B vitamins. His synthesis of riboflavin made the vitamin available in quantity and opened the way to studying its biological function as a component of the flavin coenzymes.

Later Career and Honors

Kuhn continued to direct his institute in Heidelberg after the war, and his later research dealt with enzymes and with natural substances such as the antibiotics. He held honorary doctorates from several universities. He died on 1967-08-01 in Heidelberg, Germany.

Legacy

The legacy of Richard Kuhn endures through the lasting impact of his scientific contributions. The chemistry of the carotenoids and vitamins that he established continues to be built upon by researchers across the chemical and biological sciences.

Modern Developments

Kuhn's synthesis and structural work on vitamins, including vitamin B2 and vitamin B6, helped launch the modern vitamin industry and the understanding of coenzymes. His investigations of carotenoids and natural products continue to inform pigment chemistry and nutrition science. Kuhn was also a pioneer in quantum chemistry before focusing on vitamins.

Selected Publications

- Kuhn, R., & Weygand, F. (1934). "Die Synthese des Lactoflavins." *Berichte der deutschen chemischen Gesellschaft*.

Kuhn's extensive experimental work on the carotenoids and vitamins was published in numerous papers, most of them in *Berichte der deutschen chemischen Gesellschaft* and other German journals of the period.

Chapter Summary

The 1938 prize recognized Richard Kuhn for his work on carotenoids and vitamins. He established the structures of riboflavin (vitamin B₂) and several carotenoids, including the chemistry of the vitamin B group.

Chapter 39

1939: Adolf Butenandt and Leopold Ruzicka

The Nobel Prize in Chemistry for the year 1939 was awarded jointly to Adolf Butenandt and Leopold Ruzicka.

Motivation:

for his work on sex hormones (one half, Adolf Butenandt) and for his work on polymethylenes and higher terpenes (the other half, Leopold Ruzicka)

39.1 Adolf Butenandt

Early Life and Education

Adolf Friedrich Johann Butenandt was born on 1903-03-24 in Lehe, Germany (now part of Bremerhaven). He was a German biochemist who shared the 1939 Nobel Prize in Chemistry with Leopold Ruzicka for his work on sex hormones.

Career and Major Research

Butenandt studied chemistry in Marburg and Göttingen and received his doctorate at the University of Göttingen in 1927 under Adolf Windaus. Working in Windaus's laboratory, he turned to the sex hormones and in 1929 succeeded in crystallizing the female sex hormone estrone from the urine of pregnant women, the first sex hormone to be obtained in pure form. He isolated the male sex hormone androsterone from urine and established its structure in 1934, showing that it is closely related to the female hormones. He was professor at the Technische Hochschule in Danzig from 1933 to 1936. In 1936 he became director of the Kaiser Wilhelm Institute for Biochemistry in Berlin-Dahlem, which moved to Tübingen in 1944 and later, as the Max Planck Institute for Biochemistry, to Munich. Later he studied the insect moulting hormone ecdysone and the silkworm attractant bombykol.

Nobel Prize-winning Work

The work for which Adolf Butenandt was awarded the Nobel Prize in Chemistry in 1939 was described by the Nobel Committee as: "for his work on sex hormones".

Butenandt isolated the female sex hormone estrone in pure crystalline form from pregnancy urine and later obtained the male sex hormone androsterone, establishing the structures of both and showing that the sex hormones form a single chemical family.

Significance and Impact

Butenandt's isolation of the sex hormones made them available for study and for medical use, and his structural work revealed that the male and female hormones are closely related members of the steroid family.

Later Career and Honors

In 1939 Butenandt was awarded the Nobel Prize in Chemistry but was obliged by the German government to decline it; he received the prize formally in 1949. After the war he became one of the most influential figures in German science and served as president of the Max Planck Society from 1960 to 1972. He died on 1995-01-18 in Munich, Germany.

Legacy

The legacy of Adolf Butenandt endures through the lasting impact of his scientific contributions. His isolation of the sex hormones laid the foundation of steroid endocrinology and of the pharmaceutical production of hormonal drugs.

Modern Developments

Butenandt's isolation and structural elucidation of the sex hormones testosterone, progesterone, and the estrogens founded modern steroid endocrinology and the hormonal pharmaceutical industry. These discoveries enabled hormonal contraceptives, hormone replacement therapy, and the use of corticosteroids in medicine, and steroid chemistry remains a cornerstone of the pharmaceutical industry.

Selected Publications

- Butenandt, A. (1929). "Über 'Progynon', ein krystallisiertes weibliches Sexualhormon." *Naturwissenschaften*, 17, 879.
- Butenandt, A., & Tscherning, K. (1934). "Über Androsteron, ein krystallisiertes männliches Sexualhormon."

39.2 Leopold Ruzicka

Early Life and Education

Leopold Ruzicka was born on 1887-09-13 in Vukovar, Croatia, then part of Austria-Hungary. He was a Croatian-born Swiss organic chemist who shared the 1939 Nobel Prize in Chemistry with Adolf Butenandt for his work on sex hormones.

Career and Major Research

Ruzicka studied at the Technische Hochschule in Karlsruhe, where he received his doctorate in 1910 under Hermann Staudinger, working on the chemistry of the ketenes. He followed Staudinger to the Eidgenössische Technische Hochschule in Zürich, where he habilitated and took Swiss citizenship. He was professor at the University of Utrecht from 1926 to 1929 and then professor of organic chemistry at the ETH in Zürich, where he built a large research school. He established the constitution of the natural musk compounds muscone and civetone, showing that they contain large carbon rings, and he synthesized muscone, proving that rings of thirteen and seventeen carbon atoms are stable despite earlier claims to the contrary. His investigations of the higher terpenes established the principles by which these substances are built from isoprene units. With his co-workers he synthesized the male sex hormone testosterone from cholesterol in 1934-1935, applying his knowledge of terpenoid and steroid chemistry.

Nobel Prize-winning Work

The work for which Leopold Ruzicka was awarded the Nobel Prize in Chemistry in 1939 was cited by the committee for his investigations on polymethylenes and higher terpenes. The citation preserved for this chapter reads: "for his work on polymethylenes and higher terpenes".

Ruzicka's demonstration that natural products contain large carbon rings, his synthesis of muscone, and his investigations of the higher terpenes formed the body of work for which he was honoured. Its culmination in the synthesis of the male sex hormone testosterone connected this research directly to the chemistry of the sex hormones.

Significance and Impact

Ruzicka's work overturned a widely held belief about the stability of large carbon rings and opened the field of macrocyclic chemistry, which became important for the synthesis of perfumes and pharmaceuticals. His synthesis of testosterone established a practical route to the steroid hormones.

Later Career and Honors

Ruzicka remained at the ETH in Zürich until his retirement in 1957, and his school trained many organic chemists. He held honorary doctorates from several universities and was elected a foreign member of the Royal Society. He died on 1976-09-26 in Mammern, Switzerland.

Legacy

The legacy of Leopold Ruzicka endures through the lasting impact of his scientific contributions. Macrocyclic chemistry and the chemistry of the terpenes, fields he founded, continue to be built upon by organic chemists throughout the world.

Modern Developments

Ruzicka's work on polymethylenes and higher terpenes established the chemistry of terpenes and steroids, connecting the isoprene unit to the entire family of terpenoid natural products. His synthesis of androsterone and testosterone confirmed the structural relationship between terpenes and steroid hormones, laying the groundwork for the modern steroid drug industry and for the biosynthetic understanding of cholesterol and hormones.

Selected Publications

- Ruzicka, L. (1926). "Zur Kenntnis des Kohlenstoffringes. Über die Konstitution des Muscons." *Helvetica Chimica Acta*.
- Ruzicka, L., & Wettstein, A. (1935). "Über die künstliche Herstellung des männlichen Sexualhormons Testosteron." *Helvetica Chimica Acta*.

Chapter Summary

The 1939 prize recognized Adolf Butenandt for his work on sex hormones and Leopold Ruzicka for his work on polymethylenes and higher terpenes. Their investigations established the steroid and terpene chemistry of biologically active compounds.

Chapter 40

1940:No Prize Awarded

No Nobel Prize in Chemistry was awarded in 1940, when the Second World War suspended the work of the Nobel Committees in Stockholm.

Chapter Summary

The Nobel Committee for Chemistry did not award the prize in 1940, one of the years in which the regular award cycle was interrupted. Such gaps reflect the extraordinary circumstances of world events rather than an absence of noteworthy chemical research.

Chapter 41

1941:No Prize Awarded

No Nobel Prize in Chemistry was awarded in 1941, when the Second World War suspended the work of the Nobel Committees in Stockholm.

Chapter Summary

The Nobel Committee for Chemistry did not award the prize in 1941, one of the years in which the regular award cycle was interrupted. Such gaps reflect the extraordinary circumstances of world events rather than an absence of noteworthy chemical research.

Chapter 42

1942:No Prize Awarded

No Nobel Prize in Chemistry was awarded in 1942, when the Second World War suspended the work of the Nobel Committees in Stockholm.

Chapter Summary

The Nobel Committee for Chemistry did not award the prize in 1942, one of the years in which the regular award cycle was interrupted. Such gaps reflect the extraordinary circumstances of world events rather than an absence of noteworthy chemical research.

Chapter 43

1943:George de Hevesy

The Nobel Prize in Chemistry for the year 1943 was awarded to George de Hevesy.

Motivation:

for his work on the use of isotopes as tracers in the study of chemical processes

43.1 George de Hevesy

Early Life and Education

George Charles de Hevesy, born György Hevesy, was born on 1885-08-01 in Budapest, Hungary, then part of Austria-Hungary. He was a Hungarian-Swedish radiochemist awarded the Nobel Prize in Chemistry in 1943 for his work on the use of isotopes as tracers in the study of chemical processes.

Career and Major Research

de Hevesy studied chemistry in Budapest, Freiburg, Berlin, and Zürich, and received his doctorate at the University of Freiburg in 1908. He worked with Ernest Rutherford in Manchester from 1911 to 1913, where his unsuccessful attempts to separate the radioactive isotope radium D from lead led him to realize that isotopes are chemically inseparable and could instead be used to trace the path of an element through chemical and biological processes. In 1923, together with Dirk Coster, he discovered the element hafnium (atomic number 72) in zirconium minerals, using X-ray spectroscopy. From 1920 he worked at the Niels Bohr Institute in Copenhagen, and in 1926 he became professor at the University of Freiburg; he returned to Copenhagen in 1934. He applied radioactive indicators as tracers in chemistry, biology, and medicine, measuring the absorption and distribution of elements in plants and in animals. In 1943, during the German occupation of Denmark, he fled to Sweden, where he worked at the University of Stockholm and at Uppsala until his retirement.

Nobel Prize-winning Work

The work for which George de Hevesy was awarded the Nobel Prize in Chemistry in 1943 was described by the Nobel Committee as: "for his work on the use of isotopes as tracers in the study of chemical processes".

de Hevesy showed that a radioactive isotope of an element behaves chemically exactly like the ordinary element and can therefore be used to follow that element through a chemical reaction, a plant, or a living organism.

Significance and Impact

The tracer method developed by de Hevesy has become one of the most widely used techniques in all of science. Radioisotopes are now used to follow the course of chemical reactions, the transport of substances in the body, and the fate of atoms in industrial processes.

Later Career and Honors

de Hevesy continued his research in Sweden after the war and returned to Freiburg in 1961 after his retirement. He received the Copley Medal of the Royal Society in 1966 and held honorary doctorates from many universities, and the George Hevesy Medal is awarded for achievements in radioanalytical chemistry. He died on 1966-07-05 in Freiburg im Breisgau, Germany.

Legacy

The legacy of George de Hevesy endures through the lasting impact of his scientific contributions. The isotope tracer method that he invented continues to underpin chemistry, biology, and medicine.

Modern Developments

de Hevesy's invention of radioactive tracers founded the field of radiochemistry and isotope labeling, now indispensable across biochemistry, medicine, and the life sciences. Tracer techniques are used in metabolic studies, drug development, PET imaging, and environmental science. de Hevesy was also the father of X-ray fluorescence analysis, a standard tool in materials science.

Selected Publications

- Coster, D., & Hevesy, G. (1923). "On the Missing Element of Atomic Number 72." *Nature*, 111, 79.
- Hevesy, G., & Paneth, F. (1926). "A Manual of Radioactivity." Oxford: Oxford University Press.

Chapter Summary

The 1943 prize recognized George de Hevesy for his work on the use of isotopes as tracers in the study of chemical processes. His tracer method transformed biochemistry and medicine and led to the discovery of the element hafnium.

Chapter 44

1944:Otto Hahn

The Nobel Prize in Chemistry for the year 1944 was awarded to Otto Hahn.

Motivation:

for his discovery of the fission of heavy nuclei

44.1 Otto Hahn

Early Life and Education

Otto Hahn was born on 1879-03-08 in Frankfurt am Main, Germany. He was a German radiochemist awarded the Nobel Prize in Chemistry in 1944 for his discovery of the fission of heavy nuclei.

Career and Major Research

Hahn studied chemistry at the University of Marburg, where he received his doctorate in 1901 under Theodor Zincke. He worked with William Ramsay in London, where he discovered radiothorium in 1905, and with Ernest Rutherford in Montreal from 1905 to 1906. From 1912 he headed the radioactivity department of the Kaiser Wilhelm Institute for Chemistry in Berlin-Dahlem, where he worked for many years with Lise Meitner and later with Fritz Strassmann. He discovered mesothorium and, together with Meitner, the element protactinium in 1917. In 1938, together with Fritz Strassmann, he discovered nuclear fission: bombarding uranium with neutrons, they detected barium among the reaction products and showed that the uranium nucleus had split into lighter nuclei.

Nobel Prize-winning Work

The work for which Otto Hahn was awarded the Nobel Prize in Chemistry in 1944 was described by the Nobel Committee as: "for his discovery of the fission of heavy nuclei".

The chemical evidence that uranium, on bombardment with neutrons, is converted into lighter elements such as barium established that the atomic nucleus can be split. The process was immediately explained theoretically by Lise Meitner and Otto Frisch, who gave it the name fission.

Significance and Impact

The discovery of nuclear fission changed the course of history. It made possible both the atomic bomb and the peaceful use of nuclear energy, and it earned Hahn the 1944 Nobel Prize in Chemistry, which was presented to him after the end of the Second World War.

Later Career and Honors

After the war Hahn was president of the Kaiser Wilhelm Society and, from 1946, of its successor the Max Planck Society, serving until 1960. He received the Enrico Fermi Award together with Meitner and

Strassmann in 1966, and he was an outspoken opponent of nuclear weapons. He died on 1968-07-28 in Göttingen, Germany.

Legacy

The legacy of Otto Hahn endures through the lasting impact of his scientific contributions. The discovery of nuclear fission, made in his laboratory, remains one of the defining events of twentieth-century science.

Modern Developments

Hahn's discovery of nuclear fission changed the course of science and society, enabling nuclear energy, nuclear medicine, and nuclear weapons. Fission powers modern nuclear reactors that generate about ten percent of global electricity, and fission products are used in medicine, industry, and research. Radiochemistry founded by Hahn remains essential to the nuclear industry.

Selected Publications

- Hahn, O., & Strassmann, F. (1939). "Über den Nachweis und das Verhalten der bei der Bestrahlung des Urans mittels Neutronen entstehenden Erdalkalimetalle." *Naturwissenschaften*, 27(1), 11–15.
- Hahn, O. (1962). "Vom Radiothor zur Uranspaltung." Braunschweig: Vieweg.

Chapter Summary

The 1944 prize recognized Otto Hahn for his discovery of the fission of heavy nuclei. The 1938 discovery of nuclear fission, with Fritz Strassmann, opened the nuclear age and led both to nuclear weapons and to nuclear energy.

Chapter 45

1945: Artturi Virtanen

The Nobel Prize in Chemistry for the year 1945 was awarded to Artturi Virtanen.

Motivation:

for his research and inventions in agricultural and nutrition chemistry, especially for his fodder preservation method

45.1 Artturi Virtanen

Early Life and Education

Artturi Virtanen was born on 1895-01-15 in Helsinki, Finland.

Career and Major Research

Virtanen studied chemistry at the University of Helsinki and spent his career in agricultural and food chemistry. He worked at the research laboratory of the Valio cooperative, the Finnish Butter Export Association, where he studied the preservation of green fodder. In the 1920s he developed the AIV method, named after his initials, in which fresh forage is acidified with dilute hydrochloric or sulphuric acid to a pH of about 3 to 4 to stop spoilage fermentation. He later became professor at the Helsinki University of Technology.

Nobel Prize-winning Work

The work for which Artturi Virtanen was awarded the Nobel Prize in Chemistry in 1945 was described by the Nobel Committee as: "for his research and inventions in agricultural and nutrition chemistry, especially for his fodder preservation method".

The AIV method preserved silage with far lower losses of nutrients than traditional ensiling or haymaking, while keeping the fodder palatable to livestock. The process was introduced into farm practice in Finland in 1929.

Significance and Impact

Virtanen's method solved a practical problem of northern agriculture, where the growing season is short and fodder had to be stored for winter feeding. His demonstration that a low pH stops the degradation of proteins and vitamins became the basis of modern silage science. The method spread to other countries with cold, humid climates.

Later Career and Honors

Virtanen continued research in biochemistry, including work on vitamins and on the fixation of nitrogen by the root nodules of leguminous plants. He remained at the Helsinki University of Technology for most of his career. He died in 1973.

Legacy

Virtanen's name survives in the AIV process, and the principle of acid-based fodder preservation remains in worldwide use in ensiling practice.

Modern Developments

Virtanen's work on fodder preservation and food chemistry improved agriculture and nutrition in Finland and beyond, leading to modern methods of silage and food storage that preserve vitamins and nutrients. His AIV method remains influential in animal husbandry, and his broader food chemistry research foreshadowed modern food science and the preservation of nutritional quality.

Selected Publications

- *The A.I.V. method of preserving fresh fodder*, Empire Journal of Experimental Agriculture, 1933.

Chapter Summary

The 1945 prize recognized a method for preserving cattle fodder that cut the heavy losses of nutrients suffered in traditional silage and haymaking. Artturi Virtanen's AIV process applied dilute mineral acid to fresh forage to prevent fermentative spoilage, and it came into wide use in Finland and other northern countries.

Chapter 46

1946:James B. Sumner and John H. Northrop and Wendell M. Stanley

The Nobel Prize in Chemistry for the year 1946 was awarded jointly to James B. Sumner and John H. Northrop and Wendell M. Stanley.

Motivation:

for his discovery that enzymes can be crystallized (one half, James B. Sumner) and for their preparation of enzymes and virus proteins in a pure form (the other half, jointly, John H. Northrop and Wendell M. Stanley)

46.1 James B. Sumner

Early Life and Education

James B. Sumner was born on 1887-11-19 in Canton, Massachusetts, USA.

Career and Major Research

Sumner studied at Harvard University, where he earned his doctorate in 1914. He then joined Cornell University, where he spent his career as a professor of biochemistry. In 1926 he isolated and crystallized the enzyme urease from jack beans, proving that a purified enzyme is a protein. The result was disputed for years by chemists who believed enzymes were non-protein substances.

Nobel Prize-winning Work

The work for which James B. Sumner was awarded the Nobel Prize in Chemistry in 1946 was described by the Nobel Committee as: "for his discovery that enzymes can be crystallized".

Crystallization made it possible to obtain an enzyme as a pure chemical substance and to show unambiguously that the enzyme itself was protein in nature.

Significance and Impact

Sumner's result settled the question of the chemical nature of enzymes and opened the way to their purification and detailed study. It established enzyme chemistry as a field in which individual catalysts could be isolated, characterized, and compared.

Later Career and Honors

Sumner continued his research at Cornell University until his retirement. He died in 1955.

Legacy

The crystallization of urease is regarded as the demonstration that enzymes are proteins, a cornerstone of modern biochemistry and molecular biology.

Modern Developments

Sumner's crystallization of urease proved that enzymes are proteins and founded the modern field of enzymology. This landmark discovery enabled the industrialization of enzymes, which are now used in detergents, food processing, biofuels, and pharmaceutical synthesis. Protein purification and crystallization remain fundamental techniques in structural biology and drug discovery.

Selected Publications

- *The Isolation and Crystallization of the Enzyme Urease*, Journal of Biological Chemistry, 1926.

46.2 John H. Northrop

Early Life and Education

John H. Northrop was born on 1891-07-05 in Yonkers, New York, USA.

Career and Major Research

Northrop studied chemistry at Columbia University and joined the Rockefeller Institute for Medical Research, where he worked for most of his career. In 1929 he crystallized the enzyme pepsin, and later crystallized the digestive enzymes trypsin and chymotrypsin. These results independently confirmed Sumner's conclusion that enzymes are proteins.

Nobel Prize-winning Work

The work for which John H. Northrop was awarded the Nobel Prize in Chemistry in 1946 was described by the Nobel Committee as: "for their preparation of enzymes and virus proteins in a pure form".

Northrop's crystallization of pepsin and the other protein-digesting enzymes established beyond doubt that enzymes are proteins and could be purified to crystalline form.

Significance and Impact

Northrop's work provided the decisive confirmation, from an independent laboratory, that enzymes are proteins. It also made crystalline enzyme preparations available for structural and kinetic study.

Later Career and Honors

Northrop remained at the Rockefeller Institute for Medical Research, where he directed work in enzyme chemistry. He died in 1987.

Legacy

The crystalline enzymes prepared by Northrop were among the first proteins to be studied as pure chemical substances, helping to found protein chemistry.

Modern Developments

Northrop's crystallization of pepsin and other proteolytic enzymes demonstrated the general protein nature of enzymes and founded the modern enzyme industry. His methods for protein isolation and characterization underlie current biotechnology, including the production of recombinant enzymes, therapeutic proteins, and the crystallographic analysis that powers structure-based drug design.

Selected Publications

- *Crystalline Pepsin*, Science, 1929.

46.3 Wendell M. Stanley

Early Life and Education

Wendell M. Stanley was born on 1904-08-16 in Ridgeville, Indiana, USA.

Career and Major Research

Stanley studied at Earlham College and the University of Illinois, where he received his doctorate in 1929. He worked at the Rockefeller Institute for Medical Research in Princeton, New Jersey. In 1935 he isolated tobacco mosaic virus in crystalline form and showed that the virus particle consisted largely of protein.

Nobel Prize-winning Work

The work for which Wendell M. Stanley was awarded the Nobel Prize in Chemistry in 1946 was described by the Nobel Committee as: "for their preparation of enzymes and virus proteins in a pure form".

Stanley applied crystallization to a virus rather than an enzyme, obtaining tobacco mosaic virus as a crystalline material that retained its infectivity. The work showed that a virus could behave as a well-defined chemical substance.

Significance and Impact

Stanley's crystallization of tobacco mosaic virus demonstrated that viruses, too, could be studied as pure chemical entities. It helped establish the molecular basis of virology and the protein chemistry of viruses.

Later Career and Honors

Stanley later served as professor of biochemistry at the University of California, Berkeley, where he directed a laboratory of virology. He died in 1971.

Legacy

Stanley's work joined the study of viruses to chemistry, and his crystalline tobacco mosaic virus became a model system in the early development of molecular biology.

Modern Developments

Stanley's crystallization of tobacco mosaic virus opened the field of virology and showed that viruses could be studied as molecular entities. His work pioneered modern virology, leading to vaccines against polio, influenza, and COVID-19, and to the current understanding of viruses in human and plant disease. Stanley's isolation methods also influenced the development of modern protein and particle biochemistry.

Selected Publications

- *Isolation of a Crystalline Protein Possessing the Properties of Tobacco-Mosaic Virus*, Science, 1935.

Chapter Summary

The 1946 prize recognized the crystallization of enzymes, which proved that enzymes are proteins. James B. Sumner crystallized urease, John H. Northrop crystallized pepsin and other digestive enzymes, and Wendell M. Stanley extended the approach to tobacco mosaic virus, showing that a virus could be obtained as crystalline material.

Chapter 47

1947: Sir Robert Robinson

The Nobel Prize in Chemistry for the year 1947 was awarded to Sir Robert Robinson.

Motivation:

for his investigations on plant products of biological importance, especially the alkaloids

47.1 Sir Robert Robinson

Early Life and Education

Sir Robert Robinson was born on 1886-09-13 in Rufford, England.

Career and Major Research

Robinson studied at the University of Manchester under William Henry Perkin Jr., and later held professorships at the universities of Sydney, Liverpool, and Manchester. In 1930 he became Waynflete Professor of Chemistry at Oxford, a post he held until 1955. He carried out landmark work on the structure and synthesis of natural products, including alkaloids such as morphine and strychnine, and of plant pigments such as the anthocyanins.

Nobel Prize-winning Work

The work for which Sir Robert Robinson was awarded the Nobel Prize in Chemistry in 1947 was described by the Nobel Committee as: "for his investigations on plant products of biological importance, especially the alkaloids".

In 1917 Robinson achieved a one-step synthesis of tropinone, the key ring system of the alkaloids cocaine and atropine, from succinaldehyde, methylamine, and acetone. The synthesis was a striking demonstration of biomimetic thinking in organic chemistry.

Significance and Impact

Robinson's work established the structures of major classes of plant products and showed how complex ring systems found in nature could be built in the laboratory. His electronic theory of organic reactions, presented in 1932, strongly influenced theoretical organic chemistry.

Later Career and Honors

Robinson served as president of the Royal Society from 1945 to 1950. He was knighted in 1939 and was awarded the Order of Merit. He died in 1975.

Legacy

Robinson's synthesis of tropinone remains a classic of organic synthesis, and his structural work on alkaloids and plant pigments is foundational to natural product chemistry.

Modern Developments

Robinson's structural elucidation of natural products, including alkaloids and steroids, founded the modern science of natural product chemistry and total synthesis. His investigation of the biosynthesis of alkaloids led to the concept of biomimetic synthesis and the study of how plants build complex molecules. Robinson's work underpins the design of alkaloid-derived drugs such as morphine-based analgesics and antimalarials.

Selected Publications

- *A Synthesis of Tropinone*, Journal of the Chemical Society, 1917.

Chapter Summary

The 1947 prize recognized investigations of plant products of biological importance, especially the alkaloids. Robert Robinson's studies of alkaloid structures and his 1917 synthesis of tropinone laid the groundwork for the modern chemistry of natural products.

Chapter 48

1948:Arne Tiselius

The Nobel Prize in Chemistry for the year 1948 was awarded to Arne Tiselius.

Motivation:

for his research on electrophoresis and adsorption analysis, especially for his discoveries concerning the complex nature of the serum proteins

48.1 Arne Tiselius

Early Life and Education

Arne Tiselius was born on 1902-08-10 in Stockholm, Sweden.

Career and Major Research

Tiselius studied chemistry at Uppsala University, where he received his doctorate under The Svedberg. He spent his career at Uppsala University. In 1937 he built a new apparatus for moving-boundary electrophoresis, in which proteins migrate in an electric field and can be measured as separate moving fronts.

Nobel Prize-winning Work

The work for which Arne Tiselius was awarded the Nobel Prize in Chemistry in 1948 was described by the Nobel Committee as: "for his research on electrophoresis and adsorption analysis, especially for his discoveries concerning the complex nature of the serum proteins".

With his improved electrophoresis apparatus, Tiselius showed that the serum proteins form a complex mixture of several distinct components, notably albumins and different globulins. He also developed adsorption analysis, including methods for chromatographic separation of biological substances.

Significance and Impact

Tiselius's electrophoretic separation of serum proteins revealed the complexity of the blood proteins and became a standard tool in biochemistry and clinical chemistry. Electrophoresis in its modern forms descends directly from his apparatus.

Later Career and Honors

Tiselius continued at Uppsala University, where he also served as head of the Institute of Biochemistry. He later served as president of the Nobel Foundation. He died in 1971.

Legacy

The electrophoretic methods introduced by Tiselius are fundamental to protein analysis, and his demonstration of the multiple components of serum proteins opened the field of clinical protein chemistry.

Modern Developments

Tiselius's invention of electrophoresis made possible the separation and analysis of proteins, founding modern protein chemistry and clinical diagnostics. Electrophoresis remains a cornerstone of biochemistry and medicine, from hemoglobin and serum protein analysis to DNA sequencing and forensic genetics. His moving-boundary method evolved into today's gel, capillary, and microfluidic electrophoresis.

Selected Publications

- *A New Apparatus for Electrophoretic Analysis of Colloidal Mixtures*, Transactions of the Faraday Society, 1937.

Chapter Summary

The 1948 prize recognized research on electrophoresis and adsorption analysis. Arne Tiselius's moving-boundary electrophoresis apparatus revealed that the serum proteins consist of several distinct components, and his methods became fundamental tools of protein chemistry.

Chapter 49

1949: William F. Giaque

The Nobel Prize in Chemistry for the year 1949 was awarded to William F. Giaque.

Motivation:

for his contributions in the field of chemical thermodynamics, particularly concerning the behaviour of substances at extremely low temperatures

49.1 William F. Giaque

Early Life and Education

William F. Giaque was born on 1895-05-12 in Niagara Falls, Ontario, Canada.

Career and Major Research

Giaque moved to the United States as a young man and studied at the University of California, Berkeley, where he earned his doctorate in 1922 and spent his entire scientific career. With Herrick Johnston he discovered the isotopes oxygen-17 and oxygen-18 in 1929, work that showed oxygen could no longer serve as the sole basis of the atomic weight scale.

Nobel Prize-winning Work

The work for which William F. Giaque was awarded the Nobel Prize in Chemistry in 1949 was described by the Nobel Committee as: "for his contributions in the field of chemical thermodynamics, particularly concerning the behaviour of substances at extremely low temperatures".

Working with D. P. MacDougall, Giaque applied adiabatic demagnetization of a paramagnetic salt to reach temperatures below one degree absolute, attaining about 0.25 K in 1933. The method, proposed independently by Giaque and Peter Debye, made possible the study of entropy and heat capacity of substances at the lowest attainable temperatures.

Significance and Impact

The demagnetization method and Giaque's low-temperature calorimetry provided critical tests of the third law of thermodynamics. His careful measurements of entropies helped resolve discrepancies between statistical and thermal entropies of gases.

Later Career and Honors

Giaque remained at Berkeley, where he continued research in low-temperature thermodynamics. He died in 1982.

Legacy

Adiabatic demagnetization, introduced by Giauque, remains the standard means of reaching temperatures in the millikelvin range, and his thermodynamic methods shaped modern low-temperature physics and chemistry.

Modern Developments

Giauque's low-temperature thermodynamics and discovery of adiabatic demagnetization made temperatures near absolute zero experimentally accessible. These techniques enabled the study of superconductivity, superfluidity, and quantum phenomena, and his magnetic cooling method underpins modern dilution refrigerators used in quantum computing research.

Selected Publications

- *Attainment of Temperatures Below 1° Absolute by Demagnetization of $Gd_2(SO_4)_3 \cdot 8H_2O$* , Physical Review, 1933.

Chapter Summary

The 1949 prize recognized contributions to chemical thermodynamics, especially the behaviour of substances at extremely low temperatures. William F. Giauque's adiabatic demagnetization technique reached temperatures near absolute zero and enabled precise tests of the third law of thermodynamics.

Chapter 50

1950:Otto Diels and Kurt Alder

The Nobel Prize in Chemistry for the year 1950 was awarded jointly to Otto Diels and Kurt Alder.

Motivation:

for their discovery and development of the diene synthesis

50.1 Otto Diels

Early Life and Education

Otto Diels was born on 1876-01-23 in Hamburg, Germany.

Career and Major Research

Diels studied chemistry at the University of Berlin under Emil Fischer, and in 1916 became professor and director of the Institute of Chemistry at the University of Kiel, where he remained until his retirement in 1945. With his student Kurt Alder he discovered the diene synthesis in 1928. Diels had earlier discovered the reactive ketenes (1905) and had isolated the product of a dehydrogenation of cholesterol that later proved to be a steroid.

Nobel Prize-winning Work

The work for which Otto Diels was awarded the Nobel Prize in Chemistry in 1950 was described by the Nobel Committee as: "for their discovery and development of the diene synthesis".

The diene synthesis, now called the Diels-Alder reaction, joins a conjugated diene and a dienophile in a single step to form a six-membered ring. Reported by Diels and Alder in 1928, it provides a general route to cyclic organic compounds.

Significance and Impact

The Diels-Alder reaction became one of the most important methods in synthetic organic chemistry for constructing six-membered rings. It was applied widely in the chemical industry, including the manufacture of synthetic rubber and plastics.

Later Career and Honors

Diels retired from Kiel in 1945 and remained in Kiel until his death on 1954-03-07.

Legacy

The Diels-Alder reaction, named for Diels and Alder, remains a standard and general reaction in organic synthesis, used in both academic research and industry.

Modern Developments

The Diels–Alder reaction is one of the most powerful carbon-carbon bond-forming reactions in organic chemistry, used throughout pharmaceutical synthesis, polymer chemistry, and the total synthesis of natural products. Modern applications include the synthesis of complex drug molecules, the fabrication of molecular materials, and the Diels–Alder-derived chemistry in bioorthogonal click reactions.

Selected Publications

- *Synthesen in der hydroaromatischen Reihe*, Justus Liebigs Annalen der Chemie, 1928.

50.2 Kurt Alder

Early Life and Education

Kurt Alder was born on 1902-07-10 in Königshütte, Germany (now Chorzów, Poland).

Career and Major Research

Alder studied chemistry in Berlin and took his doctorate at the University of Kiel in 1926 under Otto Diels. He worked with Diels on the diene synthesis, then joined the I. G. Farbenindustrie works at Leverkusen, where he worked on the synthetic rubber Buna. In 1940 he became professor of chemistry at the University of Cologne.

Nobel Prize-winning Work

The work for which Kurt Alder was awarded the Nobel Prize in Chemistry in 1950 was described by the Nobel Committee as: "for their discovery and development of the diene synthesis".

Alder contributed to the systematic exploration of the diene synthesis, establishing its scope and its rules of orientation, and applying it to the construction of many ring systems.

Significance and Impact

Alder's investigations extended the diene synthesis from a single observation into a general reaction of predictable outcome. The reaction proved essential in the synthesis of natural products and industrial polymers.

Later Career and Honors

Alder remained at the University of Cologne as professor and director of its Institute of Chemistry. He died on 1958-06-20 in Cologne.

Legacy

The Diels-Alder reaction and the Alder-ene reaction, both discovered in his laboratories, are named in his honor and remain widely used in organic synthesis.

Modern Developments

The Diels-Alder reaction is one of the most widely used methods for constructing carbon-carbon bonds in organic synthesis. It features prominently in the industrial and laboratory synthesis of pharmaceuticals, agrochemicals, and natural products, and its stereochemical control is a model example for modern pericyclic reaction theory developed decades later.

Selected Publications

- *Synthesen in der hydroaromatischen Reihe*, Justus Liebigs Annalen der Chemie, 1928.

Chapter Summary

The 1950 prize recognized the discovery and development of the diene synthesis. Otto Diels and Kurt Alder showed in 1928 that a conjugated diene and a dienophile combine in one step to form a six-membered ring, and the reaction became one of the most widely used methods in synthetic organic chemistry.

Chapter 51

1951:Edwin M. McMillan and Glenn T. Seaborg

The Nobel Prize in Chemistry for the year 1951 was awarded jointly to Edwin M. McMillan and Glenn T. Seaborg.

Motivation:

for their discoveries in the chemistry of the transuranium elements

51.1 Edwin M. McMillan

Early Life and Education

Edwin M. McMillan was born on 1907-09-18 in Redondo Beach, California, USA.

Career and Major Research

McMillan studied at the California Institute of Technology, where he received his doctorate in 1932, and then joined the Radiation Laboratory of the University of California, Berkeley, working with Ernest O. Lawrence's cyclotron group. In 1940 he discovered the first transuranium element, neptunium, together with Philip Abelson, by bombarding uranium with neutrons and showing that the resulting 2.3-day activity belonged to a new element of atomic number 93. He also introduced the principle of phase stability, which underlies the design of the synchrotron and of modern particle accelerators.

Nobel Prize-winning Work

The work for which Edwin M. McMillan was awarded the Nobel Prize in Chemistry in 1951 was described by the Nobel Committee as: "for their discoveries in the chemistry of the transuranium elements".

The discovery of neptunium showed that elements heavier than uranium could be produced and identified in the laboratory, opening the study of the transuranium elements and establishing that the actinide series could be extended beyond uranium.

Significance and Impact

McMillan's work demonstrated that the periodic table could be extended beyond uranium through neutron capture and beta decay. His discovery, and the plutonium work that followed from it in Berkeley, founded the chemistry of the transuranium elements and created the materials used in the first atomic weapons.

Later Career and Honors

During the Second World War McMillan worked on radar at the Massachusetts Institute of Technology and at the Los Alamos Laboratory. After the war he returned to Berkeley and served as director of the Radiation Laboratory. He died on 1991-09-07 in El Cerrito, California.

Legacy

McMillan is remembered as a discoverer of neptunium and as the originator of the principle of phase stability, which made possible the synchrotron and the large accelerators of modern particle physics.

Modern Developments

McMillan's discovery of the transuranium elements neptunium and plutonium extended the periodic table and made the nuclear era possible. Plutonium became the fuel of modern nuclear reactors and weapons, while research on transuranics continues at facilities such as Berkeley and in the synthesis of superheavy elements. McMillan's work also included early insights into the synchrotron principle used in particle accelerators worldwide.

Selected Publications

- *Radioactive Element 93*, Physical Review, 1940.

51.2 Glenn T. Seaborg

Early Life and Education

Glenn T. Seaborg was born on 1912-04-19 in Ishpeming, Michigan, USA.

Career and Major Research

Seaborg studied at the University of California, Los Angeles, and took his doctorate at the University of California, Berkeley, in 1937. He spent his career at Berkeley, where in February 1941 he and his colleagues Joseph W. Kennedy and Arthur C. Wahl produced and isolated plutonium. He went on to co-discover several further transuranium elements, including americium and curium, and in 1944 proposed the actinide concept, placing the transuranium elements in a separate series in the periodic table.

Nobel Prize-winning Work

The work for which Glenn T. Seaborg was awarded the Nobel Prize in Chemistry in 1951 was described by the Nobel Committee as: "for their discoveries in the chemistry of the transuranium elements".

Seaborg's discovery of plutonium and his isolation of the transuranium elements established a new branch of chemistry. His actinide concept correctly predicted the chemical behavior and placement of the elements beyond actinium in the periodic table.

Significance and Impact

The chemistry of the transuranium elements founded by Seaborg became essential to nuclear energy, since plutonium-239 proved to be a fissionable material usable in nuclear reactors and weapons. His actinide concept organized the heaviest elements into a coherent group and guided the search for still heavier synthetic elements.

Later Career and Honors

Seaborg was a professor at the University of California, Berkeley, and served as director of the Lawrence Berkeley Laboratory. He was also chairman of the United States Atomic Energy Commission from 1961 to 1971. Element 106, discovered in his laboratory, was named seaborgium in his honor, making him the first person to have an element named after him while still living. He died on 1999-02-25 in Lafayette, California.

Legacy

Seaborg co-discovered ten transuranium elements and transformed the far end of the periodic table, and his name is preserved in the element seaborgium. His work remains central to the chemistry of the actinides and to the technology of nuclear power.

Modern Developments

Seaborg's actinide concept reorganized the periodic table and guided the discovery of ten transuranium elements, from plutonium to seaborgium. These elements are central to nuclear energy, radiopharmaceuticals, and national security science, and Seaborg's periodic table reformulation is now standard in textbooks. Modern heavy-element and superheavy-element chemistry builds directly on his framework.

Selected Publications

- *Radioactive Element 94 from Deuterons on Uranium*, Physical Review, 1946.

Chapter Summary

The 1951 prize recognized the discovery of the transuranium elements. Edwin M. McMillan and Philip Abelson produced the first of them, neptunium, in 1940, and Glenn T. Seaborg and his colleagues produced and isolated plutonium in 1941, work that established the chemistry of the elements beyond uranium and led to the actinide concept.

Chapter 52

1952:Archer J.P. Martin and Richard L.M. Synge

The Nobel Prize in Chemistry for the year 1952 was awarded jointly to Archer J.P. Martin and Richard L.M. Synge.

Motivation:

for their invention of partition chromatography

52.1 Archer J.P. Martin

Early Life and Education

Archer J.P. Martin was born on 1910-03-01 in London, England.

Career and Major Research

Martin studied biochemistry at the University of Cambridge and obtained his doctorate there in 1936. He worked in the laboratories of the Wool Industries Research Association in Leeds, where he collaborated with Richard L.M. Synge. In 1941 they invented partition chromatography, separating amino acids between a moving solvent and a stationary water phase held on silica gel. Martin later developed gas-liquid chromatography with A.T.M. James in 1952.

Nobel Prize-winning Work

The work for which Archer J.P. Martin was awarded the Nobel Prize in Chemistry in 1952 was described by the Nobel Committee as: "for their invention of partition chromatography".

Partition chromatography separated the components of a mixture on the basis of their differing distribution between two liquid phases, making it possible to separate and identify closely related substances such as the amino acids of a protein hydrolysate.

Significance and Impact

The method gave biochemists a practical way to separate the amino acids and other small molecules in complex mixtures, transforming the analysis of proteins and natural products. It became the ancestor of paper chromatography, gas-liquid chromatography, and the high-performance liquid chromatography used widely today.

Later Career and Honors

Martin later worked at the National Institute for Medical Research in London and was a professor at the University of Cambridge. He died on 2002-07-28.

Legacy

Partition chromatography and its successors became indispensable tools in chemistry, biochemistry, and clinical analysis, and Martin's name is linked with the chromatographic methods that dominate modern analytical chemistry.

Modern Developments

Martin's partition chromatography, developed with Synge, founded modern separation science and made the development of insulin and other proteins possible. Chromatography is now the workhorse of pharmaceutical analysis, environmental testing, and the biotechnology industry, and paper chromatography became the basis for early genome sequencing efforts.

Selected Publications

- *A New Form of Chromatogram Employing Two Liquid Phases*, Biochemical Journal, 1941.

52.2 Richard L.M. Synge

Early Life and Education

Richard L.M. Synge was born on 1914-10-28 in Liverpool, England.

Career and Major Research

Synge studied at Winchester College and Trinity College, Cambridge, and took his doctorate at Cambridge in 1941. He worked at the Wool Industries Research Association in Leeds, where he and Archer J.P. Martin developed partition chromatography for the separation of amino acids. He later worked at the Rowett Research Institute in Aberdeen, where he applied the method to peptides and other biological materials.

Nobel Prize-winning Work

The work for which Richard L.M. Synge was awarded the Nobel Prize in Chemistry in 1952 was described by the Nobel Committee as: "for their invention of partition chromatography".

Synge's theoretical and experimental work with Martin turned the idea of a partition-based separation into a working chromatographic method, and his application of it to the study of peptides helped establish protein chemistry on a sound analytical basis.

Significance and Impact

The method provided a rapid, sensitive means of analyzing mixtures of amino acids and peptides, essential to the later determination of protein structure. Its influence extended into medicine, where chromatographic methods became standard for the analysis of biological fluids.

Later Career and Honors

Synge moved to the Food Research Institute in Norwich and remained active in carbohydrate chemistry and protein analysis. He died on 1994-08-18.

Legacy

Synge's work with Martin founded modern chromatographic analysis, and his contributions are remembered in the widespread laboratory use of partition and related chromatographic techniques.

Modern Developments

The Synge–Martin paper chromatography technique transformed biochemical analysis and made possible the identification of amino acids, sugars, and nucleotides in complex mixtures. Chromatography in all its modern forms, including HPLC and gas chromatography, descends from this foundational work and is used in every pharmaceutical, clinical, and environmental laboratory.

Selected Publications

- *A New Form of Chromatogram Employing Two Liquid Phases*, Biochemical Journal, 1941.

Chapter Summary

The 1952 prize recognized the invention of partition chromatography. Archer J.P. Martin and Richard L.M. Synge devised the method in 1941 while working in Leeds, using the differing distribution of substances between two liquid phases to separate amino acids and other components of complex mixtures.

Chapter 53

1953:Hermann Staudinger

The Nobel Prize in Chemistry for the year 1953 was awarded to Hermann Staudinger.

Motivation:

for his discoveries in the field of macromolecular chemistry

53.1 Hermann Staudinger

Early Life and Education

Hermann Staudinger was born on 1881-03-23 in Worms, Germany.

Career and Major Research

Staudinger studied chemistry at the universities of Halle, Darmstadt, and Munich, and received his doctorate from the University of Halle in 1903. He held professorships at the University of Karlsruhe, the ETH Zurich, and from 1926 the University of Freiburg, where he remained until his retirement. From 1920 onward he argued that substances such as rubber, cellulose, and polystyrene were macromolecules: long chains of repeating units held together by ordinary covalent bonds, rather than aggregates of small molecules. He introduced the concept of macromolecules and the term "polymer" in its modern sense.

Nobel Prize-winning Work

The work for which Hermann Staudinger was awarded the Nobel Prize in Chemistry in 1953 was described by the Nobel Committee as: "for his discoveries in the field of macromolecular chemistry".

Staudinger established that polymers are genuine large molecules with molecular weights of tens of thousands or more, built from covalently linked monomer units. He supported the idea with experiments showing that the viscosity of polymer solutions increases with molecular weight, and he demonstrated that polymerization proceeds by chain reactions.

Significance and Impact

Staudinger's macromolecular concept resolved a long controversy and created the scientific foundation of polymer chemistry. It made possible the rational design of synthetic plastics, fibers, and rubbers, and it provided the framework for understanding natural macromolecules such as proteins and cellulose.

Later Career and Honors

After retiring in 1951, Staudinger continued to work in Freiburg and remained an active advocate of the macromolecular concept until his death. He died on 1965-09-08 in Freiburg, Germany.

Legacy

Staudinger is regarded as the father of polymer chemistry, and his concept of macromolecules underpins the modern plastics and materials industries as well as the molecular understanding of biological polymers.

Modern Developments

Staudinger's concept of macromolecules, showing that polymers are long chains of covalently bonded monomers, founded modern polymer science and the plastics industry. His ideas underpin the design of plastics, synthetic rubbers, fibers, and biomaterials used throughout daily life. The field of polymer chemistry he created now spans everything from packaging and electronics to advanced composites and biomedical implants.

Selected Publications

- *Über Polymerisation*, Berichte der deutschen chemischen Gesellschaft, 1920.

Chapter Summary

The 1953 prize recognized the establishment of the macromolecular concept. Hermann Staudinger showed that polymers such as rubber and cellulose consist of very large molecules of covalently linked repeating units, a view that founded polymer science and became the basis of the synthetic materials industry.

Chapter 54

1954:Linus Pauling

The Nobel Prize in Chemistry for the year 1954 was awarded to Linus Pauling.

Motivation:

for his research into the nature of the chemical bond and its application to the elucidation of the structure of complex substances

54.1 Linus Pauling

Early Life and Education

Linus Pauling was born on 1901-02-28 in Portland, Oregon, USA.

Career and Major Research

Pauling studied chemical engineering at Oregon Agricultural College and took his doctorate at the California Institute of Technology in 1925. He spent most of his career at Caltech, where he became professor of chemistry. He applied quantum mechanics to chemistry, introducing the concepts of hybrid orbitals, electronegativity, and resonance, which he presented systematically in *The Nature of the Chemical Bond* (1939). He also used X-ray crystallography to determine the structures of minerals and, later, the structure of the alpha helix in proteins.

Nobel Prize-winning Work

The work for which Linus Pauling was awarded the Nobel Prize in Chemistry in 1954 was described by the Nobel Committee as: "for his research into the nature of the chemical bond and its application to the elucidation of the structure of complex substances".

Pauling showed that the properties of chemical bonds, including bond lengths, bond angles, and energies, could be understood and predicted from quantum mechanical principles. His concept of resonance described molecules such as benzene as hybrids of contributing structures, and his scale of electronegativity explained the polarity of bonds between different elements.

Significance and Impact

The Nature of the Chemical Bond became the standard account of chemical bonding and shaped the teaching of chemistry for decades. Pauling's methods were used to determine the structures of complex substances from minerals to proteins, and his alpha helix proposal was fundamental to the later solution of protein structures.

Later Career and Honors

Pauling left Caltech in the 1960s and later worked at the Center for the Study of Democratic Institutions in Santa Barbara and Stanford University. In 1962 he was awarded the Nobel Peace Prize for his campaign against nuclear weapons testing, making him the only person to receive two unshared Nobel Prizes. He died on 1994-08-19 in Big Sur, California.

Legacy

Pauling is remembered as one of the most influential chemists of the twentieth century, whose work defined modern structural chemistry. His name survives in concepts taught to every student of chemistry, and his double Nobel Prize remains unique.

Modern Developments

Pauling's work on the nature of the chemical bond, including his concept of hybrid orbitals and electronegativity, remains the foundation of modern quantum chemistry and the understanding of molecular structure. His later identification of the alpha-helix and beta-sheet structures of proteins founded structural biology and protein science. Pauling's legacy also extends to medicine, where his ideas shaped research on vitamins and disease, and to the broader application of chemistry to biology.

Selected Publications

- *The Nature of the Chemical Bond*, Cornell University Press, 1939.

Chapter Summary

The 1954 prize recognized research into the nature of the chemical bond. Linus Pauling explained bonding with quantum mechanics, introduced concepts such as electronegativity and resonance, and applied them to the structures of complex substances, work summed up in *The Nature of the Chemical Bond* and later honored with the 1962 Nobel Peace Prize.

Chapter 55

1955: Vincent du Vigneaud

The Nobel Prize in Chemistry for the year 1955 was awarded to Vincent du Vigneaud.

Motivation:

for his work on biochemically important sulphur compounds, especially for the first synthesis of a polypeptide hormone

55.1 Vincent du Vigneaud

Early Life and Education

Vincent du Vigneaud was born on 1901-05-18 in Chicago, Illinois, USA.

Career and Major Research

du Vigneaud studied at the University of Illinois and received his doctorate from the University of Rochester in 1927. He worked at the University of Chicago and Cornell University Medical College in New York, where he became professor of biochemistry. He studied the sulfur-containing amino acids methionine, cysteine, and cystine, and investigated the sulfur chemistry of the hormones insulin, vasopressin, and oxytocin. In 1953 his group achieved the first synthesis of the peptide hormone oxytocin, and in 1954 the synthesis of vasopressin.

Nobel Prize-winning Work

The work for which Vincent du Vigneaud was awarded the Nobel Prize in Chemistry in 1955 was described by the Nobel Committee as: "for his work on biochemically important sulphur compounds, especially for the first synthesis of a polypeptide hormone".

du Vigneaud determined the sequence of the eight amino acids in oxytocin and vasopressin, including the disulfide bridge formed by their cystine residues, and then built the complete hormone chains. The synthetic oxytocin was shown to possess the full biological activity of the natural hormone, proving the structure correct.

Significance and Impact

The synthesis of oxytocin was the first total synthesis of a peptide hormone and demonstrated that a synthetic polypeptide could reproduce the activity of a natural hormone. It established the chemistry of the sulfur-containing amino acids and opened the way to the synthesis and medical use of peptide hormones.

Later Career and Honors

du Vigneaud remained professor of biochemistry at Cornell University Medical College until his retirement, continuing research on peptide and hormone chemistry. He died on 1978-12-11 in White Plains, New York.

Legacy

The oxytocin synthesis is regarded as the founding achievement of peptide-hormone chemistry, and du Vigneaud's methods anticipated the field of peptide synthesis that later produced insulin, growth hormone, and a wide range of therapeutic peptides.

Modern Developments

du Vigneaud's total synthesis of the hormones oxytocin and vasopressin proved that complex biologically active peptides could be built in the laboratory, founding the field of peptide synthesis and peptide-based pharmaceuticals. Peptide hormones and their analogues remain essential drugs for childbirth, diabetes, and many other conditions. His work on sulfur-containing amino acids and transmethylation also contributed to modern understanding of metabolism and nutrition.

Selected Publications

- *The Synthesis of an Octapeptide Amide with the Hormonal Activity of Oxytocin*, Journal of the American Chemical Society, 1953.

Chapter Summary

The 1955 prize recognized the first synthesis of a polypeptide hormone. Vincent du Vigneaud determined the structure of oxytocin and vasopressin and synthesized them in his laboratory at Cornell, work built on his long study of the biochemically important sulfur-containing amino acids.

Chapter 56

1956: Sir Cyril Hinshelwood and Nikolay Semenov

The Nobel Prize in Chemistry for the year 1956 was awarded jointly to Sir Cyril Hinshelwood and Nikolay Semenov.

Motivation:

for their researches into the mechanism of chemical reactions

56.1 Sir Cyril Hinshelwood

Early Life and Education

Sir Cyril Hinshelwood was born on 1897-06-19 in London, England.

Career and Major Research

Hinshelwood studied at Balliol College, Oxford, and spent his career at Oxford, where he became Dr Lee's Professor of Chemistry. He studied the kinetics of gas-phase reactions and showed that many of them proceed by chain mechanisms. His investigations of the explosive reaction between hydrogen and oxygen established the concepts of chain branching and explosion limits, and he went on to study the kinetics of bacterial growth.

Nobel Prize-winning Work

The work for which Sir Cyril Hinshelwood was awarded the Nobel Prize in Chemistry in 1956 was described by the Nobel Committee as: "for their researches into the mechanism of chemical reactions".

Hinshelwood showed how the rates of chemical reactions could be interpreted in terms of the sequence of elementary steps, and he demonstrated that chain reactions with branched propagation could account for the sharp explosion limits of the hydrogen-oxygen reaction.

Significance and Impact

Hinshelwood's kinetic analysis provided a quantitative framework for understanding reaction mechanisms, especially chain reactions, and connected the theory of gas reactions to practical phenomena such as explosions. His later work extended kinetic methods to the growth of bacteria, linking chemistry to biology.

Later Career and Honors

Hinshelwood continued at Oxford and later served as president of the Royal Society of London. He was knighted in 1948 and created an officer of the Order of Merit. He died on 1967-10-09 in London.

Legacy

The chain-reaction theory developed by Hinshelwood and Semenov became a cornerstone of chemical kinetics, and Hinshelwood's name remains associated with the quantitative study of reaction mechanisms in chemistry.

Modern Developments

Hinshelwood's kinetic studies of gas-phase reactions, including chain reactions, established modern chemical kinetics and the understanding of combustion. His work underpins the design of combustion engines, flame chemistry, and atmospheric chemistry models. Hinshelwood also applied kinetics to biology, helping to explain the autocatalytic growth of bacterial populations.

Selected Publications

- *The Kinetics of Chemical Change in Gaseous Systems*, Oxford: Clarendon Press, 1926.

56.2 Nikolay Semenov

Early Life and Education

Nikolay Semenov was born on 1896-04-15 in Saratov, Russia.

Career and Major Research

Semenov studied physics and chemistry at Petrograd University and worked in Leningrad, where he founded and directed the Institute of Chemical Physics of the Soviet Academy of Sciences, later moved to Moscow. Independently of Hinshelwood, he developed the theory of chain reactions and in 1934 published his book *Chemical Kinetics and Chain Reactions*. He showed that branched chain reactions could account for the ignition and explosion limits observed in the oxidation of phosphorus and hydrogen.

Nobel Prize-winning Work

The work for which Nikolay Semenov was awarded the Nobel Prize in Chemistry in 1956 was described by the Nobel Committee as: "for their researches into the mechanism of chemical reactions".

Semenov's theory of branched chain reactions explained how a single reactive intermediate could multiply and accelerate a reaction until it becomes explosive, and it provided the mathematical treatment of the explosion limits of the hydrogen-oxygen and phosphorus-oxygen reactions.

Significance and Impact

Semenov's work established the quantitative theory of chain reactions and combustion, with direct applications to flames, explosions, and industrial chemical processes. He became a central figure in Soviet physical chemistry, training a large school of kineticists.

Later Career and Honors

Semenov directed the Institute of Chemical Physics in Moscow for the rest of his career and played a leading role in the organization of science in the Soviet Union. He died on 1986-09-25 in Moscow.

Legacy

Semenov is regarded as one of the founders of chemical kinetics, and his theory of branched chain reactions is fundamental to the modern understanding of combustion and explosions.

Modern Developments

Semenov's theory of branched chain reactions explained explosive and combustion chemistry, founding the modern science of combustion kinetics and explosion safety. His work underpins the design of engines, industrial reactors, and the understanding of atmospheric radical chain chemistry. Semenov's principles are also applied in predicting and preventing industrial explosions.

Selected Publications

- *Chemical Kinetics and Chain Reactions*, Oxford: Clarendon Press, 1935.

Chapter Summary

The 1956 prize recognized research into the mechanism of chemical reactions. Working independently, Sir Cyril Hinshelwood at Oxford and Nikolay Semenov in the Soviet Union developed the theory of chain reactions and showed how branched chain mechanisms explain the explosion limits of gas-phase reactions such as the oxidation of hydrogen.

Chapter 57

1957:Lord Todd

The Nobel Prize in Chemistry for the year 1957 was awarded to Lord Todd.

Motivation:

for his work on nucleotides and nucleotide co-enzymes

57.1 Lord Todd

Early Life and Education

Lord Todd was born on 1907-10-02 in Glasgow, Scotland.

Career and Major Research

Alexander Robertus Todd studied at the University of Glasgow, took his doctorate at the University of Frankfurt, and earned a doctorate at the University of Oxford. He held the chair of chemistry at the University of Manchester and in 1944 became professor of organic chemistry at the University of Cambridge, where he remained for the rest of his career. He worked out the structures of the nucleotides and of the nucleoside coenzymes, and his group achieved the first syntheses of adenosine triphosphate (ATP), adenosine diphosphate (ADP), and the coenzyme flavin-adenine dinucleotide (FAD).

Nobel Prize-winning Work

The work for which Lord Todd was awarded the Nobel Prize in Chemistry in 1957 was described by the Nobel Committee as: "for his work on nucleotides and nucleotide co-enzymes".

Todd established how nucleotides are linked through their sugar and phosphate units and showed that the coenzymes are built from nucleotides joined in specific arrangements. He also contributed to the chemistry of vitamin B12 and of the nucleic acids, predicting a linear phosphate-sugar backbone for RNA.

Significance and Impact

Todd's nucleotide chemistry provided the structural basis for understanding ATP as the energy currency of cells and for the later understanding of nucleic acid structure. His syntheses of the coenzymes proved their structures beyond doubt and made them available for biochemical study.

Later Career and Honors

Todd was raised to the peerage as Lord Todd of Trumpington in 1962 and served as master of Christ's College, Cambridge, and president of the Royal Society of London. He died on 1997-01-10 in Cambridge, England.

Legacy

Lord Todd is regarded as the founder of modern nucleotide chemistry, and his work laid the structural foundations for the discovery of the double helix and for the chemistry of nucleic acids.

Modern Developments

Todd's syntheses of nucleotides, including ATP, NAD, and the coenzymes, established the chemistry of nucleic acids and bioenergetics. His work directly laid the foundations for the understanding of DNA replication and for the modern biotechnology and pharmaceutical industries. Todd's phosphate ester chemistry is central to the synthesis of oligonucleotide drugs and modern genetics.

Selected Publications

- *Synthesis of Adenosine Triphosphate*, Nature, 1948.

Chapter Summary

The 1957 prize recognized work on nucleotides and nucleotide coenzymes. Lord Todd determined the structures of the nucleotides and coenzymes and achieved their chemical synthesis at Cambridge, providing the structural basis for understanding ATP and the nucleic acids.

Chapter 58

1958:Frederick Sanger

The Nobel Prize in Chemistry for the year 1958 was awarded to Frederick Sanger.

Motivation:

for his work on the structure of proteins, especially that of insulin

58.1 Frederick Sanger

Early Life and Education

Frederick Sanger was born on 1918-08-13 in Rendcomb, England.

Career and Major Research

Sanger studied at St John's College, Cambridge, and received his doctorate from Cambridge in 1943. He worked in the department of biochemistry at Cambridge and later at the Medical Research Council Laboratory of Molecular Biology. He developed methods, using the reagent dinitrofluorobenzene, to label the terminal amino groups of protein chains, and applied them to determine the complete amino acid sequence of the hormone insulin, which he published in 1955.

Nobel Prize-winning Work

The work for which Frederick Sanger was awarded the Nobel Prize in Chemistry in 1958 was described by the Nobel Committee as: "for his work on the structure of proteins, especially that of insulin".

Sanger determined that insulin consists of two polypeptide chains, connected by disulfide bridges, and established the exact order of the amino acids in both chains. This was the first time the complete primary structure of a protein had been established.

Significance and Impact

The sequencing of insulin demonstrated that proteins have definite, precise amino acid sequences, and it founded the field of protein sequencing. The methods Sanger developed became the standard approach for determining the order of amino acids in proteins.

Later Career and Honors

Sanger continued at the Medical Research Council Laboratory of Molecular Biology, where he later developed methods for sequencing DNA, for which he shared the 1980 Nobel Prize in Chemistry with Walter Gilbert and Paul Berg. He is one of only four people to have received two Nobel Prizes. He died on 2013-11-19 in Cambridge, England.

Legacy

Sanger established protein sequencing and later invented the chain-termination method of DNA sequencing, the technique used for the first complete sequencing of the human genome. His name is carried by the Sanger Institute in his honor.

Modern Developments

Sanger's determination of the amino acid sequence of insulin proved that proteins have defined sequences and founded protein sequencing, which became the basis of molecular biology and the biotechnology industry. His methods evolved through DNA sequencing, earning Sanger a second Nobel Prize, and his name now graces the Sanger sequencing technology that powered the Human Genome Project.

Selected Publications

- *The Free Amino Groups of Insulin*, Biochemical Journal, 1945.

Chapter Summary

The 1958 prize recognized the determination of protein structure. Frederick Sanger established the complete amino acid sequence of insulin at Cambridge, the first protein sequence ever determined, and his later work on DNA sequencing earned him a second Nobel Prize in 1980.

Chapter 59

1959:Jaroslav Heyrovsky

The Nobel Prize in Chemistry for the year 1959 was awarded to Jaroslav Heyrovsky.

Motivation:

for his discovery and development of the polarographic methods of analysis

59.1 Jaroslav Heyrovsky

Early Life and Education

Jaroslav Heyrovsky was born on 1890-12-20 in Prague.

Career and Major Research

Heyrovsky studied at Charles University in Prague and at University College London, where he earned his doctorate in 1918 under the physical chemist Frederick G. Donnan. He returned to Prague and spent his career at Charles University, becoming professor of physical chemistry. In 1922 he invented polarography, an analytical method in which current is measured as a function of the voltage applied to a dropping mercury electrode. He developed the method into a general technique of electroanalysis, and in 1950 he founded the Polarographic Institute of the Czechoslovak Academy of Sciences in Prague.

Nobel Prize-winning Work

The work for which Jaroslav Heyrovsky was awarded the Nobel Prize in Chemistry in 1959 was described by the Nobel Committee as: "for his discovery and development of the polarographic methods of analysis".

Polarography records the current-voltage curves obtained when solutions are electrolyzed with a dropping mercury electrode, and the position and height of the resulting waves identify the substance and measure its concentration. The method made possible the quantitative analysis of reducible and oxidizable substances down to very small concentrations.

Significance and Impact

Polarography became one of the most widely used techniques of analytical chemistry, applied to trace metals, organic compounds, and biological materials in industry and medicine. It stimulated the development of the whole field of modern electroanalytical chemistry.

Later Career and Honors

Heyrovsky remained in Prague and directed the Polarographic Institute until his death. He died on 1967-03-27 in Prague.

Legacy

Heyrovsky is regarded as the founder of polarography and of modern electroanalysis, and the Jaroslav Heyrovsky Institute of Physical Chemistry in Prague continues the school he created.

Modern Developments

Heyrovsky's polarography founded modern electroanalytical chemistry, enabling the detection and measurement of trace metals in solution. Polarography and its descendants, including voltammetry and stripping analysis, are standard tools in environmental monitoring, clinical chemistry, and battery research. The polarograph concept also influenced modern electrochemical sensors.

Selected Publications

- *A Polarographic Study of the Electro-Kinetic Phenomena of Absorption, Electro-Reduction and Overpotential Displayed at the Dropping Mercury Cathode*, 1934.

Chapter Summary

The 1959 prize recognized the discovery and development of polarography. Jaroslav Heyrovsky invented the method at Charles University in Prague in 1922, using the dropping mercury electrode to obtain current-voltage curves, and developed it into a widely used technique of analytical chemistry.

Chapter 60

1960: Willard F. Libby

The Nobel Prize in Chemistry for the year 1960 was awarded to Willard F. Libby.

Motivation:

for his method to use carbon-14 for age determination in archaeology, geology, geophysics, and other branches of science

60.1 Willard F. Libby

Early Life and Education

Willard F. Libby was born on 1908-12-17 in Grand Valley, Colorado, USA.

Career and Major Research

Libby studied at the University of California, Berkeley, where he earned his doctorate in 1933, and then taught there. During the Second World War he worked on the Manhattan Project in the isotopic separation of uranium. After the war, at the University of Chicago, he developed the method of radiocarbon dating, announcing in 1949 the first determinations of the age of archaeological samples based on the decay of carbon-14.

Nobel Prize-winning Work

The work for which Willard F. Libby was awarded the Nobel Prize in Chemistry in 1960 was described by the Nobel Committee as: "for his method to use carbon-14 for age determination in archaeology, geology, geophysics, and other branches of science".

Libby realized that carbon-14, produced continuously in the atmosphere by cosmic radiation and incorporated into living organisms through photosynthesis, decays with a known half-life after death. By measuring the residual carbon-14 in ancient wood, charcoal, and bone, he could calculate the time elapsed since the material stopped exchanging carbon with the atmosphere.

Significance and Impact

Radiocarbon dating provided, for the first time, a general method of absolute dating for organic remains up to about fifty thousand years old. It revolutionized archaeology, geology, and Quaternary science, and its checks against objects of known age proved the method reliable.

Later Career and Honors

Libby left Chicago to become a member of the United States Atomic Energy Commission in 1955 and later was a professor of chemistry at the University of California, Los Angeles. He died on 1980-09-08 in Los Angeles.

Legacy

Radiocarbon dating remains one of the most important dating methods in archaeology and earth science, and Libby's discovery founded the field of radiometric dating with cosmogenic isotopes.

Modern Developments

Libby's radiocarbon dating revolutionized archaeology, geology, and paleoclimatology, providing a universal clock for organic materials up to about 50,000 years old. Radiocarbon dating is now routine in dating ancient artifacts, studying climate change, and forensic science, and Libby's methods extended to other cosmogenic isotopes.

Selected Publications

- *Radiocarbon Dating*, University of Chicago Press, 1952.

Chapter Summary

The 1960 prize recognized the development of radiocarbon dating. Willard F. Libby showed at the University of Chicago that the decay of carbon-14 in organic material could be used to determine its age, providing archaeologists and geologists with a method of absolute dating for organic remains.

Chapter 61

1961:Melvin Calvin

The Nobel Prize in Chemistry for the year 1961 was awarded to Melvin Calvin.

Motivation:

for his research on the carbon dioxide assimilation in plants

61.1 Melvin Calvin

Early Life and Education

Melvin Calvin was born on 1911-04-08 in St. Paul, Minnesota, USA.

Career and Major Research

Calvin studied at the Michigan College of Mining and Technology and received his doctorate from the University of California, Berkeley, in 1935. He spent his career at Berkeley, where he worked at the Lawrence Radiation Laboratory and became professor of chemistry. Using the radioactive tracer carbon-14, introduced into photosynthesis research by his colleague Andrew Benson, Calvin and his group, including James Bassham, traced the path of carbon in photosynthesis and worked out the Calvin cycle.

Nobel Prize-winning Work

The work for which Melvin Calvin was awarded the Nobel Prize in Chemistry in 1961 was described by the Nobel Committee as: "for his research on the carbon dioxide assimilation in plants".

By exposing algae to carbon-14 labeled carbon dioxide for brief periods and analyzing the products, Calvin's group identified the sequence of reactions by which carbon dioxide is fixed into organic compounds. They showed that the first product is 3-phosphoglycerate and that the cycle regenerates its five-carbon acceptor, ribulose-1,5-bisphosphate, so that carbon can be assimilated continuously.

Significance and Impact

The Calvin cycle, as it came to be called, was the first complete description of a major metabolic pathway, and it explained how plants convert carbon dioxide into the sugars on which all life depends. The tracer methods used became standard in biochemistry.

Later Career and Honors

Calvin remained at the University of California, Berkeley, where he directed the Laboratory of Chemical Biodynamics. He died on 1997-01-08 in Berkeley, California.

Legacy

The Calvin cycle is one of the foundations of biochemistry and plant science, and Calvin's use of isotopic tracers set the pattern for the elucidation of metabolic pathways throughout the biological sciences.

Modern Developments

Calvin's elucidation of the photosynthetic carbon cycle (the Calvin cycle) revealed how plants convert carbon dioxide into sugar and remains the basis of our understanding of carbon fixation. His work underpins modern efforts to improve crop yields, engineer carbon capture, and develop artificial photosynthesis for renewable fuels. Radioactive tracer techniques used in his research are now standard in biology.

Selected Publications

- *The Path of Carbon in Photosynthesis*, Science, 1962.

Chapter Summary

The 1961 prize recognized research on carbon dioxide assimilation in plants. Melvin Calvin and his colleagues at Berkeley used carbon-14 tracers to trace the path of carbon in photosynthesis and established the cycle of reactions by which plants fix carbon dioxide into carbohydrates.

Chapter 62

1962:Max F. Perutz and John C. Kendrew

The Nobel Prize in Chemistry for the year 1962 was awarded jointly to Max F. Perutz and John C. Kendrew.

Motivation:

for their studies of the structures of globular proteins

62.1 Max F. Perutz

Early Life and Education

Max F. Perutz was born on 1914-05-19 in Vienna, Austria.

Career and Major Research

Perutz studied chemistry at the University of Vienna and came to Cambridge in 1936, where he worked at the Cavendish Laboratory and took his doctorate in 1940. He devoted his career to the X-ray analysis of haemoglobin, developing the method of isomorphous replacement, in which heavy atoms are attached to the protein to solve the phase problem. He founded and directed the Medical Research Council Unit for Molecular Biology in Cambridge, which later became the Laboratory of Molecular Biology. In 1959 he and his colleagues obtained a structure of haemoglobin at a resolution of 5.5 angstroms.

Nobel Prize-winning Work

The work for which Max F. Perutz was awarded the Nobel Prize in Chemistry in 1962 was described by the Nobel Committee as: "for their studies of the structures of globular proteins".

Perutz determined the three-dimensional structure of the globular protein haemoglobin, showing how its four subunits are arranged. The method of isomorphous replacement he developed became the key technique for determining the structures of proteins by X-ray crystallography.

Significance and Impact

The solution of the haemoglobin structure made it possible to relate the behavior of a protein molecule, including its oxygen uptake and release, to its three-dimensional architecture. Perutz's technique laid the foundation for the entire field of protein crystallography.

Later Career and Honors

Perutz remained at the Medical Research Council Laboratory of Molecular Biology, which he directed for many years, and continued studies of haemoglobin throughout his life. He died on 2002-02-06 in Cambridge,

England.

Legacy

Perutz founded protein crystallography and the Medical Research Council Laboratory of Molecular Biology, one of the most influential research institutions in modern biology, and his work opened the era of molecular structure determination for proteins.

Modern Developments

Perutz's use of X-ray crystallography to solve the structure of hemoglobin founded structural biology and revealed how proteins use conformational change to function cooperatively. Hemoglobin remains a paradigm for allosteric regulation, informing drug design and the treatment of sickle-cell disease. Perutz's methods evolved into the modern structural genomics and cryo-EM era.

Selected Publications

- *Structure of Haemoglobin: A Three-Dimensional Fourier Synthesis at 5.5 Å Resolution*, Proceedings of the Royal Society of London, 1960.

62.2 John C. Kendrew

Early Life and Education

John C. Kendrew was born on 1917-03-24 in Oxford, England.

Career and Major Research

Kendrew studied chemistry at Trinity College, Cambridge, and during the war worked on operational research for the Royal Air Force. He returned to Cambridge, where he joined the Medical Research Council Unit under Max F. Perutz and applied X-ray crystallography to the smaller globular protein myoglobin. In 1957 he and his colleagues computed a low-resolution structure of myoglobin, and in 1959 a structure at 2 angstroms resolution, the first atomic-level structure of a protein.

Nobel Prize-winning Work

The work for which John C. Kendrew was awarded the Nobel Prize in Chemistry in 1962 was described by the Nobel Committee as: "for their studies of the structures of globular proteins".

Kendrew determined the complete three-dimensional structure of the globular protein myoglobin, showing in detail how the polypeptide chain is folded around the heme group that carries oxygen. It was the first time the atomic arrangement of a protein had been established.

Significance and Impact

The myoglobin structure showed for the first time what a folded protein looks like at atomic resolution, revealing the positions of every amino acid side chain. It demonstrated that the methods of X-ray crystallography could reach atomic detail for proteins and set the pattern for the structural biology that followed.

Later Career and Honors

Kendrew continued at the Medical Research Council Laboratory of Molecular Biology and later served as director-general of the European Molecular Biology Laboratory in Heidelberg. He died on 1997-08-30 in Cambridge, England.

Legacy

Kendrew's myoglobin structure was the first protein structure solved at atomic resolution and stands as a founding achievement of structural biology.

Modern Developments

Kendrew's determination of the three-dimensional structure of myoglobin was the first protein structure ever solved, founding structural biology and the field that today determines thousands of protein structures annually. His work demonstrated that proteins fold into unique, detailed conformations and opened the door to structure-based drug design, enzymology, and protein engineering.

Selected Publications

- *A Three-Dimensional Model of the Myoglobin Molecule Obtained by X-Ray Analysis*, Nature, 1958.

Chapter Summary

The 1962 prize recognized the determination of the structures of globular proteins. Max F. Perutz solved the structure of haemoglobin and developed the method of isomorphous replacement, while John C. Kendrew used it to determine the structure of myoglobin, the first protein structure obtained at atomic resolution.

Chapter 63

1963:Karl Ziegler and Giulio Natta

The Nobel Prize in Chemistry for the year 1963 was awarded jointly to Karl Ziegler and Giulio Natta.

Motivation:

for their discoveries in the field of the chemistry and technology of high polymers

63.1 Karl Ziegler

Early Life and Education

Karl Ziegler was born on 1898-11-26 in Helsa, Germany.

Career and Major Research

Ziegler studied chemistry at the universities of Marburg and Munich and received his doctorate from the University of Marburg in 1920. He became director of the Kaiser Wilhelm Institute, later the Max Planck Institute, for Coal Research in Mülheim an der Ruhr, where he spent most of his career. His research on organometallic compounds, especially the aluminium alkyls, led in 1953 to the discovery that combinations of an aluminium alkyl with a titanium compound could polymerize ethylene at ordinary pressure to linear polyethylene.

Nobel Prize-winning Work

The work for which Karl Ziegler was awarded the Nobel Prize in Chemistry in 1963 was described by the Nobel Committee as: "for their discoveries in the field of the chemistry and technology of high polymers".

Ziegler's organometallic catalysts polymerized ethylene at low pressure to a linear, high-density polyethylene, in contrast to the high-pressure radical process then in use. The discovery provided a general catalytic route to polyolefins and became the basis of an entire industrial technology.

Significance and Impact

The low-pressure polymerization of ethylene transformed the plastics industry, making polyethylene cheaper and giving it new properties. The catalyst systems that Ziegler discovered, later combined with Natta's stereospecific polymers as Ziegler-Natta catalysts, produce the majority of the world's polyethylene and polypropylene.

Later Career and Honors

Ziegler remained director of the Max Planck Institute for Coal Research in Mülheim until 1969. He died on 1973-08-11 in Mülheim, Germany.

Legacy

Ziegler's work founded the chemistry of organoaluminium polymerization catalysts, and the Ziegler-Natta catalysts that bear his name dominate the industrial production of polyolefin plastics.

Modern Developments

Ziegler's discovery of organometallic catalysts for ethylene polymerization created the foundation of the modern plastics industry and of organometallic catalysis. Ziegler-Natta catalysts produce polyethylene and polypropylene, among the most widely used materials on Earth. Ziegler's organoaluminum chemistry also underpins industrial aluminum alkyl synthesis and modern polymer science.

Selected Publications

- *Das Mülheimer Normaldruck-Polyäthylen-Verfahren*, Angewandte Chemie, 1955.

63.2 Giulio Natta

Early Life and Education

Giulio Natta was born on 1903-02-26 in Imperia, Italy.

Career and Major Research

Natta studied chemical engineering at the Polytechnic of Milan, where he received his doctorate in 1924. He held professorships at the universities of Pavia, Rome, and Turin before returning to the Polytechnic of Milan in 1938. Learning of Ziegler's catalysts, he applied them to the polymerization of propylene and in 1954 produced isotactic polypropylene, in which the methyl groups are arranged regularly along the chain, giving a crystalline, strong plastic.

Nobel Prize-winning Work

The work for which Giulio Natta was awarded the Nobel Prize in Chemistry in 1963 was described by the Nobel Committee as: "for their discoveries in the field of the chemistry and technology of high polymers".

Natta showed that Ziegler-type catalysts could control the spatial arrangement of monomer units during polymerization, producing stereospecific polymers with regular structure. His discovery of isotactic polypropylene established stereospecific polymerization and the field of polymer stereochemistry.

Significance and Impact

Isotactic polypropylene became one of the world's most important plastics, and Natta's concept of stereospecific polymerization explained how the regular arrangement of groups along a polymer chain determines its crystallinity and properties. He introduced the terms isotactic, syndiotactic, and atactic.

Later Career and Honors

Natta remained professor at the Polytechnic of Milan and continued research in polymer chemistry until ill health forced his retirement in 1963. He died on 1979-05-02 in Bergamo, Italy.

Legacy

Natta is regarded as the founder of stereospecific polymerization, and the Ziegler-Natta catalysts are among the most important catalysts in industrial chemistry, producing polypropylene and polyethylene on a massive scale.

Modern Developments

Natta's stereospecific polymerization using Ziegler-Natta catalysts produced the first isotactic polypropylene, founding the modern stereoregular polymer industry. Polypropylene is now one of the world's most produced plastics, used in packaging, textiles, and medical products. Natta's work pioneered the study of polymer tacticity and its control of material properties.

Selected Publications

- *Une Nouvelle Classe de Polymères d'Alpha-Oléfines Ayant une Régularité de Structure Exceptionnelle*, Journal of Polymer Science, 1955.

Chapter Summary

The 1963 prize recognized discoveries in the chemistry and technology of high polymers. Karl Ziegler discovered organoaluminium-titanium catalysts that polymerize ethylene at low pressure, and Giulio Natta applied them to produce stereospecific polymers such as isotactic polypropylene, founding the industrial polyolefin technology.

Chapter 64

1964:Dorothy Crowfoot Hodgkin

The Nobel Prize in Chemistry for the year 1964 was awarded to Dorothy Crowfoot Hodgkin.

Motivation:

for her determinations by X-ray techniques of the structures of important biochemical substances

64.1 Dorothy Crowfoot Hodgkin

Early Life and Education

Dorothy Crowfoot Hodgkin was born on 1910-05-12 in Cairo, Egypt.

Career and Major Research

Hodgkin studied chemistry at Somerville College, Oxford, and took her doctorate at the University of Cambridge in 1937 under J.D. Bernal. She returned to Oxford, where she spent her career, becoming Wolfson Research Professor of the Royal Society. She applied X-ray crystallography to increasingly complex biochemical substances, determining the structure of cholesterol in 1937, of penicillin in 1945, and of vitamin B12 in 1955.

Nobel Prize-winning Work

The work for which Dorothy Crowfoot Hodgkin was awarded the Nobel Prize in Chemistry in 1964 was described by the Nobel Committee as: "for her determinations by X-ray techniques of the structures of important biochemical substances".

Hodgkin's determination of the structure of penicillin confirmed the previously disputed beta-lactam ring, and her solution of the structure of vitamin B12 revealed the large corrin ring and its central cobalt atom. Both structures were among the most complex molecules ever solved by X-ray analysis at the time.

Significance and Impact

Hodgkin's structures of penicillin and vitamin B12 demonstrated that X-ray crystallography could solve molecules of biochemical and pharmaceutical importance, guiding later work on the synthesis and medical use of these substances. Her achievement inspired the use of the method throughout biochemistry.

Later Career and Honors

Hodgkin continued X-ray analysis of insulin, work begun in the 1930s, and her group finally established the full insulin structure in 1969. She received many honors, including the Order of Merit in 1965. She died on 1994-07-29 in Shipston-on-Stour, England.

Legacy

Hodgkin was one of the pioneers of macromolecular crystallography and one of the few women to receive the Nobel Prize in Chemistry. Her structures of cholesterol, penicillin, vitamin B12, and insulin remain landmarks of structural biology.

Modern Developments

Hodgkin's X-ray analyses of penicillin, vitamin B12, and insulin revealed the three-dimensional structures of complex biological molecules and established X-ray crystallography as the definitive tool of structural biology. Her insulin structure guided the development of insulin therapies, and her methods paved the way for the protein-structure revolution that now powers drug discovery.

Selected Publications

- *Structure of Vitamin B12*, Nature, 1955.

Chapter Summary

The 1964 prize recognized determinations of the structures of important biochemical substances by X-ray techniques. Dorothy Crowfoot Hodgkin solved the structures of cholesterol, penicillin, and vitamin B12 at Oxford, and her later work established the structure of insulin.

Chapter 65

1965:Robert B. Woodward

The Nobel Prize in Chemistry for the year 1965 was awarded to Robert B. Woodward.

Motivation:

for his outstanding achievements in the art of organic synthesis

65.1 Robert B. Woodward

Early Life and Education

Robert B. Woodward was born on 1917-04-10 in Boston, Massachusetts, USA.

Career and Major Research

Woodward studied at the Massachusetts Institute of Technology, where he received his doctorate in 1937, and joined Harvard University, where he was professor of chemistry from 1950. He completed the total synthesis of quinine in 1944 and later led teams that synthesized cholesterol, cortisone, strychnine, chlorophyll, and vitamin B12. Together with Roald Hoffmann he formulated the Woodward-Hoffmann rules on the conservation of orbital symmetry.

Nobel Prize-winning Work

The work for which Robert B. Woodward was awarded the Nobel Prize in Chemistry in 1965 was described by the Nobel Committee as: "for his outstanding achievements in the art of organic synthesis".

Woodward's syntheses of complex natural products, beginning with quinine in 1944, established new standards for planning and executing multistep syntheses. His approach combined rigorous structural reasoning with carefully designed reaction sequences and physical methods of analysis.

Significance and Impact

The total syntheses of quinine, cholesterol, cortisone, strychnine, chlorophyll, and vitamin B12 demonstrated that complex natural molecules could be built from simple starting materials. These achievements made him one of the most influential organic chemists of the twentieth century and shaped modern synthetic methodology.

Later Career and Honors

Woodward remained at Harvard for his entire academic career and also served as a director of research at the Woodward Research Institute in Basel. He died on 1979-07-08 in Cambridge, Massachusetts.

Legacy

The Woodward-Hoffmann rules and Woodward's total syntheses remain cornerstones of organic chemistry education and research. The many students trained in his laboratory carried his approach into the wider field.

Modern Developments

Woodward's total syntheses, including cholesterol, chlorophyll, and vitamin B12, set the standard for organic synthesis and founded modern synthetic chemistry. His ideas, formalized in the Woodward-Hoffmann rules, guide the design of countless reactions used in the pharmaceutical and agrochemical industries. Woodward's approach to synthesis remains the blueprint for complex molecule construction today.

Selected Publications

"The Total Synthesis of Quinine" (Journal of the American Chemical Society, 1944)

Chapter Summary

The 1965 prize recognized Robert B. Woodward for his total syntheses of complex natural products, from quinine in 1944 to vitamin B12. His work established the standards of modern synthetic organic chemistry.

Chapter 66

1966:Robert S. Mulliken

The Nobel Prize in Chemistry for the year 1966 was awarded to Robert S. Mulliken.

Motivation:

for his fundamental work concerning chemical bonds and the electronic structure of molecules by the molecular orbital method

66.1 Robert S. Mulliken

Early Life and Education

Robert S. Mulliken was born on 1896-06-07 in Newburyport, Massachusetts, USA.

Career and Major Research

Mulliken studied at the Massachusetts Institute of Technology and received his doctorate from the University of Chicago in 1921. He spent most of his career at the University of Chicago, where he developed the molecular orbital method for describing chemical bonds and the electronic structure of molecules. He also studied molecular spectra and introduced an electronegativity scale based on ionization energies and electron affinities.

Nobel Prize-winning Work

The work for which Robert S. Mulliken was awarded the Nobel Prize in Chemistry in 1966 was described by the Nobel Committee as: "for his fundamental work concerning chemical bonds and the electronic structure of molecules by the molecular orbital method".

Mulliken assigned electrons to molecular orbitals that extend over an entire molecule, providing a quantum-mechanical description of bonding that complemented valence bond theory. His approach explained the spectra, ionization energies, and bond properties of many molecules.

Significance and Impact

The molecular orbital method became one of the two principal frameworks of quantum chemistry and is the basis of most modern computational chemistry. Mulliken's orbital diagrams and notation are still used in chemistry education.

Later Career and Honors

Mulliken remained at the University of Chicago after his retirement in 1961 and held a research professorship at Florida State University from 1985. He died on 1986-10-31 in Arlington, Virginia, USA.

Legacy

Mulliken's molecular orbital theory is central to modern chemistry and underpins much of computational and physical chemistry.

Modern Developments

Mulliken's molecular orbital theory, for which he developed the concept of electronegativity and orbital overlapping, is the foundation of modern computational chemistry. Molecular orbital theory is used in nearly every quantum chemical calculation today, from drug design to materials science, and underlies our understanding of bonding, spectroscopy, and reactivity.

Selected Publications

"The Assignment of Quantum Numbers for Electrons in Molecules" (Physical Review, 1928)

"Spectroscopy, Molecular Orbitals, and Chemical Bonding" (Science, 1967)

Chapter Summary

The 1966 prize recognized Robert S. Mulliken for the development of the molecular orbital method. His theory of chemical bonding and molecular electronic structure remains fundamental to chemistry.

Chapter 67

1967:Manfred Eigen and Ronald G.W. Norrish and George Porter

The Nobel Prize in Chemistry for the year 1967 was awarded jointly to Manfred Eigen and Ronald G.W. Norrish and George Porter.

Motivation:

for their studies of extremely fast chemical reactions, effected by disturbing the equilibrium by means of very short pulses of energy

67.1 Manfred Eigen

Early Life and Education

Manfred Eigen was born on 1927-05-09 in Bochum, Germany.

Career and Major Research

Eigen studied physics and chemistry at the University of Göttingen, where he received his doctorate in 1951. He joined the Max Planck Institute for Physical Chemistry in Göttingen, later renamed the Max Planck Institute for Biophysical Chemistry, and became its director. He developed relaxation methods, including temperature-jump and pressure-jump techniques, to measure the rates of very fast reactions in solution.

Nobel Prize-winning Work

The work for which Manfred Eigen was awarded the Nobel Prize in Chemistry in 1967 was described by the Nobel Committee as: "for their studies of extremely fast chemical reactions, effected by disturbing the equilibrium by means of very short pulses of energy".

Eigen disturbed a reaction equilibrium with a rapid change of temperature, pressure, or electric field and then followed the return to equilibrium, measuring rate constants for processes such as proton transfer that occur in microseconds.

Significance and Impact

Eigen's relaxation methods opened the study of elementary chemical reactions in solution, including acid-base and enzyme-catalyzed reactions, to direct measurement.

Later Career and Honors

Eigen later turned to the origin of life and to self-organization in biological systems. He died on 2019-02-06 in Göttingen, Germany.

Legacy

Relaxation techniques remain standard tools in chemical kinetics, and Eigen's later work influenced research on the molecular basis of early evolution.

Modern Developments

Eigen's use of relaxation techniques to measure ultra-fast chemical reactions revealed the kinetics of proton transfer and enzyme catalysis. His work founded fast-reaction chemistry and, through his later studies of self-organizing molecular systems, influenced the understanding of the origins of life and the development of the concept of RNA-like early replicators.

Selected Publications

"Immeasurably Fast Reactions" (Nobel Lecture, 1967)

67.2 Ronald G.W. Norrish

Early Life and Education

Ronald G.W. Norrish was born on 1897-11-09 in Cambridge, England.

Career and Major Research

Norrish studied at the University of Cambridge, where he spent his academic career and became professor of physical chemistry. With George Porter he developed flash photolysis, in which an intense flash of light generates short-lived chemical species that are then analyzed by a second flash.

Nobel Prize-winning Work

The work for which Ronald G.W. Norrish was awarded the Nobel Prize in Chemistry in 1967 was described by the Nobel Committee as: "for their studies of extremely fast chemical reactions, effected by disturbing the equilibrium by means of very short pulses of energy".

Flash photolysis made it possible to observe free radicals and other transient species directly and to measure the rates of their reactions.

Significance and Impact

Flash photolysis transformed the study of photochemistry and provided the first direct observations of many reactive intermediates in the gas phase.

Later Career and Honors

Norrish remained at Cambridge until his retirement in 1965. He died on 1978-06-04 in Cambridge, England.

Legacy

The Norrish type I and type II photochemical reactions of carbonyl compounds remain standard topics in photochemistry.

Modern Developments

Norrish's photochemistry of radicals and his flash photolysis technique, developed with Porter, founded modern photochemistry and made it possible to observe short-lived reactive intermediates. Flash photolysis evolved into femtosecond spectroscopy and today's study of the fastest chemical processes, with applications in vision, photosynthesis, and materials science.

Selected Publications

"Chemical Reactions Produced by Very High Light Intensities" (Nature, 1949)

67.3 George Porter

Early Life and Education

George Porter was born on 1920-12-06 in Stainforth, England.

Career and Major Research

Porter studied at the University of Leeds and then at the University of Cambridge, where he received his doctorate under Norrish. He was professor at the University of Sheffield and later director of the Royal Institution in London. He extended flash photolysis to study very fast reactions in gases and liquids.

Nobel Prize-winning Work

The work for which George Porter was awarded the Nobel Prize in Chemistry in 1967 was described by the Nobel Committee as: "for their studies of extremely fast chemical reactions, effected by disturbing the equilibrium by means of very short pulses of energy".

Porter's refinements of flash photolysis allowed the spectra and kinetics of short-lived molecules to be recorded, and he applied the method to fields from photochemistry to photosynthesis.

Significance and Impact

Flash photolysis became a cornerstone of photochemistry and photobiology and led directly to later laser-based techniques for studying fast processes.

Later Career and Honors

Porter became director of the Royal Institution and president of the Royal Society and was created Baron Porter of Luddenham. He died on 2002-08-31 in Canterbury, England.

Legacy

Porter's work established the experimental study of fast photochemical processes and influenced fields from solar energy research to atmospheric chemistry.

Modern Developments

Porter's flash photolysis technique, developed with Norrish, revolutionized the study of fast reactions and enabled the direct observation of free radicals and excited states. His method evolved into the modern field of ultrafast spectroscopy, which today probes chemical reactions on the femtosecond timescale and underpins research in vision, photovoltaics, and photocatalysis.

Selected Publications

"Chemical Reactions Produced by Very High Light Intensities" (Nature, 1949)

Chapter Summary

The 1967 prize recognized Eigen, Norrish, and Porter for their studies of extremely fast chemical reactions. Eigen's relaxation techniques and Norrish and Porter's flash photolysis made the kinetics of elementary processes measurable.

Chapter 68

1968:Lars Onsager

The Nobel Prize in Chemistry for the year 1968 was awarded to Lars Onsager.

Motivation:

for the discovery of the reciprocal relations bearing his name, which are fundamental for the thermodynamics of irreversible processes

68.1 Lars Onsager

Early Life and Education

Lars Onsager was born on 1903-11-27 in Oslo, Norway.

Career and Major Research

Onsager studied chemical engineering at the Norwegian Institute of Technology in Trondheim. After positions in Europe and the United States, he joined Yale University, where he was professor of theoretical chemistry. He derived the reciprocal relations of irreversible thermodynamics and later solved the two-dimensional Ising model exactly.

Nobel Prize-winning Work

The work for which Lars Onsager was awarded the Nobel Prize in Chemistry in 1968 was described by the Nobel Committee as: "for the discovery of the reciprocal relations bearing his name, which are fundamental for the thermodynamics of irreversible processes".

Onsager showed that the coefficients linking fluxes and forces in non-equilibrium systems obey symmetry relations, providing a general foundation for the thermodynamics of irreversible processes.

Significance and Impact

The Onsager reciprocal relations underlie the description of transport phenomena such as diffusion, heat conduction, and thermoelectric effects and are fundamental to non-equilibrium thermodynamics.

Later Career and Honors

Onsager remained at Yale until 1972, then held a research professorship at the University of Miami in Coral Gables, Florida. He died on 1976-10-05 in Coral Gables, Florida, USA.

Legacy

Onsager's reciprocal relations and his exact solution of the Ising model are landmarks of twentieth-century statistical mechanics.

Modern Developments

Onsager's reciprocal relations in irreversible thermodynamics provided the mathematical foundation for the modern study of non-equilibrium processes, including heat conduction, diffusion, and electrical transport. His exact solution of the two-dimensional Ising model remains a landmark of statistical mechanics, and his theories underpin modern modeling of complex fluids, electrochemical systems, and biological transport.

Selected Publications

"Reciprocal Relations in Irreversible Processes" (Physical Review, 1931)

"Crystal Statistics. I. A Two-Dimensional Model with an Order-Disorder Transition" (Physical Review, 1944)

Chapter Summary

The 1968 prize honored Lars Onsager for the discovery of the reciprocal relations of irreversible thermodynamics. His work provides the foundation for understanding transport processes in chemistry and physics.

Chapter 69

1969:Derek Barton and Odd Hassel

The Nobel Prize in Chemistry for the year 1969 was awarded jointly to Derek Barton and Odd Hassel.

Motivation:

for their contributions to the development of the concept of conformation and its application in chemistry

69.1 Derek Barton

Early Life and Education

Derek Barton was born on 1918-09-08 in Gravesend, England.

Career and Major Research

Barton studied at Imperial College London, where he received his doctorate and later became a professor. He introduced conformational analysis, showing that the three-dimensional shape of a molecule determines its chemical reactivity, and applied it to the chemistry of steroids and other natural products. He also discovered the photochemical reaction that bears his name.

Nobel Prize-winning Work

The work for which Derek Barton was awarded the Nobel Prize in Chemistry in 1969 was described by the Nobel Committee as: "for their contributions to the development of the concept of conformation and its application in chemistry".

Building on Hassel's structural studies of cyclohexane, Barton showed that the preferred conformation of a steroid controls how its substituents react, providing a predictive framework for organic stereochemistry.

Significance and Impact

Conformational analysis became a fundamental tool in organic chemistry, natural product synthesis, and drug design.

Later Career and Honors

Barton later directed a laboratory of the CNRS at Gif-sur-Yvette in France. He died on 1998-03-16 in College Station, Texas, USA.

Legacy

The concepts of axial and equatorial substituents and of conformational control of reactivity remain central to organic chemistry.

Modern Developments

Barton's conformational analysis explained how molecular shape controls reaction rates and selectivity, becoming a cornerstone of modern organic and medicinal chemistry. His ideas guide the rational design of drugs, natural product syntheses, and stereoselective reactions. Barton's later radical chemistry, including the Barton reaction, remains widely used in synthesis.

Selected Publications

"The Conformation of the Steroid Nucleus" (Experientia, 1950)

"The Principles of Conformational Analysis" (Nobel Lecture, 1969)

69.2 Odd Hassel

Early Life and Education

Odd Hassel was born on 1897-05-17 in Oslo, Norway.

Career and Major Research

Hassel studied at the University of Oslo and earned his doctorate in Berlin, then returned to Norway, where he was professor at the University of Oslo. Using electron diffraction and dipole moment measurements, he established that cyclohexane exists preferentially in the chair conformation with distinct axial and equatorial positions.

Nobel Prize-winning Work

The work for which Odd Hassel was awarded the Nobel Prize in Chemistry in 1969 was described by the Nobel Committee as: "for their contributions to the development of the concept of conformation and its application in chemistry".

Hassel's structural determinations of cyclohexane and its derivatives provided the experimental foundation for Barton's conformational analysis of organic molecules.

Significance and Impact

Hassel's results showed that saturated six-membered rings adopt well-defined conformations, a fact essential to understanding the structure and reactivity of alicyclic compounds.

Later Career and Honors

Hassel remained at the University of Oslo for his entire career. He died on 1981-05-11 in Oslo, Norway.

Legacy

The chair and boat description of cyclohexane introduced through Hassel's work is standard content in every organic chemistry course.

Modern Developments

Hassel's studies of cyclohexane conformation established the concept of axial and equatorial positions that is now fundamental to understanding the shape and reactivity of organic molecules. Conformational analysis underpins the design of pharmaceuticals, polymers, and molecular machines, and Hassel's insights remain central to organic stereochemistry.

Selected Publications

"Structural Aspects of Interatomic Charge-Transfer Bonding" (Nobel Lecture, 1969)

Chapter Summary

The 1969 prize recognized Barton and Hassel for developing the concept of conformation and applying it in chemistry. Their work linked molecular shape to chemical reactivity and founded conformational analysis.

Chapter 70

1970:Luis Leloir

The Nobel Prize in Chemistry for the year 1970 was awarded to Luis Leloir.

Motivation:

for his discovery of sugar nucleotides and their role in the biosynthesis of carbohydrates

70.1 Luis Leloir

Early Life and Education

Luis Leloir was born on 1906-09-06 in Paris, France.

Career and Major Research

Leloir studied medicine at the University of Buenos Aires, where he later worked as a researcher. He established an institute in Buenos Aires, later named the Leloir Institute, and studied the metabolism of carbohydrates. He discovered uridine diphosphate glucose and other sugar nucleotides and showed that they act as intermediates in the biosynthesis of glycogen and other carbohydrates.

Nobel Prize-winning Work

The work for which Luis Leloir was awarded the Nobel Prize in Chemistry in 1970 was described by the Nobel Committee as: "for his discovery of sugar nucleotides and their role in the biosynthesis of carbohydrates".

Leloir found that sugars must be activated by attachment to nucleotides such as uridine diphosphate before they can be incorporated into polysaccharides, revealing the general mechanism of carbohydrate biosynthesis.

Significance and Impact

The discovery of sugar nucleotides explained how glycogen, starch, and other carbohydrates are synthesized in cells and opened the field of glycobiology.

Later Career and Honors

Leloir continued to direct his institute in Buenos Aires until his death. He died on 1987-12-02 in Buenos Aires, Argentina.

Legacy

Sugar nucleotides are recognized as fundamental building blocks of carbohydrate metabolism in all organisms.

Modern Developments

Leloir's discovery of sugar nucleotides and his work on carbohydrate metabolism founded modern glycobiology and the understanding of how cells build polysaccharides. His research underpins the production of oligosaccharides and glycoconjugates used in medicine, including vaccines and therapeutic antibodies, and continues to guide research on metabolic diseases.

Selected Publications

"A Coenzyme Concerned in the Transformation of Glucose-1-Phosphate to Galactose-1-Phosphate" (Journal of Biological Chemistry, 1950)

"Two Decades of Research on the Biosynthesis of Saccharides" (Nobel Lecture, 1970)

Chapter Summary

The 1970 prize honored Luis Leloir for the discovery of sugar nucleotides and their role in carbohydrate biosynthesis. His work revealed the molecular mechanism by which cells build polysaccharides.

Chapter 71

1971:Gerhard Herzberg

The Nobel Prize in Chemistry for the year 1971 was awarded to Gerhard Herzberg.

Motivation:

for his contributions to the knowledge of electronic structure and geometry of molecules, particularly free radicals

71.1 Gerhard Herzberg

Early Life and Education

Gerhard Herzberg was born on 1904-12-25 in Hamburg, Germany.

Career and Major Research

Herzberg studied at the Technische Hochschule Darmstadt, where he received his doctorate in 1928. After emigrating to Canada, he worked at the University of Saskatchewan and then at the National Research Council in Ottawa. He used molecular spectroscopy to determine the electronic structure and geometry of molecules, including short-lived free radicals.

Nobel Prize-winning Work

The work for which Gerhard Herzberg was awarded the Nobel Prize in Chemistry in 1971 was described by the Nobel Committee as: "for his contributions to the knowledge of electronic structure and geometry of molecules, particularly free radicals".

Herzberg measured and analyzed the spectra of diatomic and polyatomic molecules and of radicals such as CH, OH, and CH₂, establishing their structures and energy levels with high precision.

Significance and Impact

Herzberg's spectroscopic data provided the experimental basis for understanding molecular structure and for identifying molecules in planetary atmospheres and interstellar space.

Later Career and Honors

Herzberg remained at the National Research Council in Ottawa for the rest of his career. He died on 1999-03-03 in Ottawa, Canada.

Legacy

Herzberg's books on molecular spectra and structure are classic references that trained generations of spectroscopists.

Modern Developments

Herzberg's spectroscopic studies of free radicals and diatomic molecules established molecular spectroscopy and provided the foundational data for understanding chemical bonds. His work opened the modern field of molecular astrophysics, enabling astronomers to identify molecules in interstellar space, and his techniques are used to study reaction intermediates and atmospheric chemistry.

Selected Publications

"Molecular Spectra and Molecular Structure, Vol. I: Spectra of Diatomic Molecules" (1939)

"The Spectra and Structures of Simple Free Radicals" (1971)

Chapter Summary

The 1971 prize recognized Gerhard Herzberg for determining the electronic structure and geometry of molecules, especially free radicals, by spectroscopy. His measurements are fundamental to molecular physics and astronomy.

Chapter 72

1972:Christian Anfinsen and Stanford Moore and William H. Stein

The Nobel Prize in Chemistry for the year 1972 was awarded jointly to Christian Anfinsen and Stanford Moore and William H. Stein.

Motivation:

for his work on ribonuclease, especially concerning the connection between the amino acid sequence and the biologically active conformation

72.1 Christian Anfinsen

Early Life and Education

Christian Anfinsen was born on 1916-03-26 in Monessen, Pennsylvania, USA.

Career and Major Research

Anfinsen studied at Swarthmore College and received his doctorate from Harvard University in 1943. He worked at the National Institutes of Health in Bethesda, where he studied the pancreatic enzyme ribonuclease. He showed that the enzyme could be unfolded and then refolded spontaneously to its active form.

Nobel Prize-winning Work

The work for which Christian Anfinsen was awarded the Nobel Prize in Chemistry in 1972 was described by the Nobel Committee as: "for his work on ribonuclease, especially concerning the connection between the amino acid sequence and the biologically active conformation".

Anfinsen demonstrated that the amino acid sequence of ribonuclease determines its biologically active three-dimensional structure, establishing the thermodynamic principle of protein folding.

Significance and Impact

The finding that a protein folds spontaneously according to its sequence laid the foundation for protein chemistry and for understanding protein folding and misfolding diseases.

Later Career and Honors

Anfinsen remained at the National Institutes of Health for most of his career. He died on 1995-05-14 in Randallstown, Maryland, USA.

Legacy

Anfinsen's principle that sequence determines structure is a cornerstone of molecular biology and bioinformatics.

Modern Developments

Anfinsen's demonstration that the amino acid sequence determines protein folding established the thermodynamic view of proteins and the protein-folding problem. His principle underpins modern protein engineering, the prediction of protein structure from sequence, and the design of new proteins. Anfinsen's ideas culminated in the AlphaFold revolution in structural biology.

Selected Publications

"Principles That Govern the Folding of Protein Chains" (Science, 1973)

72.2 Stanford Moore

Early Life and Education

Stanford Moore was born on 1913-09-04 in Chicago, Illinois, USA.

Career and Major Research

Moore studied at Vanderbilt University and received his doctorate from the University of Wisconsin. He worked at the Rockefeller Institute, later Rockefeller University, in New York for his entire career. With William H. Stein he developed methods of ion-exchange chromatography for the quantitative analysis of amino acids.

Nobel Prize-winning Work

The work for which Stanford Moore was awarded the Nobel Prize in Chemistry in 1972 was described by the Nobel Committee as: "for his work on ribonuclease, especially concerning the connection between the amino acid sequence and the biologically active conformation".

Moore and Stein determined the complete amino acid sequence of ribonuclease, providing the chemical basis for Anfinsen's studies of its folding.

Significance and Impact

Their chromatographic methods and automatic amino acid analyzers made protein sequence determination practical and transformed biochemistry.

Later Career and Honors

Moore remained at Rockefeller University until his retirement. He died on 1982-08-23 in New York City, USA.

Legacy

Moore and Stein's analytical methods are the forerunners of modern protein analysis and proteomics.

Modern Developments

Moore and Stein developed methods for the quantitative analysis of amino acids and the determination of the amino acid sequence of proteins, including the first complete sequence of ribonuclease. Their automated amino acid analyzers and sequencing methods laid the groundwork for the modern biotechnology industry and for the large-scale protein analysis used in proteomics today.

Selected Publications

"Photometric Ninhydrin Method for Use in the Chromatography of Amino Acids" (Journal of Biological Chemistry, 1948)

"Automatic Recording Apparatus for Use in the Chromatography of Amino Acids" (Analytical Chemistry, 1958)

72.3 William H. Stein

Early Life and Education

William H. Stein was born on 1911-06-25 in New York City, New York, USA.

Career and Major Research

Stein studied at Harvard College and received his doctorate from Columbia University. He joined the Rockefeller Institute in 1938 and remained there for his entire career. With Stanford Moore he developed chromatographic methods for analyzing amino acids and proteins.

Nobel Prize-winning Work

The work for which William H. Stein was awarded the Nobel Prize in Chemistry in 1972 was described by the Nobel Committee as: "for his work on ribonuclease, especially concerning the connection between the amino acid sequence and the biologically active conformation".

Stein and Moore established the amino acid sequence of ribonuclease and related the structure of the enzyme to its catalytic activity.

Significance and Impact

Their determination of a complete enzyme sequence, achieved through quantitative chromatography, was a landmark in protein chemistry.

Later Career and Honors

Stein remained at Rockefeller University for the rest of his career. He died on 1980-02-02 in New York City, USA.

Legacy

Stein's analytical techniques helped establish the modern era of protein science.

Modern Developments

Moore and Stein's methods for analyzing amino acid composition and sequence made modern protein biochemistry possible and led to the first determination of the complete sequence of an enzyme, ribonuclease. Their automated analyzers evolved into today's protein sequencers and mass spectrometry-based proteomics, which are central to biology and medicine.

Selected Publications

"Photometric Ninhydrin Method for Use in the Chromatography of Amino Acids" (Journal of Biological Chemistry, 1948)

"Automatic Recording Apparatus for Use in the Chromatography of Amino Acids" (Analytical Chemistry, 1958)

Chapter Summary

The 1972 prize recognized Anfinsen, Moore, and Stein for their work on ribonuclease. Their research connected the amino acid sequence of an enzyme to its biologically active three-dimensional structure.

Chapter 73

1973:Ernst Otto Fischer and Geoffrey Wilkinson

The Nobel Prize in Chemistry for the year 1973 was awarded jointly to Ernst Otto Fischer and Geoffrey Wilkinson.

Motivation:

for their pioneering work, performed independently, on the chemistry of the organometallic, so called sandwich compounds

73.1 Ernst Otto Fischer

Early Life and Education

Ernst Otto Fischer was born on 1918-11-10 in Munich, Germany.

Career and Major Research

Fischer studied at the Technical University of Munich, where he received his doctorate and later became a professor. He investigated the structure of ferrocene and other cyclopentadienyl complexes of the transition metals, and he discovered the transition-metal carbene complexes that bear his name.

Nobel Prize-winning Work

The work for which Ernst Otto Fischer was awarded the Nobel Prize in Chemistry in 1973 was described by the Nobel Committee as: "for their pioneering work, performed independently, on the chemistry of the organometallic, so called sandwich compounds".

Working independently of Wilkinson, Fischer established by X-ray crystallography that ferrocene has a sandwich structure in which an iron atom is held between two cyclopentadienyl rings, and he extended this chemistry to other metals.

Significance and Impact

Sandwich compounds opened a major field of organometallic chemistry with wide applications in catalysis and materials science.

Later Career and Honors

Fischer remained at the Technical University of Munich for the rest of his career. He died on 2007-07-23 in Munich, Germany.

Legacy

Metallocenes and related sandwich complexes are now fundamental building blocks in inorganic and organometallic chemistry.

Modern Developments

Fischer's synthesis of coordination compounds and his understanding of metal-carbon bonds founded organometallic chemistry. The ferrocene sandwich compounds he and others explored became a cornerstone of organometallic catalysis, and modern homogeneous catalysis, including hydroformylation and cross-coupling reactions used in drug and materials synthesis, descends from his work.

Selected Publications

"Zur Kristallstruktur des Ferrocens" (*Zeitschrift für Naturforschung*, 1952)

73.2 Geoffrey Wilkinson

Early Life and Education

Geoffrey Wilkinson was born on 1921-07-14 in Todmorden, England.

Career and Major Research

Wilkinson studied at Imperial College London, where he received his doctorate. After research positions in Canada and at Harvard University, he became professor at Imperial College. He established the sandwich structure of ferrocene and later introduced the rhodium complex known as Wilkinson's catalyst, widely used for hydrogenating alkenes.

Nobel Prize-winning Work

The work for which Geoffrey Wilkinson was awarded the Nobel Prize in Chemistry in 1973 was described by the Nobel Committee as: "for their pioneering work, performed independently, on the chemistry of the organometallic, so called sandwich compounds".

Working with collaborators at Harvard, Wilkinson recognized that ferrocene, then called iron bis-cyclopentadienyl, consists of an iron atom sandwiched between two cyclopentadienyl rings.

Significance and Impact

The discovery of the sandwich structure transformed the understanding of metal-carbon bonding and stimulated the growth of homogeneous catalysis.

Later Career and Honors

Wilkinson remained at Imperial College until his retirement. He died on 1996-09-26 in London, England.

Legacy

Wilkinson's catalyst remains a model for homogeneous hydrogenation, and metallocene chemistry is central to modern inorganic chemistry.

Modern Developments

Wilkinson's work on organometallic chemistry, including the discovery and characterization of ferrocene, opened the modern era of homogeneous catalysis. The Wilkinson catalyst for hydrogenation and the countless transition-metal catalysts developed since are now indispensable in pharmaceutical synthesis, agrochemical production, and the manufacture of fine chemicals.

Selected Publications

"The Structure of Iron Bis-Cyclopentadienyl" (Journal of the American Chemical Society, 1952)

Chapter Summary

The 1973 prize recognized Fischer and Wilkinson for their independent work on sandwich compounds. Their elucidation of the structure of ferrocene founded the modern chemistry of metallocenes.

Chapter 74

1974:Paul J. Flory

The Nobel Prize in Chemistry for the year 1974 was awarded to Paul J. Flory.

Motivation:

for his fundamental achievements, both theoretical and experimental, in the physical chemistry of the macromolecules

74.1 Paul J. Flory

Early Life and Education

Paul J. Flory was born on 1910-06-19 in Sterling, Illinois, USA.

Career and Major Research

Flory studied at Manchester College in Indiana and received his doctorate from Ohio State University in 1934. He worked at DuPont, Standard Oil of New Jersey, and other industrial laboratories, then held professorships at Cornell University, the Mellon Institute, and Stanford University. He developed the theory of polymer solutions, including the Flory-Huggins theory, and studied the kinetics of polymerization and the statistical mechanics of polymer chains.

Nobel Prize-winning Work

The work for which Paul J. Flory was awarded the Nobel Prize in Chemistry in 1974 was described by the Nobel Committee as: "for his fundamental achievements, both theoretical and experimental, in the physical chemistry of the macromolecules".

Flory showed that the physical properties of polymers follow from the statistical behavior of their long chains, and his equations describe polymer solutions, gels, and the thermodynamics of mixing.

Significance and Impact

Flory's theories are the foundation of modern polymer science and have guided the design of plastics, fibers, and rubber.

Later Career and Honors

Flory remained at Stanford University for the rest of his career. He died on 1985-09-09 in Big Sur, California, USA.

Legacy

The Flory-Huggins theory and Flory's other results remain standard tools in polymer physics and materials science.

Modern Developments

Flory's theories of polymer chain statistics, excluded volume, and rubber elasticity remain the mathematical foundation of modern polymer and soft-matter science. His work underpins the design and engineering of plastics, elastomers, fibers, and gels used across industry and medicine. Flory's quantitative approach to polymer thermodynamics is still the standard framework for understanding how long-chain molecules behave in solution, melts, and networks.

Selected Publications

"Thermodynamics of High Polymer Solutions" (Journal of Chemical Physics, 1942)

"Principles of Polymer Chemistry" (Cornell University Press, 1953)

Chapter Summary

The 1974 prize honored Paul J. Flory for his theoretical and experimental achievements in the physical chemistry of macromolecules. His work established polymer science as a quantitative discipline.

Chapter 75

1975: John Cornforth and Vladimir Prelog

The Nobel Prize in Chemistry for the year 1975 was awarded jointly to John Cornforth and Vladimir Prelog.

Motivation:

for his work on the stereochemistry of enzyme-catalyzed reactions (one half, John Cornforth) and for his research into the stereochemistry of organic molecules and reactions (the other half, Vladimir Prelog)

75.1 John Cornforth

Early Life and Education

John Cornforth was born on 1917-09-07 in Sydney, Australia.

Career and Major Research

Cornforth studied at the University of Sydney and at Oxford, where he received his doctorate. He worked at the National Institute for Medical Research in London and later became a professor at the University of Sussex. He was deaf from an early age. Using isotopic labeling, he worked out the stereochemistry of many enzyme-catalyzed reactions, particularly in the biosynthesis of cholesterol.

Nobel Prize-winning Work

The work for which John Cornforth was awarded the Nobel Prize in Chemistry in 1975 was described by the Nobel Committee as: "for his work on the stereochemistry of enzyme-catalyzed reactions".

Cornforth traced the fate of individual hydrogen atoms and methyl groups through biosynthetic pathways, establishing the three-dimensional course of reactions catalyzed by enzymes.

Significance and Impact

His methods for following stereochemistry with isotopic labels provided deep insight into how enzymes control the geometry of reactions.

Later Career and Honors

Cornforth was knighted in 1977. He died on 2013-12-08 in Sussex, England.

Legacy

Cornforth's stereochemical analyses are fundamental to mechanistic biochemistry and to the understanding of enzyme catalysis.

Modern Developments

Cornforth's synthesis of isotopically labeled cholesterol and his stereochemical analysis of enzyme-catalyzed reactions established the field of stereochemistry of biosynthesis. His work revealed how enzymes control stereochemistry and continues to inform drug design and the understanding of metabolic pathways. Cornforth's methods are central to modern mechanistic enzymology.

Selected Publications

"Asymmetry and Enzyme Action" (Nobel Lecture, 1975)

75.2 Vladimir Prelog

Early Life and Education

Vladimir Prelog was born on 1906-07-23 in Sarajevo, Bosnia.

Career and Major Research

Prelog studied at the Czech Technical University in Prague, where he received his doctorate. He worked in the chemical industry in Zagreb and then joined the Swiss Federal Institute of Technology in Zurich, where he was professor of organic chemistry. He studied the stereochemistry of organic molecules, including medium-sized rings, and with Robert Cahn and Christopher Ingold developed the system for specifying the absolute configuration of chiral molecules.

Nobel Prize-winning Work

The work for which Vladimir Prelog was awarded the Nobel Prize in Chemistry in 1975 was described by the Nobel Committee as: "for his research into the stereochemistry of organic molecules and reactions".

Prelog's research clarified the relationship between molecular structure and chirality, and the Cahn-Ingold-Prelog rules he helped devise are used to describe stereoisomers.

Significance and Impact

The Cahn-Ingold-Prelog priority rules became the international standard for describing the configuration of stereocenters.

Later Career and Honors

Prelog remained at the Swiss Federal Institute of Technology in Zurich until his retirement. He died on 1998-01-07 in Zurich, Switzerland.

Legacy

The R/S system of stereochemical nomenclature developed with Prelog's participation is used throughout chemistry.

Modern Developments

Prelog's studies of stereochemistry and his work on medium-ring compounds deepened the understanding of molecular chirality and its consequences. The Cahn-Ingold-Prelog priority rules, which he co-developed, are now used worldwide to name and describe the three-dimensional configuration of chiral molecules, a fundamental tool in organic chemistry, pharmacology, and biology.

Selected Publications

"Specification of Molecular Chirality" (Angewandte Chemie International Edition, 1966)

Chapter Summary

The 1975 prize recognized Cornforth and Prelog for their work on the stereochemistry of enzyme-catalyzed reactions and of organic molecules. Their research established how molecular geometry governs chemical reactivity.

Chapter 76

1976: William Lipscomb

The Nobel Prize in Chemistry for the year 1976 was awarded to William Lipscomb.

Motivation:

for his studies on the structure of boranes illuminating problems of chemical bonding

76.1 William Lipscomb

Early Life and Education

William Lipscomb was born on 1919-12-09 in Cleveland, Ohio, USA.

Career and Major Research

Lipscomb studied at the University of Kentucky and received his doctorate from the California Institute of Technology in 1946. He taught at the University of Minnesota and then at Harvard University, where he was professor of chemistry. He determined the structures of the boranes, including diborane and the higher boron hydrides, and developed theories of their electron-deficient bonding.

Nobel Prize-winning Work

The work for which William Lipscomb was awarded the Nobel Prize in Chemistry in 1976 was described by the Nobel Committee as: "for his studies on the structure of boranes illuminating problems of chemical bonding".

Lipscomb used X-ray and electron diffraction to determine the structures of the boranes and showed how these electron-deficient molecules are held together by multicenter bonds.

Significance and Impact

His structural work clarified the nature of the chemical bond in compounds that cannot be described by ordinary two-center two-electron bonds.

Later Career and Honors

Lipscomb remained at Harvard University until his retirement. He died on 2011-04-14 in Cambridge, Massachusetts, USA.

Legacy

Lipscomb's borane structures and bonding theories underpin modern boron cluster chemistry.

Modern Developments

Lipscomb's boron hydride studies revealed three-center, two-electron bonds and founded the modern chemistry of electron-deficient compounds. His methods extended to the elucidation of enzyme mechanisms, including carboxypeptidase. Boron chemistry is now used in drug discovery, including boron neutron capture therapy for cancer, and in the synthesis of complex organic molecules.

Selected Publications

"Boron Hydrides" (1963)

"The Boranes and Their Relatives" (Nobel Lecture, 1976)

Chapter Summary

The 1976 prize honored William Lipscomb for his structural studies of the boranes. His work illuminated the nature of chemical bonding in electron-deficient molecules.

Chapter 77

1977:Ilya Prigogine

The Nobel Prize in Chemistry for the year 1977 was awarded to Ilya Prigogine.

Motivation:

for his contributions to non-equilibrium thermodynamics, particularly the theory of dissipative structures

77.1 Ilya Prigogine

Early Life and Education

Ilya Prigogine was born on 1917-01-25 in Moscow, Russia.

Career and Major Research

Prigogine studied at the Free University of Brussels, where he received his doctorate and later became a professor. He also held a professorship at the University of Texas at Austin. He developed the thermodynamics of irreversible processes and introduced the concept of dissipative structures, ordered states that form in systems far from equilibrium.

Nobel Prize-winning Work

The work for which Ilya Prigogine was awarded the Nobel Prize in Chemistry in 1977 was described by the Nobel Committee as: "for his contributions to non-equilibrium thermodynamics, particularly the theory of dissipative structures".

Prigogine showed that open systems exchanging energy and matter with their surroundings can spontaneously organize into stable, ordered structures such as those seen in oscillating chemical reactions.

Significance and Impact

Dissipative structures connected chemistry to biology by explaining how order can arise in non-equilibrium systems, influencing fields from chemical kinetics to ecology.

Later Career and Honors

Prigogine worked in both Brussels and Austin for the rest of his career. He died on 2003-05-28 in Brussels, Belgium.

Legacy

Prigogine's ideas on self-organization in far-from-equilibrium systems remain influential across the natural and social sciences.

Modern Developments

Prigogine's theory of dissipative structures explained how order can emerge from chaos in far-from-equilibrium systems, providing a scientific foundation for understanding self-organization in chemistry and biology. His ideas are applied in pattern formation, chemical oscillations such as the Belousov–Zhabotinsky reaction, and the thermodynamics of life itself.

Selected Publications

"Introduction to Thermodynamics of Irreversible Processes" (1955)

"From Being to Becoming: Time and Complexity in the Physical Sciences" (1980)

Chapter Summary

The 1977 prize recognized Ilya Prigogine for his contributions to non-equilibrium thermodynamics. His theory of dissipative structures explains how order can emerge in systems far from equilibrium.

Chapter 78

1978:Peter Mitchell

The Nobel Prize in Chemistry for the year 1978 was awarded to Peter Mitchell.

Motivation:

for his contribution to the understanding of biological energy transfer through the formulation of the chemiosmotic theory

78.1 Peter Mitchell

Early Life and Education

Peter Mitchell was born on 1920-09-29 in Mitcham, England.

Career and Major Research

Mitchell studied at the University of Cambridge, where he received his doctorate. He worked at the University of Edinburgh and later, with Jennifer Moyle, established the Glynn Research Institute in Bodmin, Cornwall, to carry out independent research. He formulated the chemiosmotic theory of biological energy transduction.

Nobel Prize-winning Work

The work for which Peter Mitchell was awarded the Nobel Prize in Chemistry in 1978 was described by the Nobel Committee as: "for his contribution to the understanding of biological energy transfer through the formulation of the chemiosmotic theory".

Mitchell proposed that electron transport in respiration and photosynthesis builds up a gradient of protons across a membrane, and that the flow of protons back through the membrane drives the synthesis of ATP.

Significance and Impact

The chemiosmotic theory explained how energy is stored and used in cells and became the accepted mechanism of oxidative phosphorylation and photophosphorylation.

Later Career and Honors

Mitchell worked at the Glynn Research Institute until his death. He died on 1992-04-10 in Bodmin, Cornwall, England.

Legacy

The chemiosmotic mechanism is now a cornerstone of bioenergetics and cellular biology.

Modern Developments

Mitchell's chemiosmotic theory revealed that cells generate energy by building a proton gradient across membranes, a concept now recognized as universal in bioenergetics. Chemiosmosis explains the mechanism of ATP synthesis in mitochondria, chloroplasts, and bacteria and underpins the design of artificial photosynthetic systems and the understanding of drug action on energy metabolism.

Selected Publications

"Coupling of Phosphorylation to Electron and Hydrogen Transfer by a Chemi-Osmotic Type of Mechanism"
(Nature, 1961)

Chapter Summary

The 1978 prize honored Peter Mitchell for formulating the chemiosmotic theory of biological energy transfer. His work explained how proton gradients drive ATP synthesis in cells.

Chapter 79

1979:Herbert C. Brown and Georg Wittig

The Nobel Prize in Chemistry for the year 1979 was awarded jointly to Herbert C. Brown and Georg Wittig.

Motivation:

for their development of the use of boron- and phosphorus-containing compounds, respectively, into important reagents in organic synthesis

79.1 Herbert C. Brown

Early Life and Education

Herbert C. Brown was born on 1912-05-22 in London, England.

Career and Major Research

Brown immigrated to the United States with his family as a child and grew up in Chicago. He studied at Wright Junior College and the University of Chicago, where he received his doctorate. He taught at Wayne State University, now part of Wayne State University, and then at Purdue University, where he remained for the rest of his career. He discovered hydroboration, the addition of borane to alkenes, and developed the chemistry of organoboranes as versatile synthetic intermediates.

Nobel Prize-winning Work

The work for which Herbert C. Brown was awarded the Nobel Prize in Chemistry in 1979 was described by the Nobel Committee as: "for their development of the use of boron- and phosphorus-containing compounds, respectively, into important reagents in organic synthesis".

Brown showed that hydroboration gives organoboranes that can be converted into a wide variety of functional groups with high selectivity and stereochemical control.

Significance and Impact

Hydroboration and organoborane chemistry became among the most widely used methods in organic synthesis.

Later Career and Honors

Brown remained at Purdue University until his retirement. He died on 2004-12-19 in West Lafayette, Indiana, USA.

Legacy

Brown's boron chemistry is a standard toolkit of synthetic organic chemistry.

Modern Developments

Brown's hydroboration reaction revolutionized organic synthesis by providing a mild, regio- and stereoselective way to convert alkenes into alcohols and other functional groups. Hydroboration and the related organoborane chemistry are now standard tools in the synthesis of pharmaceuticals and natural products, and boron reagents feature prominently in modern cross-coupling and borylation chemistry.

Selected Publications

"Hydroboration" (1962)

79.2 Georg Wittig

Early Life and Education

Georg Wittig was born on 1897-06-16 in Berlin, Germany.

Career and Major Research

Wittig studied at the University of Marburg, where he received his doctorate. He held professorships at Marburg, Braunschweig, Tübingen, and Heidelberg. He discovered the rearrangement of ylides and the reaction that bears his name, in which a phosphorus ylide reacts with a carbonyl compound to form an alkene.

Nobel Prize-winning Work

The work for which Georg Wittig was awarded the Nobel Prize in Chemistry in 1979 was described by the Nobel Committee as: "for their development of the use of boron- and phosphorus-containing compounds, respectively, into important reagents in organic synthesis".

The Wittig reaction allows chemists to place a carbon-carbon double bond at a defined position in a molecule, with predictable stereochemistry.

Significance and Impact

The Wittig reaction became one of the most valuable methods for synthesizing alkenes, including the natural product vitamin A.

Later Career and Honors

Wittig remained at the University of Heidelberg until his retirement. He died on 1987-08-26 in Heidelberg, Germany.

Legacy

The Wittig reaction is a fundamental tool for constructing carbon-carbon double bonds in organic synthesis.

Modern Developments

Wittig's discovery of the olefination reaction that bears his name provided a reliable method for constructing carbon-carbon double bonds, a transformation now used routinely in the synthesis of pharmaceuticals, natural products, and materials. The Wittig reaction remains a cornerstone of organic synthesis and has been extended into modern variants such as the Horner–Wadsworth–Emmons reaction.

Selected Publications

"Zur Reaktionsweise des Pentaphenyl-phosphors und einiger Derivate" (Liebigs Annalen der Chemie, 1953)

Chapter Summary

The 1979 prize recognized Brown and Wittig for developing boron- and phosphorus-containing reagents into important tools of organic synthesis. Their methods are essential to modern synthetic chemistry.

Chapter 80

1980:Paul Berg and Walter Gilbert and Frederick Sanger

The Nobel Prize in Chemistry for the year 1980 was awarded half to Paul Berg and half to Walter Gilbert and Frederick Sanger.

Motivation:

for his fundamental studies of the biochemistry of nucleic acids, with particular regard to recombinant-DNA (one half, Paul Berg) and for their respective contributions concerning the determination of base sequences in nucleic acids (the other half, jointly, Walter Gilbert and Frederick Sanger)

80.1 Paul Berg

Early Life and Education

Paul Berg was born on 1926-06-30 in New York City, New York, USA.

Career and Major Research

Berg studied at Pennsylvania State University and received his doctorate from Case Western Reserve University in 1952. He worked at the National Institutes of Health and at Stanford University, where he was professor of biochemistry. He developed methods for combining the DNA of different organisms and helped pioneer recombinant-DNA technology.

Nobel Prize-winning Work

The work for which Paul Berg was awarded the Nobel Prize in Chemistry in 1980 was described by the Nobel Committee as: "for his fundamental studies of the biochemistry of nucleic acids, with particular regard to recombinant-DNA".

Berg constructed the first recombinant DNA molecules by splicing DNA from different species, creating the basis for genetic engineering.

Significance and Impact

Recombinant DNA technology gave scientists the ability to isolate and amplify genes and proteins, transforming biology, medicine, and the biotechnology industry.

Later Career and Honors

Berg remained at Stanford University until his retirement and later became a leading voice on the responsible use of genetic engineering. He died on 2023-02-15 in Stanford, California, USA.

Legacy

Berg is regarded as one of the founders of genetic engineering and of the modern era of molecular biology.

Modern Developments

Berg's development of recombinant DNA technology founded genetic engineering and the modern biotechnology industry. His techniques enable the production of therapeutic proteins, including insulin and monoclonal antibodies, and underpinned the sequencing and manipulation of genomes. Berg's early advocacy for biosafety regulation also shaped modern governance of genetic research.

Selected Publications

"Biochemical Method for Inserting New Genetic Information into DNA of Simian Virus 40: Circular SV40 DNA Molecules Containing Lambda Phage Genes and the Galactose Operon of Escherichia coli" (Proceedings of the National Academy of Sciences, 1972)

80.2 Walter Gilbert

Early Life and Education

Walter Gilbert was born on 1932-03-21 in Boston, Massachusetts, USA.

Career and Major Research

Gilbert studied at Harvard University and received his doctorate from the University of Cambridge. He taught physics and later biology at Harvard University, where he was professor of molecular biology. With Allan Maxam he developed a chemical method for sequencing DNA.

Nobel Prize-winning Work

The work for which Walter Gilbert was awarded the Nobel Prize in Chemistry in 1980 was described by the Nobel Committee as: "for their respective contributions concerning the determination of base sequences in nucleic acids".

Gilbert developed the Maxam-Gilbert method, in which DNA is labeled and cut with base-specific chemicals to determine its sequence, and he also discovered the mechanism of repression by which gene regulation is controlled.

Significance and Impact

His DNA sequencing methods and his work on gene regulation were central to the rise of molecular biology and genomics.

Later Career and Honors

Gilbert later worked in biotechnology and in the private sector before returning to Harvard. He is still alive.

Legacy

Gilbert's sequencing methods helped make genome analysis possible and contributed to the early genomics industry.

Modern Developments

Gilbert's methods for DNA sequencing and his discovery of the lac repressor helped establish molecular biology and the technology of reading genetic information. His sequencing method, developed alongside Sanger's, contributed directly to the Human Genome Project and the modern genomics and personalized medicine industries.

Selected Publications

"A New Method for Sequencing DNA" (Proceedings of the National Academy of Sciences, 1977)

80.3 Frederick Sanger

Early Life and Education

Frederick Sanger was born on 1918-08-13 in Rendcomb, England.

Career and Major Research

Sanger studied at the University of Cambridge, where he received his doctorate. He worked for his entire career at Cambridge, at the Department of Biochemistry and later at the Laboratory of Molecular Biology. He determined the amino acid sequence of insulin and later developed chain-termination methods for sequencing DNA.

Nobel Prize-winning Work

The work for which Frederick Sanger was awarded the Nobel Prize in Chemistry in 1980 was described by the Nobel Committee as: "for their respective contributions concerning the determination of base sequences in nucleic acids".

Sanger developed the dideoxy chain-termination method, in which DNA synthesis is stopped at each base to read the sequence. He had earlier received the 1958 Nobel Prize for determining the structure of insulin.

Significance and Impact

Sanger's sequencing method became the standard technology of genomics and was used to determine the human genome.

Later Career and Honors

Sanger retired in 1983. He is the only person to have been awarded the Nobel Prize in Chemistry twice. He died on 2013-11-19 in Cambridge, England.

Legacy

Sanger's methods for sequencing proteins and DNA laid the foundations of the genomics revolution.

Modern Developments

Sanger's second Nobel Prize recognized his invention of chain-termination DNA sequencing, the method that made reading genomes practical and affordable. Sanger sequencing powered the Human Genome Project and remains the gold standard for verifying DNA sequences, while his earlier protein sequencing established the molecular biology paradigm.

Selected Publications

"The Free Amino Groups of Insulin" (Biochemical Journal, 1945)

"DNA Sequencing with Chain-Terminating Inhibitors" (Proceedings of the National Academy of Sciences, 1977)

Chapter Summary

The 1980 prize recognized Paul Berg, Walter Gilbert, and Frederick Sanger for their contributions to nucleic acid biochemistry. Their work established recombinant DNA technology and methods for determining DNA sequences.

Chapter 81

1981:Kenichi Fukui and Roald Hoffmann

The Nobel Prize in Chemistry for the year 1981 was awarded jointly to Kenichi Fukui and Roald Hoffmann.

Motivation:

for their theories, developed independently, concerning the course of chemical reactions

81.1 Kenichi Fukui

Early Life and Education

Kenichi Fukui was born on 1918-10-04 in Nara, Japan.

Career and Major Research

Fukui studied at the Kyoto Imperial University, where he received his doctorate and later became professor of chemistry. He developed the frontier orbital theory, which describes how the highest occupied and lowest unoccupied molecular orbitals control the reactivity of molecules.

Nobel Prize-winning Work

The work for which Kenichi Fukui was awarded the Nobel Prize in Chemistry in 1981 was described by the Nobel Committee as: "for their theories, developed independently, concerning the course of chemical reactions".

Fukui showed that chemical reactions are governed by the interaction of the frontier orbitals, explaining the regio- and stereoselectivity of many reactions.

Significance and Impact

Frontier orbital theory provided a simple and powerful way to predict the course of chemical reactions and became a standard tool for chemists.

Later Career and Honors

Fukui remained at Kyoto University until his retirement. He was the first Japanese Nobel laureate in chemistry. He died on 1998-01-09 in Kyoto, Japan.

Legacy

Fukui's frontier orbital concept underlies modern computational chemistry and the understanding of reactivity.

Modern Developments

Fukui's frontier orbital theory explained how the highest occupied and lowest unoccupied molecular orbitals control chemical reactivity, founding modern theoretical chemistry and the prediction of reaction pathways. Frontier orbital thinking underpins computational drug design, catalysis research, and the understanding of pericyclic and photochemical reactions.

Selected Publications

"A Molecular Orbital Theory of Reactivity in Aromatic Hydrocarbons" (Journal of Chemical Physics, 1952)

81.2 Roald Hoffmann

Early Life and Education

Roald Hoffmann was born on 1937-07-18 in Zloczów, Poland.

Career and Major Research

Hoffmann survived the Holocaust and emigrated to the United States, studying at Columbia University and Harvard University, where he received his doctorate. He worked at Harvard with Robert B. Woodward and then became professor at Cornell University. He developed extended Hückel calculations and, with Woodward, formulated the Woodward-Hoffmann rules for the conservation of orbital symmetry.

Nobel Prize-winning Work

The work for which Roald Hoffmann was awarded the Nobel Prize in Chemistry in 1981 was described by the Nobel Committee as: "for their theories, developed independently, concerning the course of chemical reactions".

The Woodward-Hoffmann rules predict the course of pericyclic reactions on the basis of the symmetry of molecular orbitals.

Significance and Impact

The rules explained how stereochemistry is controlled in pericyclic reactions and helped unify organic reaction theory.

Later Career and Honors

Hoffmann remained at Cornell University and is also a published writer of poetry and essays. He is still alive.

Legacy

The Woodward-Hoffmann rules remain a fundamental principle of physical organic chemistry.

Modern Developments

Hoffmann's extended Hückel theory and the Woodward-Hoffmann rules, developed with Robert Woodward, provide the framework for understanding and predicting the stereochemistry of pericyclic reactions. His rules are used daily in planning synthetic routes and are taught in every advanced organic chemistry course, while his theoretical methods remain tools for analyzing molecular electronic structure.

Selected Publications

"The Conservation of Orbital Symmetry" (Angewandte Chemie International Edition, 1969)

Chapter Summary

The 1981 prize recognized Fukui and Hoffmann for their independently developed theories of the course of chemical reactions. Their orbital-based theories explained and predicted reactivity in organic chemistry.

Chapter 82

1982: Aaron Klug

The Nobel Prize in Chemistry for the year 1982 was awarded to Aaron Klug.

Motivation:

for his development of crystallographic electron microscopy and his structural elucidation of biologically important nucleic acid-protein complexes

82.1 Aaron Klug

Early Life and Education

Aaron Klug was born on 1926-08-11 in Zelva, Lithuania.

Career and Major Research

Klug's family moved to South Africa when he was a child. He studied at the University of the Witwatersrand and received his doctorate from the University of Cambridge. He worked at Birkbeck College in London and then at the Laboratory of Molecular Biology in Cambridge, where he developed crystallographic electron microscopy and used it to study the structures of viruses and other biological assemblies.

Nobel Prize-winning Work

The work for which Aaron Klug was awarded the Nobel Prize in Chemistry in 1982 was described by the Nobel Committee as: "for his development of crystallographic electron microscopy and his structural elucidation of biologically important nucleic acid-protein complexes".

Klug combined electron microscopy with mathematical analysis to reconstruct three-dimensional structures from images, and he used these methods to determine the structure of the tobacco mosaic virus and, with his colleagues, the nucleosome.

Significance and Impact

His imaging methods allowed the structures of large biological assemblies to be determined at high resolution, opening a new field of structural biology.

Later Career and Honors

Klug worked at the Laboratory of Molecular Biology in Cambridge for most of his career and was awarded the Order of Merit in 1995. He died on 2018-11-20 in Cambridge, England.

Legacy

The techniques Klug pioneered are the forerunners of modern cryo-electron microscopy.

Modern Developments

Klug's development of crystallographic electron microscopy made it possible to visualize the structures of viruses and large macromolecular assemblies. His methods evolved into modern cryo-electron microscopy, which now routinely determines the structures of proteins and viruses at atomic resolution and is central to the development of vaccines and drugs, including for COVID-19.

Selected Publications

"From Macromolecules to Biological Assemblies" (Nobel Lecture, 1982)

"Structure of the Nucleosome Core Particle at 7 Å Resolution" (Nature, 1984)

Chapter Summary

The 1982 prize honored Aaron Klug for developing crystallographic electron microscopy and for determining the structures of nucleic acid-protein complexes. His methods founded modern structural biology of large assemblies.

Chapter 83

1983:Henry Taube

The Nobel Prize in Chemistry for the year 1983 was awarded to Henry Taube.

Motivation:

for his work on the mechanisms of electron transfer reactions, especially in metal complexes

83.1 Henry Taube

Early Life and Education

Henry Taube was born on 1915-11-30 in Neudorf, Saskatchewan, Canada.

Career and Major Research

Taube studied at the University of Saskatchewan and received his doctorate from the University of California, Berkeley. He taught at Cornell University and at Stanford University, where he remained for the rest of his career. He established the mechanistic principles of electron transfer in transition-metal complexes.

Nobel Prize-winning Work

The work for which Henry Taube was awarded the Nobel Prize in Chemistry in 1983 was described by the Nobel Committee as: "for his work on the mechanisms of electron transfer reactions, especially in metal complexes".

Taube distinguished inner-sphere and outer-sphere electron transfer mechanisms and showed how the coordination sphere of a metal controls the rate and pathway of electron transfer.

Significance and Impact

His work explained how electrons move between metal centers and clarified the role of bridging ligands, influencing bioinorganic chemistry and redox catalysis.

Later Career and Honors

Taube remained at Stanford University until his retirement. He died on 2005-11-16 in Palo Alto, California, USA.

Legacy

Taube's mechanistic framework for electron transfer is fundamental to inorganic chemistry and redox biology.

Modern Developments

Taube's studies of electron transfer in metal complexes established the principles of inner-sphere and outer-sphere electron transfer, which underlie all of redox chemistry and bioinorganic chemistry. His concepts explain how metalloenzymes, corrosion, and batteries work and guide the design of catalysts, electron transfer proteins, and modern energy storage materials.

Selected Publications

"Electron Transfer Between Metal Complexes - A Retrospective" (Nobel Lecture, 1983)

Chapter Summary

The 1983 prize recognized Henry Taube for his work on the mechanisms of electron transfer reactions in metal complexes. His inner- and outer-sphere classifications became fundamental to coordination chemistry.

Chapter 84

1984:Robert Bruce Merrifield

The Nobel Prize in Chemistry for the year 1984 was awarded to Robert Bruce Merrifield.

Motivation:

for his development of methodology for chemical synthesis on a solid matrix

84.1 Robert Bruce Merrifield

Early Life and Education

Robert Bruce Merrifield was born on 1921-07-15 in Fort Worth, Texas, USA.

Career and Major Research

Merrifield studied at the University of California, Los Angeles, where he received his doctorate in 1949. He joined the Rockefeller Institute in New York, where he developed solid-phase synthesis, a method for building peptides step by step on insoluble polymer beads.

Nobel Prize-winning Work

The work for which Robert Bruce Merrifield was awarded the Nobel Prize in Chemistry in 1984 was described by the Nobel Committee as: "for his development of methodology for chemical synthesis on a solid matrix".

In solid-phase peptide synthesis, amino acids are attached to a solid support and coupled one at a time with washing in between, making peptide assembly rapid and amenable to automation.

Significance and Impact

The method made it practical to synthesize peptides and proteins, including insulin and enzymes, and became the basis of automated peptide synthesizers.

Later Career and Honors

Merrifield remained at Rockefeller University until his retirement. He died on 2006-05-14 in Cresskill, New Jersey, USA.

Legacy

Solid-phase synthesis is the foundation of modern peptide chemistry and peptide-based drugs.

Modern Developments

Merrifield's solid-phase peptide synthesis automated the construction of peptides and proteins, founding the modern peptide industry and the era of combinatorial chemistry. Solid-phase methods are now used to synthesize peptide drugs, hormones, and vaccines and evolved into the DNA chip and solid-phase oligonucleotide technologies central to genomics and molecular diagnostics.

Selected Publications

"Solid Phase Peptide Synthesis. I. The Synthesis of a Tetrapeptide" (Journal of the American Chemical Society, 1963)

Chapter Summary

The 1984 prize honored Robert Bruce Merrifield for developing methodology for chemical synthesis on a solid matrix. His solid-phase peptide synthesis transformed the production of peptides and proteins.

Chapter 85

1985:Herbert A. Hauptman and Jerome Karle

The Nobel Prize in Chemistry for the year 1985 was awarded jointly to Herbert A. Hauptman and Jerome Karle.

Motivation:

for their outstanding achievements in the development of direct methods for the determination of crystal structures

85.1 Herbert A. Hauptman

Early Life and Education

Herbert A. Hauptman was born in 1917 in New York City, New York, USA.

Career and Major Research

Hauptman studied at the City College of New York (B.S., 1937) and Columbia University (M.A., 1939), earning his doctorate in physics from the University of Maryland in 1955. He worked at the Naval Research Laboratory in Washington, D.C., and later led the Medical Foundation of Buffalo, which was renamed the Hauptman-Woodward Medical Research Institute in his honor.

Nobel Prize-winning Work

The work for which Herbert A. Hauptman was awarded the Nobel Prize in Chemistry in 1985 was described by the Nobel Committee as: "for their outstanding achievements in the development of direct methods for the determination of crystal structures".

Together with Jerome Karle, he developed a mathematical framework for deriving the phases of X-ray reflections directly from the measured diffraction intensities, thereby solving the phase problem that had limited crystal structure analysis.

Significance and Impact

Direct methods turned X-ray crystallography of small molecules into a routine and largely automated procedure, replacing the trial-and-error and heavy-atom approaches used before. The technique has been used to determine tens of thousands of crystal structures, including many important drugs and biological molecules.

Later Career and Honors

Hauptman served as president of the Hauptman-Woodward Medical Research Institute in Buffalo for many years. He received the Nobel Prize in Chemistry in 1985 and died on 2011-10-23 in Buffalo, New York.

Legacy

Direct methods remain the standard technique for small-molecule crystal structure determination, and the Hauptman-Woodward Medical Research Institute continues to bear his name.

Modern Developments

Hauptman's direct methods for solving the phase problem in X-ray crystallography automated the determination of crystal structures, enabling tens of thousands of small-molecule structures to be solved each year. His mathematical approach underpins modern structural chemistry, drug development, and materials science and evolved into the phasing algorithms used in macromolecular crystallography.

Selected Publications

"Solution of the Phase Problem I. The Centrosymmetric Crystal" (ACA Monograph, 1953) "Crystal Structure Determination: The Role of the Cosine Seminvariants" (Plenum Press, 1972)

85.2 Jerome Karle

Early Life and Education

Jerome Karle was born in 1918 in Brooklyn, New York, USA.

Career and Major Research

Karle studied at the City College of New York (B.S., 1937) and Harvard University (A.M., 1938), earning his doctorate in physical chemistry from the University of Michigan in 1944. He worked at the Naval Research Laboratory in Washington, D.C., beginning in 1944, for his entire career.

Nobel Prize-winning Work

The work for which Jerome Karle was awarded the Nobel Prize in Chemistry in 1985 was described by the Nobel Committee as: "for their outstanding achievements in the development of direct methods for the determination of crystal structures".

Working with Herbert A. Hauptman, he developed the statistical and mathematical methods that allowed the phases of X-ray reflections to be computed directly from measured intensities, solving the phase problem of X-ray crystallography.

Significance and Impact

The direct methods developed by Hauptman and Karle made crystal structure determination faster and fully objective, and they were adopted by crystallographers worldwide. Karle also applied anomalous scattering of X-rays to determine the absolute configurations of molecules.

Later Career and Honors

Karle remained at the Naval Research Laboratory until his retirement and continued research there. He received the Nobel Prize in Chemistry in 1985 and died on 2013-06-06 in Annandale, Virginia.

Legacy

Karle's direct methods and his work on anomalous scattering remain foundational to modern X-ray crystallography, used in both chemistry and structural biology.

Modern Developments

Karle's direct methods and his work on anomalous scattering enabled the automated solution of crystal structures and the determination of absolute configurations. These methods are used daily in structural chemistry and pharmaceutical development, while his diffraction techniques underlie modern crystallography of small molecules and of proteins.

Selected Publications

"Solution of the Phase Problem I. The Centrosymmetric Crystal" (ACA Monograph, 1953)

Chapter Summary

The 1985 Nobel Prize in Chemistry recognized the solution of the phase problem in X-ray crystallography. The direct methods developed by Hauptman and Karle turned structure determination from measured diffraction intensities into a routine, automated procedure used in laboratories worldwide.

Chapter 86

1986:Dudley R. Herschbach and Yuan T. Lee and John C. Polanyi

The Nobel Prize in Chemistry for the year 1986 was awarded jointly to Dudley R. Herschbach and Yuan T. Lee and John C. Polanyi.

Motivation:

for their contributions concerning the dynamics of chemical elementary processes

86.1 Dudley R. Herschbach

Early Life and Education

Dudley R. Herschbach was born in 1932 in San Jose, California, USA.

Career and Major Research

Herschbach studied at Stanford University (B.S., 1954; M.S., 1955) and earned his Ph.D. in chemical physics from Harvard University in 1958. He taught at the University of California, Berkeley, before joining Harvard University in 1963, where he remained for his career.

Nobel Prize-winning Work

The work for which Dudley R. Herschbach was awarded the Nobel Prize in Chemistry in 1986 was described by the Nobel Committee as: "for their contributions concerning the dynamics of chemical elementary processes".

He developed the crossed molecular beam technique, in which beams of reactant molecules are directed at one another to study individual reactive collisions and the forces that govern them.

Significance and Impact

Crossed molecular beams provided the first detailed experimental view of how reactant energy and orientation determine the outcome of elementary chemical reactions. The method launched the modern field of reaction dynamics.

Later Career and Honors

Herschbach continued teaching and research at Harvard University, where he is professor emeritus. He shared the 1986 Nobel Prize in Chemistry and was awarded the National Medal of Science in 1991.

Legacy

His crossed-beam experiments, extended by Yuan T. Lee, remain a standard approach for studying elementary reaction mechanisms.

Modern Developments

Herschbach's crossed molecular beam studies revealed the detailed dynamics of chemical reactions, founding the field of molecular reaction dynamics. His techniques evolved into femtosecond spectroscopy and state-resolved chemistry, enabling scientists to watch chemical bonds break and form in real time. These insights underpin modern combustion, atmospheric, and laser chemistry.

Selected Publications

"Molecular Dynamics of Elementary Chemical Reactions" (Science, 1987)

86.2 Yuan T. Lee

Early Life and Education

Yuan T. Lee was born in 1936 in Hsinchu, Taiwan.

Career and Major Research

Lee received his B.S. from National Taiwan University (1959) and his Ph.D. from the University of California, Berkeley (1965), where he studied under Dudley R. Herschbach. He served on the faculty of the University of Chicago before joining the University of California, Berkeley, in 1974.

Nobel Prize-winning Work

The work for which Yuan T. Lee was awarded the Nobel Prize in Chemistry in 1986 was described by the Nobel Committee as: "for their contributions concerning the dynamics of chemical elementary processes".

He built a universal crossed molecular beam apparatus that could detect virtually every product of a bimolecular reaction, allowing extremely detailed studies of reaction dynamics.

Significance and Impact

Lee's instrument extended crossed-beam studies to complex reactions, including hydrocarbon combustion and photodissociation. It produced the most complete experimental picture of elementary chemical processes at the time.

Later Career and Honors

Lee returned to Taiwan in 1994 to serve as president of Academia Sinica until 2006. He shared the 1986 Nobel Prize in Chemistry.

Legacy

His universal detection method remains a cornerstone of experimental reaction dynamics.

Modern Developments

Lee's crossed molecular beam experiments, including his studies of the $O(3P) + H_2$ reaction and the photodissociation of molecules, provided detailed pictures of how chemical reactions occur. His work founded modern reaction dynamics and physical chemistry, with applications in atmospheric chemistry, combustion, and the design of lasers and sensors.

Selected Publications

"Molecular Beam Studies of Elementary Chemical Processes" (Science, 1987)

86.3 John C. Polanyi

Early Life and Education

John C. Polanyi was born in 1929 in Berlin, Germany.

Career and Major Research

Polanyi studied at the University of Manchester (B.Sc., 1949; Ph.D., 1952), then worked at the National Research Council in Ottawa and at Princeton University. He joined the University of Toronto in 1956, where he was appointed University Professor in 1974.

Nobel Prize-winning Work

The work for which John C. Polanyi was awarded the Nobel Prize in Chemistry in 1986 was described by the Nobel Committee as: "for their contributions concerning the dynamics of chemical elementary processes".

He developed infrared chemiluminescence, a technique that detects the infrared light emitted by newly formed product molecules to reveal how energy is distributed among their vibrations.

Significance and Impact

Infrared chemiluminescence made it possible to map the energy released in exothermic reactions, providing direct experimental evidence for how energy is partitioned into new chemical bonds.

Later Career and Honors

Polanyi continued his research at the University of Toronto, and he later studied the dynamics of reactions at surfaces. He shared the 1986 Nobel Prize in Chemistry.

Legacy

His findings on reaction energy disposal have fundamental importance for chemiluminescence, lasers, and the spectroscopy of reaction products.

Modern Developments

Polanyi's infrared chemiluminescence technique revealed how energy is distributed in the products of chemical reactions, founding the field of reaction dynamics. His work showed that newly formed molecules emit characteristic infrared light, a principle now used in chemical lasers and in understanding combustion and atmospheric chemistry.

Selected Publications

"Some Concepts in Reaction Dynamics" (Science, 1987)

Chapter Summary

The 1986 Nobel Prize in Chemistry was awarded for the study of chemical elementary processes at the molecular level. Herschbach and Lee provided detailed experimental methods using crossed molecular beams, while Polanyi contributed infrared chemiluminescence for measuring reaction energy disposal.

Chapter 87

1987:Donald J. Cram and Jean-Marie Lehn and Charles J. Pedersen

The Nobel Prize in Chemistry for the year 1987 was awarded jointly to Donald J. Cram and Jean-Marie Lehn and Charles J. Pedersen.

Motivation:

for their development and use of molecules with structure-specific interactions of high selectivity

87.1 Donald J. Cram

Early Life and Education

Donald J. Cram was born in 1919 in Chester, Vermont, USA.

Career and Major Research

Cram studied at Rollins College (B.S., 1941) and the University of Nebraska (M.S., 1942), and he received his Ph.D. from the University of California, Los Angeles, in 1947. He spent his career at UCLA, where he worked on host-guest chemistry and molecular recognition.

Nobel Prize-winning Work

The work for which Donald J. Cram was awarded the Nobel Prize in Chemistry in 1987 was described by the Nobel Committee as: "for their development and use of molecules with structure-specific interactions of high selectivity".

He built three-dimensional host molecules, such as spherands, with cavities that bind specific guest ions with high selectivity, and he formulated the principles of host-guest chemistry.

Significance and Impact

Cram's synthetic hosts demonstrated that molecules could recognize and bind one another with enzyme-like selectivity, laying the foundation for molecular recognition research.

Later Career and Honors

Cram remained at UCLA until his retirement, and he shared the 1987 Nobel Prize in Chemistry. He died on 2001-06-19.

Legacy

The concept of the host-guest complex is now basic to supramolecular chemistry.

Modern Developments

Cram's synthesis of molecules with designed three-dimensional structures, including spherands and cyclophanes, founded the field of host-guest chemistry and molecular recognition. His preorganization concept underpins the modern design of sensors, drug delivery systems, and self-assembling materials, and his work inspired the field of molecular machines.

Selected Publications

"Host-Guest Chemistry: Complexation between Simple Classes of Organic Compounds" (Science, 1988)

87.2 Jean-Marie Lehn

Early Life and Education

Jean-Marie Lehn was born in 1939 in Rosheim, France.

Career and Major Research

Lehn studied at the University of Strasbourg (Ph.D., 1963) and conducted postdoctoral work with Robert Burns Woodward at Harvard University. He was a professor at the University of Strasbourg from 1970 and at the Collège de France from 1980.

Nobel Prize-winning Work

The work for which Jean-Marie Lehn was awarded the Nobel Prize in Chemistry in 1987 was described by the Nobel Committee as: "for their development and use of molecules with structure-specific interactions of high selectivity".

He synthesized cryptands, three-dimensional cage molecules with cavities that bind ions and small molecules with high selectivity, and he established the field he named supramolecular chemistry.

Significance and Impact

Cryptands extended molecular recognition to three dimensions and helped create supramolecular chemistry as a discipline at the interface of chemistry and biology.

Later Career and Honors

Lehn continued his research at the University of Strasbourg Institute of Advanced Study, and he shared the 1987 Nobel Prize in Chemistry.

Legacy

His concepts of self-assembly and self-organization, together with his syntheses, define much of modern supramolecular chemistry.

Modern Developments

Lehn's research on cryptates and supramolecular chemistry founded the field of supramolecular chemistry, where molecules assemble through non-covalent interactions. His work underpins the design of molecular sensors, drug delivery systems, self-healing materials, and the emerging field of molecular machines, and his concept of molecular information is a foundation of nanotechnology.

Selected Publications

"Diaza-Polyoxa-Macrocycles et Macrobicycles" (Tetrahedron Letters, 1969)

87.3 Charles J. Pedersen

Early Life and Education

Charles J. Pedersen was born in 1904 in Busan, Korea.

Career and Major Research

Pedersen studied at the University of Dayton and the Massachusetts Institute of Technology (M.S., 1927), and he joined the DuPont Company in 1927, where he worked until 1969.

Nobel Prize-winning Work

The work for which Charles J. Pedersen was awarded the Nobel Prize in Chemistry in 1987 was described by the Nobel Committee as: "for their development and use of molecules with structure-specific interactions of high selectivity".

He synthesized crown ethers, cyclic polyethers that selectively bind metal cations in their ring cavities, while seeking new catalysts at DuPont.

Significance and Impact

The first crown ethers, reported in 1967, showed that small synthetic macrocycles could mimic natural ion carriers, opening the field of macrocyclic host chemistry.

Later Career and Honors

Pedersen retired from DuPont in 1969, and he shared the 1987 Nobel Prize in Chemistry. He died on 1989-10-26.

Legacy

Crown ethers are now widely used in synthesis, separation science, and materials chemistry.

Modern Developments

Pedersen's discovery of crown ethers opened the field of macrocyclic host-guest chemistry and molecular recognition. Crown ethers are now used in phase-transfer catalysis, ion sensing, and the selective extraction of metal ions, and they provided the template for the development of cryptands, calixarenes, and modern supramolecular chemistry.

Selected Publications

"Cyclic Polyethers and Their Complexes with Metal Salts" (Journal of the American Chemical Society, 1967)

Chapter Summary

The 1987 Nobel Prize in Chemistry recognized molecules that interact with one another with high selectivity. Pedersen discovered the crown ethers, Cram built refined host-guest systems, and Lehn created the cryptands and founded supramolecular chemistry.

Chapter 88

1988:Johann Deisenhofer and Robert Huber and Hartmut Michel

The Nobel Prize in Chemistry for the year 1988 was awarded jointly to Johann Deisenhofer and Robert Huber and Hartmut Michel.

Motivation:

for the determination of the three-dimensional structure of a photosynthetic reaction centre

88.1 Johann Deisenhofer

Early Life and Education

Johann Deisenhofer was born in 1943 in Zusamalthheim, Germany.

Career and Major Research

Deisenhofer studied physics at the Technical University of Munich and earned his Ph.D. in 1974 at the Max Planck Institute of Biochemistry in Martinsried under Robert Huber. He later joined the University of Texas Southwestern Medical Center in Dallas.

Nobel Prize-winning Work

The work for which Johann Deisenhofer was awarded the Nobel Prize in Chemistry in 1988 was described by the Nobel Committee as: "for the determination of the three-dimensional structure of a photosynthetic reaction centre".

He determined the three-dimensional structure of the photosynthetic reaction centre of the purple bacterium *Rhodospseudomonas viridis* by X-ray crystallography, showing how light is captured and converted into a charge separation across the membrane.

Significance and Impact

This was the first structure of an integral membrane protein determined at atomic resolution, and it revealed in detail how photosynthesis works at the molecular level.

Later Career and Honors

Deisenhofer continued his research as an investigator of the Howard Hughes Medical Institute at the University of Texas Southwestern Medical Center. He shared the 1988 Nobel Prize in Chemistry.

Legacy

The reaction-centre structure became the reference model for membrane protein crystallography.

Modern Developments

Deisenhofer's X-ray crystallographic determination of the photosynthetic reaction center revealed the three-dimensional architecture of photosynthesis and earned the Nobel Prize for explaining how light energy is converted into chemical energy. His structure remains the paradigm for understanding photosynthesis and guides the design of artificial photosynthetic systems for renewable energy.

Selected Publications

"Structure of the Protein Subunits in the Photosynthetic Reaction Centre of *Rhodospseudomonas viridis* at 3 Å Resolution" (Nature, 1985)

88.2 Robert Huber

Early Life and Education

Robert Huber was born in 1937 in Munich, Germany.

Career and Major Research

Huber studied chemistry at the Technical University of Munich, where he received his Ph.D. in 1963. He became a director at the Max Planck Institute of Biochemistry in Martinsried.

Nobel Prize-winning Work

The work for which Robert Huber was awarded the Nobel Prize in Chemistry in 1988 was described by the Nobel Committee as: "for the determination of the three-dimensional structure of a photosynthetic reaction centre".

He led the crystallographic work, using the crystals prepared by Hartmut Michel, that solved the structure of the photosynthetic reaction centre.

Significance and Impact

The structure determined how the protein complex converts absorbed light into separated charges, and it established X-ray crystallography as a method for studying membrane proteins.

Later Career and Honors

Huber remained at the Max Planck Institute of Biochemistry in Martinsried. He shared the 1988 Nobel Prize in Chemistry.

Legacy

His structural studies extended across many proteins and made Martinsried a center of structural biology.

Modern Developments

Huber's crystallographic studies of proteins, including the photosynthetic reaction center and serine proteases, provided some of the first high-resolution views of membrane protein complexes and enzymes. His work on trypsin-like proteases remains a foundation of enzyme mechanism and drug design, underpinning the development of anticoagulant and antiviral drugs.

Selected Publications

"Structure of the Protein Subunits in the Photosynthetic Reaction Centre of *Rhodospseudomonas viridis* at 3 Å Resolution" (Nature, 1985)

88.3 Hartmut Michel

Early Life and Education

Hartmut Michel was born in 1948 in Ludwigsburg, Germany.

Career and Major Research

Michel studied at the University of Tübingen and received his Ph.D. from the University of Würzburg in 1977. He conducted research at the Max Planck Institute of Biochemistry in Martinsried and, from 1987, became a director at the Max Planck Institute of Biophysics in Frankfurt.

Nobel Prize-winning Work

The work for which Hartmut Michel was awarded the Nobel Prize in Chemistry in 1988 was described by the Nobel Committee as: "for the determination of the three-dimensional structure of a photosynthetic reaction centre".

He succeeded in crystallizing the photosynthetic reaction centre of *Rhodospseudomonas viridis*, a membrane protein that had resisted crystallization, which made the structure determination possible.

Significance and Impact

Michel's crystallization of the reaction centre was a breakthrough for structural biology and pioneered methods for crystallizing membrane proteins.

Later Career and Honors

Michel continued his research at the Max Planck Institute of Biophysics in Frankfurt. He shared the 1988 Nobel Prize in Chemistry.

Legacy

His approach to membrane protein crystallization opened the way to the atomic structures of many other membrane proteins.

Modern Developments

Michel's crystallization and structural determination of membrane proteins, including the photosynthetic reaction center, overcame one of the hardest problems in structural biology. His work enabled the study of membrane proteins at atomic resolution, opening the door to the modern structure determination of G protein-coupled receptors and ion channels that are targets for many drugs.

Selected Publications

"Structure of the Protein Subunits in the Photosynthetic Reaction Centre of *Rhodospseudomonas viridis* at 3 Å Resolution" (Nature, 1985)

Chapter Summary

The 1988 Nobel Prize in Chemistry recognized the first atomic structure of a photosynthetic reaction centre. Michel prepared crystals of the membrane protein complex, and Huber and Deisenhofer used them to reveal how light energy is converted into a separated charge across the membrane.

Chapter 89

1989: Sidney Altman and Thomas R. Cech

The Nobel Prize in Chemistry for the year 1989 was awarded jointly to Sidney Altman and Thomas R. Cech.

Motivation:

for their discovery of catalytic properties of RNA

89.1 Sidney Altman

Early Life and Education

Sidney Altman was born in 1939 in Montreal, Canada.

Career and Major Research

Altman studied at the Massachusetts Institute of Technology (B.Sc., 1960) and earned his Ph.D. from the University of Colorado in 1967. He joined Yale University in 1971, where he spent his career.

Nobel Prize-winning Work

The work for which Sidney Altman was awarded the Nobel Prize in Chemistry in 1989 was described by the Nobel Committee as: "for their discovery of catalytic properties of RNA".

He discovered that the RNA component of ribonuclease P, not its protein part, is the catalytic subunit that processes transfer RNA precursors.

Significance and Impact

The finding that RNA itself is the catalyst overturned the long-held assumption that only proteins could be enzymes.

Later Career and Honors

Altman remained at Yale University, where he served as chairman of the biology department. He shared the 1989 Nobel Prize in Chemistry, and he died in 2022.

Legacy

The discovery of catalytic RNA supported the idea of an early RNA world and founded the field of ribozyme research.

Modern Developments

Altman's discovery that RNA can act as a catalyst, together with Cech's, established the RNA world hypothesis and founded the field of ribozyme research. Catalytic RNA is now used in biotechnology and medicine, and the concept underpins our understanding of the origins of life and the design of ribozyme-based tools and therapeutics.

Selected Publications

"Ribonuclease P: An Enzyme with an Essential RNA Component" (Proceedings of the National Academy of Sciences, 1978)

89.2 Thomas R. Cech

Early Life and Education

Thomas R. Cech was born in 1947 in Chicago, Illinois, USA.

Career and Major Research

Cech studied at Grinnell College (B.A., 1970) and earned his Ph.D. from the University of California, Berkeley, in 1975. He joined the University of Colorado Boulder in 1978.

Nobel Prize-winning Work

The work for which Thomas R. Cech was awarded the Nobel Prize in Chemistry in 1989 was described by the Nobel Committee as: "for their discovery of catalytic properties of RNA".

He discovered that the ribosomal RNA precursor of the ciliate *Tetrahymena thermophila* removes its own intron without any protein, showing that RNA can catalyze its own splicing.

Significance and Impact

Self-splicing RNA demonstrated RNA catalysis directly and had broad implications for understanding early evolution.

Later Career and Honors

Cech remained at the University of Colorado Boulder, and he served as president of the Howard Hughes Medical Institute from 2000 to 2008. He shared the 1989 Nobel Prize in Chemistry.

Legacy

The ribozymes he characterized became tools for biochemistry and inspired the RNA world hypothesis.

Modern Developments

Cech's discovery of catalytic RNA (ribozymes) in *Tetrahymena* revealed that RNA is not merely a carrier of genetic information but a biological catalyst. This discovery reshaped the understanding of the origins of life, supported the RNA world hypothesis, and founded the field of RNA enzymology, which now informs RNA-based therapeutics and genome-editing tools.

Selected Publications

"Self-splicing RNA: Autoexcision and Autocyclization of the Ribosomal RNA Intervening Sequence of *Tetrahymena*" (*Cell*, 1982)

Chapter Summary

The 1989 Nobel Prize in Chemistry recognized the discovery that RNA can act as a catalyst. Altman and Cech independently showed that RNA possesses enzymatic activity, a finding that reshaped molecular biology and theories of early life.

Chapter 90

1990:Elias James Corey

The Nobel Prize in Chemistry for the year 1990 was awarded to Elias James Corey.

Motivation:

for his development of the theory and methodology of organic synthesis

90.1 Elias James Corey

Early Life and Education

Elias James Corey was born in 1928 in Methuen, Massachusetts, USA.

Career and Major Research

Corey studied at the Massachusetts Institute of Technology (B.S., 1948; Ph.D., 1951). He was a professor at the University of Illinois from 1951 to 1959, and then joined Harvard University, where he spent his career.

Nobel Prize-winning Work

The work for which Elias James Corey was awarded the Nobel Prize in Chemistry in 1990 was described by the Nobel Committee as: "for his development of the theory and methodology of organic synthesis".

He developed retrosynthetic analysis, a systematic method for planning complex organic syntheses by disconnecting target molecules into simpler precursor structures.

Significance and Impact

Retrosynthetic analysis became the standard conceptual framework of organic synthesis and guided the synthesis of hundreds of natural products.

Later Career and Honors

Corey continued teaching and research at Harvard University, where he is professor emeritus. He received the 1990 Nobel Prize in Chemistry.

Legacy

His methodology, which includes the Corey-Bakshi-Shibata asymmetric reduction, remains central to synthetic organic chemistry.

Modern Developments

Corey's development of retrosynthetic analysis transformed organic synthesis from an empirical art into a systematic, logical discipline. His approach, together with his synthesis of prostaglandins and other complex natural products, underpins the computer-assisted synthesis planning used in modern pharmaceutical research and drug discovery.

Selected Publications

"The Logic of Chemical Synthesis" (Wiley, 1989)

"Computer-Assisted Design of Complex Organic Syntheses" (Science, 1969)

Chapter Summary

The 1990 Nobel Prize in Chemistry recognized the theory and methodology of organic synthesis. Corey's retrosynthetic analysis gave chemists a rational and teachable strategy for constructing complex molecules.

Chapter 91

1991:Richard R. Ernst

The Nobel Prize in Chemistry for the year 1991 was awarded to Richard R. Ernst.

Motivation:

for his contributions to the development of the methodology of high resolution nuclear magnetic resonance (NMR) spectroscopy

91.1 Richard R. Ernst

Early Life and Education

Richard R. Ernst was born in 1933 in Winterthur, Switzerland.

Career and Major Research

Ernst earned his Ph.D. from the ETH Zurich in 1962, and he worked at Varian Associates in Palo Alto from 1962 to 1968. He returned to the ETH Zurich as a professor in 1968, where he spent the rest of his career.

Nobel Prize-winning Work

The work for which Richard R. Ernst was awarded the Nobel Prize in Chemistry in 1991 was described by the Nobel Committee as: "for his contributions to the development of the methodology of high resolution nuclear magnetic resonance (NMR) spectroscopy".

He introduced Fourier transform NMR, which made spectra dramatically more sensitive, and he developed two-dimensional NMR, which separates overlapping signals into two frequency dimensions.

Significance and Impact

These innovations transformed NMR into a general tool of chemistry and made possible the determination of structures of complex molecules in solution.

Later Career and Honors

Ernst remained at the ETH Zurich, where he was appointed an emeritus professor. He received the 1991 Nobel Prize in Chemistry, and he died in 2021.

Legacy

Fourier transform and two-dimensional NMR are the basis of modern NMR spectroscopy in chemistry, biology, and medicine.

Modern Developments

Ernst's development of Fourier transform and two-dimensional NMR spectroscopy revolutionized molecular structure determination. Modern NMR, including multidimensional and solid-state methods, is now indispensable in chemistry, biology, and medicine, from the structure determination of proteins to magnetic resonance imaging in hospitals worldwide.

Selected Publications

"Application of Fourier Transform Spectroscopy to Magnetic Resonance" (Review of Scientific Instruments, 1966)

Chapter Summary

The 1991 Nobel Prize in Chemistry recognized methodological advances in high-resolution NMR spectroscopy. Ernst's Fourier transform and two-dimensional techniques increased the sensitivity and resolving power of NMR to the point that the structures of complex biomolecules could be determined in solution.

Chapter 92

1992:Rudolph A. Marcus

The Nobel Prize in Chemistry for the year 1992 was awarded to Rudolph A. Marcus.

Motivation:

for his contributions to the theory of electron transfer reactions in chemical systems

92.1 Rudolph A. Marcus

Early Life and Education

Rudolph A. Marcus was born on 1923-07-21 in Montreal, Canada.

Career and Major Research

Marcus studied at McGill University (B.Sc., 1943; Ph.D., 1946). He held positions at the Polytechnic Institute of Brooklyn and the University of Illinois, and from 1978 he was a professor at the California Institute of Technology.

Nobel Prize-winning Work

The work for which Rudolph A. Marcus was awarded the Nobel Prize in Chemistry in 1992 was described by the Nobel Committee as: "for his contributions to the theory of electron transfer reactions in chemical systems".

He developed the Marcus theory of electron transfer, which relates the rate of an electron transfer reaction to the free energy driving force and the reorganization of the surrounding medium.

Significance and Impact

The theory, which includes the prediction of an inverted region, explained electron transfer across chemistry, electrochemistry, and biology, including the reactions of photosynthesis and respiration.

Later Career and Honors

Marcus remained at the California Institute of Technology, where he was the Arthur Amos Noyes Professor of Chemistry. He received the 1992 Nobel Prize in Chemistry, and he died in 2026.

Legacy

Marcus theory remains central to understanding electron transfer in molecular systems.

Modern Developments

Marcus's theory of electron transfer, developed in the 1950s and 1960s, explains the rates of electron transfer reactions in chemistry and biology. The Marcus theory is used to understand photosynthesis, respiration, corrosion, and the function of solar cells and batteries, and remains a cornerstone of modern physical chemistry and electrochemistry.

Selected Publications

"On the Theory of Oxidation-Reduction Reactions Involving Electron Transfer" (Journal of Chemical Physics, 1956)

Chapter Summary

The 1992 Nobel Prize in Chemistry recognized the theory of electron transfer reactions. Marcus provided a quantitative description of how electrons move between chemical species, with broad applications from electrochemistry to biological energy conversion.

Chapter 93

1993:Kary B. Mullis and Michael Smith

The Nobel Prize in Chemistry for the year 1993 was awarded jointly to Kary B. Mullis and Michael Smith.

Motivation:

for his invention of the polymerase chain reaction (PCR) method (one half, Kary B. Mullis) and for his fundamental contributions to the establishment of oligonucleotide-based, site-directed mutagenesis and its development for protein studies (the other half, Michael Smith)

93.1 Kary B. Mullis

Early Life and Education

Kary B. Mullis was born in 1944 in Lenoir, North Carolina, USA.

Career and Major Research

Mullis studied at the Georgia Institute of Technology (B.S., 1966) and earned his Ph.D. in biochemistry from the University of California, Berkeley, in 1973. He joined the Cetus Corporation in 1979.

Nobel Prize-winning Work

The work for which Kary B. Mullis was awarded the Nobel Prize in Chemistry in 1993 was described by the Nobel Committee as: "for his invention of the polymerase chain reaction (PCR) method".

He invented the polymerase chain reaction, an enzymatic technique that amplifies a specific segment of DNA by making billions of copies automatically.

Significance and Impact

PCR transformed molecular biology and medicine, enabling DNA cloning, sequencing, diagnosis of diseases, and forensic identification.

Later Career and Honors

Mullis left Cetus in 1986 and later worked as an independent consultant. He shared the 1993 Nobel Prize in Chemistry, and he died in 2019.

Legacy

PCR is one of the most widely used techniques in biology.

Modern Developments

Mullis's invention of the polymerase chain reaction (PCR) made it possible to amplify specific DNA sequences a billion-fold, transforming molecular biology and medicine. PCR underpins DNA fingerprinting, genetic testing, prenatal diagnostics, and the detection of infectious diseases, and was the essential enabling technology behind genomics, including the Human Genome Project.

Selected Publications

"Specific Enzymatic Amplification of DNA In Vitro: The Polymerase Chain Reaction" (Cold Spring Harbor Symposia on Quantitative Biology, 1986)

93.2 Michael Smith

Early Life and Education

Michael Smith was born in 1932 in Blackpool, England.

Career and Major Research

Smith studied at the University of Manchester (B.Sc., 1953; Ph.D., 1956). He emigrated to Canada and joined the University of British Columbia in 1956, where he spent his career.

Nobel Prize-winning Work

The work for which Michael Smith was awarded the Nobel Prize in Chemistry in 1993 was recognized for his fundamental contributions to the establishment of oligonucleotide-based, site-directed mutagenesis and its development for protein studies.

He developed site-directed mutagenesis, a method to introduce designed mutations at specific positions in a gene using synthetic oligonucleotides, enabling the function of proteins to be studied.

Significance and Impact

Site-directed mutagenesis became an essential tool in protein engineering and molecular biology.

Later Career and Honors

Smith remained at the University of British Columbia, where he founded the Biotechnology Laboratory. He shared the 1993 Nobel Prize in Chemistry, and he died in 2000.

Legacy

The method laid the basis for modern genetic engineering and protein research.

Modern Developments

Smith's site-directed mutagenesis, the technique of introducing specific changes into DNA, enabled the precise study of gene function and protein structure-activity relationships. His method founded modern protein engineering and is used to create enzymes, antibodies, and industrial biocatalysts, while also underpinning the directed evolution methods honored in 2018.

Selected Publications

"Oligonucleotide-Directed Mutagenesis Using M13-Derived Vectors: An Efficient and General Procedure for the Production of Point Mutations in Any Fragment of DNA" (Nucleic Acids Research, 1982)

Chapter Summary

The 1993 Nobel Prize in Chemistry recognized two DNA-based methods that transformed molecular biology. Mullis invented the polymerase chain reaction for amplifying DNA, while Smith developed site-directed mutagenesis for making designed changes in genes.

Chapter 94

1994:George A. Olah

The Nobel Prize in Chemistry for the year 1994 was awarded to George A. Olah.

Motivation:

for his contribution to carbocation chemistry

94.1 George A. Olah

Early Life and Education

George A. Olah was born in 1927 in Budapest, Hungary.

Career and Major Research

Olah earned his Ph.D. from the Technical University of Budapest in 1949. After leaving Hungary in 1956, he worked at Dow Chemical in Canada and then at Case Western Reserve University. From 1977 he was a professor at the University of Southern California, where he founded the Loker Hydrocarbon Research Institute.

Nobel Prize-winning Work

The work for which George A. Olah was awarded the Nobel Prize in Chemistry in 1994 was described by the Nobel Committee as: "for his contribution to carbocation chemistry".

He used superacids to prepare stable, long-lived carbocations, and he showed that carbocations are real intermediates in many chemical reactions, resolving a long-standing debate in organic chemistry.

Significance and Impact

His work underpinned understanding of reactions that are central to petroleum chemistry and refined synthetic organic chemistry.

Later Career and Honors

Olah remained at the University of Southern California. He received the 1994 Nobel Prize in Chemistry, and he died in 2017.

Legacy

The study of carbocations is now fundamental to organic chemistry, and Olah's later concept of a methanol economy influenced energy research.

Modern Developments

Olah's studies of carbocations and his development of superacid chemistry revealed the structure and reactivity of long-lived carbocations, founding the field of carbocation chemistry. His work underpins modern petroleum refining, catalysis, and the understanding of reaction intermediates, and his concept of electrophilic chemistry informs the design of new acid catalysts.

Selected Publications

"My Search for Carbocations and Their Role in Chemistry" (Angewandte Chemie International Edition in English, 1995)

Chapter Summary

The 1994 Nobel Prize in Chemistry recognized contributions to carbocation chemistry. Olah demonstrated that carbocations can be prepared and studied as stable species in superacid media, establishing their role as reaction intermediates.

Chapter 95

1995: Paul J. Crutzen and Mario J. Molina and F. Sherwood Rowland

The Nobel Prize in Chemistry for the year 1995 was awarded jointly to Paul J. Crutzen and Mario J. Molina and F. Sherwood Rowland.

Motivation:

for their work in atmospheric chemistry, particularly concerning the formation and decomposition of ozone

95.1 Paul J. Crutzen

Early Life and Education

Paul J. Crutzen was born in 1933 in Amsterdam, the Netherlands.

Career and Major Research

Crutzen earned his Ph.D. from the University of Stockholm in 1973. He became a director at the Max Planck Institute for Chemistry in Mainz, where he worked from 1980 to 2000.

Nobel Prize-winning Work

The work for which Paul J. Crutzen was awarded the Nobel Prize in Chemistry in 1995 was described by the Nobel Committee as: "for their work in atmospheric chemistry, particularly concerning the formation and decomposition of ozone".

He showed that nitrogen oxides catalyze the destruction of stratospheric ozone, clarifying the chemical cycles that control the ozone layer.

Significance and Impact

Establishing the role of nitrogen oxides in ozone chemistry was key to predicting how human activity could affect the ozone layer.

Later Career and Honors

Crutzen continued his research at the Max Planck Institute for Chemistry in Mainz. He shared the 1995 Nobel Prize in Chemistry, and he died in 2021.

Legacy

His work is fundamental to atmospheric chemistry, and he helped introduce the term Anthropocene for the current epoch of human influence on the Earth system.

Modern Developments

Crutzen's discovery that nitrogen oxides from fertilizers, combustion, and aircraft exhaust destroy stratospheric ozone explained how human activities can alter the global ozone layer and climate. His research underpinned the Montreal Protocol, which banned ozone-depleting substances and allowed the ozone layer to recover, and established him as a pioneer of the concept of the Anthropocene.

Selected Publications

"The Influence of Nitrogen Oxides on the Atmospheric Ozone Content" (Quarterly Journal of the Royal Meteorological Society, 1970)

95.2 Mario J. Molina

Early Life and Education

Mario J. Molina was born in 1943 in Mexico City, Mexico.

Career and Major Research

Molina studied chemical engineering at the Universidad Nacional Autonoma de Mexico (1965) and earned his Ph.D. from the University of California, Berkeley, in 1972. He was a postdoctoral researcher with F. Sherwood Rowland at the University of California, Irvine, and later held professorships at the Massachusetts Institute of Technology and the University of California, San Diego.

Nobel Prize-winning Work

The work for which Mario J. Molina was awarded the Nobel Prize in Chemistry in 1995 was described by the Nobel Committee as: "for their work in atmospheric chemistry, particularly concerning the formation and decomposition of ozone".

He showed that chlorofluorocarbons are photolyzed in the stratosphere, releasing chlorine atoms that catalytically destroy ozone.

Significance and Impact

The CFC-ozone hypothesis, developed with F. Sherwood Rowland, led to international regulation of chlorofluorocarbons under the Montreal Protocol.

Later Career and Honors

Molina continued research at the University of California, San Diego, and later founded the Molina Center for Energy and the Environment in Mexico City. He shared the 1995 Nobel Prize in Chemistry, and he died in 2020.

Legacy

His work linked atmospheric science directly to international policy and contributed to the recovery of the ozone layer.

Modern Developments

Molina's demonstration that chlorofluorocarbons (CFCs) are destroyed by ultraviolet light in the stratosphere, releasing chlorine that catalytically destroys ozone, identified the chemical mechanism of ozone depletion. His research provided the scientific basis for the Montreal Protocol, widely regarded as the most successful international environmental agreement, which has enabled the ozone layer to heal.

Selected Publications

"Stratospheric Sink for Chlorofluoromethanes: Chlorine Atom-Catalysed Destruction of Ozone" (Nature, 1974)

95.3 F. Sherwood Rowland

Early Life and Education

F. Sherwood Rowland was born in 1927 in Delaware, Ohio, USA.

Career and Major Research

Rowland studied at Ohio Wesleyan University (B.A., 1948) and earned his Ph.D. from the University of Chicago in 1952. He joined the University of California, Irvine, in 1964.

Nobel Prize-winning Work

The work for which F. Sherwood Rowland was awarded the Nobel Prize in Chemistry in 1995 was described by the Nobel Committee as: "for their work in atmospheric chemistry, particularly concerning the formation and decomposition of ozone".

With Mario J. Molina, he established that chlorofluorocarbons accumulate in the stratosphere and decompose to release chlorine that destroys ozone.

Significance and Impact

The prediction was confirmed by measurements of the Antarctic ozone hole, which led to the Montreal Protocol banning ozone-depleting substances.

Later Career and Honors

Rowland remained at the University of California, Irvine, where he was Donald Bren Professor of Chemistry. He shared the 1995 Nobel Prize in Chemistry, and he died in 2012.

Legacy

His atmospheric research demonstrated the global consequences of long-lived industrial gases.

Modern Developments

Rowland's prediction that chlorofluorocarbons (CFCs) would destroy stratospheric ozone, with Molina, identified one of the first global environmental threats caused by human chemistry. His research led to the Montreal Protocol and the eventual ban of CFCs, and his methods of studying atmospheric chemical cycles remain models for understanding climate change.

Selected Publications

"Stratospheric Sink for Chlorofluoromethanes: Chlorine Atom-Catalysed Destruction of Ozone" (Nature, 1974)

Chapter Summary

The 1995 Nobel Prize in Chemistry recognized atmospheric chemistry and the formation and decomposition of ozone. Crutzen identified the role of nitrogen oxides, while Molina and Rowland showed that chlorofluorocarbons deplete stratospheric ozone.

Chapter 96

1996: Robert F. Curl Jr. and Sir Harold Kroto and Richard E. Smalley

The Nobel Prize in Chemistry for the year 1996 was awarded jointly to Robert F. Curl Jr. and Sir Harold Kroto and Richard E. Smalley.

Motivation:

for their discovery of fullerenes

96.1 Robert F. Curl Jr.

Early Life and Education

Robert F. Curl Jr. was born in 1933 in Alice, Texas, USA.

Career and Major Research

Curl studied at Rice University (B.A., 1954) and earned his Ph.D. from the University of California, Berkeley, in 1957. He joined the faculty of Rice University in 1958, where he spent his career.

Nobel Prize-winning Work

The work for which Robert F. Curl Jr. was awarded the Nobel Prize in Chemistry in 1996 was described by the Nobel Committee as: "for their discovery of fullerenes".

In experiments with Kroto and Smalley, laser vaporization of graphite produced clusters of carbon, among which the 60-atom molecule C₆₀ was identified as a truncated icosahedron.

Significance and Impact

The discovery of buckminsterfullerene opened the field of fullerene chemistry and nanostructured carbon materials.

Later Career and Honors

Curl was a professor emeritus at Rice University. He shared the 1996 Nobel Prize in Chemistry, and he died in 2022.

Legacy

Fullerenes became a new branch of carbon chemistry and materials science.

Modern Developments

Curl's cocrystallization studies of fullerenes, and his contribution to the discovery of C₆₀ with Kroto and Smalley, opened the era of carbon nanotechnology. Fullerenes and their relatives, carbon nanotubes and graphene, are now studied for applications in electronics, energy storage, drug delivery, and advanced materials.

Selected Publications

"C₆₀: Buckminsterfullerene" (Nature, 1985)

96.2 Sir Harold Kroto

Early Life and Education

Sir Harold Kroto was born in 1939 in Wisbech, England.

Career and Major Research

Kroto studied at the University of Sheffield (B.Sc., 1961; Ph.D., 1964). He conducted research at the University of Sussex for most of his career and later held a position at Florida State University.

Nobel Prize-winning Work

The work for which Sir Harold Kroto was awarded the Nobel Prize in Chemistry in 1996 was described by the Nobel Committee as: "for their discovery of fullerenes".

With Curl and Smalley he identified C₆₀, whose hollow cage structure was suggested by the geodesic domes designed by Buckminster Fuller.

Significance and Impact

The identification of C₆₀ revealed an entirely new molecular form of carbon.

Later Career and Honors

Kroto was knighted for services to science and education. He shared the 1996 Nobel Prize in Chemistry, and he died in 2016.

Legacy

His advocacy for science education, and the discovery itself, left a lasting mark on chemistry and science communication.

Modern Developments

Kroto's discovery of buckminsterfullerene (C₆₀), the third allotropic form of carbon, opened the field of carbon nanomaterials that now includes nanotubes and graphene. These materials are used in electronics, energy storage, composites, and medicine, and Kroto's work inspired the entire modern nanoscience industry.

Selected Publications

"C₆₀: Buckminsterfullerene" (Nature, 1985)

96.3 Richard E. Smalley

Early Life and Education

Richard E. Smalley was born in 1943 in Akron, Ohio, USA.

Career and Major Research

Smalley studied at the University of Michigan (B.S., 1965) and earned his Ph.D. from Princeton University in 1973. He joined Rice University in 1976.

Nobel Prize-winning Work

The work for which Richard E. Smalley was awarded the Nobel Prize in Chemistry in 1996 was described by the Nobel Committee as: "for their discovery of fullerenes".

He built the laser-vaporization apparatus in which fullerenes were discovered and helped establish the structure of C₆₀.

Significance and Impact

The discovery demonstrated that carbon forms stable hollow cage molecules, launching the field of nanoscale carbon science.

Later Career and Honors

Smalley later directed the Center for Nanoscale Science and Technology at Rice University. He shared the 1996 Nobel Prize in Chemistry, and he died in 2005.

Legacy

His vision of carbon nanotubes and nanoscale materials helped establish nanotechnology as a research field.

Modern Developments

Smalley's discovery of C₆₀ and the co-founding of fullerene chemistry launched carbon nanoscience and the nanotechnology revolution. His insights and advocacy shaped the modern field of nanomaterials, and fullerenes, nanotubes, and graphene now impact electronics, energy, and medicine.

Selected Publications

"C₆₀: Buckminsterfullerene" (Nature, 1985)

Chapter Summary

The 1996 Nobel Prize in Chemistry recognized the discovery of fullerenes. Curl, Kroto, and Smalley found that carbon can form hollow cage molecules such as C_{60} , opening up a new field of carbon chemistry.

Chapter 97

1997: Paul D. Boyer and John E. Walker and Jens C. Skou

The Nobel Prize in Chemistry for the year 1997 was awarded jointly to Paul D. Boyer and John E. Walker and Jens C. Skou.

Motivation:

for their elucidation of the enzymatic mechanism underlying the synthesis of adenosine triphosphate (ATP)

97.1 Paul D. Boyer

Early Life and Education

Paul D. Boyer was born in 1918 in Provo, Utah, USA.

Career and Major Research

Boyer studied at Brigham Young University (B.S., 1939) and earned his Ph.D. from the University of Wisconsin in 1943. He joined the University of California, Los Angeles, in 1963, where he worked until his retirement.

Nobel Prize-winning Work

The work for which Paul D. Boyer was awarded the Nobel Prize in Chemistry in 1997 was described by the Nobel Committee as: "for their elucidation of the enzymatic mechanism underlying the synthesis of adenosine triphosphate (ATP)".

He proposed the binding-change mechanism of ATP synthase, in which the energy of a proton gradient drives the release of ATP from the enzyme rather than its formation directly.

Significance and Impact

The mechanism explained how ATP synthase couples proton flow to ATP synthesis in all living cells.

Later Career and Honors

Boyer was professor emeritus at the University of California, Los Angeles. He shared the 1997 Nobel Prize in Chemistry, and he died in 2018.

Legacy

The binding-change mechanism is now a central concept in bioenergetics.

Modern Developments

Boyer's binding change mechanism of ATP synthase, together with Walker's structural work, explained how cells generate ATP, the universal energy currency of life. ATP synthase is now recognized as a nanoscale rotary motor, and its mechanism is central to bioenergetics, drug development, and the design of molecular machines.

Selected Publications

"The Binding Change Mechanism for ATP Synthase: Some Probabilities and Possibilities" (*Biochimica et Biophysica Acta*, 1993)

97.2 John E. Walker

Early Life and Education

John E. Walker was born in 1941 in Halifax, England.

Career and Major Research

Walker studied at the University of Oxford (B.A., 1964; Ph.D., 1969). He joined the Medical Research Council Laboratory of Molecular Biology in Cambridge in 1974, where he spent his career.

Nobel Prize-winning Work

The work for which John E. Walker was awarded the Nobel Prize in Chemistry in 1997 was described by the Nobel Committee as: "for their elucidation of the enzymatic mechanism underlying the synthesis of adenosine triphosphate (ATP)".

He determined the amino acid sequences and the three-dimensional structure of the catalytic F1 part of ATP synthase, confirming the rotary binding-change mechanism.

Significance and Impact

The atomic structure provided direct evidence for how the enzyme synthesizes ATP by rotation.

Later Career and Honors

Walker continued at the Medical Research Council Laboratory of Molecular Biology in Cambridge. He shared the 1997 Nobel Prize in Chemistry.

Legacy

His structure of F1-ATPase remains a landmark in molecular biology and bioenergetics.

Modern Developments

Walker's X-ray crystallographic structure of ATP synthase revealed how this remarkable enzyme rotates to synthesize ATP, the energy currency of cells. His work, with Boyer's mechanism, explained the fundamental process of biological energy conversion and remains the basis for understanding bioenergetics, mitochondrial medicine, and the design of synthetic molecular motors.

Selected Publications

"Structure at 2.8 Å Resolution of F1-ATPase from Bovine Heart Mitochondria" (Nature, 1994)

97.3 Jens C. Skou

Early Life and Education

Jens C. Skou was born in 1918 in Lemvig, Denmark.

Career and Major Research

Skou earned his M.D. from the University of Copenhagen in 1944. He was a professor of biophysics at the University of Aarhus.

Nobel Prize-winning Work

The work for which Jens C. Skou was awarded the Nobel Prize in Chemistry in 1997 was described by the Nobel Committee as: "for their elucidation of the enzymatic mechanism underlying the synthesis of adenosine triphosphate (ATP)".

He discovered the sodium-potassium ATPase, the enzyme that pumps sodium and potassium ions across cell membranes using energy from ATP.

Significance and Impact

This was the first known ion pump, and it is essential for nerve and muscle function.

Later Career and Honors

Skou remained at the University of Aarhus. He shared the 1997 Nobel Prize in Chemistry, and he died in 2018.

Legacy

His discovery founded the study of transport ATPases, a major family of membrane proteins.

Modern Developments

Skou's discovery of the Na^+/K^+ -ATPase, the enzyme that pumps sodium and potassium ions across cell membranes, founded the field of ion transport and revealed how cells maintain their electrical and osmotic balance. The sodium pump is the target of digitalis drugs used to treat heart failure and remains a central model for understanding all ion-transporting ATPases.

Selected Publications

"The Influence of Some Cations on an Adenosine Triphosphatase from Peripheral Nerves" (Biochimica et Biophysica Acta, 1957)

Chapter Summary

The 1997 Nobel Prize in Chemistry recognized the enzymes that synthesize ATP and use its energy. Boyer and Walker explained the mechanism of ATP synthase, while Skou discovered the sodium-potassium pump.

Chapter 98

1998:Walter Kohn and John Pople

The Nobel Prize in Chemistry for the year 1998 was awarded jointly to Walter Kohn and John Pople.

Motivation:

for his development of the density-functional theory (one half, Walter Kohn) and for his development of computational methods in quantum chemistry (the other half, John Pople)

98.1 Walter Kohn

Early Life and Education

Walter Kohn was born in 1923 in Vienna, Austria.

Career and Major Research

Kohn studied at the University of Toronto (B.A., 1945; M.A., 1946) and earned his Ph.D. from Harvard University in 1948. He held positions at the Carnegie Institute of Technology and the University of California, San Diego, before joining the University of California, Santa Barbara.

Nobel Prize-winning Work

The work for which Walter Kohn was awarded the Nobel Prize in Chemistry in 1998 was described by the Nobel Committee as: "for his development of the density-functional theory".

He formulated density functional theory, which describes the energy of a system of electrons in terms of the electron density rather than individual wavefunctions, and he devised the Kohn-Sham equations that made the method practical.

Significance and Impact

Density functional theory became the most widely used computational method for the electronic structure of molecules and solids.

Later Career and Honors

Kohn continued his research at the University of California, Santa Barbara. He received the 1998 Nobel Prize in Chemistry, and he died in 2016.

Legacy

The Kohn-Sham method is the foundation of modern computational chemistry and materials science.

Modern Developments

Kohn's development of density functional theory (DFT) made it possible to calculate the electronic structure of atoms, molecules, and solids with practical accuracy, founding modern computational chemistry and materials science. DFT is now used daily to design drugs, catalysts, batteries, and semiconductors and is among the most cited scientific theories in history.

Selected Publications

"Self-Consistent Equations Including Exchange and Correlation Effects" (Physical Review, 1965)

98.2 John Pople

Early Life and Education

John Pople was born in 1925 in Burnham-on-Sea, England.

Career and Major Research

Pople studied at the University of Cambridge (B.A., 1946; Ph.D., 1951). He worked at the National Physical Laboratory and Carnegie Mellon University, and from 1986 he was a professor at Northwestern University.

Nobel Prize-winning Work

The work for which John Pople was awarded the Nobel Prize in Chemistry in 1998 was recognized for his development of computational methods in quantum chemistry.

He created the Gaussian series of computer programs and developed model chemistries that made ab initio quantum chemistry widely accessible.

Significance and Impact

His computational methods allowed chemists to calculate molecular structures and reaction energies accurately as a matter of routine.

Later Career and Honors

Pople was a professor at Northwestern University. He shared the 1998 Nobel Prize in Chemistry, and he died in 2004.

Legacy

The Gaussian software he created remains one of the most used programs in computational chemistry.

Modern Developments

Pople's development of computational methods, including the Gaussian series of programs, made quantum chemical calculations accessible and standardized for the entire scientific community. His methods underpin the modern use of computational chemistry in drug design, materials science, and catalysis, and Gaussian remains a cornerstone of research worldwide.

Selected Publications

"Approximate Molecular Orbital Theory" (McGraw-Hill, 1970)

Chapter Summary

The 1998 Nobel Prize in Chemistry recognized computational methods in quantum chemistry. Kohn developed density functional theory, while Pople created practical computational methods and software that made such calculations routine.

Chapter 99

1999: Ahmed Zewail

The Nobel Prize in Chemistry for the year 1999 was awarded to Ahmed Zewail.

Motivation:

for his studies of the transition states of chemical reactions using femtosecond spectroscopy

99.1 Ahmed Zewail

Early Life and Education

Ahmed Zewail was born in 1946 in Damanhur, Egypt.

Career and Major Research

Zewail studied at Alexandria University (B.S., M.S.) and earned his Ph.D. from the University of Pennsylvania in 1974. He joined the California Institute of Technology in 1976, where he became the Linus Pauling Professor of Chemistry and Physics.

Nobel Prize-winning Work

The work for which Ahmed Zewail was awarded the Nobel Prize in Chemistry in 1999 was described by the Nobel Committee as: "for his studies of the transition states of chemical reactions using femtosecond spectroscopy".

He developed femtosecond spectroscopy, using laser pulses of a few femtoseconds to observe the breaking and forming of chemical bonds in real time, thereby directly studying the transition states of reactions.

Significance and Impact

The new field of femtochemistry allowed the first direct observation of the short-lived transition states of chemical reactions.

Later Career and Honors

Zewail remained at the California Institute of Technology. He received the 1999 Nobel Prize in Chemistry, and he died in 2016.

Legacy

Femtochemistry opened a window onto the fastest steps of chemical change and influenced fields from molecular biology to materials science.

Modern Developments

Zewail's femtosecond spectroscopy, which he called femtochemistry, enabled scientists to observe chemical reactions in real time as bonds break and form. His techniques are now used to study photosynthesis, vision, and DNA photochemistry, and femtosecond lasers underpin modern microscopy, materials science, and manufacturing.

Selected Publications

"Femtochemistry: Atomic-Scale Dynamics of the Chemical Bond" (Journal of Physical Chemistry, 1993)

Chapter Summary

The 1999 Nobel Prize in Chemistry recognized the study of the transition states of chemical reactions. Zewail's femtosecond spectroscopy made it possible to observe chemical bond making and breaking in real time.

Chapter 100

2000: Alan Heeger and Alan MacDiarmid and Hideki Shirakawa

The Nobel Prize in Chemistry for the year 2000 was awarded jointly to Alan Heeger and Alan MacDiarmid and Hideki Shirakawa.

Motivation:

for the discovery and development of conductive polymers

100.1 Alan Heeger

Early Life and Education

Alan Heeger was born in 1936 in Sioux City, Iowa, USA.

Career and Major Research

Heeger studied at the University of Nebraska (B.S., 1957) and earned his Ph.D. from the University of California, Berkeley, in 1961. He taught at the University of Pennsylvania before joining the University of California, Santa Barbara, in 1982.

Nobel Prize-winning Work

The work for which Alan Heeger was awarded the Nobel Prize in Chemistry in 2000 was described by the Nobel Committee as: "for the discovery and development of conductive polymers".

With MacDiarmid and Shirakawa he discovered that doping the polymer polyacetylene with iodine increases its electrical conductivity dramatically, by up to a million-fold.

Significance and Impact

The discovery created the field of conducting polymers and organic electronic materials.

Later Career and Honors

Heeger continued his research at the University of California, Santa Barbara, where he is professor emeritus. He shared the 2000 Nobel Prize in Chemistry.

Legacy

Conductive polymers are now used in organic light-emitting diodes, solar cells, and other electronic devices.

Modern Developments

Heeger's conductive polymers, including doped polyacetylene, created the field of organic electronics. Organic semiconductors now power OLED displays, flexible solar cells, and wearable electronics, and his discovery launched the modern industry of plastic electronics found in smartphones, televisions, and medical devices.

Selected Publications

"Electrical Conductivity in Doped Polyacetylene" (Physical Review Letters, 1977)

100.2 Alan MacDiarmid

Early Life and Education

Alan MacDiarmid was born in 1927 in Masterton, New Zealand.

Career and Major Research

MacDiarmid studied at the University of New Zealand (B.Sc., M.Sc.) and earned his Ph.D. from the University of Cambridge in 1955. He joined the University of Pennsylvania in 1955, where he spent his career.

Nobel Prize-winning Work

The work for which Alan MacDiarmid was awarded the Nobel Prize in Chemistry in 2000 was described by the Nobel Committee as: "for the discovery and development of conductive polymers".

He and his colleagues oxidized polyacetylene with halogen vapors, producing the first electrically conducting organic polymer.

Significance and Impact

The demonstration that a plastic could conduct electricity overturned the assumption that polymers were insulators.

Later Career and Honors

MacDiarmid remained at the University of Pennsylvania. He shared the 2000 Nobel Prize in Chemistry, and he died in 2007.

Legacy

His work established the chemistry of conducting organic polymers.

Modern Developments

MacDiarmid's development of conductive polymers, including polyaniline and iodine-doped polyacetylene, founded the field of organic electronics. Conducting polymers are now used in antistatic coatings, sensors, and energy storage, and MacDiarmid's work helped launch the flexible electronics and wearable technology industries.

Selected Publications

"Electrical Conductivity in Doped Polyacetylene" (Physical Review Letters, 1977)

100.3 Hideki Shirakawa

Early Life and Education

Hideki Shirakawa was born in 1936 in Tokyo, Japan.

Career and Major Research

Shirakawa studied at the Tokyo Institute of Technology (B.S., 1961; Ph.D., 1966). He was a professor at the University of Tsukuba from 1979.

Nobel Prize-winning Work

The work for which Hideki Shirakawa was awarded the Nobel Prize in Chemistry in 2000 was described by the Nobel Committee as: "for the discovery and development of conductive polymers".

He developed the method for preparing polyacetylene as silvery films, the material whose iodination produced the first conducting polymer.

Significance and Impact

His reproducible synthesis of polyacetylene films made the discovery of polymer conductivity possible.

Later Career and Honors

Shirakawa continued his research at the University of Tsukuba. He shared the 2000 Nobel Prize in Chemistry.

Legacy

His polymerization methods remain fundamental to conducting polymer chemistry.

Modern Developments

Shirakawa's synthesis of the first conductive polymer, doped polyacetylene, revealed that plastics could conduct electricity, founding the field of organic electronics. His discovery underpins modern OLED displays, organic solar cells, and flexible circuits, and inspired the use of conjugated polymers throughout materials science.

Selected Publications

"Electrical Conductivity in Doped Polyacetylene" (Physical Review Letters, 1977)

Chapter Summary

The 2000 Nobel Prize in Chemistry recognized the discovery that polymers can be made electrically conductive by doping. Heeger, MacDiarmid, and Shirakawa's work on doped polyacetylene founded the science of conducting polymers.

Chapter 101

2001: William Knowles and Ryoji Noyori and K. Barry Sharpless

The Nobel Prize in Chemistry for the year 2001 was awarded jointly to William Knowles and Ryoji Noyori and K. Barry Sharpless.

Motivation:

for their work on chirally catalysed hydrogenation reactions (one half, jointly, William Knowles and Ryoji Noyori) and for his work on chirally catalysed oxidation reactions (the other half, K. Barry Sharpless)

101.1 William Knowles

Early Life and Education

William Knowles was born in 1917 in Taunton, Massachusetts, USA.

Career and Major Research

Knowles studied at Harvard University (B.A., 1939) and earned his Ph.D. from Columbia University in 1942. He worked at the Monsanto Company, where he remained until his retirement in 1986.

Nobel Prize-winning Work

The work for which William Knowles was awarded the Nobel Prize in Chemistry in 2001 was described by the Nobel Committee as: "for their work on chirally catalysed hydrogenation reactions".

He developed the first catalytic asymmetric hydrogenation, using a chiral rhodium catalyst, and applied it industrially to make L-DOPA, a drug used to treat Parkinson's disease.

Significance and Impact

This was the first industrial example of asymmetric catalysis, producing a single enantiomer of a pharmaceutical.

Later Career and Honors

Knowles retired from Monsanto in 1986. He shared the 2001 Nobel Prize in Chemistry, and he died in 2012.

Legacy

His work showed that chiral catalysts could produce enantiopure compounds, and it opened the field of industrial asymmetric catalysis.

Modern Developments

Knowles's development of catalytic asymmetric hydrogenation using chiral phosphine ligands enabled the industrial production of the anti-Parkinson drug L-DOPA and founded the field of asymmetric catalysis. His work opened the way to the modern synthesis of chiral drugs, agrochemicals, and fine chemicals with defined handedness.

Selected Publications

"Asymmetric Hydrogenation" (Accounts of Chemical Research, 1983)

101.2 Ryoji Noyori

Early Life and Education

Ryoji Noyori was born in 1938 in Kobe, Japan.

Career and Major Research

Noyori studied at Kyoto University (B.S., 1961; Ph.D., 1967). He was a professor at Nagoya University and later served as president of RIKEN from 2003 to 2015.

Nobel Prize-winning Work

The work for which Ryoji Noyori was awarded the Nobel Prize in Chemistry in 2001 was described by the Nobel Committee as: "for their work on chirally catalysed hydrogenation reactions".

He developed the BINAP ligand and chiral ruthenium catalysts for highly enantioselective hydrogenation of a broad range of substrates.

Significance and Impact

BINAP-based catalysis provided widely used methods for making chiral compounds, including pharmaceuticals and fragrances.

Later Career and Honors

Noyori continued his research at Nagoya University and RIKEN. He shared the 2001 Nobel Prize in Chemistry.

Legacy

His chiral molecular catalysts are among the most important tools in asymmetric synthesis.

Modern Developments

Noyori's development of chiral catalysts for asymmetric hydrogenation and his BINAP-based catalysts made the practical synthesis of single-handed molecules possible. His work enabled the industrial production of chiral pharmaceuticals and fragrances and remains central to asymmetric catalysis, one of the most important tools in modern drug synthesis.

Selected Publications

"Asymmetric Catalysis by Architectural and Functional Molecular Engineering" (Angewandte Chemie International Edition, 2001)

101.3 K. Barry Sharpless

Early Life and Education

K. Barry Sharpless was born in 1941 in Philadelphia, Pennsylvania, USA.

Career and Major Research

Sharpless studied at Dartmouth College (B.A., 1963) and earned his Ph.D. from Stanford University in 1968. He held professorships at the Massachusetts Institute of Technology and Stanford University, and he joined the Scripps Research Institute in 1990.

Nobel Prize-winning Work

The work for which K. Barry Sharpless was awarded the Nobel Prize in Chemistry in 2001 was recognized for his work on chirally catalysed oxidation reactions.

He developed catalytic asymmetric epoxidation and dihydroxylation, reliable methods for converting alkenes into chiral epoxides and diols with high enantioselectivity.

Significance and Impact

The Sharpless oxidations became some of the most widely used reactions in asymmetric synthesis.

Later Career and Honors

Sharpless continued his research at the Scripps Research Institute, and he received a second Nobel Prize in Chemistry in 2022 for the development of click chemistry.

Legacy

His asymmetric oxidation reactions are standard tools for building chiral molecules, and his click chemistry concept transformed how chemists assemble molecules.

Modern Developments

Sharpless's asymmetric epoxidation and his concept of click chemistry, the simple, reliable, and selective joining of molecular building blocks, transformed how chemists build molecules. Sharpless chemistry is used in drug discovery, materials science, and bioconjugation, and click chemistry is now a cornerstone of modern chemical biology.

Selected Publications

"The First Practical Method for Asymmetric Epoxidation" (Journal of the American Chemical Society, 1980)

Chapter Summary

The 2001 Nobel Prize in Chemistry recognized catalytic methods for producing chiral molecules. Knowles and Noyori developed asymmetric hydrogenation, while Sharpless created asymmetric oxidation reactions.

Chapter 102

2002: John B. Fenn and Koichi Tanaka and Kurt Wüthrich

The Nobel Prize in Chemistry for the year 2002 was awarded jointly to John B. Fenn and Koichi Tanaka and Kurt Wüthrich.

Motivation:

for their development of soft desorption ionisation methods for mass spectrometric analyses of biological macromolecules (one half, jointly, John B. Fenn and Koichi Tanaka) and for his development of nuclear magnetic resonance spectroscopy for determining the three-dimensional structure of biological macromolecules in solution (the other half, Kurt Wüthrich)

102.1 John B. Fenn

Early Life and Education

John B. Fenn was born in 1917 in New York City, New York, USA.

Career and Major Research

Fenn studied at Berea College (B.S., 1937) and earned his Ph.D. from Princeton University in 1940. He was a professor at Yale University from 1967 to 1994, and he later held a position at Virginia Commonwealth University.

Nobel Prize-winning Work

The work for which John B. Fenn was awarded the Nobel Prize in Chemistry in 2002 was described by the Nobel Committee as: "for their development of soft desorption ionisation methods for mass spectrometric analyses of biological macromolecules".

He developed electrospray ionization, which produces ions from large biomolecules in solution for analysis by mass spectrometry.

Significance and Impact

Together with the work of Koichi Tanaka, electrospray ionization made mass spectrometry applicable to proteins and other macromolecules.

Later Career and Honors

Fenn shared the 2002 Nobel Prize in Chemistry, and he died in 2010.

Legacy

Electrospray ionization is now a standard tool for analyzing proteins and other biomolecules.

Modern Developments

Fenn's development of electrospray ionization mass spectrometry made it possible to analyze large biomolecules, including proteins and DNA, transforming biology and medicine. Electrospray mass spectrometry is now a standard tool for protein identification, drug discovery, and clinical diagnostics, and underpins modern proteomics.

Selected Publications

"Electrospray Ionization for Mass Spectrometry of Large Biomolecules" (Science, 1989)

102.2 Koichi Tanaka

Early Life and Education

Koichi Tanaka was born in 1959 in Toyama, Japan.

Career and Major Research

Tanaka joined the Shimadzu Corporation in 1983 as a research engineer, and he remained with the company.

Nobel Prize-winning Work

The work for which Koichi Tanaka was awarded the Nobel Prize in Chemistry in 2002 was described by the Nobel Committee as: "for their development of soft desorption ionisation methods for mass spectrometric analyses of biological macromolecules".

He developed soft laser desorption mass spectrometry, using a laser with a metal powder and glycerol matrix to ionize large intact biomolecules.

Significance and Impact

His method allowed proteins and other macromolecules to be analyzed without fragmentation, making mass spectrometry useful for biology.

Later Career and Honors

Tanaka remained at the Shimadzu Corporation. He shared the 2002 Nobel Prize in Chemistry.

Legacy

His approach contributed to the methods that now underlie the mass spectrometry of biological macromolecules.

Modern Developments

Tanaka's soft laser desorption method enabled mass spectrometry of large biomolecules, founding the field of biological mass spectrometry. His techniques evolved into MALDI mass spectrometry, now used to identify proteins, analyze polymers, and detect pathogens in clinical and biotechnological laboratories worldwide.

Selected Publications

"Protein and Polymer Analyses up to m/z 100 000 by Laser Ionization Time-of-Flight Mass Spectrometry" (Rapid Communications in Mass Spectrometry, 1988)

102.3 Kurt Wüthrich

Early Life and Education

Kurt Wüthrich was born in 1938 in Aarberg, Switzerland.

Career and Major Research

Wüthrich studied at the University of Bern and earned his Ph.D. from the University of Basel in 1964. He was a professor at the ETH Zurich from 1969, and he also held a position at the Scripps Research Institute in La Jolla.

Nobel Prize-winning Work

The work for which Kurt Wüthrich was awarded the Nobel Prize in Chemistry in 2002 was recognized for his development of nuclear magnetic resonance spectroscopy for determining the three-dimensional structure of biological macromolecules in solution.

He developed methods for determining complete three-dimensional structures of proteins in solution by NMR, using the nuclear Overhauser effect to measure distances between atoms.

Significance and Impact

The methods made NMR a second technique, alongside X-ray crystallography, for protein structure determination.

Later Career and Honors

Wüthrich continued his research at the ETH Zurich and the Scripps Research Institute. He shared the 2002 Nobel Prize in Chemistry.

Legacy

His structural NMR methods are now used routinely to study proteins in solution.

Modern Developments

Wüthrich's development of multidimensional NMR methods for determining protein structures in solution founded the field of protein NMR and enabled the study of biomolecules under physiological conditions. NMR-based structural biology remains essential for understanding protein dynamics, drug binding, and the interactions of intrinsically disordered proteins.

Selected Publications

"NMR of Proteins and Nucleic Acids" (Wiley, 1986)

Chapter Summary

The 2002 Nobel Prize in Chemistry recognized methods for analyzing biological macromolecules. Fenn and Tanaka developed ionization methods for mass spectrometry, while Wüthrich developed NMR for structure determination in solution.

Chapter 103

2003: Peter Agre and Roderick MacKinnon

The Nobel Prize in Chemistry for the year 2003 was awarded jointly to Peter Agre and Roderick MacKinnon.

Motivation:

for the discovery of water channels (one half, Peter Agre) and for structural and mechanistic studies of ion channels (the other half, Roderick MacKinnon)

103.1 Peter Agre

Early Life and Education

Peter Agre was born in 1949 in Northfield, Minnesota, USA.

Career and Major Research

Agre studied at Augsburg College (B.A., 1970) and earned his M.D. from the Johns Hopkins University in 1974. He was a professor at the Johns Hopkins University and later at the Johns Hopkins Bloomberg School of Public Health.

Nobel Prize-winning Work

The work for which Peter Agre was awarded the Nobel Prize in Chemistry in 2003 was described by the Nobel Committee as: "for the discovery of water channels".

He discovered aquaporins, integral membrane proteins that form water channels and allow rapid water transport across cell membranes.

Significance and Impact

The discovery explained how water passes through biological membranes much faster than simple diffusion would permit; aquaporins are essential in the kidneys and many tissues.

Later Career and Honors

Agre continued his research at the Johns Hopkins University, where he directed the Malaria Research Institute. He shared the 2003 Nobel Prize in Chemistry.

Legacy

Aquaporins are now known to be present in virtually all living organisms.

Modern Developments

Agre's discovery of aquaporins, the water channels that let water cross cell membranes rapidly, solved the long-standing puzzle of water transport and founded the field of aquaporin biology. Aquaporins are now targets for diuretic and anti-cancer drugs and inspire the design of biomimetic membranes for water purification.

Selected Publications

"Appearance of Water Channels in *Xenopus* Oocytes Expressing Red Cell CHIP28 Protein" (Science, 1992)

103.2 Roderick MacKinnon

Early Life and Education

Roderick MacKinnon was born in 1956 in Burlington, Massachusetts, USA.

Career and Major Research

MacKinnon studied at Brandeis University (B.A., 1978) and earned his M.D. from Tufts University in 1982. He joined Rockefeller University in 1996.

Nobel Prize-winning Work

The work for which Roderick MacKinnon was awarded the Nobel Prize in Chemistry in 2003 was recognized for structural and mechanistic studies of ion channels.

He determined the three-dimensional structure of a potassium channel by X-ray crystallography, revealing how the selectivity filter distinguishes potassium ions from sodium ions.

Significance and Impact

The structure explained the molecular basis of ion selectivity and conduction that underlies nerve signaling.

Later Career and Honors

MacKinnon remained at Rockefeller University, where he is an investigator of the Howard Hughes Medical Institute. He shared the 2003 Nobel Prize in Chemistry.

Legacy

His structures of potassium channels clarified how the nervous system transmits electrical signals.

Modern Developments

MacKinnon's determination of the three-dimensional structures of potassium channels revealed how ion channels select and conduct ions with astonishing specificity and speed. His work explained the molecular basis of nerve and muscle signaling and underpins the pharmacology of ion-channel drugs used to treat heart disease, epilepsy, and pain.

Selected Publications

"The Structure of the Potassium Channel: Molecular Basis of K⁺ Conduction and Selectivity" (Science, 1998)

Chapter Summary

The 2003 Nobel Prize in Chemistry recognized the channels that control water and ion movement across cell membranes. Agre discovered the aquaporin water channels, and MacKinnon solved the structure of potassium channels.

Chapter 104

2004: Aaron Ciechanover and Avram Hershko and Irwin Rose

The Nobel Prize in Chemistry for the year 2004 was awarded jointly to Aaron Ciechanover and Avram Hershko and Irwin Rose.

Motivation:

for the discovery of ubiquitin-mediated protein degradation

104.1 Aaron Ciechanover

Early Life and Education

Aaron Ciechanover was born in 1947 in Haifa, Israel.

Career and Major Research

Ciechanover earned his M.D. from the Hebrew University-Hadassah Medical School in 1974 and his D.Sc. from the Technion-Israel Institute of Technology in 1982, where he is a professor.

Nobel Prize-winning Work

The work for which Aaron Ciechanover was awarded the Nobel Prize in Chemistry in 2004 was described by the Nobel Committee as: "for the discovery of ubiquitin-mediated protein degradation".

With Hershko and Rose he established that ubiquitin is covalently attached to proteins, marking them for degradation by the proteasome.

Significance and Impact

Ubiquitin-mediated degradation controls the levels of most cellular proteins and is essential for cell division, DNA repair, and immunity.

Later Career and Honors

Ciechanover continued his research at the Technion-Israel Institute of Technology. He shared the 2004 Nobel Prize in Chemistry.

Legacy

The ubiquitin system is now known to regulate almost every cellular process.

Modern Developments

Ciechanover's discovery of ubiquitin-mediated protein degradation revealed how cells tag proteins for destruction, a process that controls almost every cellular function, including the cell cycle, DNA repair, and immune responses. Ubiquitin biology is now the basis for the proteasome-inhibitor cancer drugs bortezomib and carfilzomib and for the protein-degradation approach called targeted protein degradation (PROTACs).

Selected Publications

"Proposed Role of ATP in Protein Breakdown: Conjugation of Proteins with Multiple Chains of the Polypeptide of ATP-Dependent Proteolysis" (Proceedings of the National Academy of Sciences, 1980)

104.2 Avram Hershko

Early Life and Education

Avram Hershko was born in 1937 in Karcag, Hungary.

Career and Major Research

Hershko earned his M.D. and Ph.D. from the Hebrew University-Hadassah Medical School, and he is a professor at the Technion-Israel Institute of Technology.

Nobel Prize-winning Work

The work for which Avram Hershko was awarded the Nobel Prize in Chemistry in 2004 was described by the Nobel Committee as: "for the discovery of ubiquitin-mediated protein degradation".

He worked out the enzymatic cascade that conjugates ubiquitin to proteins, including the role of ATP in the reaction.

Significance and Impact

His biochemical dissection identified the components of the ubiquitin system.

Later Career and Honors

Hershko continued his research at the Technion-Israel Institute of Technology. He shared the 2004 Nobel Prize in Chemistry.

Legacy

The reaction sequence he characterized is now a textbook chapter of cell biology.

Modern Developments

Hershko's discovery of the ubiquitin proteolytic system, with Ciechanover and Rose, established the pathway by which cells destroy misfolded and short-lived proteins. This work explained how the cell cycle, gene expression, and immune responses are regulated, and it led to the development of proteasome inhibitor drugs used to treat multiple myeloma.

Selected Publications

"Proposed Role of ATP in Protein Breakdown: Conjugation of Proteins with Multiple Chains of the Polypeptide of ATP-Dependent Proteolysis" (Proceedings of the National Academy of Sciences, 1980)

104.3 Irwin Rose

Early Life and Education

Irwin Rose was born in 1926 in Brooklyn, New York, USA.

Career and Major Research

Rose earned his B.A. and Ph.D. from the University of Chicago (Ph.D., 1952). He worked at the Institute for Cancer Research in Philadelphia, and he later became a professor at the University of California, Irvine.

Nobel Prize-winning Work

The work for which Irwin Rose was awarded the Nobel Prize in Chemistry in 2004 was described by the Nobel Committee as: "for the discovery of ubiquitin-mediated protein degradation".

He contributed the biochemical analysis that identified the ATP-dependent conjugation of ubiquitin as the signal for protein breakdown.

Significance and Impact

The demonstration that ATP drives ubiquitin conjugation revealed how cells control protein lifetimes.

Later Career and Honors

Rose moved to the University of California, Irvine, in 1997. He shared the 2004 Nobel Prize in Chemistry, and he died in 2015.

Legacy

His work helped establish the biochemistry of the ubiquitin pathway.

Modern Developments

Rose's discovery that ubiquitin-dependent degradation requires ATP revealed the energetics of protein turnover and helped establish the ubiquitin-proteasome system as the cell's main protein destruction machinery. His work underpins modern research on protein quality control, neurodegeneration, and the development of PROTAC drugs.

Selected Publications

"Proposed Role of ATP in Protein Breakdown: Conjugation of Proteins with Multiple Chains of the Polypeptide of ATP-Dependent Proteolysis" (Proceedings of the National Academy of Sciences, 1980)

Chapter Summary

The 2004 Nobel Prize in Chemistry recognized the discovery of ubiquitin-mediated protein degradation. Ciechanover, Hershko, and Rose showed how cells tag proteins for destruction, a system now known to regulate nearly all cellular processes.

Chapter 105

2005: Yves Chauvin and Robert H. Grubbs and Richard R. Schrock

The Nobel Prize in Chemistry for the year 2005 was awarded jointly to Yves Chauvin and Robert H. Grubbs and Richard R. Schrock.

Motivation:

for the development of the metathesis method in organic synthesis

105.1 Yves Chauvin

Early Life and Education

Yves Chauvin was born in 1930 in Menen, Belgium.

Career and Major Research

Chauvin studied at the École supérieure de chimie, physique, électronique de Lyon and spent his career at the Institut Français du Pétrole (IFP) in Rueil-Malmaison, France, where he became honorary research director. In 1971 he proposed the mechanism of olefin metathesis, in which a metal carbene and a metallacyclobutane intermediate carry the breaking and re-forming of carbon-carbon double bonds.

Nobel Prize-winning Work

The work for which Yves Chauvin was awarded the Nobel Prize in Chemistry in 2005 was described by the Nobel Committee as: "for the development of the metathesis method in organic synthesis".

Chauvin's mechanism explained how the double bonds of alkenes are broken and reorganized in the presence of certain metal complexes, giving the reaction a rational basis that other chemists could build upon.

Significance and Impact

By explaining the metathesis mechanism, Chauvin's work enabled the later development of practical catalysts that made metathesis a standard reaction in organic synthesis. The reaction is now used industrially to make polymers, fine chemicals, and pharmaceuticals.

Later Career and Honors

Chauvin retired from the Institut Français du Pétrole and was elected to the French Academy of Sciences. He died on 2015-01-27 in Tours, France.

Legacy

Chauvin's mechanistic insight underlies every modern metathesis catalyst, including the ruthenium and molybdenum systems recognized in the same prize.

Modern Developments

Chauvin's elucidation of the mechanism of olefin metathesis, based on metallacyclobutane intermediates, provided the theoretical foundation for the metathesis reaction. His mechanism guided the development of the Grubbs and Schrock catalysts that now enable the efficient synthesis of pharmaceuticals, polymers, and fine chemicals.

Selected Publications

"Catalyse de Transformation des Oléfines par les Complexes du Tungstène" (Die Makromolekulare Chemie, 1971)

105.2 Robert H. Grubbs

Early Life and Education

Robert H. Grubbs was born in 1942 in Possum Trot, Kentucky, USA.

Career and Major Research

Grubbs earned a bachelor's degree from the University of Florida and a doctorate from Columbia University in 1968, followed by postdoctoral work at Stanford University. He became a professor at the California Institute of Technology, where in the early 1990s he discovered ruthenium-based metathesis catalysts that are stable in air and tolerant of many functional groups.

Nobel Prize-winning Work

The work for which Robert H. Grubbs was awarded the Nobel Prize in Chemistry in 2005 was described by the Nobel Committee as: "for the development of the metathesis method in organic synthesis".

The Grubbs catalysts brought olefin metathesis out of the specialist laboratory, making the reaction easy to run under ordinary conditions.

Significance and Impact

Because the ruthenium catalysts work in air and tolerate a wide range of chemical groups, they allowed metathesis to be applied broadly in organic synthesis. The reaction is now used to produce pharmaceuticals, polymers, and agrochemicals.

Later Career and Honors

Grubbs remained professor of chemistry at the California Institute of Technology for the rest of his career. He died on 2021-12-19 in Duarte, California, USA.

Legacy

The Grubbs catalyst is one of the most widely used catalysts in synthetic chemistry and is standard equipment in research and industry.

Modern Developments

Grubbs's metathesis catalysts, including the ruthenium-based Grubbs catalyst, made olefin metathesis a practical, widely used reaction in organic synthesis. Metathesis is now used in the industrial production of pharmaceuticals, agrochemicals, and specialty polymers, and his catalysts transformed the field of green chemistry.

Selected Publications

"The Development of L2X2Ru=CHR Olefin Metathesis Catalysts" (Accounts of Chemical Research, 2001)

105.3 Richard R. Schrock

Early Life and Education

Richard R. Schrock was born in 1945 in Berne, Indiana, USA.

Career and Major Research

Schrock earned bachelor's and doctoral degrees from Harvard University, in 1967 and 1971, and completed postdoctoral work at the University of Cambridge. After a period at DuPont Central Research he joined the faculty of the Massachusetts Institute of Technology in 1975. He developed highly active molybdenum and tungsten alkylidene metathesis catalysts.

Nobel Prize-winning Work

The work for which Richard R. Schrock was awarded the Nobel Prize in Chemistry in 2005 was described by the Nobel Committee as: "for the development of the metathesis method in organic synthesis".

Schrock's molybdenum alkylidene catalysts were the first to show exceptional activity in olefin metathesis, revealing the power of the reaction.

Significance and Impact

Schrock's catalysts made it possible to conduct metathesis efficiently at low temperatures and to control the products precisely. They remain important for reactions such as ring-opening metathesis polymerization.

Later Career and Honors

Schrock remains a professor of chemistry at the Massachusetts Institute of Technology, where he continues research on organometallic chemistry and catalysis.

Legacy

Together with the catalysts of Grubbs, Schrock's alkylidene systems made olefin metathesis one of the central reactions of modern organic synthesis.

Modern Developments

Schrock's molybdenum and tungsten metathesis catalysts, the first to achieve controlled olefin metathesis, transformed the reaction from a curiosity into a powerful synthetic tool. Metathesis is now used to make pharmaceuticals, advanced polymers, and fuels, and Schrock's work pioneered the field of high-oxidation-state organometallic catalysis.

Selected Publications

"Preparation and Characterization of Molybdenum and Tungsten Alkylidene Complexes" (Journal of the American Chemical Society, 1990)

Chapter Summary

The 2005 prize recognized the three chemists who turned olefin metathesis into a practical tool of organic synthesis: Chauvin's mechanism, Schrock's molybdenum alkylidene catalysts, and Grubbs's air-stable ruthenium catalysts. Metathesis is now used to make polymers, agrochemicals, and pharmaceuticals.

Chapter 106

2006:Roger D. Kornberg

The Nobel Prize in Chemistry for the year 2006 was awarded to Roger D. Kornberg.

Motivation:

for his studies of the molecular basis of eukaryotic transcription

106.1 Roger D. Kornberg

Early Life and Education

Roger D. Kornberg was born in 1947 in St. Louis, Missouri, USA.

Career and Major Research

Kornberg earned a bachelor's degree from Harvard University and a doctorate from Stanford University in 1972. After postdoctoral work at the MRC Laboratory of Molecular Biology in Cambridge with Francis Crick, he joined the faculty of Stanford University, where he remains. His father, Arthur Kornberg, received the Nobel Prize in Physiology or Medicine in 1959.

Nobel Prize-winning Work

The work for which Roger D. Kornberg was awarded the Nobel Prize in Chemistry in 2006 was described by the Nobel Committee as: "for his studies of the molecular basis of eukaryotic transcription".

Kornberg determined the three-dimensional structure of RNA polymerase II, the enzyme that copies DNA into messenger RNA in the cells of plants, animals, and fungi, and showed how it works together with transcription factors to read the genetic information.

Significance and Impact

His structural work revealed at atomic resolution how the transcription machinery selects and reads genes and how gene expression is regulated. It explained why transcription in complex organisms is controlled at the molecular level.

Later Career and Honors

Kornberg continues to direct research on transcription and chromatin at the Stanford University School of Medicine.

Legacy

The structure of RNA polymerase II stands as a foundation of modern molecular biology, clarifying how genetic information is expressed in all higher organisms.

Modern Developments

Kornberg's X-ray crystallographic structure of RNA polymerase II revealed how DNA is transcribed into RNA, the first step in gene expression. His structure, showing the enzyme at work in an intact transcription complex, underpins the modern understanding of gene regulation and the design of transcription-targeting drugs.

Selected Publications

"Structural Basis of Transcription: An RNA Polymerase II Elongation Complex at 3.3 Å Resolution" (Science, 2001)

Chapter Summary

The 2006 prize honored work that revealed, at atomic resolution, how RNA polymerase II transcribes DNA into RNA in eukaryotic cells. It established the molecular basis of transcription and its regulation.

Chapter 107

2007:Gerhard Ertl

The Nobel Prize in Chemistry for the year 2007 was awarded to Gerhard Ertl.

Motivation:

for his studies of chemical processes on solid surfaces

107.1 Gerhard Ertl

Early Life and Education

Gerhard Ertl was born in 1936 in Stuttgart, Germany.

Career and Major Research

Ertl studied physics at the University of Stuttgart and received his doctorate from the Technical University of Munich in 1965. He became director of the Fritz Haber Institute of the Max Planck Society in Berlin. He established the field of surface chemistry and worked out the elementary steps of reactions on metal surfaces, notably the mechanism of ammonia synthesis on iron and the oxidation of carbon monoxide on platinum.

Nobel Prize-winning Work

The work for which Gerhard Ertl was awarded the Nobel Prize in Chemistry in 2007 was described by the Nobel Committee as: "for his studies of chemical processes on solid surfaces".

Ertl showed at the level of individual atoms how gases react on well-defined crystal surfaces, identifying the separate steps of catalytic reactions such as adsorption, reaction, and desorption.

Significance and Impact

His work turned heterogeneous catalysis from an empirical art into a molecular science. The understanding of the Haber–Bosch ammonia synthesis he provided has guided the design of the industrial catalysts used to make fertilizer.

Later Career and Honors

Ertl headed the physical chemistry department of the Fritz Haber Institute and later served as its director, becoming director emeritus. He received numerous honors, including the Wolf Prize in Chemistry in 1998.

Legacy

Ertl's surface science methods are now standard tools in catalysis research, and his studies are the reference point for understanding how catalysts work at the atomic level.

Modern Developments

Ertl's surface chemistry studies, including his visualization of the Haber–Bosch ammonia synthesis at the atomic level, established modern surface science and heterogeneous catalysis. His methods are now used to optimize industrial catalysts for fertilizers, fuel cells, and pollution control, and his work underpins the design of catalytic converters and clean-energy technologies.

Selected Publications

"Reactions at Surfaces: From Atoms to Complexity" (Angewandte Chemie International Edition, 2008)

Chapter Summary

The 2007 prize recognized the founding of modern surface chemistry by Gerhard Ertl, who described the elementary steps of reactions on solid surfaces. This work provided the molecular basis for understanding heterogeneous catalysis, including the industrial synthesis of ammonia.

Chapter 108

2008:Osamu Shimomura and Martin Chalfie and Roger Y. Tsien

The Nobel Prize in Chemistry for the year 2008 was awarded jointly to Osamu Shimomura and Martin Chalfie and Roger Y. Tsien.

Motivation:

for the discovery and development of the green fluorescent protein, GFP

108.1 Osamu Shimomura

Early Life and Education

Osamu Shimomura was born in 1928 in Fukuchiyama, Japan.

Career and Major Research

Shimomura studied at Nagasaki Medical College (later part of Nagasaki University) and moved to the United States, where he worked at Princeton University and Boston University and carried out research at the Marine Biological Laboratory in Woods Hole. In the early 1960s he isolated the bioluminescent protein aequorin and the green fluorescent protein (GFP) from the jellyfish *Aequorea victoria*.

Nobel Prize-winning Work

The work for which Osamu Shimomura was awarded the Nobel Prize in Chemistry in 2008 was described by the Nobel Committee as: "for the discovery and development of the green fluorescent protein, GFP".

Shimomura discovered GFP and showed that it glows green under ultraviolet light on its own, without any additional protein or cofactor.

Significance and Impact

Because GFP's light-emitting group is self-contained, the protein made it possible to attach a visible tag to other proteins inside living cells. This discovery provided the tool that later developments turned into a universal marker for biology.

Later Career and Honors

Shimomura was professor emeritus of Boston University and continued his work on bioluminescent proteins. He died on 2018-10-19 in Nagasaki, Japan.

Legacy

The discovery of GFP opened the way for imaging gene expression and protein behavior in living organisms, a technique now used across biology and medicine.

Modern Developments

Shimomura's discovery of the green fluorescent protein (GFP) and its mechanism of luminescence founded the field of in vivo imaging. GFP and its engineered variants are now used in thousands of laboratories to visualize proteins, cells, and gene expression in living organisms, transforming cell biology and neuroscience.

Selected Publications

"Extraction, Purification and Properties of Aequorin, a Bioluminescent Protein from the Luminous Hydromedusan, Aequorea" (Journal of Cellular and Comparative Physiology, 1962)

108.2 Martin Chalfie

Early Life and Education

Martin Chalfie was born in 1947 in Chicago, Illinois, USA.

Career and Major Research

Chalfie earned bachelor's and doctoral degrees from Harvard University, in 1969 and 1977, and did post-doctoral work with Sydney Brenner at the MRC Laboratory of Molecular Biology in Cambridge, where he studied the nervous system of the nematode *Caenorhabditis elegans*. He became a professor at Columbia University.

Nobel Prize-winning Work

The work for which Martin Chalfie was awarded the Nobel Prize in Chemistry in 2008 was described by the Nobel Committee as: "for the discovery and development of the green fluorescent protein, GFP".

Chalfie demonstrated that the gene for GFP could be inserted into other organisms and that the protein would light up wherever the gene was active, providing a direct readout of gene expression.

Significance and Impact

His demonstration made GFP the marker of choice for following gene expression and protein location in living cells. It showed that GFP works as a genetic marker in a wide range of organisms.

Later Career and Honors

Chalfie continues his research on the nematode nervous system at Columbia University.

Legacy

Chalfie's work established the standard way of visualizing gene activity in living organisms, from bacteria to mammals.

Modern Developments

Chalfie's demonstration that GFP could be used as a genetic marker to visualize gene expression in living organisms established GFP as the tool of choice for biological imaging. His work made possible the study of neurons, development, and disease in transparent organisms and founded modern optogenetics and live-cell imaging.

Selected Publications

"Green Fluorescent Protein as a Marker for Gene Expression" (Science, 1994)

108.3 Roger Y. Tsien

Early Life and Education

Roger Y. Tsien was born in 1952 in New York City, New York, USA.

Career and Major Research

Tsien earned a bachelor's degree from Harvard University in 1972 and a doctorate from the University of Cambridge in 1977. He became professor of pharmacology, chemistry, and biochemistry at the University of California, San Diego, and an investigator of the Howard Hughes Medical Institute.

Nobel Prize-winning Work

The work for which Roger Y. Tsien was awarded the Nobel Prize in Chemistry in 2008 was described by the Nobel Committee as: "for the discovery and development of the green fluorescent protein, GFP".

Tsien worked out how the GFP chromophore forms and engineered variants that glow in different colors, from blue to red, and designed fluorescent indicators that report the activity of cells.

Significance and Impact

His colored variants allowed several different proteins to be tracked at the same time inside one cell. Tsien's palette of fluorescent proteins is now used throughout cell biology and neuroscience.

Later Career and Honors

Tsien worked at the University of California, San Diego, for most of his career. He died on 2016-08-24 in Eugene, Oregon, USA.

Legacy

The rainbow of fluorescent proteins he created underlies modern imaging of the interior of living cells, including studies of the brain.

Modern Developments

Tsien's engineering of GFP variants with new colors and improved properties created a palette of fluorescent proteins that revolutionized biological imaging. His techniques underpin modern super-resolution microscopy, cell tracking, and the study of neuronal activity, and his fluorescent indicators remain essential tools in neuroscience and cell biology.

Selected Publications

"The Green Fluorescent Protein" (Annual Review of Biochemistry, 1998)

"Improved Green Fluorescence" (Nature, 1995)

Chapter Summary

The 2008 prize recognized the discovery and development of the green fluorescent protein: Shimomura found GFP in a jellyfish, Chalfie made it a marker of gene expression, and Tsien created color variants for imaging. GFP-based imaging is now among the most widely used tools in biology.

Chapter 109

2009: Venkatraman Ramakrishnan and Thomas A. Steitz and Ada E. Yonath

The Nobel Prize in Chemistry for the year 2009 was awarded jointly to Venkatraman Ramakrishnan and Thomas A. Steitz and Ada E. Yonath.

Motivation:

for studies of the structure and function of the ribosome

109.1 Venkatraman Ramakrishnan

Early Life and Education

Venkatraman Ramakrishnan was born in 1952 in Chidambaram, India.

Career and Major Research

Ramakrishnan earned a bachelor's degree from the Maharaja Sayajirao University of Baroda and a doctorate in physics from the University of California, San Diego, in 1976. After work at the Brookhaven National Laboratory he became a group leader at the MRC Laboratory of Molecular Biology in Cambridge, where he determined the atomic structure of the small (30S) subunit of the ribosome.

Nobel Prize-winning Work

The work for which Venkatraman Ramakrishnan was awarded the Nobel Prize in Chemistry in 2009 was described by the Nobel Committee as: "for studies of the structure and function of the ribosome".

Ramakrishnan's structure of the 30S subunit showed how the ribosome reads the genetic message and how antibiotics interfere with that process.

Significance and Impact

The structure explained how the small subunit ensures the accuracy of protein synthesis and revealed the binding sites of several antibiotics, guiding the design of improved drugs.

Later Career and Honors

Ramakrishnan later served as president of the Royal Society from 2015 to 2020 and continues his research in Cambridge.

Legacy

His work is central to the modern understanding of how the ribosome decodes messenger RNA and makes proteins.

Modern Developments

Ramakrishnan's X-ray crystallographic structures of the ribosome, together with Steitz and Yonath, revealed the atomic architecture of the machine that translates mRNA into protein. His structures explain how antibiotics target the ribosome, guiding the design of new antibacterial drugs, and his work underpins modern understanding of protein synthesis.

Selected Publications

"Structure of the 30S Ribosomal Subunit" (Nature, 2000)

109.2 Thomas A. Steitz

Early Life and Education

Thomas A. Steitz was born in 1940 in Milwaukee, Wisconsin, USA.

Career and Major Research

Steitz earned bachelor's and doctoral degrees from Harvard University, in 1962 and 1966. He joined Yale University in 1970 and became a Sterling Professor of Molecular Biophysics and Biochemistry and an investigator of the Howard Hughes Medical Institute. With Peter Moore, he determined the atomic structure of the large (50S) subunit of the ribosome in 2000.

Nobel Prize-winning Work

The work for which Thomas A. Steitz was awarded the Nobel Prize in Chemistry in 2009 was described by the Nobel Committee as: "for studies of the structure and function of the ribosome".

Steitz's structure of the 50S subunit showed where the peptide bond is formed and revealed the site at which many antibiotics stop protein synthesis.

Significance and Impact

Because the large subunit carries out the chemistry of protein synthesis, its structure explained how proteins are made and how the ribosome can be blocked by drugs.

Later Career and Honors

Steitz remained at Yale University for the rest of his career. He died on 2018-10-09 in Branford, Connecticut, USA.

Legacy

His ribosome structures created molecular targets for a new generation of antibiotics.

Modern Developments

Steitz's crystallographic structures of the ribosome revealed how the ribosome catalyzes protein synthesis and how antibiotics bind and block it. His work provided the atomic basis for understanding translation and for the rational design of antibiotics that treat infections such as tuberculosis, and it is foundational to structural biology.

Selected Publications

"The Complete Atomic Structure of the Large Ribosomal Subunit at 2.4 Å Resolution" (Science, 2000)

109.3 Ada E. Yonath

Early Life and Education

Ada E. Yonath was born in 1939 in Jerusalem, Israel.

Career and Major Research

Yonath earned bachelor's, master's, and doctoral degrees at the Weizmann Institute of Science, completing her doctorate in 1968, followed by postdoctoral work at the Massachusetts Institute of Technology and the University of Chicago. She became a professor at the Weizmann Institute and director of its Kimmelman Center for Biomolecular Structure and Assembly.

Nobel Prize-winning Work

The work for which Ada E. Yonath was awarded the Nobel Prize in Chemistry in 2009 was described by the Nobel Committee as: "for studies of the structure and function of the ribosome".

Yonath produced the first crystals of ribosomes from extremophilic bacteria that diffracted X-rays well, opening the way to the atomic structures determined by the prize.

Significance and Impact

Her crystallization of the ribosome, a complex assembly of RNA and proteins, was a landmark that made high-resolution structure determination of the protein factory possible.

Later Career and Honors

Yonath continues her research on the ribosome and its interaction with antibiotics at the Weizmann Institute of Science.

Legacy

Her pioneering crystallization methods remain the basis for structural studies of the ribosome and of other large biological assemblies.

Modern Developments

Yonath's determination of the three-dimensional structures of ribosomes, including those of antibiotic-resistant bacteria, revealed the molecular machinery of protein synthesis. Her work explains how antibiotics interfere with the ribosome and has guided the development of new antibacterial drugs to combat resistance.

Selected Publications

"The Complete Atomic Structure of the Large Ribosomal Subunit at 2.4 Å Resolution" (Science, 2000)

Chapter Summary

The 2009 prize recognized the determination of the structure and function of the ribosome, the cellular machine that translates the genetic code into proteins. The atomic structures of its subunits explain how proteins are made and how antibiotics act, and they have guided drug development.

Chapter 110

2010: Richard F. Heck and Ei-ichi Negishi and Akira Suzuki

The Nobel Prize in Chemistry for the year 2010 was awarded jointly to Richard F. Heck and Ei-ichi Negishi and Akira Suzuki.

Motivation:

for palladium-catalyzed cross couplings in organic synthesis

110.1 Richard F. Heck

Early Life and Education

Richard F. Heck was born in 1931 in Springfield, Massachusetts, USA.

Career and Major Research

Heck received his doctorate from the University of California, Los Angeles, in 1954. He worked for the Hercules Company and later became a professor at the University of Delaware, where he developed the palladium-catalyzed coupling of organic halides with alkenes now known as the Heck reaction.

Nobel Prize-winning Work

The work for which Richard F. Heck was awarded the Nobel Prize in Chemistry in 2010 was described by the Nobel Committee as: "for palladium-catalyzed cross couplings in organic synthesis".

The Heck reaction joins an aryl or vinyl halide to an alkene in the presence of a palladium catalyst, forming a new carbon-carbon bond.

Significance and Impact

The Heck reaction was the first practical way to form carbon-carbon bonds between complex molecules under mild conditions. It is used throughout the pharmaceutical and materials industries.

Later Career and Honors

Heck retired from the University of Delaware and later moved to the Philippines. He died on 2015-10-09 in Manila, Philippines.

Legacy

The Heck reaction remains one of the most widely used transformations in organic synthesis and is the basis of many industrial processes.

Modern Developments

Heck's palladium-catalyzed coupling of aryl halides with alkenes, the Heck reaction, became one of the most widely used methods for forming carbon-carbon bonds in organic synthesis. The Heck reaction and the related cross-coupling reactions are now essential tools in the pharmaceutical and agrochemical industries, used to make drugs such as montelukast and several anti-inflammatory agents.

Selected Publications

"Palladium-Catalyzed Reactions of Organic Halides with Olefins" (Accounts of Chemical Research, 1979)

110.2 Ei-ichi Negishi

Early Life and Education

Ei-ichi Negishi was born in 1935 in Changchun, China.

Career and Major Research

Negishi earned a bachelor's degree from the University of Tokyo in 1958 and a doctorate from the University of Pennsylvania in 1963. He worked as a postdoctoral researcher with Herbert C. Brown at Purdue University and later became a professor there. In the late 1970s he developed the palladium- or nickel-catalyzed cross coupling of organozinc compounds with organic halides.

Nobel Prize-winning Work

The work for which Ei-ichi Negishi was awarded the Nobel Prize in Chemistry in 2010 was described by the Nobel Committee as: "for palladium-catalyzed cross couplings in organic synthesis".

The Negishi coupling uses an organozinc reagent to transfer a carbon group to a palladium or nickel catalyst, which then delivers it to an organic halide.

Significance and Impact

The Negishi coupling allows carbon-carbon bonds to be made with high selectivity and is widely used in the synthesis of pharmaceuticals, agrochemicals, and materials.

Later Career and Honors

Negishi remained a professor at Purdue University for four decades. He died on 2021-06-06 in Indianapolis, Indiana, USA.

Legacy

The Negishi coupling is a standard reaction of modern cross-coupling chemistry and is taught and used in laboratories worldwide.

Modern Developments

Negishi's palladium- or nickel-catalyzed cross-coupling of organozinc reagents, the Negishi coupling, provided a general and selective method for forming carbon-carbon bonds. Cross-coupling chemistry is now a pillar of modern organic synthesis, used in the industrial production of pharmaceuticals, agrochemicals, and organic electronic materials.

Selected Publications

"Palladium- or Nickel-Catalyzed Cross Coupling: A New Selective Method for Carbon-Carbon Bond Formation" (Accounts of Chemical Research, 1982)

110.3 Akira Suzuki

Early Life and Education

Akira Suzuki was born in 1930 in Mukawa, Japan.

Career and Major Research

Suzuki received his doctorate from Hokkaido University in 1959 and became a professor there. He developed the cross coupling of organoboron compounds, such as boronic acids, with organic halides in the presence of a palladium catalyst, a reaction now known as the Suzuki coupling.

Nobel Prize-winning Work

The work for which Akira Suzuki was awarded the Nobel Prize in Chemistry in 2010 was described by the Nobel Committee as: "for palladium-catalyzed cross couplings in organic synthesis".

The Suzuki coupling forms carbon-carbon bonds between an organoboron compound and an organic halide, with the boron group transferred under mild conditions.

Significance and Impact

Because organoboron reagents are stable and easy to handle, the Suzuki coupling is one of the most widely used reactions for building complex molecules, including pharmaceuticals and electronic materials.

Later Career and Honors

Suzuki is professor emeritus of Hokkaido University and continues to be active in chemistry.

Legacy

The Suzuki coupling is a cornerstone of cross-coupling chemistry and is used in industrial processes ranging from drug manufacture to the production of liquid crystal displays.

Modern Developments

Suzuki's palladium-catalyzed cross-coupling of organoboron compounds, the Suzuki reaction, is now one of the most powerful and widely used reactions in organic synthesis. It is used throughout the pharmaceutical industry, in the synthesis of natural products, and in the manufacture of OLED materials, and its mild conditions and tolerance of functional groups made it a favorite of industrial chemists.

Selected Publications

"Stereoselective Synthesis of Arylated (E)-Alkenes by the Reaction of Alk-1-enylboranes with Aryl Halides in the Presence of Palladium Catalyst" (Journal of the Chemical Society, Chemical Communications, 1979)

Chapter Summary

The 2010 prize recognized the palladium-catalyzed cross-coupling reactions developed by Heck, Negishi, and Suzuki, which make it possible to join carbon atoms selectively. These reactions are now indispensable in the production of pharmaceuticals, agrochemicals, and advanced materials.

Chapter 111

2011: Dan Shechtman

The Nobel Prize in Chemistry for the year 2011 was awarded to Dan Shechtman.

Motivation:

for the discovery of quasicrystals

111.1 Dan Shechtman

Early Life and Education

Dan Shechtman was born in 1941 in Tel Aviv, Israel.

Career and Major Research

Shechtman earned bachelor's, master's, and doctoral degrees in materials engineering from the Technion–Israel Institute of Technology in Haifa. In 1982, while on sabbatical at the National Bureau of Standards in the United States, he observed an electron diffraction pattern with fivefold symmetry from a rapidly cooled aluminum–manganese alloy, a symmetry forbidden for ordinary crystals.

Nobel Prize-winning Work

The work for which Dan Shechtman was awarded the Nobel Prize in Chemistry in 2011 was described by the Nobel Committee as: "for the discovery of quasicrystals".

Shechtman's diffraction patterns showed a solid with long-range order but no translational periodicity, a phase that could not be described as a conventional crystal. The new material was named a quasicrystal, and his findings were published with colleagues in 1984.

Significance and Impact

The discovery overturned the long-standing belief that all crystals must have repeating translational symmetry. Quasicrystals possess order, yet their atoms are arranged with rotational symmetries such as fivefold that crystals were thought to forbid.

Later Career and Honors

Shechtman was professor of materials science at the Technion, where he is now professor emeritus, and he has held positions at research institutions in the United States and Europe.

Legacy

Since his discovery, hundreds of quasicrystalline materials have been identified, and quasicrystals are now recognized as a distinct class of solid matter with applications in coatings and alloys.

Modern Developments

Shechtman's discovery of quasicrystals, materials with forbidden fivefold rotational symmetry, overturned the long-held assumption that crystals must be periodic and founded a new class of solid materials. Quasicrystals are now used in applications such as non-stick coatings and high-strength alloys, and they appear naturally in meteorites and some minerals. His work reshaped the very definition of a crystal and continues to inspire research in materials science and mathematics.

Selected Publications

"Metallic Phase with Long-Range Orientational Order and No Translational Symmetry" (Physical Review Letters, 1984)

Chapter Summary

The 2011 prize recognized Dan Shechtman's discovery of quasicrystals, materials with ordered but non-periodic atomic arrangements. The finding changed the definition of a crystal and opened a new branch of materials science.

Chapter 112

2012:Robert J. Lefkowitz and Brian Kobilka

The Nobel Prize in Chemistry for the year 2012 was awarded jointly to Robert J. Lefkowitz and Brian Kobilka.

Motivation:

for studies of G-protein-coupled receptors

112.1 Robert J. Lefkowitz

Early Life and Education

Robert J. Lefkowitz was born in 1943 in New York City, New York, USA.

Career and Major Research

Lefkowitz earned his medical degree from Columbia University in 1966. He became professor of medicine and biochemistry at Duke University and an investigator of the Howard Hughes Medical Institute. He identified the receptors for adrenaline as G-protein-coupled receptors, cloned several of them, and discovered the arrestin proteins that regulate them.

Nobel Prize-winning Work

The work for which Robert J. Lefkowitz was awarded the Nobel Prize in Chemistry in 2012 was described by the Nobel Committee as: "for studies of G-protein-coupled receptors".

Lefkowitz showed that a large family of receptors acts through G proteins, explaining how hormones and neurotransmitters signal across the cell membrane, and he uncovered the molecules that switch these signals off.

Significance and Impact

G-protein-coupled receptors mediate the responses of cells to light, odor, and most hormones, and they are the targets of a large fraction of approved drugs. Lefkowitz's work established how these receptors recognize their signals and are regulated.

Later Career and Honors

Lefkowitz continues his research at the Duke University School of Medicine and the Howard Hughes Medical Institute.

Legacy

His discovery of arrestins and his account of receptor regulation are central to pharmacology, and his laboratory trained a generation of researchers in receptor biology.

Modern Developments

Lefkowitz's discovery of the beta-adrenergic receptor and his characterization of G protein-coupled receptors (GPCRs) revealed the largest family of drug targets in medicine. Over a third of all approved drugs act on GPCRs, including treatments for asthma, heart disease, and hypertension, and his work founded the modern field of receptor pharmacology.

Selected Publications

"Beta-Arrestins: A New Family of Regulators of G-Protein-Coupled Receptors" (Journal of Biological Chemistry, 1990)

112.2 Brian Kobilka

Early Life and Education

Brian Kobilka was born in 1955 in Little Falls, Minnesota, USA.

Career and Major Research

Kobilka earned a bachelor's degree from the University of Minnesota Duluth in 1977 and a medical degree from Washington University in St. Louis in 1981. He worked in Robert Lefkowitz's laboratory at Duke University and then became a professor at Stanford University, where in 2007 he determined the crystal structure of the beta-2 adrenergic receptor.

Nobel Prize-winning Work

The work for which Brian Kobilka was awarded the Nobel Prize in Chemistry in 2012 was described by the Nobel Committee as: "for studies of G-protein-coupled receptors".

Kobilka obtained the first high-resolution structure of a G-protein-coupled receptor bound to a diffusible signal, and later captured the receptor in action with the G protein it activates.

Significance and Impact

The structures revealed at atomic detail how a drug molecule binds the receptor and how the receptor changes shape to transmit its signal, explaining how GPCRs work as molecular switches.

Later Career and Honors

Kobilka continues his research on receptor structure at the Stanford University School of Medicine.

Legacy

Kobilka's structural work provided the framework for understanding GPCR activation and has aided the design of drugs that act on this family of receptors.

Modern Developments

Kobilka's determination of the three-dimensional structure of the beta2-adrenergic receptor in its active state revealed how GPCRs are activated and how drugs stabilize them. His structures underpin modern structure-based drug design for receptors that are targets of a large fraction of pharmaceuticals, including beta-blockers and antihistamines.

Selected Publications

"Crystal Structure of the Human β_2 Adrenergic G-Protein-Coupled Receptor" (Nature, 2007)

Chapter Summary

The 2012 prize recognized the studies of G-protein-coupled receptors by Lefkowitz and Kobilka, which explained how these receptors transmit signals into cells. Their work is the foundation of much of modern pharmacology, since GPCRs are targets of many medicines.

Chapter 113

2013: Martin Karplus and Michael Levitt and Arieh Warshel

The Nobel Prize in Chemistry for the year 2013 was awarded jointly to Martin Karplus and Michael Levitt and Arieh Warshel.

Motivation:

for the development of multiscale models for complex chemical systems

113.1 Martin Karplus

Early Life and Education

Martin Karplus was born in 1930 in Vienna, Austria.

Career and Major Research

Karplus earned a bachelor's degree from Harvard University in 1950 and a doctorate from the California Institute of Technology in 1953. He became a professor at Harvard University, where he pioneered the use of molecular dynamics to simulate the motion of macromolecules and co-developed methods that combine quantum and classical mechanics.

Nobel Prize-winning Work

The work for which Martin Karplus was awarded the Nobel Prize in Chemistry in 2013 was described by the Nobel Committee as: "for the development of multiscale models for complex chemical systems".

Karplus showed how computer simulation could follow the movement of every atom in a large molecule, and his work underlies the molecular dynamics programs used across chemistry and biology.

Significance and Impact

His simulations made it possible to watch how proteins move and fold and how they interact with other molecules, connecting theory with experiment at the atomic level.

Later Career and Honors

Karplus was the Theodore William Richards Professor of Chemistry at Harvard University, where he worked for over five decades. He died on 2024-12-28 in Cambridge, Massachusetts, USA.

Legacy

The computational methods he developed, including the CHARMM program, are standard tools for studying biomolecules and are used in drug design.

Modern Developments

Karplus's molecular dynamics simulations of proteins, combining quantum and classical mechanics, founded the field of computational biochemistry. His methods are now used to study enzyme mechanisms, drug binding, and protein folding, and hybrid QM/MM techniques based on his work are standard tools in drug design and biophysics.

Selected Publications

"CHARMM: A Program for Macromolecular Energy, Minimization, and Dynamics Calculations" (Journal of Computational Chemistry, 1983)

113.2 Michael Levitt

Early Life and Education

Michael Levitt was born in 1947 in Pretoria, South Africa.

Career and Major Research

Levitt studied at King's College London and received his doctorate from the University of Cambridge in 1971. He became a professor at Stanford University. He carried out some of the first molecular dynamics simulations of proteins and, with Arieh Warshel, developed a combined quantum and classical description of enzyme reactions.

Nobel Prize-winning Work

The work for which Michael Levitt was awarded the Nobel Prize in Chemistry in 2013 was described by the Nobel Committee as: "for the development of multiscale models for complex chemical systems".

Levitt developed simplified representations of protein structure that made simulation of large systems feasible, and helped establish the multiscale approach that treats different parts of a molecule with different levels of theory.

Significance and Impact

His simulations of protein dynamics and folding were among the first to show that computers could describe the behavior of biological macromolecules realistically.

Later Career and Honors

Levitt continues his research on computational structural biology at Stanford University.

Legacy

Levitt's methods are part of the foundation of computational structural biology, used for predicting protein structure and simulating biomolecular systems.

Modern Developments

Levitt's coarse-grained models and his work on simulating protein dynamics and folding helped found the field of computational structural biology. His methods underpin the modern prediction and design of protein structures and were early steps toward the AlphaFold and Rosetta approaches used in biotechnology and medicine.

Selected Publications

"Theoretical Studies of Enzymic Reactions: Dielectric, Electrostatic and Steric Stabilization of the Carbo-
nium Ion in the Reaction of Lysozyme" (Journal of Molecular Biology, 1976)

113.3 Arie Warshel

Early Life and Education

Arie Warshel was born in 1940 in Sde Nahum, Israel.

Career and Major Research

Warshel earned bachelor's and master's degrees at the Technion–Israel Institute of Technology and a doctorate from the Weizmann Institute of Science in 1969. He became a professor at the University of Southern California, where he pioneered methods that combine quantum mechanics for reacting bonds with classical mechanics for the surrounding atoms.

Nobel Prize-winning Work

The work for which Arie Warshel was awarded the Nobel Prize in Chemistry in 2013 was described by the Nobel Committee as: "for the development of multiscale models for complex chemical systems".

Warshel developed the QM/MM approach, in which the part of a molecule where a chemical reaction occurs is treated quantum mechanically while the rest is treated classically, allowing enzyme reactions to be simulated.

Significance and Impact

His calculations explained how enzymes accelerate reactions by stabilizing transition states, providing a quantitative link between enzyme structure and catalytic power.

Later Career and Honors

Warshel continues his research on enzyme mechanisms and computational chemistry at the University of Southern California.

Legacy

The QM/MM method is now used routinely to model reactions in enzymes, materials, and solution, and it is a cornerstone of modern computational chemistry.

Modern Developments

Warshel's development of QM/MM methods, combining quantum mechanics with molecular mechanics, made it possible to simulate chemical reactions in enzymes and complex molecular environments. His methods are now used worldwide to study enzyme catalysis, drug metabolism, and photochemistry and remain central tools in computational chemistry.

Selected Publications

"Theoretical Studies of Enzymic Reactions: Dielectric, Electrostatic and Steric Stabilization of the Carbo-
nium Ion in the Reaction of Lysozyme" (Journal of Molecular Biology, 1976)

Chapter Summary

The 2013 prize recognized the development of multiscale models that combine quantum and classical mechanics to simulate complex chemical systems. These methods allow chemists to compute how reactions proceed in enzymes and large molecules, a foundation of modern computational chemistry.

Chapter 114

2014:Eric Betzig and Stefan W. Hell and William E. Moerner

The Nobel Prize in Chemistry for the year 2014 was awarded jointly to Eric Betzig and Stefan W. Hell and William E. Moerner.

Motivation:

for the development of super-resolved fluorescence microscopy

114.1 Eric Betzig

Early Life and Education

Eric Betzig was born in 1960 in Ann Arbor, Michigan, USA.

Career and Major Research

Betzig earned a bachelor's degree from the California Institute of Technology in 1983 and a doctorate in applied physics from Cornell University in 1988. He worked at Bell Laboratories and later at the Janelia Research Campus of the Howard Hughes Medical Institute, where he developed photoactivated localization microscopy (PALM) together with Harald Hess.

Nobel Prize-winning Work

The work for which Eric Betzig was awarded the Nobel Prize in Chemistry in 2014 was described by the Nobel Committee as: "for the development of super-resolved fluorescence microscopy".

PALM switches on fluorescent molecules one at a time and records their positions, building up a high-resolution image of a cell from many individual localizations, far below the diffraction limit of light.

Significance and Impact

Betzig's method allows cellular structures to be imaged at nanometer resolution, revealing details that optical microscopy could not previously resolve.

Later Career and Honors

Betzig continues his research on high-resolution imaging at the Janelia Research Campus.

Legacy

PALM and related single-molecule methods have made nanoscale imaging of living cells possible and are used across cell biology.

Modern Developments

Betzig's development of super-resolution fluorescence microscopy, including PALM, broke the diffraction limit and made it possible to image molecules and structures at the nanometer scale inside living cells. His techniques transformed cell biology and neuroscience, enabling the direct observation of individual proteins, synapses, and organelles.

Selected Publications

"Imaging Intracellular Fluorescent Proteins at Nanometer Resolution" (Science, 2006)

114.2 Stefan W. Hell

Early Life and Education

Stefan W. Hell was born in 1962 in Arad, Romania.

Career and Major Research

Hell studied physics and received his doctorate from the University of Heidelberg in 1990. He became a director at the Max Planck Institute for Biophysical Chemistry in Göttingen, where he developed stimulated emission depletion (STED) microscopy.

Nobel Prize-winning Work

The work for which Stefan W. Hell was awarded the Nobel Prize in Chemistry in 2014 was described by the Nobel Committee as: "for the development of super-resolved fluorescence microscopy".

STED microscopy uses a second laser beam to switch off fluorescence everywhere except at the center of the spot, shrinking the effective focus far below the diffraction limit.

Significance and Impact

STED gave scientists a way to image structures inside cells with resolution beyond the classical limit of light microscopy, and it works on living samples.

Later Career and Honors

Hell continues his research on super-resolution microscopy at the Max Planck Institute for Biophysical Chemistry in Göttingen.

Legacy

STED microscopy is now a commercialized and widely used technique for nanoscale imaging in the life sciences.

Modern Developments

Hell's development of STED super-resolution microscopy, together with his work on stimulated emission, broke the Abbe diffraction limit and allowed imaging of living cells with nanometer resolution. STED and related nanoscopy methods are now standard tools in cell biology, enabling the study of protein organization, synapses, and cellular dynamics.

Selected Publications

"Fluorescence Microscopy with Diffraction Resolution Barrier Broken by Stimulated Emission" (Proceedings of the National Academy of Sciences, 2000)

114.3 William E. Moerner

Early Life and Education

William E. Moerner was born in 1953 in Pleasanton, California, USA.

Career and Major Research

Moerner earned a bachelor's degree from Washington University in St. Louis in 1975 and a doctorate from Cornell University in 1982. He worked at IBM Almaden Research Center and later became a professor at Stanford University, where he studied the optics of single molecules.

Nobel Prize-winning Work

The work for which William E. Moerner was awarded the Nobel Prize in Chemistry in 2014 was described by the Nobel Committee as: "for the development of super-resolved fluorescence microscopy".

Moerner achieved the first optical detection and spectroscopy of a single molecule in a solid, demonstrating that individual molecules could be observed and studied.

Significance and Impact

His single-molecule measurements showed that individual molecules differ in their behavior, and they provided the physical basis for the localization methods used in super-resolution microscopy.

Later Career and Honors

Moerner continues his research on single-molecule imaging and spectroscopy at Stanford University.

Legacy

Moerner's single-molecule techniques opened the field of single-molecule chemistry and are a foundation of modern super-resolution imaging.

Modern Developments

Moerner's discovery that a single molecule could be detected and its fluorescence imaged at the single-molecule level founded the field of single-molecule spectroscopy and super-resolution imaging. His work enables the tracking of individual proteins, DNA, and enzymes in real time, underpinning modern biophysics and the super-resolution microscopes now used in biology.

Selected Publications

"Optical Detection and Spectroscopy of Single Molecules in a Solid" (Physical Review Letters, 1989)

Chapter Summary

The 2014 prize recognized three methods that break the diffraction barrier of light microscopy: STED by Hell, PALM by Betzig, and the single-molecule detection by Moerner on which localization methods rely. Together they allow biological structures to be imaged at the nanometer scale.

Chapter 115

2015:Tomas Lindahl and Paul Modrich and Aziz Sancar

The Nobel Prize in Chemistry for the year 2015 was awarded jointly to Tomas Lindahl and Paul Modrich and Aziz Sancar.

Motivation:

for mechanistic studies of DNA repair

115.1 Tomas Lindahl

Early Life and Education

Tomas Lindahl was born in 1938 in Stockholm, Sweden.

Career and Major Research

Lindahl received his doctorate from the Karolinska Institutet in Stockholm in 1967. He later moved to Britain, where he led a laboratory at the MRC laboratories at Clare Hall and became an emeritus scientist at the Francis Crick Institute in London. He demonstrated that DNA is chemically unstable and identified the repair enzymes that remove damaged bases.

Nobel Prize-winning Work

The work for which Tomas Lindahl was awarded the Nobel Prize in Chemistry in 2015 was described by the Nobel Committee as: "for mechanistic studies of DNA repair".

Lindahl showed that bases are lost or damaged in DNA even without external agents, and he discovered the base excision repair pathway that removes and replaces such lesions.

Significance and Impact

His discovery of base excision repair established that cells constantly protect their DNA against spontaneous damage, which would otherwise cause mutations and cancer.

Later Career and Honors

Lindahl was director of the Cancer Research UK London Research Institute at Clare Hall and is now an emeritus group leader at the Francis Crick Institute.

Legacy

Base excision repair is recognized as one of the essential safeguards of genetic information in all living cells.

Modern Developments

Lindahl's discovery of base excision repair revealed the fundamental mechanism by which cells repair the constant chemical damage to DNA. His work established that DNA repair systems maintain the integrity of the genome and underpins modern cancer research, since defects in repair pathways are linked to hereditary cancers and drive many targeted cancer therapies.

Selected Publications

"An N-Glycosidase from *Escherichia coli* That Releases Free Uracil from DNA Containing Deaminated Cytosine Residues" (Proceedings of the National Academy of Sciences, 1974)

115.2 Paul Modrich

Early Life and Education

Paul Modrich was born in 1946 in Raton, New Mexico, USA.

Career and Major Research

Modrich earned a bachelor's degree from the Massachusetts Institute of Technology in 1968 and a doctorate from Stanford University in 1973. He became a professor of biochemistry at Duke University and an investigator of the Howard Hughes Medical Institute, where he worked out the mechanism of mismatch repair.

Nobel Prize-winning Work

The work for which Paul Modrich was awarded the Nobel Prize in Chemistry in 2015 was described by the Nobel Committee as: "for mechanistic studies of DNA repair".

Modrich identified the proteins that recognize and correct errors made when DNA is copied, and he reconstructed the mismatch repair process in detail.

Significance and Impact

Mismatch repair catches the mistakes introduced during DNA replication, reducing the mutation rate enormously. Defects in this system predispose people to hereditary colorectal cancer.

Later Career and Honors

Modrich continues his research on DNA mismatch repair at Duke University.

Legacy

His work established mismatch repair as a central guardian of the genome and linked its failure to human disease.

Modern Developments

Modrich's elucidation of mismatch repair, the DNA repair pathway that corrects replication errors, revealed how cells achieve the astonishing fidelity of DNA replication. His work explains the hereditary colon cancer syndrome Lynch syndrome and underpins modern cancer genetics and the development of immunotherapy for mismatch repair-deficient tumors.

Selected Publications

"DNA Mismatch Correction" (Annual Review of Biochemistry, 1987)

115.3 Aziz Sancar

Early Life and Education

Aziz Sancar was born in 1946 in Savur, Turkey.

Career and Major Research

Sancar earned a medical degree from Istanbul University and a doctorate in molecular biology from the University of Texas at Dallas in 1977. He became a professor at the University of North Carolina at Chapel Hill, where he characterized the nucleotide excision repair system.

Nobel Prize-winning Work

The work for which Aziz Sancar was awarded the Nobel Prize in Chemistry in 2015 was described by the Nobel Committee as: "for mechanistic studies of DNA repair".

Sancar identified the UvrABC enzyme system that cuts damaged DNA on both sides of a lesion and showed how the damaged piece is removed and replaced, and he studied the human equivalent of this process.

Significance and Impact

Nucleotide excision repair removes damage caused by ultraviolet light and many chemicals. Sancar's work explained how cells remove such bulky lesions and why defects in the system cause skin cancer.

Later Career and Honors

Sancar continues his research on DNA repair and its links to cancer at the University of North Carolina at Chapel Hill.

Legacy

The UvrABC system he characterized is a paradigm for repair processes, and his studies connect DNA repair to cancer and chemotherapy.

Modern Developments

Sancar's mapping of nucleotide excision repair, the pathway that repairs DNA damage caused by ultraviolet light, explained how cells protect the genome from sunlight and environmental mutagens. His work revealed the molecular clockwork of DNA repair and underpins the understanding of skin cancer risk and the design of chemoprevention strategies.

Selected Publications

"DNA Excision Repair" (Annual Review of Biochemistry, 1996)

Chapter Summary

The 2015 prize recognized the mechanistic understanding of the three repair systems that protect the genetic code: base excision repair by Lindahl, mismatch repair by Modrich, and nucleotide excision repair by Sancar. Together they explained how cells correct the damage that would otherwise lead to mutations and disease.

Chapter 116

2016: Jean-Pierre Sauvage and Sir J. Fraser Stoddart and Bernard L. Feringa

The Nobel Prize in Chemistry for the year 2016 was awarded jointly to Jean-Pierre Sauvage and Sir J. Fraser Stoddart and Bernard L. Feringa.

Motivation:

for the design and synthesis of molecular machines

116.1 Jean-Pierre Sauvage

Early Life and Education

Jean-Pierre Sauvage was born in 1944 in Paris, France.

Career and Major Research

Sauvage received his doctorate from the University of Strasbourg and became a director of research of the CNRS at the University of Strasbourg. In 1983 he achieved the first synthesis of a catenane, two interlocked molecular rings, by threading one ring through a template and joining the second ring around it.

Nobel Prize-winning Work

The work for which Jean-Pierre Sauvage was awarded the Nobel Prize in Chemistry in 2016 was described by the Nobel Committee as: "for the design and synthesis of molecular machines".

Sauvage's template method made it possible to construct molecules whose parts are mechanically linked yet free to move relative to one another.

Significance and Impact

Interlocked molecules such as catenanes are the basis of molecular machines, since their linked parts can be made to move in a controlled way.

Later Career and Honors

Sauvage is professor emeritus at the University of Strasbourg, where he continues research on molecular knots and interlocked structures.

Legacy

His synthesis of catenanes opened the field of mechanically interlocked molecules, now central to molecular nanotechnology.

Modern Developments

Sauvage's synthesis of catenanes and rotaxanes, molecular structures in which rings are mechanically interlocked, founded the field of mechanically interlocked molecules and molecular machines. His work laid the groundwork for molecular motors, switches, and pumps that are now used in molecular nanotechnology, drug delivery, and the design of artificial molecular machines.

Selected Publications

"Une Nouvelle Famille de Molécules: les Métallo-Caténanes" (Tetrahedron Letters, 1983)

116.2 Sir J. Fraser Stoddart

Early Life and Education

Sir J. Fraser Stoddart was born in 1942 in Edinburgh, Scotland.

Career and Major Research

Stoddart earned bachelor's and doctoral degrees from the University of Edinburgh, receiving his doctorate in 1966. He was a professor at Northwestern University and later at the University of California, Los Angeles, and the University of Hong Kong. He developed methods for making rotaxanes, rings threaded on molecular axes with stoppers at each end, and used them to build molecular switches.

Nobel Prize-winning Work

The work for which Sir J. Fraser Stoddart was awarded the Nobel Prize in Chemistry in 2016 was described by the Nobel Committee as: "for the design and synthesis of molecular machines".

Stoddart showed that a ring in a rotaxane can be moved back and forth along its axle in response to a stimulus, producing a controllable molecular shuttle.

Significance and Impact

His molecular shuttles and switches demonstrated that molecules could perform mechanical tasks on command, laying the groundwork for nanoscale machines.

Later Career and Honors

Stoddart was chair professor of chemistry at the University of Hong Kong at the end of his career. He died on 2024-12-30 in Melbourne, Australia.

Legacy

Stoddart's interlocked molecules underpin molecular electronics and have inspired proposals for molecular computers and drug-delivery systems.

Modern Developments

Stoddart's synthesis of rotaxanes and his construction of molecular switches and motors, including a molecular elevator and a molecular chain, demonstrated that molecules can perform mechanical functions. His work founded the field of artificial molecular machines, now being developed for use in nanoelectronics, drug delivery, and adaptive materials.

Selected Publications

"Molecular Meccano. 1. [2]Rotaxanes and a [2]Catenane Made to Order" (Journal of the American Chemical Society, 1992)

116.3 Bernard L. Feringa

Early Life and Education

Bernard L. Feringa was born in 1951 in Barger-Compascuum, the Netherlands.

Career and Major Research

Feringa received his doctorate from the University of Groningen in 1978 and became a professor there. In 1999 he built the first molecular motor driven by light, a molecule that rotates in one direction only when illuminated.

Nobel Prize-winning Work

The work for which Bernard L. Feringa was awarded the Nobel Prize in Chemistry in 2016 was described by the Nobel Committee as: "for the design and synthesis of molecular machines".

Feringa's light-driven motor converts light energy into a continuous unidirectional rotation of one part of the molecule relative to the other.

Significance and Impact

His molecular motors were the first molecules that could be made to rotate continuously in a controlled direction, a requirement for useful machines at the molecular scale.

Later Career and Honors

Feringa continues his research on molecular motors and rotary machines at the University of Groningen.

Legacy

Feringa's motors are being developed for applications such as responsive materials, and they established that molecular machines can do real work.

Modern Developments

Feringa's design of the first molecular motor, a molecule that rotates continuously when illuminated, demonstrated that molecules can act as tiny engines. His motors are now used to power nanoscale machines, create responsive materials, and enable molecular systems with applications in medicine, catalysis, and smart materials.

Selected Publications

"Light-Driven Monodirectional Molecular Rotor" (Nature, 1999)

Chapter Summary

The 2016 prize recognized the design and synthesis of molecular machines: Sauvage built interlocked catenanes, Stoddart constructed rotaxane shuttles, and Feringa made light-driven molecular motors. Together they showed that molecules can be engineered to perform controlled mechanical movements.

Chapter 117

2017: Jacques Dubochet and Joachim Frank and Richard Henderson

The Nobel Prize in Chemistry for the year 2017 was awarded jointly to Jacques Dubochet and Joachim Frank and Richard Henderson.

Motivation:

for developing cryo-electron microscopy for the high-resolution structure determination of biomolecules in solution

117.1 Jacques Dubochet

Early Life and Education

Jacques Dubochet was born in 1942 in Aigle, Switzerland.

Career and Major Research

Dubochet studied at the University of Lausanne and received a doctorate in biophysics from the University of Geneva and the University of Basel in 1973. After work at the European Molecular Biology Laboratory in Heidelberg he became a professor at the University of Lausanne. He developed the method of vitrifying water by cooling it so fast that it solidifies without forming ice crystals, preserving biological molecules in their natural state in the electron microscope.

Nobel Prize-winning Work

The work for which Jacques Dubochet was awarded the Nobel Prize in Chemistry in 2017 was described by the Nobel Committee as: "for developing cryo-electron microscopy for the high-resolution structure determination of biomolecules in solution".

Dubochet's vitrified water made it possible to image intact biomolecules in a thin layer of glass-like ice under the electron beam.

Significance and Impact

His plunge-freezing method protects delicate biological specimens from damage and dehydration, and it is an essential step in every modern cryo-EM experiment.

Later Career and Honors

Dubochet is professor emeritus at the University of Lausanne and has spoken widely about science and society.

Legacy

Water vitrification is the technical foundation of cryo-electron microscopy, now a routine method for determining biomolecular structures.

Modern Developments

Dubochet's development of vitrified (frozen-hydrated) specimen preparation made it possible to image biological samples in their native state with cryo-electron microscopy. His method, which vitrifies water so it does not form damaging ice crystals, is the foundation of modern cryo-EM, now a routine technique for determining the structures of proteins and viruses.

Selected Publications

"Cryo-Electron Microscopy of Viruses" (Nature, 1984)

117.2 Joachim Frank

Early Life and Education

Joachim Frank was born in 1940 in Siegen, Germany.

Career and Major Research

Frank studied physics and received his doctorate from the Technical University of Munich in 1970. He became a professor at Columbia University, where he developed the computational methods that combine noisy electron microscope images of single molecules into three-dimensional structures.

Nobel Prize-winning Work

The work for which Joachim Frank was awarded the Nobel Prize in Chemistry in 2017 was described by the Nobel Committee as: "for developing cryo-electron microscopy for the high-resolution structure determination of biomolecules in solution".

Frank's image-processing procedures aligned and averaged thousands of individual molecular images, reconstructing detailed three-dimensional structures of machines such as the ribosome.

Significance and Impact

His software made it possible to extract high-resolution information from low-dose images of molecules that cannot be crystallized.

Later Career and Honors

Frank continues his work on image processing and the structure of the ribosome at Columbia University.

Legacy

Frank's computational approach is the core of single-particle cryo-EM, which has revolutionized structural biology.

Modern Developments

Frank's computational method for merging noisy 2D electron microscopy images into 3D reconstructions made cryo-electron microscopy practical. His algorithms transformed cryo-EM into a technique capable of atomic-resolution structures and underpin the modern revolution in structural biology that now routinely solves structures of membrane proteins and viruses.

Selected Publications

"SPIDER and WEB: Processing and Visualization of Images in 3D Electron Microscopy and Related Fields" (Journal of Structural Biology, 1996)

117.3 Richard Henderson

Early Life and Education

Richard Henderson was born in 1945 in Edinburgh, Scotland.

Career and Major Research

Henderson earned a bachelor's degree from the University of Edinburgh and a doctorate from the University of Cambridge in 1969. He became a group leader at the MRC Laboratory of Molecular Biology in Cambridge, where in 1975 he and Nigel Unwin produced the first three-dimensional structure of a membrane protein, bacteriorhodopsin, using electron microscopy.

Nobel Prize-winning Work

The work for which Richard Henderson was awarded the Nobel Prize in Chemistry in 2017 was described by the Nobel Committee as: "for developing cryo-electron microscopy for the high-resolution structure determination of biomolecules in solution".

Henderson showed that electron microscopy could reach atomic resolution by improving the detectors and sample preparation of cryo-EM, and he demonstrated this by imaging a protein at near-atomic resolution.

Significance and Impact

His work proved that cryo-EM could determine structures as accurately as X-ray crystallography, a milestone that established the method for routine use.

Later Career and Honors

Henderson was a long-serving group leader at the MRC Laboratory of Molecular Biology and helped lead the development of direct electron detectors.

Legacy

Henderson's demonstration that cryo-EM reaches atomic resolution transformed structural biology and earned the method widespread adoption.

Modern Developments

Henderson's pioneering cryo-electron microscopy of proteins, including his demonstration that electron diffraction could yield atomic-resolution structures, proved that EM could rival X-ray crystallography. His work led to the modern cryo-EM revolution that now determines protein structures without crystals, reshaping structural biology and drug discovery.

Selected Publications

"Three-Dimensional Model of Purple Membrane Obtained by Electron Microscopy" (Nature, 1975)

Chapter Summary

The 2017 prize recognized the development of cryo-electron microscopy, which determines the three-dimensional structure of biomolecules in solution without crystallization. Dubochet contributed the vitrification of water, Frank the image processing, and Henderson the path to atomic resolution, together making cryo-EM a cornerstone of structural biology.

Chapter 118

2018: Frances H. Arnold and George P. Smith and Sir Gregory P. Winter

The Nobel Prize in Chemistry for the year 2018 was awarded jointly to Frances H. Arnold and George P. Smith and Sir Gregory P. Winter.

Motivation:

for the directed evolution of enzymes (one half, Frances H. Arnold) and for the phage display of peptides and antibodies (the other half, jointly, George P. Smith and Sir Gregory P. Winter)

118.1 Frances H. Arnold

Early Life and Education

Frances H. Arnold was born in 1956 in Edgewood, Pennsylvania, USA.

Career and Major Research

Arnold earned a bachelor's degree from Princeton University in 1979 and a doctorate in chemical engineering from the University of California, Berkeley, in 1985. She became a professor at the California Institute of Technology, where she pioneered directed evolution, the repeated round of random mutation and screening that steers enzymes toward desired properties. She received one half of the prize.

Nobel Prize-winning Work

The work for which Frances H. Arnold was awarded the Nobel Prize in Chemistry in 2018 was described by the Nobel Committee as: "for the directed evolution of enzymes".

Directed evolution harnesses natural selection in the laboratory, generating many enzyme variants and selecting those with improved stability, activity, or entirely new functions.

Significance and Impact

Her methods have produced enzymes with properties not found in nature, now used to make pharmaceuticals, fuels, and chemicals by routes that are cheaper and more environmentally friendly.

Later Career and Honors

Arnold continues her research on directed evolution and protein engineering at the California Institute of Technology.

Legacy

Directed evolution has become a standard industrial tool and has expanded the reactions available to synthetic chemistry.

Modern Developments

Arnold's directed evolution of enzymes, which uses iterative rounds of mutation and selection to improve proteins, founded the field of protein engineering by evolutionary design. Her methods are now used industrially to create enzymes for pharmaceuticals, biofuels, and consumer products, and directed evolution is a cornerstone of modern synthetic biology and green chemistry.

Selected Publications

"Design by Directed Evolution" (Accounts of Chemical Research, 1998)

118.2 George P. Smith

Early Life and Education

George P. Smith was born in 1941 in Norwalk, Connecticut, USA.

Career and Major Research

Smith earned a bachelor's degree from Haverford College in 1963 and a doctorate from Harvard University in 1970. He became a professor at the University of Missouri, where in 1985 he invented phage display, inserting foreign genetic material into a filamentous phage so that the encoded protein appears on the surface of the virus particle.

Nobel Prize-winning Work

The work for which George P. Smith was awarded the Nobel Prize in Chemistry in 2018 was described by the Nobel Committee as: "for the phage display of peptides and antibodies".

In phage display, the protein on the phage surface is physically linked to the gene that encodes it, so that proteins with desired binding properties can be selected from huge libraries and their genes recovered. He shared the other half of the prize with Gregory Winter.

Significance and Impact

Phage display made it possible to select proteins and peptides with desired properties by simple binding, a process of evolution in the laboratory.

Later Career and Honors

Smith is professor emeritus at the University of Missouri, where he worked for his entire career.

Legacy

Phage display became the basis of protein engineering and antibody discovery, and it is used to find molecules that bind specific targets.

Modern Developments

Smith's invention of phage display, a method for displaying proteins and peptides on the surface of bacteriophages, made it possible to select proteins that bind specific targets. Phage display underpins modern antibody engineering, including the development of many therapeutic antibodies, and remains a central tool in protein and peptide discovery.

Selected Publications

"Filamentous Fusion Phage: Novel Expression Vectors That Display Cloned Antigens on the Virion Surface" (Science, 1985)

118.3 Sir Gregory P. Winter

Early Life and Education

Sir Gregory P. Winter was born in 1951 in Leicester, England.

Career and Major Research

Winter earned bachelor's and doctoral degrees from the University of Cambridge, receiving his doctorate in 1976 while at the MRC Laboratory of Molecular Biology. There he applied phage display to antibodies, so that antibody fragments with chosen specificities could be selected and further evolved. He shared the other half of the prize with George Smith.

Nobel Prize-winning Work

The work for which Sir Gregory P. Winter was awarded the Nobel Prize in Chemistry in 2018 was described by the Nobel Committee as: "for the phage display of peptides and antibodies".

Winter's antibody phage display allowed human antibodies and antibody fragments with improved properties to be produced by selection, and it underpinned the development of the first fully human therapeutic antibody, adalimumab.

Significance and Impact

His work made it possible to humanize and optimize antibodies in the laboratory, giving rise to a generation of therapeutic antibodies used in medicine.

Later Career and Honors

Winter is a former deputy director of the MRC Laboratory of Molecular Biology and was a founder of the company Cambridge Antibody Technology.

Legacy

Antibody phage display is a foundation of modern biopharmaceutical research and has produced many approved drugs.

Modern Developments

Winter's application of phage display to engineer human antibodies founded the field of antibody engineering and enabled the development of therapeutic antibodies. The humanized and fully human antibodies produced by his methods now treat cancer, autoimmune disease, and infections, and constitute a multi-billion-dollar class of biopharmaceuticals.

Selected Publications

"Making Antibodies by Phage Display Technology" (Annual Review of Immunology, 1994)

Chapter Summary

The 2018 prize recognized methods that use evolution in the laboratory to create useful proteins: Arnold received one half for the directed evolution of enzymes, and Smith and Winter shared the other half for the phage display of proteins and antibodies. Together these methods are central to modern biotechnology and drug development.

Chapter 119

2019: John B. Goodenough and M. Stanley Whittingham and Akira Yoshino

The Nobel Prize in Chemistry for the year 2019 was awarded jointly to John B. Goodenough and M. Stanley Whittingham and Akira Yoshino.

Motivation:

for the development of lithium-ion batteries

119.1 John B. Goodenough

Early Life and Education

John B. Goodenough was born in 1922 in Jena, Germany.

Career and Major Research

Goodenough earned a bachelor's degree in mathematics from Yale University in 1944 and a doctorate in physics from the University of Chicago in 1952. He worked at the Massachusetts Institute of Technology and the University of Oxford before becoming a professor at the University of Texas at Austin in 1986. In 1980 he identified lithium cobalt oxide (LiCoO_2) as a cathode material that stores lithium ions at a higher voltage than earlier materials.

Nobel Prize-winning Work

The work for which John B. Goodenough was awarded the Nobel Prize in Chemistry in 2019 was described by the Nobel Committee as: "for the development of lithium-ion batteries".

Goodenough's cobalt oxide cathode doubled the voltage of the battery and made a practical, high-energy rechargeable battery possible.

Significance and Impact

His cathode was the key step that allowed lithium-ion batteries to store enough energy for portable electronics and electric vehicles. At the age of 97, Goodenough became the oldest recipient of a Nobel Prize.

Later Career and Honors

Goodenough continued to work on battery materials at the University of Texas at Austin into his nineties. He died on 2023-06-25 in Austin, Texas, USA.

Legacy

The rechargeable lithium-ion battery powers the world's mobile phones, laptops, and electric cars, and the layered-oxide cathodes he pioneered remain in widespread use.

Modern Developments

Goodenough's lithium-ion battery research, including his identification of the layered oxide cathodes used in commercial cells, enabled the rechargeable batteries that power smartphones, laptops, and electric vehicles. His work underlies the global transition to electric mobility and renewable energy storage, and his later iron phosphate and high-voltage materials continue to shape battery technology.

Selected Publications

" Li_xCoO_2 ($0 < x \leq 1$): A New Cathode Material for Batteries of High Energy Density" (Materials Research Bulletin, 1980)

119.2 M. Stanley Whittingham

Early Life and Education

M. Stanley Whittingham was born in 1941 in Nottingham, England.

Career and Major Research

Whittingham received his doctorate from the University of Oxford in 1968. He worked at the Exxon research laboratories and later became a professor at the Binghamton University of the State University of New York. In the early 1970s he showed that lithium ions can move in and out of layered titanium disulfide, creating a rechargeable cathode with the intercalation mechanism used in lithium-ion batteries.

Nobel Prize-winning Work

The work for which M. Stanley Whittingham was awarded the Nobel Prize in Chemistry in 2019 was described by the Nobel Committee as: "for the development of lithium-ion batteries".

Whittingham demonstrated the principle of electrochemical intercalation, in which lithium ions are stored inside the layered structure of an electrode without destroying it.

Significance and Impact

His demonstration that lithium could be reversibly inserted into a host material established the rechargeable battery chemistry on which later lithium-ion batteries were built.

Later Career and Honors

Whittingham became professor of chemistry and materials science at Binghamton University, where he continues research on energy storage.

Legacy

Intercalation electrodes, introduced by Whittingham, are the basis of every rechargeable lithium-ion battery in use today.

Modern Developments

Whittingham's discovery of the reversible lithium intercalation in titanium disulfide demonstrated that lithium could be stored and released reversibly in a solid host, founding the lithium-ion battery. His work established the concept of intercalation electrodes that now power modern portable electronics and electric vehicles.

Selected Publications

"Electrical Energy Storage and Intercalation Chemistry" (Science, 1976)

119.3 Akira Yoshino

Early Life and Education

Akira Yoshino was born in 1948 in Suita, Japan.

Career and Major Research

Yoshino studied chemistry at Kyoto University, earning bachelor's and master's degrees in 1970 and 1972. He joined the Asahi Kasei Corporation, where he developed the carbon anode for lithium-ion batteries, using petroleum coke as the material that stores lithium.

Nobel Prize-winning Work

The work for which Akira Yoshino was awarded the Nobel Prize in Chemistry in 2019 was described by the Nobel Committee as: "for the development of lithium-ion batteries".

Yoshino combined his carbon anode with Goodenough's cathode to build the first lithium-ion battery, and the technology was commercialized by Sony in 1991.

Significance and Impact

His carbon anode removed the safety problems of metallic lithium and gave the battery its practical, rechargeable form, leading to the batteries used in consumer electronics.

Later Career and Honors

Yoshino worked at Asahi Kasei for most of his career and later became a professor at Meijo University.

Legacy

The lithium-ion battery he assembled now powers mobile phones, laptops, and electric vehicles across the world.

Modern Developments

Yoshino's development of the first commercially viable lithium-ion battery, using carbon anodes and lithium cobalt oxide cathodes, created the technology that now powers smartphones, laptops, and electric vehicles. Lithium-ion batteries are central to the global transition to clean energy, and Yoshino's safety innovations enabled their mass production.

Selected Publications

"The Birth of the Lithium-Ion Battery" (Angewandte Chemie International Edition, 2012)

Chapter Summary

The 2019 prize recognized the development of the lithium-ion battery by three researchers: Whittingham established intercalation, Goodenough created the high-voltage cobalt oxide cathode, and Yoshino built the first practical cell with a carbon anode. Their rechargeable battery now powers portable electronics and electric vehicles worldwide.

Chapter 120

2020:Emmanuelle Charpentier and Jennifer A. Doudna

The Nobel Prize in Chemistry for the year 2020 was awarded jointly to Emmanuelle Charpentier and Jennifer A. Doudna.

Motivation:

for the development of a method for genome editing

120.1 Emmanuelle Charpentier

Early Life and Education

Emmanuelle Charpentier was born in 1968 in Juvisy-sur-Orge, France.

Career and Major Research

Charpentier received her doctorate from the Institut Pasteur in Paris in 1995. She held research positions in Europe and the United States and became scientific director at the Max Planck Unit for the Science of Pathogens in Berlin. Her work on the bacterial immune system CRISPR showed that it could be programmed by RNA to cut DNA at specific sites.

Nobel Prize-winning Work

The work for which Emmanuelle Charpentier was awarded the Nobel Prize in Chemistry in 2020 was described by the Nobel Committee as: "for the development of a method for genome editing".

In 2012 Charpentier and Jennifer Doudna, with Martin Jinek, showed that the protein Cas9, guided by a dual RNA, could cut any DNA sequence chosen by the researcher.

Significance and Impact

This reprogrammable RNA-guided nuclease became the basis of the CRISPR-Cas9 method, which has made genome editing fast, simple, and available to laboratories worldwide.

Later Career and Honors

Charpentier continues to direct research on RNA biology and bacterial infection at her institute in Berlin.

Legacy

CRISPR-Cas9 genome editing is used in fundamental research, agriculture, and the development of therapies for genetic disease.

Modern Developments

Charpentier's discovery of the role of tracrRNA in the CRISPR-Cas9 system, with Doudna, identified the mechanism that became the basis of gene editing. CRISPR technology is now used worldwide to edit genomes in research, agriculture, and medicine, including approved therapies for sickle-cell disease, and her work founded the modern era of genetic engineering.

Selected Publications

"A Programmable Dual-RNA-Guided DNA Endonuclease in Adaptive Bacterial Immunity" (Science, 2012)

120.2 Jennifer A. Doudna

Early Life and Education

Jennifer A. Doudna was born in 1964 in Washington, D.C., USA.

Career and Major Research

Doudna earned a bachelor's degree from Pomona College in 1985 and a doctorate from Harvard University in 1989. She became a professor at the University of California, Berkeley, and an investigator of the Howard Hughes Medical Institute, working on the structure and function of RNA.

Nobel Prize-winning Work

The work for which Jennifer A. Doudna was awarded the Nobel Prize in Chemistry in 2020 was described by the Nobel Committee as: "for the development of a method for genome editing".

With Emmanuelle Charpentier, Doudna established how the CRISPR-associated protein Cas9 is programmed by guide RNA and demonstrated its programmable DNA cleavage in 2012.

Significance and Impact

Her biochemical characterization explained how Cas9 recognizes and cuts its target, providing the understanding needed to turn the system into a general genome-editing tool.

Later Career and Honors

Doudna continues her research at the University of California, Berkeley, and is a co-founder of companies that apply CRISPR technology.

Legacy

CRISPR-Cas9 has transformed genetic research and is being developed as a treatment for inherited diseases and other conditions.

Modern Developments

Doudna's biochemical characterization and repurposing of the CRISPR-Cas9 system into a programmable genome-editing tool transformed biology and medicine. CRISPR is now used to create gene-edited crops, model human diseases, and develop therapies for genetic disorders, and her work launched the fastest-growing field in modern biotechnology.

Selected Publications

"A Programmable Dual-RNA-Guided DNA Endonuclease in Adaptive Bacterial Immunity" (Science, 2012)

Chapter Summary

The 2020 prize recognized the development of CRISPR-Cas9, a method for genome editing that originated in a bacterial immune system. Charpentier and Doudna showed that a single RNA-guided protein can cut any chosen DNA sequence, and the technique is now used across biology and medicine.

Chapter 121

2021: Benjamin List and David W.C. MacMillan

The Nobel Prize in Chemistry for the year 2021 was awarded jointly to Benjamin List and David W.C. MacMillan.

Motivation:

for the development of asymmetric organocatalysis

121.1 Benjamin List

Early Life and Education

Benjamin List was born in 1968 in Frankfurt am Main, Germany.

Career and Major Research

List received his doctorate from the Goethe University Frankfurt in 1997 and did postdoctoral work with Dieter Seebach at the ETH Zurich. He became director of the Max Planck Institute for Coal Research in Mülheim an der Ruhr. In 2000 he showed that the amino acid proline catalyzes asymmetric aldol reactions, adding a carbon group to a carbonyl compound with high control of stereochemistry.

Nobel Prize-winning Work

The work for which Benjamin List was awarded the Nobel Prize in Chemistry in 2021 was described by the Nobel Committee as: "for the development of asymmetric organocatalysis".

List's proline catalysis, based on the enamine mechanism used by enzymes, showed that a simple organic molecule could carry out asymmetric synthesis without any metal.

Significance and Impact

His discovery demonstrated that small organic molecules are powerful asymmetric catalysts, opening the field that is now called organocatalysis.

Later Career and Honors

List continues his research on organocatalysis and related reactions at the Max Planck Institute for Coal Research in Mülheim.

Legacy

Organocatalysis has become a standard method for producing enantiomerically pure molecules, used in the synthesis of pharmaceuticals and natural products.

Modern Developments

List's discovery of asymmetric organocatalysis, showing that small organic molecules can catalyze reactions with high enantioselectivity, founded a new and general field of catalysis. Organocatalysis is now a standard tool in the pharmaceutical and agrochemical industries, enabling the green and cost-effective synthesis of chiral molecules.

Selected Publications

"Proline-Catalyzed Direct Asymmetric Aldol Reactions" (Journal of the American Chemical Society, 2000)

121.2 David W.C. MacMillan

Early Life and Education

David W.C. MacMillan was born in 1968 in Bellshill, Scotland.

Career and Major Research

MacMillan earned a bachelor's degree from the University of Glasgow in 1991 and a doctorate from the University of California, Irvine, in 1996. He became a professor at Princeton University. In 2000 he developed iminium catalysis, in which an amine catalyst activates a carbonyl compound through an iminium ion, and he coined the term organocatalysis for catalysis by small organic molecules.

Nobel Prize-winning Work

The work for which David W.C. MacMillan was awarded the Nobel Prize in Chemistry in 2021 was described by the Nobel Committee as: "for the development of asymmetric organocatalysis".

MacMillan showed that iminium ions, formed between an amine catalyst and an enal, undergo selective reactions such as cycloadditions with high stereocontrol, and he demonstrated the generality of metal-free asymmetric catalysis.

Significance and Impact

His methods extended organocatalysis to a broad range of reactions and made it a practical alternative to metal-based asymmetric catalysts.

Later Career and Honors

MacMillan continues his research on catalysis at Princeton University.

Legacy

Together with List, MacMillan established organocatalysis as a third pillar of asymmetric synthesis, alongside enzymes and transition-metal catalysts, and his chemistry is now used widely in drug synthesis.

Modern Developments

MacMillan's development of iminium-enamine organocatalysis, and his coinage of the term organocatalysis, established small-molecule catalysis as a third pillar of catalysis alongside metals and enzymes. Organocatalysis is now widely used in asymmetric synthesis, and his methods enabled efficient routes to chiral drugs and natural products.

Selected Publications

"New Strategies for Organic Catalysis: The First Highly Enantioselective Organocatalytic Diels-Alder Reaction" (Journal of the American Chemical Society, 2000)

Chapter Summary

The 2021 prize recognized the development of asymmetric organocatalysis by Benjamin List and David MacMillan, who showed that small organic molecules can catalyze enantioselective reactions. Organocatalysis is now a major tool for making chiral molecules in the pharmaceutical industry.

Chapter 122

2022: Carolyn Bertozzi and Morten Meldal and K. Barry Sharpless

The Nobel Prize in Chemistry for the year 2022 was awarded jointly to Carolyn Bertozzi and Morten Meldal and K. Barry Sharpless.

Motivation:

for the development of click chemistry and bioorthogonal chemistry

122.1 Carolyn Bertozzi

Early Life and Education

Carolyn Bertozzi was born in 1966 in Boston, Massachusetts, USA.

Career and Major Research

Bertozzi earned a bachelor's degree from Harvard University in 1988 and a doctorate from the University of California, Berkeley, in 1993. She became a professor at Stanford University and an investigator of the Howard Hughes Medical Institute. She developed bioorthogonal chemistry, reactions that proceed inside living organisms without disturbing the surrounding chemistry, and used them to study the sugars on cell surfaces.

Nobel Prize-winning Work

The work for which Carolyn Bertozzi was awarded the Nobel Prize in Chemistry in 2022 was described by the Nobel Committee as: "for the development of click chemistry and bioorthogonal chemistry".

Bertozzi's modified Staudinger reaction joined a chemical tag on a cell surface sugar to a probe inside a live organism, demonstrating that bioorthogonal reactions can be carried out in living systems.

Significance and Impact

Her methods allow researchers to follow biomolecules such as glycans in living cells and animals, and they have been applied to the delivery of drugs and imaging agents.

Later Career and Honors

Bertozzi continues her research on chemical biology at Stanford University, where she directs the Sarafan ChEM-H institute.

Legacy

Bioorthogonal chemistry has become a standard tool for labeling and imaging molecules in vivo and is used in the design of medicines.

Modern Developments

Bertozzi's development of bioorthogonal chemistry and its application to imaging glycans on living cells founded the field of bioorthogonal chemistry and modern chemical biology. Her methods, including click chemistry in living systems, are now used to study glycosylation, develop drug conjugates, and enable in vivo imaging, with applications in medicine and materials.

Selected Publications

"Cell Surface Engineering by a Modified Staudinger Reaction" (Science, 2000)

122.2 Morten Meldal

Early Life and Education

Morten Meldal was born in 1954 in Copenhagen, Denmark.

Career and Major Research

Meldal received his doctorate in chemistry from the University of Copenhagen in 1982. He worked at the Carlsberg Laboratory in Copenhagen and became a professor at the University of Copenhagen. In 2002, working with solid-phase peptide chemistry, he developed the copper-catalyzed azide-alkyne cycloaddition (CuAAC), which joins an azide and an alkyne rapidly and selectively to form a triazole ring.

Nobel Prize-winning Work

The work for which Morten Meldal was awarded the Nobel Prize in Chemistry in 2022 was described by the Nobel Committee as: "for the development of click chemistry and bioorthogonal chemistry".

Meldal's CuAAC reaction, developed on solid support, turned the azide-alkyne cycloaddition into a fast, reliable, and water-compatible reaction, the archetypal click reaction.

Significance and Impact

Because the reaction works quickly, cleanly, and with nearly any partners, it is used to join molecules in combinatorial chemistry, drug discovery, and materials science.

Later Career and Honors

Meldal continues his research in chemistry and chemical biology at the University of Copenhagen.

Legacy

CuAAC is one of the most widely used reactions in modern chemistry, applied from the modification of biomolecules to the synthesis of new materials.

Modern Developments

Meldal's independent development of copper-catalyzed azide-alkyne cycloaddition, the reaction at the heart of click chemistry, provided a robust tool for joining molecules together. Click chemistry is now used in drug discovery, materials science, bioconjugation, and the synthesis of large molecular libraries.

Selected Publications

"Peptidotriazoles on Solid Phase: [1,2,3]-Triazoles by Regiospecific Copper(I)-Catalyzed 1,3-Dipolar Cycloaddition of Terminal Alkynes to Azides" (Journal of Organic Chemistry, 2002)

122.3 K. Barry Sharpless

Early Life and Education

K. Barry Sharpless was born in 1941 in Philadelphia, Pennsylvania, USA.

Career and Major Research

Sharpless earned a bachelor's degree from Dartmouth College in 1963 and a doctorate from Stanford University in 1968. He became a professor at the Scripps Research Institute in La Jolla. He coined the term click chemistry for modular reactions that join molecular building blocks quickly and reliably, and with his colleagues he developed the copper-catalyzed azide-alkyne cycloaddition in 2002. He had earlier received the 2001 Nobel Prize in Chemistry for his work on asymmetric oxidation.

Nobel Prize-winning Work

The work for which K. Barry Sharpless was awarded the Nobel Prize in Chemistry in 2022 was described by the Nobel Committee as: "for the development of click chemistry and bioorthogonal chemistry".

Sharpless defined click chemistry as reactions that are fast, selective, and high-yielding under simple conditions, and the azide-alkyne reaction he helped develop became its most famous example.

Significance and Impact

Click chemistry gave chemists a simple way to connect molecules, and it is now used across drug discovery, materials, and biology. The 2022 award made Sharpless a two-time Nobel recipient.

Later Career and Honors

Sharpless continues his research on click chemistry at the Scripps Research Institute.

Legacy

Click chemistry and the CuAAC reaction are foundational to modern chemical biology and have been used to build libraries of drug candidates and new materials.

Modern Developments

Sharpless's formulation of click chemistry, and his independent discovery of the copper-catalyzed azide-alkyne cycloaddition with Meldal, provided chemistry with a general method for rapidly and reliably joining molecular fragments. Click chemistry is now ubiquitous in drug development, chemical biology, and materials science, and his earlier asymmetric epoxidation remains a landmark of enantioselective synthesis.

Selected Publications

"The First Practical Method for Asymmetric Epoxidation" (Journal of the American Chemical Society, 1980)

Chapter Summary

The 2022 prize recognized the development of click chemistry and bioorthogonal chemistry: Sharpless conceived click chemistry and helped develop the copper-catalyzed azide-alkyne reaction, Meldal developed the reaction on solid phase, and Bertozzi made such reactions work inside living organisms. These methods allow molecules to be joined quickly and specifically and are now used in drug discovery and cell biology.

Chapter 123

2023:Moungi Bawendi and Louis Brus and Aleksey Yekimov

The Nobel Prize in Chemistry for the year 2023 was awarded jointly to Moungi Bawendi and Louis Brus and Aleksey Yekimov.

Motivation:

for the discovery and synthesis of quantum dots

123.1 Moungi Bawendi

Early Life and Education

Moungi Bawendi was born in 1961 in Paris, France.

Career and Major Research

Bawendi earned a bachelor's degree from Harvard University in 1982 and a doctorate in chemistry from the University of Chicago in 1988. He became a professor at the Massachusetts Institute of Technology, where in 1993 he developed a method for producing near-identical, size-controlled quantum dots in large quantities.

Nobel Prize-winning Work

The work for which Moungi Bawendi was awarded the Nobel Prize in Chemistry in 2023 was described by the Nobel Committee as: "for the discovery and synthesis of quantum dots".

Bawendi's synthesis of monodisperse quantum dots made the particles available in precise, reproducible sizes, so that their color could be controlled by size alone.

Significance and Impact

His method turned quantum dots from an object of study into a practical material, enabling their use in displays, lasers, and biomedical imaging.

Later Career and Honors

Bawendi continues his research on semiconductor nanocrystals at the Massachusetts Institute of Technology.

Legacy

The synthetic chemistry he developed is the basis of the commercial quantum dot technology used in televisions and screens.

Modern Developments

Bawendi's development of methods for producing high-quality, monodisperse semiconductor nanocrystals (quantum dots) with controlled size and properties made quantum dots practical for research and industry. Quantum dots now illuminate displays, enable biomedical imaging, and are used in solar cells and photodetectors.

Selected Publications

"Synthesis and Characterization of Nearly Monodisperse CdE (E = S, Se, Te) Semiconductor Nanocrystallites" (Journal of the American Chemical Society, 1993)

123.2 Louis Brus

Early Life and Education

Louis Brus was born in 1943 in Cleveland, Ohio, USA.

Career and Major Research

Brus earned a bachelor's degree from Rice University in 1965 and a doctorate from Columbia University in 1969. He worked at Bell Laboratories and later became a professor at Columbia University. In the early 1980s he discovered that the optical properties of colloidal semiconductor particles change with their size, the quantum size effect now known as quantum confinement.

Nobel Prize-winning Work

The work for which Louis Brus was awarded the Nobel Prize in Chemistry in 2023 was described by the Nobel Committee as: "for the discovery and synthesis of quantum dots".

Brus showed that in nanometer-sized semiconductor crystallites, the energy levels and color of light absorption depend on the particle size, the defining property of quantum dots.

Significance and Impact

His discovery established that the same material emits different colors depending on its size, and it founded the chemistry of colloidal quantum dots.

Later Career and Honors

Brus continues his research on nanocrystals and their spectroscopy at Columbia University.

Legacy

The size-dependence that Brus explained is the basis of quantum dot technology, from displays to biological imaging.

Modern Developments

Brus's discovery that the optical properties of semiconductor nanocrystals depend on their size, now known as quantum confinement, founded the field of colloidal quantum dots. Quantum dots are now used in televisions and displays, biomedical imaging, and optoelectronics, and his work opened the field of nanoscience.

Selected Publications

"Electron-Electron and Electron-Hole Interactions in Small Semiconductor Crystallites" (Journal of Chemical Physics, 1984)

123.3 Aleksey Yekimov

Early Life and Education

Aleksey Yekimov was born in 1945 in Saint Petersburg, Russia.

Career and Major Research

Yekimov worked as a physicist at the Vavilov State Optical Institute in Leningrad. In 1981 he produced copper chloride nanocrystals in molten glass and showed that their absorption of light depended on their size, demonstrating the quantum size effect in a semiconductor.

Nobel Prize-winning Work

The work for which Aleksey Yekimov was awarded the Nobel Prize in Chemistry in 2023 was described by the Nobel Committee as: "for the discovery and synthesis of quantum dots".

Yekimov's experiments provided the first clear evidence that the electronic properties of semiconductor nanocrystals change with their dimensions, the phenomenon underlying quantum dots.

Significance and Impact

His observation in glass-hosted crystals was the starting point for the entire field of size-tunable semiconductor nanoparticles.

Later Career and Honors

Yekimov worked at the Vavilov State Optical Institute in Leningrad and, since 1999, at Nanocrystals Technology in the United States.

Legacy

The discovery Yekimov made in glass established quantum dots as a new class of material whose properties are set by size.

Modern Developments

Yekimov's discovery of quantum size effects in glass-doped copper chloride crystals, the first clear demonstration of quantum confinement, founded the field of quantum dots. His work underpins the modern quantum dot industry used in displays, lighting, and biomedical imaging, and it opened the era of size-controlled semiconductor nanocrystals.

Selected Publications

"Quantum Size Effect in Semiconductor Microcrystals" (Solid State Communications, 1985)

Chapter Summary

The 2023 prize recognized the discovery and synthesis of quantum dots: Yekimov first observed size-dependent properties in glass, Brus explained the quantum size effect in colloidal particles, and Bawendi developed the synthesis of uniform nanocrystals. Quantum dots are now used in displays and biomedical imaging.

Chapter 124

2024:David Baker and Demis Hassabis and John Jumper

The Nobel Prize in Chemistry for the year 2024 was awarded jointly to David Baker and Demis Hassabis and John Jumper.

Motivation:

for computational protein design (one half, David Baker) and for protein structure prediction (the other half, jointly, Demis Hassabis and John Jumper)

124.1 David Baker

Early Life and Education

David Baker was born in 1962 in Seattle, Washington, USA.

Career and Major Research

Baker received his doctorate from the University of California, Berkeley, in 1989. He became a professor of biochemistry at the University of Washington, an investigator of the Howard Hughes Medical Institute, and director of the Institute for Protein Design. He led the development of the Rosetta software for modeling and designing proteins, and in 2003 his group designed a novel protein fold with atomic-level accuracy.

Nobel Prize-winning Work

The work for which David Baker was awarded the Nobel Prize in Chemistry in 2024 was described by the Nobel Committee as: "for computational protein design".

Baker's de novo design methods allowed proteins with new sequences, folds, and functions to be created from first principles, starting only from the rules of protein structure.

Significance and Impact

His computational protein design makes it possible to create proteins on demand for medicine, biotechnology, and materials, including designed vaccines and nanomachines.

Later Career and Honors

Baker continues to direct research in protein design at the University of Washington and leads spin-off companies that apply designed proteins.

Legacy

Rosetta and the Institute for Protein Design have made computational protein design a practical route to custom biomolecules.

Modern Developments

Baker's computational design of new proteins, using the Rosetta software he developed, founded the field of computational protein design. His methods have created enzymes, vaccines, and novel protein therapeutics, and his software underpins the modern biotechnology of designed proteins, including the engineered antigens now used in vaccines.

Selected Publications

"Design of a Novel Globular Protein Fold with Atomic-Level Accuracy" (Science, 2003)

124.2 Demis Hassabis

Early Life and Education

Demis Hassabis was born in 1976 in London, England.

Career and Major Research

Hassabis studied computer science at the University of Cambridge and earned a doctorate in cognitive neuroscience from University College London in 2009. He co-founded the artificial intelligence company DeepMind in 2010, where he led the development of AlphaFold, a system that predicts protein structures from amino acid sequences.

Nobel Prize-winning Work

The work for which Demis Hassabis was awarded the Nobel Prize in Chemistry in 2024 was described by the Nobel Committee as: "for computational protein design (one half, David Baker) and for protein structure prediction (the other half, jointly, Demis Hassabis and John Jumper)".

AlphaFold achieved near-experimental accuracy in predicting protein structures, and in 2020 it solved in days what experimental methods required years to determine.

Significance and Impact

AlphaFold made reliable structure prediction available for millions of proteins, transforming structural biology and accelerating research in medicine and biology.

Later Career and Honors

Hassabis is chief executive of Google DeepMind, where he continues to lead research in artificial intelligence.

Legacy

AlphaFold ended a fifty-year problem in biology and demonstrated the power of machine learning for scientific discovery.

Modern Developments

Hassabis's development of AlphaFold, an artificial intelligence system that predicts protein structures from amino acid sequences, transformed structural biology and computational biology. AlphaFold has predicted the structures of hundreds of millions of proteins, accelerating drug discovery, vaccine design, and the understanding of disease, and his work opened the era of AI-driven science.

Selected Publications

"Highly Accurate Protein Structure Prediction with AlphaFold" (Nature, 2021)

124.3 John Jumper

Early Life and Education

John Jumper was born in 1985 in Little Rock, Arkansas, USA.

Career and Major Research

Jumper earned a bachelor's degree from Vanderbilt University in 2007 and a doctorate in theoretical chemistry from the University of Chicago in 2017. He joined the artificial intelligence company DeepMind, where he designed the AlphaFold2 system that predicts protein structures with atomic-level accuracy.

Nobel Prize-winning Work

The work for which John Jumper was awarded the Nobel Prize in Chemistry in 2024 was described by the Nobel Committee as: "for computational protein design (one half, David Baker) and for protein structure prediction (the other half, jointly, Demis Hassabis and John Jumper)".

Jumper's architecture for AlphaFold2 combined machine learning with protein structure geometry, producing predictions that rival experimental structures.

Significance and Impact

AlphaFold2 provided accurate structures for nearly all proteins of known organisms, a resource that has accelerated research across biology.

Later Career and Honors

Jumper continues to work on machine learning for biology at Google DeepMind.

Legacy

The AlphaFold2 method he designed is now a standard tool in structural biology and is used worldwide.

Modern Developments

Jumper's development of the AlphaFold2 neural network, which predicts protein structures at experimental accuracy, revolutionized the field of protein structure prediction. AlphaFold2 has been used by over a million researchers to model proteins for medicine, agriculture, and biotechnology, making structure prediction a routine tool of modern biology.

Selected Publications

"Highly Accurate Protein Structure Prediction with AlphaFold" (Nature, 2021)

Chapter Summary

The 2024 prize recognized two advances in protein science: David Baker received one half for computational protein design, and Demis Hassabis and John Jumper shared the other half for the protein structure prediction achieved by AlphaFold. Together these methods let scientists create new proteins and predict the structures of existing ones with unprecedented accuracy.

Chapter 125

2025: Susumu Kitagawa and Richard Robson and Omar M. Yaghi

The Nobel Prize in Chemistry for the year 2025 was awarded jointly to Susumu Kitagawa and Richard Robson and Omar M. Yaghi.

Motivation:

for the development of metal-organic frameworks

125.1 Susumu Kitagawa

Early Life and Education

Susumu Kitagawa was born in 1951 in Kyoto, Japan.

Career and Major Research

Kitagawa studied chemistry at Kyoto University and became a professor there. He pioneered porous coordination polymers, the family of materials now known as metal-organic frameworks, and showed that their cavities can reversibly take up and release gases and other small molecules.

Nobel Prize-winning Work

The work for which Susumu Kitagawa was awarded the Nobel Prize in Chemistry in 2025 was described by the Nobel Committee as: "for the development of metal-organic frameworks".

Kitagawa demonstrated that coordination polymers can form crystalline frameworks with permanent porosity, and his studies established how their flexible structures respond to guests such as gas molecules.

Significance and Impact

His work on porous coordination polymers revealed that metal-organic frameworks could be tuned for applications such as gas storage and separation.

Later Career and Honors

Kitagawa is professor emeritus of Kyoto University, where he continues research on porous materials.

Legacy

Kitagawa's concept of porous coordination polymers is a foundation of metal-organic framework chemistry and of modern porous materials science.

Modern Developments

Kitagawa's discovery of porous coordination polymers and metal-organic frameworks (MOFs) that could store and separate gases established the field of MOF chemistry. MOFs now promise applications in hydrogen and carbon dioxide storage, gas separation, catalysis, and drug delivery, and his work founded a major new class of porous materials.

Selected Publications

"Functional Porous Coordination Polymers" (Angewandte Chemie International Edition, 2004)

125.2 Richard Robson

Early Life and Education

Richard Robson was born in 1937 in Glusburn, England.

Career and Major Research

Robson earned bachelor's and doctoral degrees from the University of Oxford, receiving his doctorate in 1962. He became a professor of inorganic chemistry at the University of Melbourne, where in the late 1980s he designed coordination polymer networks with predictable topologies by linking metal centers with rigid organic units.

Nobel Prize-winning Work

The work for which Richard Robson was awarded the Nobel Prize in Chemistry in 2025 was described by the Nobel Committee as: "for the development of metal-organic frameworks".

Robson showed how to assemble metal ions and organic linkers into crystalline networks with large channels, establishing the design principles for what became metal-organic frameworks.

Significance and Impact

His networks demonstrated that the architecture of a porous solid could be planned in advance from its building blocks, a central idea of the field.

Later Career and Honors

Robson was professor of inorganic chemistry at the University of Melbourne until his retirement in 2004.

Legacy

Robson's design rules for infinite framework structures are a cornerstone of the chemistry of coordination polymers and metal-organic frameworks.

Modern Developments

Robson's early synthesis and characterization of coordination networks with permanent porosity laid the foundations of metal-organic framework (MOF) chemistry. MOFs are now among the most studied materials in chemistry, with applications in gas storage, carbon capture, separation, sensing, and drug delivery.

Selected Publications

"Assembly of Porphyrin Building Blocks into Network Structures with Large Channels" (Nature, 1994)

125.3 Omar M. Yaghi

Early Life and Education

Omar M. Yaghi was born in 1965 in Amman, Jordan.

Career and Major Research

Yaghi received his doctorate from the University of Illinois at Urbana-Champaign in 1990. He held professorships at Arizona State University, the University of Michigan, and the University of California, Los Angeles, before becoming a professor at the University of California, Berkeley. He synthesized the highly porous metal-organic framework MOF-5 in 1999 and founded the field of reticular chemistry.

Nobel Prize-winning Work

The work for which Omar M. Yaghi was awarded the Nobel Prize in Chemistry in 2025 was described by the Nobel Committee as: "for the development of metal-organic frameworks".

Yaghi's metal-organic frameworks combine metal clusters and organic linkers into extended porous crystals with record surface areas, and his reticular chemistry approach makes their structures predictable and tunable.

Significance and Impact

His robust, high-surface-area frameworks are used for storing gases such as hydrogen and methane, capturing carbon dioxide, and harvesting water from desert air.

Later Career and Honors

Yaghi continues his research on reticular chemistry at the University of California, Berkeley.

Legacy

The metal-organic frameworks Yaghi developed are among the most porous materials known and are being commercialized for energy and environmental applications.

Modern Developments

Yaghi's synthesis of metal-organic frameworks (MOFs) with record surface areas and his development of reticular chemistry created the largest class of porous materials ever assembled. MOFs are now investigated for carbon capture, hydrogen storage, catalysis, and drug delivery, and his work founded the modern field of reticular chemistry.

Selected Publications

"Design and Synthesis of an Exceptionally Stable and Highly Porous Metal-Organic Framework" (Nature, 1999)

Chapter Summary

The 2025 prize recognized the development of metal-organic frameworks, porous crystals built from metal ions and organic linkers. Kitagawa pioneered porous coordination polymers, Robson established the design of network structures, and Yaghi synthesized high-porosity frameworks with predictable architectures, materials now used for gas storage, carbon capture, and catalysis.